

GRADE 6 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 7) • Forward 2 (Grade 8)
52 question sets • 1625 questions

CONTENTS

Set 1	6.RP.A.1	Understand ratios and use ratio language	6
Set 2	6.RP.A.2	Understand unit rate in the context of a ratio	43
Set 3	6.RP.A.3.a	Tables of equivalent ratios; plot pairs on the coordinate plane	80
Set 4	6.RP.A.3.b	Unit rate problems: unit pricing and constant speed	117
Set 5	6.RP.A.3.c	Percent of a quantity as a rate per 100	160
Set 6	6.RP.A.3.c	Find the whole, given a part and the percent	203
Set 7	6.RP.A.3.d	Convert measurement units using ratio reasoning	240
Set 8	6.NS.A.1	Divide fractions by fractions	277
Set 9	6.NS.A.1	Interpret and solve word problems with fraction division	326
Set 10	6.NS.B.2	Divide multi-digit numbers using the standard algorithm	369
Set 11	6.NS.B.3	Add and subtract multi-digit decimals	412
Set 12	6.NS.B.3	Multiply multi-digit decimals	461
Set 13	6.NS.B.3	Divide multi-digit decimals	504
Set 14	6.NS.B.4	Greatest common factor and least common multiple	541
Set 15	6.NS.B.4	Use the distributive property to factor a sum with the GCF	578
Set 16	6.NS.C.5	Positive and negative numbers in real-world contexts	615
Set 17	6.NS.C.6.a	Opposite signs as opposite sides of 0; the opposite of the opposite	652

CONTENTS (continued)

Set 18	6.NS.C.6.b	Signs of coordinates in the quadrants; reflections across the axes	683
Set 19	6.NS.C.6.c	Find and position rational numbers on a number line	726
Set 20	6.NS.C.6.c	Find and position pairs of numbers on the coordinate plane	763
Set 21	6.NS.C.7.a	Interpret inequalities as relative position on a number line	800
Set 22	6.NS.C.7.b	Write, interpret, and explain order statements in real-world contexts	837
Set 23	6.NS.C.7.c	Absolute value as distance from 0	874
Set 24	6.NS.C.7.d	Distinguish comparisons of absolute value from statements about order	911
Set 25	6.NS.C.8	Distances between points on the coordinate plane	948
Set 26	6.EE.A.1	Write and evaluate numerical expressions with whole-number exponents	991
Set 27	6.EE.A.2.a	Write expressions in which letters stand for numbers	1028
Set 28	6.EE.A.2.b	Identify parts of an expression: term, sum, product, factor, quotient, coefficient	1065
Set 29	6.EE.A.2.c	Evaluate expressions and formulas for given values	1102
Set 30	6.EE.A.3	Apply properties of operations to generate equivalent expressions	1145
Set 31	6.EE.A.4	Identify when two expressions are equivalent	1182
Set 32	6.EE.B.5	Use substitution to decide whether a number makes an equation or inequality true	1219
Set 33	6.EE.B.6	Use variables to write expressions for real-world problems	1256
Set 34	6.EE.B.7	Solve one-step equations of the form $x + p = q$ and $px = q$	1293

CONTENTS (continued)

Set 35	6.EE.B.8	Write inequalities $x > c$ or $x < c$ and graph their solutions	1336
Set 36	6.EE.C.9	Dependent and independent variables: tables, graphs, and equations	1373
Set 37	6.G.A.1	Area of right triangles and other triangles	1410
Set 38	6.G.A.1	Area of parallelograms and trapezoids	1447
Set 39	6.G.A.1	Area of polygons by composing and decomposing	1490
Set 40	6.G.A.2	Volume of right rectangular prisms with fractional edge lengths	1527
Set 41	6.G.A.3	Polygons in the coordinate plane; side lengths from coordinates	1564
Set 42	6.G.A.4	Nets of three-dimensional figures and surface area	1607
Set 43	6.SP.A.1	Recognize statistical questions	1644
Set 44	6.SP.A.2	Describe a distribution by its center, spread, and shape	1675
Set 45	6.SP.A.3	A measure of center vs. a measure of variation	1712
Set 46	6.SP.B.4	Display data in dot plots and histograms	1749
Set 47	6.SP.B.4	Display data in box plots	1786
Set 48	6.SP.B.5.a	Report the number of observations	1823
Set 49	6.SP.B.5.b	Describe the attribute measured and its units	1860
Set 50	6.SP.B.5.c	Find the mean and median of a data set	1891
Set 51	6.SP.B.5.c	Find the interquartile range and mean absolute deviation	1934

CONTENTS (continued)

Set 52	6.SP.B.5.d	Choose measures of center and variability based on the shape of the data	1971
---------------	-------------------	--	------

Understand ratios and use ratio language

MAIN	Grade 6 main question	6.RP.A.1
BACKWARD 1	Multiplicative comparison	4.OA.A.1
BACKWARD 2	Fraction as part of a whole	3.NF.A.1
BACKWARD 3	Reading counts from a table or graph	3.MD.B.3
FORWARD 1	Decide whether quantities are proportional	7.RP.A.2.a
FORWARD 2	Graph proportional relationships; slope as unit rate	8.EE.B.5

SET 1 • 6.RP.A.1

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.1

A class has 12 girls and 15 boys.

Write the ratio of girls to boys.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q1/5

The table shows the pets owned by students in a class.
Write the ratio of cats to fish.

Pet	Dogs	Cats	Fish
Students	8	6	3

Ratio: _____

A recipe uses 3 cups of flour for every 2 cups of sugar.

Which statement describes the same ratio?

- A.** For every 2 cups of flour, there are 3 cups of sugar.
- B.** The ratio of sugar to flour is 2 : 3.
- C.** The ratio of flour to sugar is 2 : 3.
- D.** For every 5 cups of sugar, there are 3 cups of flour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q3/5

At a zoo, the ratio of penguins to seals is 5 : 2.

Which statement is true?

- A.** For every 5 seals, there are 2 penguins.
- B.** For every 2 seals, there are 5 penguins.
- C.** For every 7 penguins, there are 2 seals.
- D.** There are 3 more seals than penguins.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q4/5

A soccer team won 9 games and lost 4 games. There were no ties.

Write the ratio of games lost to games played.

Ratio: _____

SET 1 • 6.RP.A.1

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.1

True or false?

24 is 3 times as many as 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • BACKWARD 1 • Q1/5

Maya has 5 stickers. Leo has 4 times as many stickers as Maya.

How many stickers does Leo have?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • BACKWARD 1 • Q2/5

Which equation shows "35 is 5 times as many as 7"?

A. $35 = 5 \times 7$

B. $35 = 5 + 7$

C. $7 = 5 \times 35$

D. $35 - 7 = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • BACKWARD 1 • Q3/5

True or false?

18 is 6 times as many as 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • BACKWARD 1 • Q4/5

A tree is 6 feet tall. A building is 9 times as tall as the tree.

How tall is the building?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 6.RP.A.1

BACKWARD 2

Fraction as part of a whole • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

The bar is divided into equal parts.

What fraction of the bar is shaded?



Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 2 • Q1/5

A pizza is cut into 8 equal slices. Sam eats 3 slices.
What fraction of the pizza does Sam eat?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 2 • Q2/5

True or false?

A bar is cut into 5 equal parts and 2 parts are shaded. The shaded part is $\frac{2}{5}$ of the bar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 2 • Q3/5

What fraction of the bar is shaded?



A. $\frac{1}{6}$

B. $\frac{1}{5}$

C. $\frac{6}{1}$

D. $\frac{5}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 2 • Q4/5

A garden has 4 equal sections. Tomatoes grow in 3 of the sections.

What fraction of the garden has tomatoes?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 2 • Q5/5

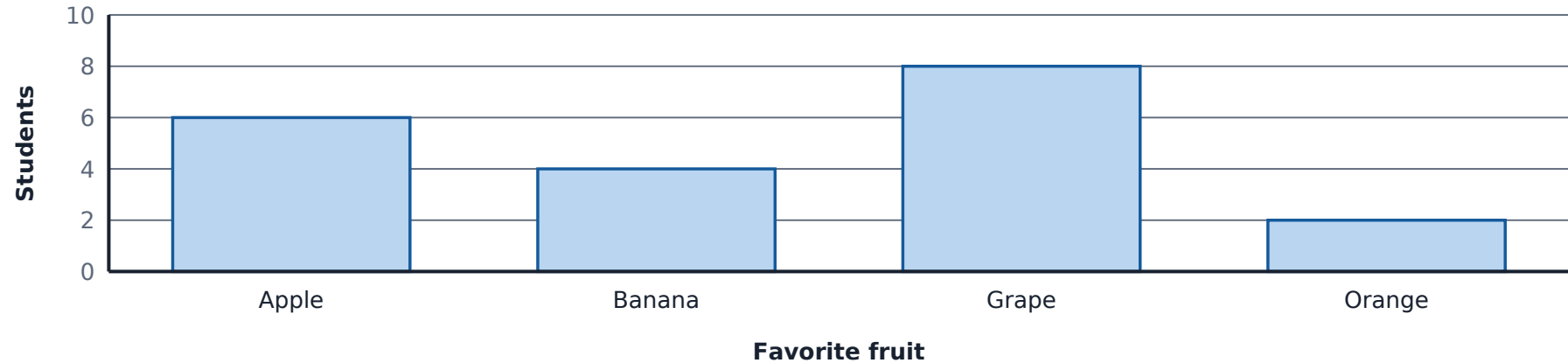
SET 1 • 6.RP.A.1

BACKWARD 3

Reading counts from a table or graph • Grade 3 • Measurement & Data •
3.MD.B.3

The graph shows the favorite fruits of a class.

How many students chose grape?



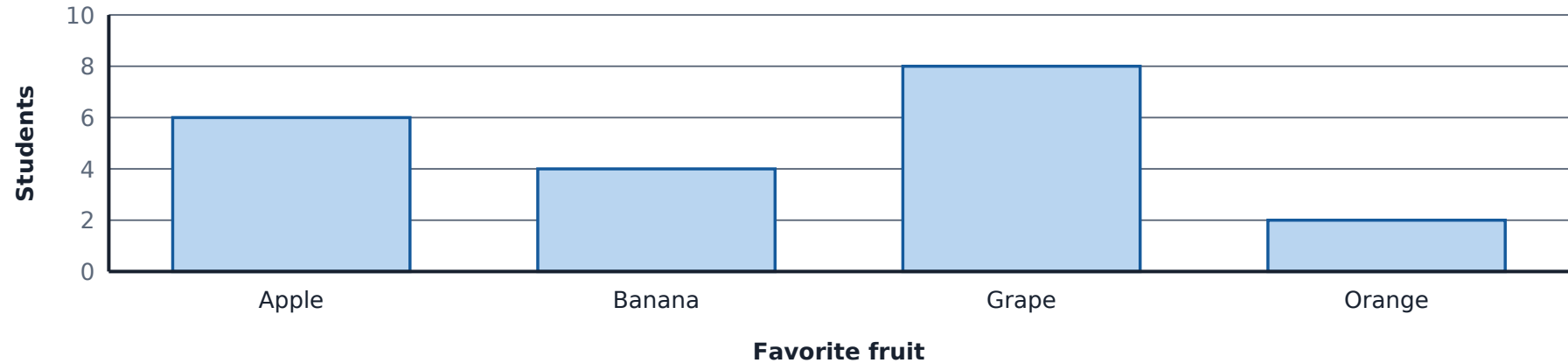
Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 3 • Q1/5

The graph shows the favorite fruits of a class.

How many more students chose apple than banana?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 3 • Q2/5

The table shows the pets owned by students.
How many students own fish?

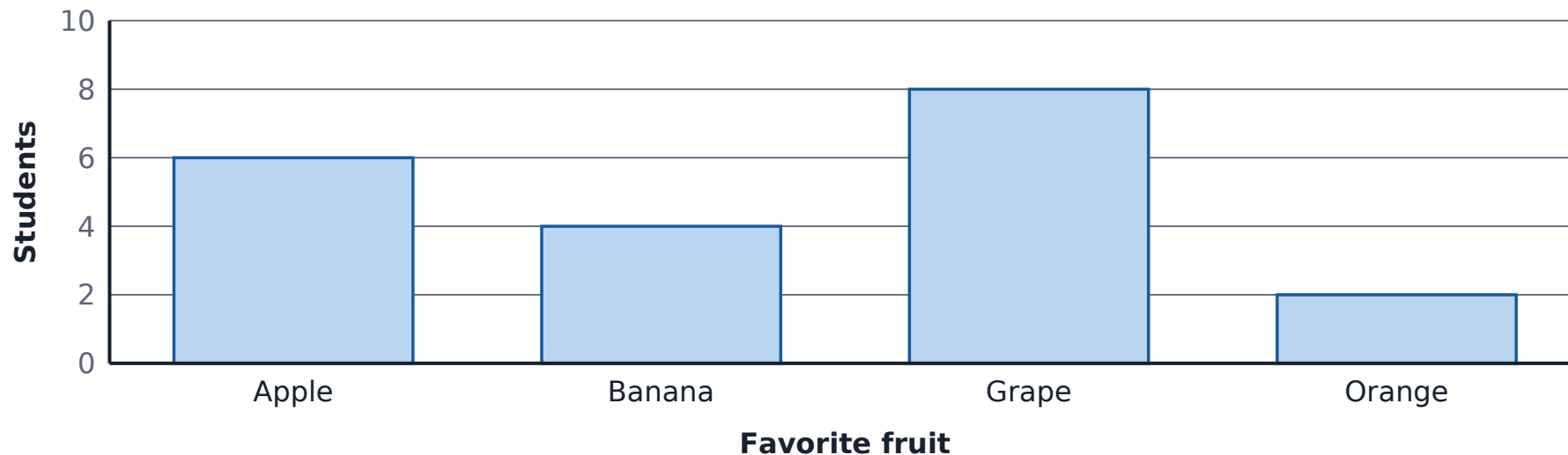
Pet	Dogs	Cats	Fish
Students	8	6	3

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 3 • Q3/5

Which fruit was chosen by 4 students?



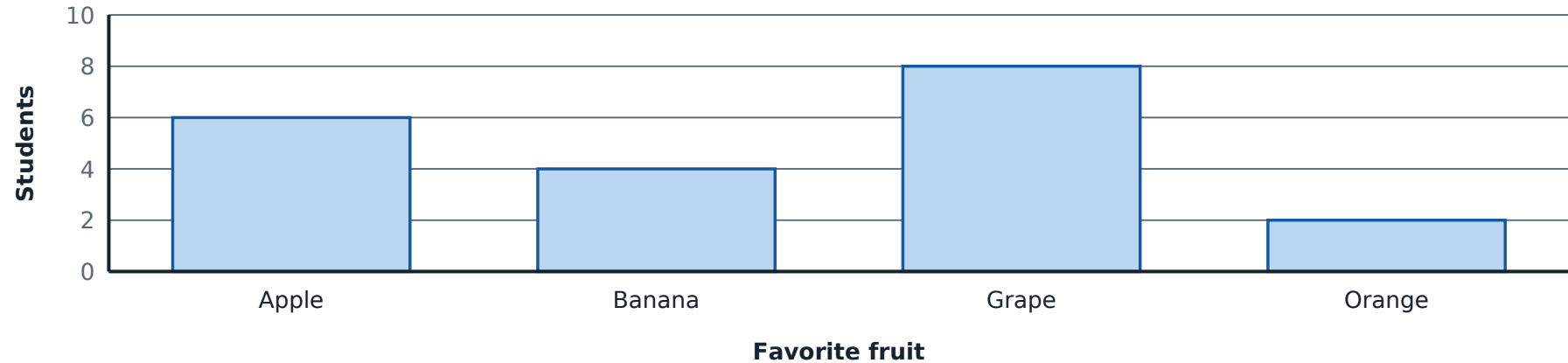
- A.** Apple **B.** Banana **C.** Grape **D.** Orange

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 3 • Q4/5

The graph shows the favorite fruits of a class.

How many students chose orange?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • 6.RP.A.1

FORWARD 1

Decide whether quantities are proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship between hours worked and pay.

Hours	1	2	3	4
Pay (\$)	12	24	36	48

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 1 • FORWARD 1 • Q1/5

True or false?

The table shows a proportional relationship between tickets and cost.

Tickets	2	4	6
Cost (\$)	10	18	26

A. True

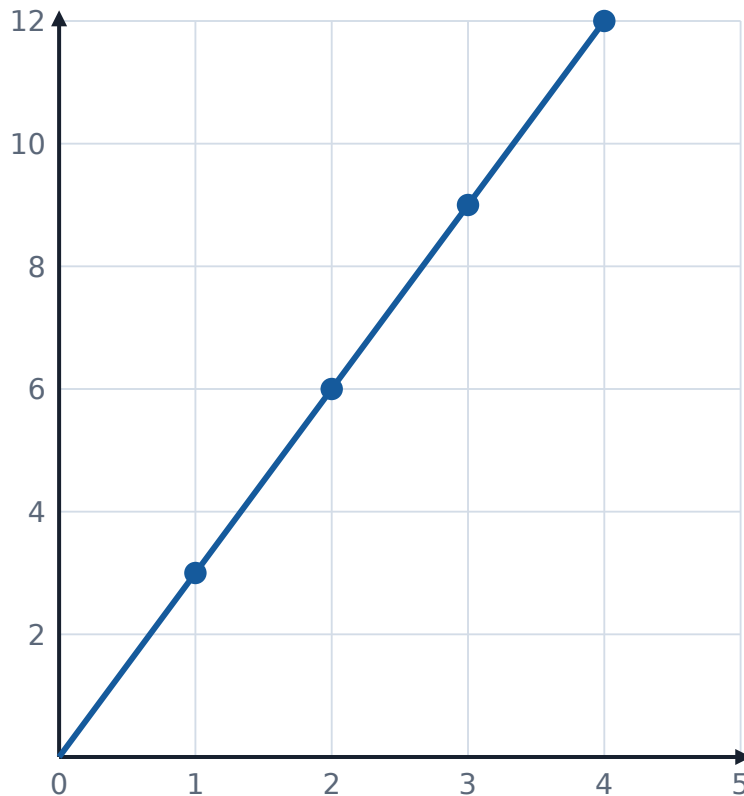
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 1 • FORWARD 1 • Q2/5

Does the graph show a proportional relationship?

- A.** Yes, because it is a straight line through $(0, 0)$.
- B.** No, because it does not pass through $(0, 0)$.
- C.** No, because the points are not on a line.
- D.** Yes, because y increases.

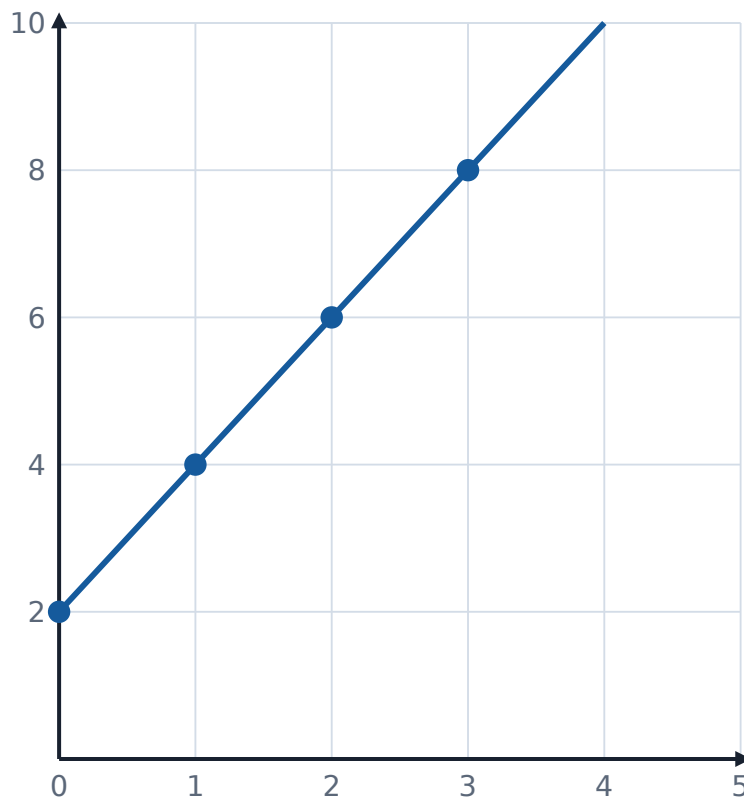


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 1 • FORWARD 1 • Q3/5

Which statement about the graph is true?

- A.** It is proportional because it is a straight line.
- B.** It is not proportional because it does not pass through $(0, 0)$.
- C.** It is proportional because y increases by 2.
- D.** It is not proportional because the points are not on a line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 1 • FORWARD 1 • Q4/5

Which set of ordered pairs shows a proportional relationship?

A. (1, 4), (2, 8), (3, 12)

B. (1, 5), (2, 7), (3, 9)

C. (2, 3), (4, 5), (6, 7)

D. (1, 2), (3, 5), (5, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 1 • FORWARD 1 • Q5/5

SET 1 • 6.RP.A.1

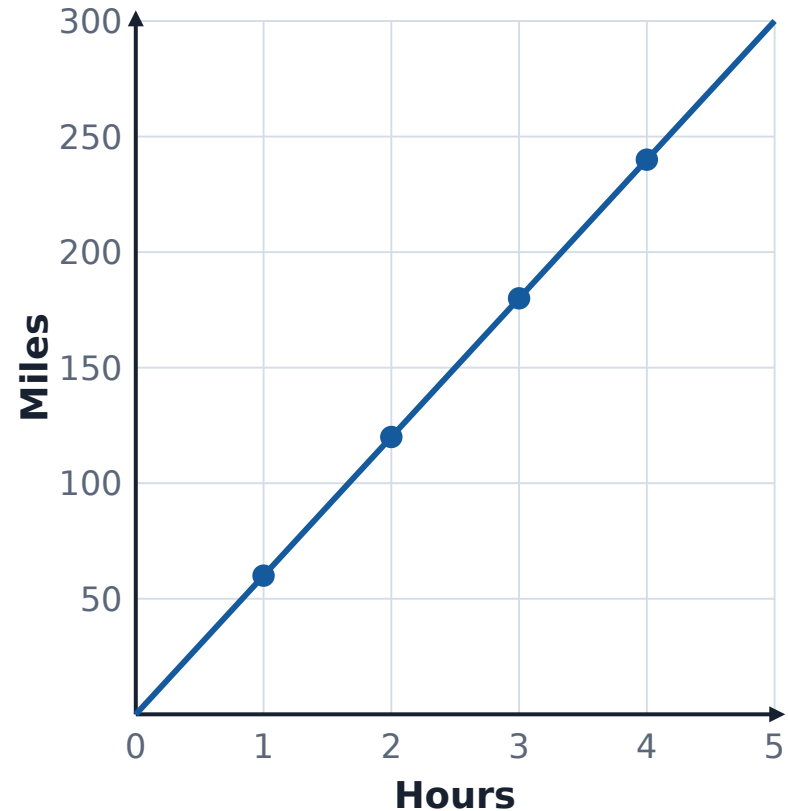
FORWARD 2

Graph proportional relationships; slope as unit rate • Grade 8 • Expressions & Equations • 8.EE.B.5

The graph shows the distance a car travels over time.

What is the slope of the line in miles per hour?

Slope: _____



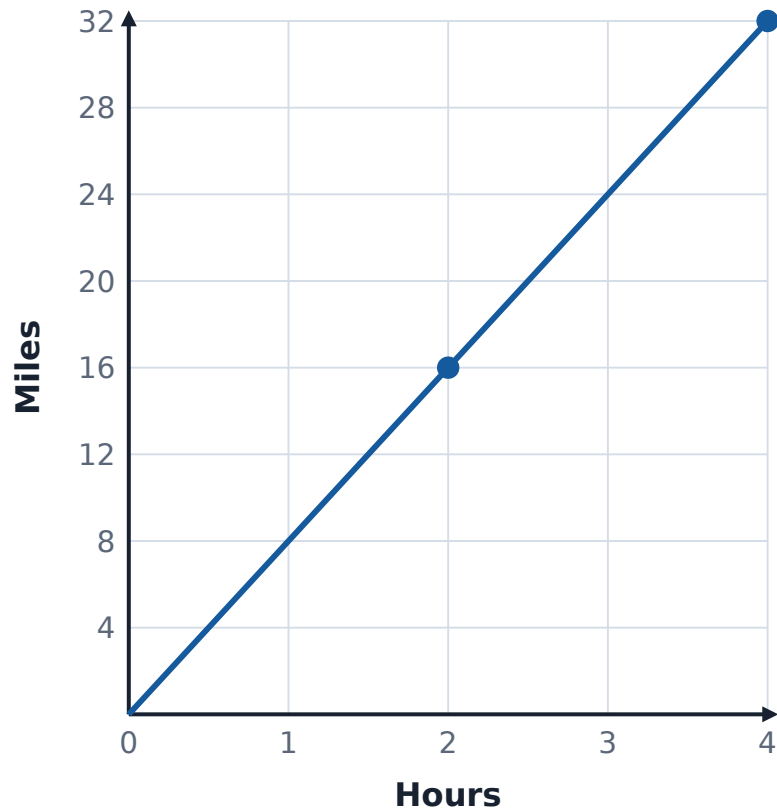
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q1/5

Cyclist A's distance is shown in the graph. Cyclist B rides according to $d = 7.5t$, where d is miles and t is hours.

Which cyclist is faster?

- A.** Cyclist A
- B.** Cyclist B
- C.** They ride at the same speed.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q2/5

Tank A fills according to $y = 12x$, where x is minutes and y is gallons. The graph of Tank B is a line through $(0, 0)$ and $(3, 45)$.

Which tank fills faster, and by how many gallons per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q3/5

The graph of a proportional relationship passes through $(0, 0)$ and $(6, 9)$.

What is the slope of the line?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q4/5

Store A: 2 pounds of apples cost \$3.50 and 4 pounds cost \$7.00.

Store B: the cost is $c = 1.6p$ for p pounds.

Which store charges less per pound?

A. Store A

B. Store B

C. Both charge the same.

Understand unit rate in the context of a ratio

MAIN	Grade 6 main question	6.RP.A.2
BACKWARD 1	Whole-number division	4.NBT.B.6
BACKWARD 2	Fraction as division	5.NF.B.3
BACKWARD 3	Meaning of "per" (amount in each group)	3.OA.A.2
FORWARD 1	Unit rates with fractions	7.RP.A.1
FORWARD 2	Rate of change and initial value	8.F.B.4

SET 2 • 6.RP.A.2

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.2

A car travels 150 miles in 3 hours at a constant speed.
What is the unit rate in miles per hour?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 2 • MAIN • Q1/5

A 4-pound bag of rice costs \$12.

What is the price per pound?

Price per pound: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 2 • MAIN • Q2/5

A recipe uses 3 cups of flour for every 4 cups of sugar.
How many cups of flour are used for each cup of sugar?

Cups of flour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 2 • MAIN • Q3/5

There are 2 teachers for every 30 students on a field trip.

Which statement uses the unit rate correctly?

- A.** There are 15 students per teacher.
- B.** There are 28 students per teacher.
- C.** There are 30 students per teacher.
- D.** There are 60 students per teacher.

A printer prints 90 pages in 6 minutes.

How many pages does it print per minute?

Pages per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 2 • MAIN • Q5/5

SET 2 • 6.RP.A.2

BACKWARD 1

Whole-number division • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Find the quotient.

$$150 \div 3$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 2 • BACKWARD 1 • Q1/5

Which is equal to $96 \div 4$?

A. 24

B. 22

C. 26

D. 23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 2 • BACKWARD 1 • Q2/5

True or false?

$$84 \div 6 = 14$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 2 • BACKWARD 1 • Q3/5

Find the quotient.

$$212 \div 4$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 2 • BACKWARD 1 • Q4/5

Find the quotient.

$$135 \div 5$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 6.RP.A.2

BACKWARD 2

Fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $3 \div 4$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 2 • BACKWARD 2 • Q1/5

True or false?

$\frac{2}{5}$ means the same as $2 \div 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 2 • BACKWARD 2 • Q2/5

Four friends share 3 sandwiches equally.

How much sandwich does each friend get?

Each friend gets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 2 • BACKWARD 2 • Q3/5

Which division expression is equal to $\frac{7}{8}$?

A. $7 \div 8$

B. $8 \div 7$

C. 7×8

D. $8 - 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 2 • BACKWARD 2 • Q4/5

A 5-yard ribbon is cut into 6 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 6.RP.A.2

BACKWARD 3

Meaning of "per" (amount in each group) • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

12 apples are packed equally into 4 bags.

How many apples are in each bag?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 2 • BACKWARD 3 • Q1/5

True or false?

"5 miles per hour" means 5 miles for each 1 hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 2 • BACKWARD 3 • Q2/5

Which phrase means the same as "\$2 per pound"?

- A.** \$2 for each pound
- B.** \$2 for all of the pounds
- C.** 2 pounds for each dollar
- D.** \$2 more than a pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 2 • BACKWARD 3 • Q3/5

24 eggs are put into cartons, with 6 eggs per carton.

How many cartons are filled?

Cartons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 2 • BACKWARD 3 • Q4/5

Jen reads 40 pages in 5 days. She reads the same number of pages each day.

How many pages does she read per day?

Pages: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • 6.RP.A.2

FORWARD 1

Unit rates with fractions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.1

Kim walks $\frac{1}{2}$ mile in $\frac{1}{4}$ hour.

What is her speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 2 • FORWARD 1 • Q1/5

A painter paints $\frac{2}{3}$ of a wall in $\frac{1}{3}$ hour.

At this rate, how many walls can the painter paint per hour?

Walls per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 2 • FORWARD 1 • Q2/5

A snail crawls $\frac{5}{6}$ foot in $\frac{1}{3}$ minute.

How many feet does it crawl per minute?

Feet per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 2 • FORWARD 1 • Q3/5

A machine fills $\frac{3}{4}$ of a tank in $\frac{1}{2}$ hour.

How many tanks does it fill per hour?

A. $1\frac{1}{2}$

B. $\frac{3}{8}$

C. $\frac{1}{4}$

D. $1\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 2 • FORWARD 1 • Q4/5

Leo mows $\frac{7}{8}$ of an acre in $\frac{1}{2}$ hour.

How many acres does he mow per hour?

Acres per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 6.RP.A.2

FORWARD 2

Rate of change and initial value • Grade 8 • Functions • 8.F.B.4

The table shows a linear function.

What is the rate of change?

x	0	1	2	3
y	5	8	11	14

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 2 • FORWARD 2 • Q1/5

A plumber charges a \$40 fee plus \$65 per hour.

Write a function for the cost C after h hours.

$C =$ _____

A line passes through $(2, 7)$ and $(6, 19)$.

What is the rate of change?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 2 • FORWARD 2 • Q3/5

A tank is draining at a constant rate.

What is the rate of change in gallons per minute?

Minutes	0	5	10
Gallons	200	170	140

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 2 • FORWARD 2 • Q4/5

A taxi ride costs \$3 plus \$2.50 per mile.

Which function gives the cost y for a ride of x miles?

A. $y = 2.5x + 3$

B. $y = 3x + 2.5$

C. $y = 5.5x$

D. $y = 2.5x - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 2 • FORWARD 2 • Q5/5

Tables of equivalent ratios; plot pairs on the coordinate plane

MAIN	Grade 6 main question	6.RP.A.3.a
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Two related number patterns	5.OA.B.3
BACKWARD 3	Plotting points in the first quadrant	5.G.A.2
FORWARD 1	Constant of proportionality	7.RP.A.2.b
FORWARD 2	Linear and nonlinear functions	8.F.A.3

SET 3 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

The table shows equivalent ratios.

Find the missing value.

Cups of juice	2	4	6	10
Cups of water	3	6	9	?

Missing value: _____

Paint is mixed in the same ratio in every column.

Find the missing value.

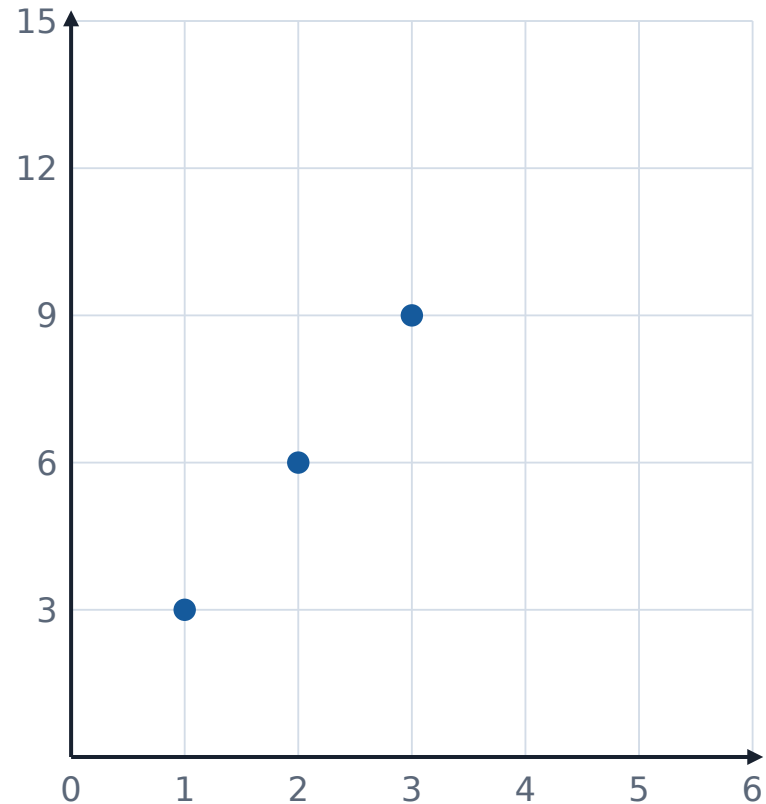
Blue (cups)	3	6	?	15
Yellow (cups)	2	4	8	10

Missing value: _____

The points show pairs from a ratio table.

Which point also belongs to this ratio table?

- A.** (4, 12) **B.** (4, 10)
C. (5, 12) **D.** (12, 4)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • MAIN • Q3/5

Class A has 4 boys for every 6 girls.

Class B has 6 boys for every 9 girls.

Which statement is true?

- A.** The ratios are equivalent.
- B.** Class A has a greater ratio of boys to girls.
- C.** Class B has a greater ratio of boys to girls.
- D.** The ratios cannot be compared.

A recipe uses 5 cups of oats for every 2 cups of raisins.
Find the missing values A and B.

Oats (cups)	5	10	A	25
Raisins (cups)	2	B	6	10

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • MAIN • Q5/5

SET 3 • 6.RP.A.3.a

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

$\frac{2}{3}$ is equivalent to $\frac{8}{12}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 3 • BACKWARD 1 • Q1/5

Find the missing number.

$$\frac{3}{4} = \frac{?}{20}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 3 • BACKWARD 1 • Q2/5

Which fraction is equivalent to $\frac{5}{6}$?

A. $\frac{10}{12}$

B. $\frac{10}{11}$

C. $\frac{6}{7}$

D. $\frac{5}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$\frac{4}{5}$ is equivalent to $\frac{6}{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 3 • BACKWARD 1 • Q4/5

Find the missing number.

$$\frac{2}{5} = \frac{6}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 6.RP.A.3.a

BACKWARD 2

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

Pattern A starts at 0 and adds 3.

Pattern B starts at 0 and adds 6.

What is the 5th number in Pattern B?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 3 • BACKWARD 2 • Q1/5

True or false?

Each number in Pattern B is 2 times the matching number in Pattern A.

Pattern A (add 4)	0	4	8	12
Pattern B (add 8)	0	8	16	24

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 3 • BACKWARD 2 • Q2/5

Pattern A starts at 0 and adds 5.

Pattern B starts at 0 and adds 10.

What is the 4th number in Pattern A?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 3 • BACKWARD 2 • Q3/5

Pattern A: 0, 2, 4, 6, 8

Pattern B: 0, 6, 12, 18, 24

How does each number in Pattern B compare to the matching number in Pattern A?

A. 3 times as large

B. 4 more

C. 6 times as large

D. 2 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 3 • BACKWARD 2 • Q4/5

Pattern A: 0, 1, 2, 3, ...

Pattern B: 0, 4, 8, 12, ...

Write the ordered pair formed by the 4th numbers of the patterns.

Ordered pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 3 • BACKWARD 2 • Q5/5

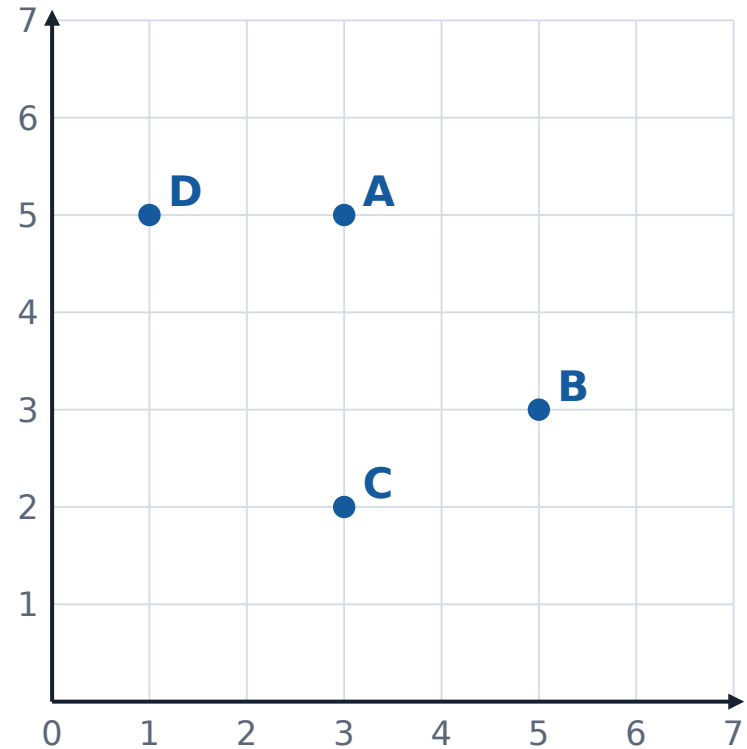
SET 3 • 6.RP.A.3.a

BACKWARD 3

Plotting points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

Which point is located at (3, 5)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

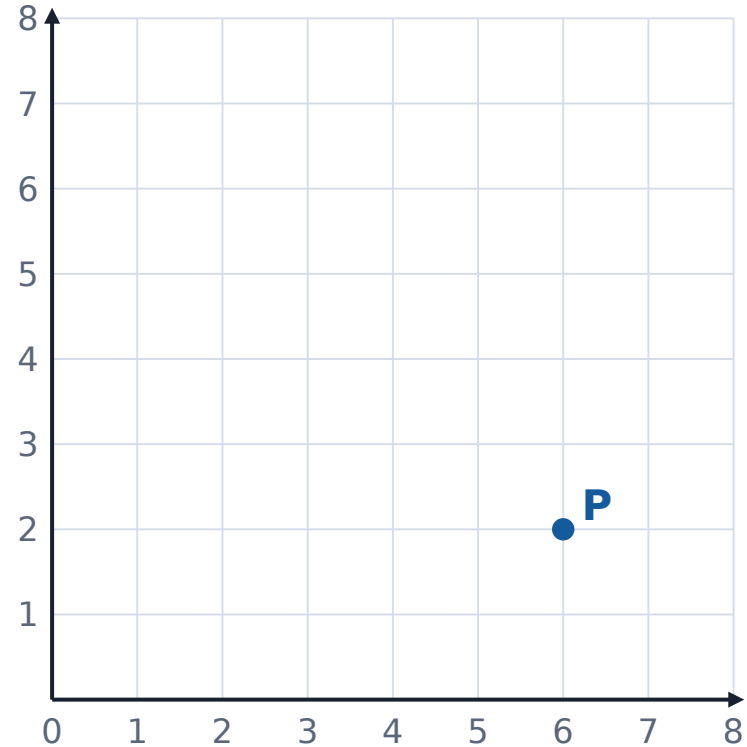


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 3 • BACKWARD 3 • Q1/5

What are the coordinates of point P?

P = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 3 • BACKWARD 3 • Q2/5

True or false?

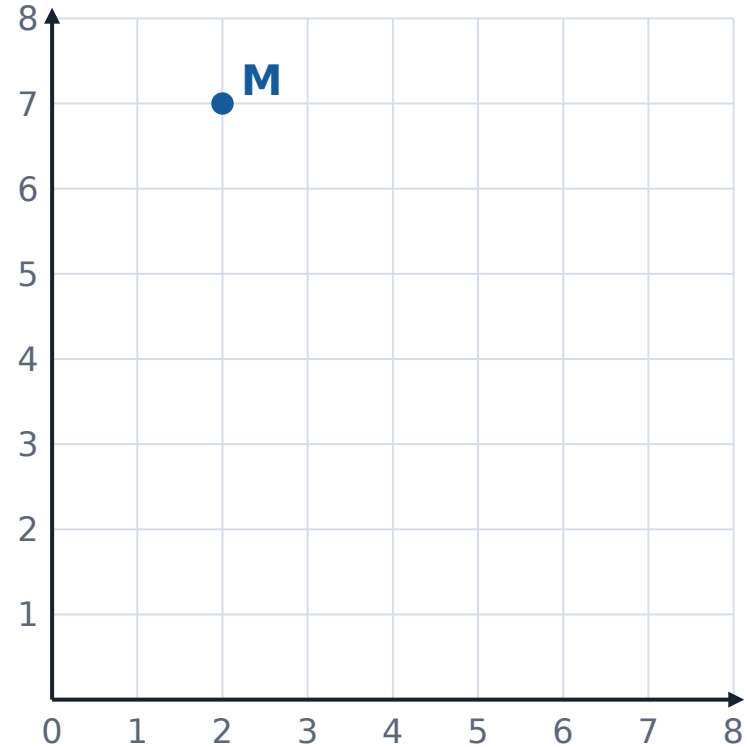
To plot $(4, 1)$, start at $(0, 0)$, move 4 units right, then move 1 unit up.

A. True

B. False

Which ordered pair names point M?

- A.** (2, 7)
- B.** (7, 2)
- C.** (2, 2)
- D.** (7, 7)

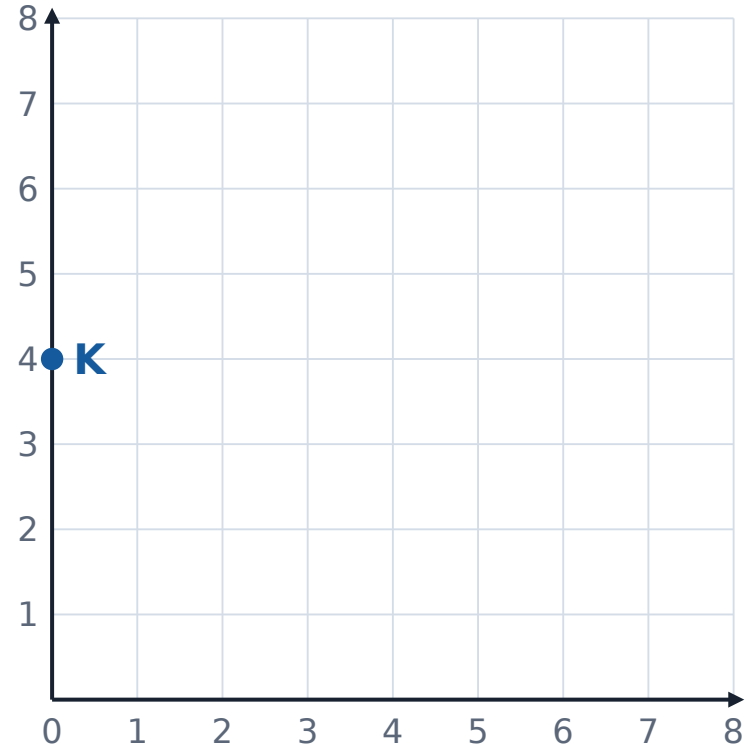


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 3 • BACKWARD 3 • Q4/5

What are the coordinates of point K?

K = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • 6.RP.A.3.a

FORWARD 1

Constant of proportionality • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality ($\frac{y}{x}$)?

x	2	5	8
y	7	17.5	28

k = _____

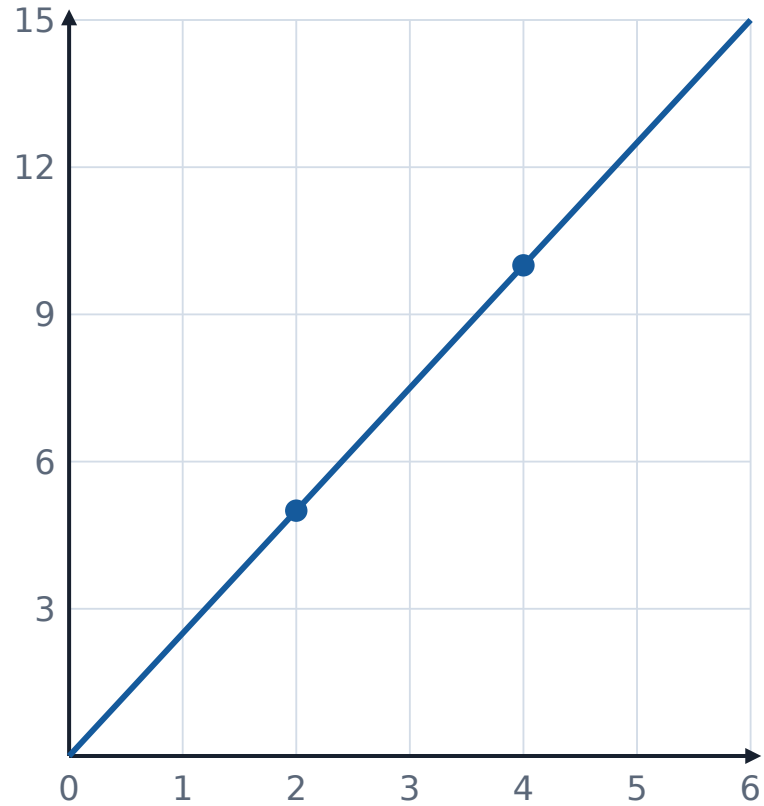
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 3 • FORWARD 1 • Q1/5

The graph shows a proportional relationship.

What is the constant of proportionality?

$k =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 3 • FORWARD 1 • Q2/5

The table shows the cost of cheese.

What is the constant of proportionality in dollars per pound?

Pounds	3	6	9
Cost (\$)	7.50	15.00	22.50

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 3 • FORWARD 1 • Q3/5

The graph of a proportional relationship passes through (0, 0) and (5, 2).

What is the constant of proportionality?

A. $\frac{2}{5}$

B. $\frac{5}{2}$

C. 2

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 3 • FORWARD 1 • Q4/5

A faucet fills 15 liters every 6 minutes at a constant rate.

What is the constant of proportionality in liters per minute?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 3 • FORWARD 1 • Q5/5

SET 3 • 6.RP.A.3.a

FORWARD 2

Linear and nonlinear functions • Grade 8 • Functions • 8.F.A.3

True or false?

The table represents a linear function.

x	0	1	2	3
y	1	4	7	10

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 2 • Q1/5

Which equation does NOT represent a linear function?

A. $y = 3x + 2$

B. $y = x^2$

C. $y = \frac{1}{2}x$

D. $y = -4x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 2 • Q2/5

True or false?

The table represents a linear function.

x	1	2	3	4
y	1	4	9	16

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 2 • Q3/5

Which set of ordered pairs lies on a linear function?

- A.** $(0, 2), (1, 4), (2, 8), (3, 16)$
- B.** $(0, 5), (1, 8), (2, 11), (3, 14)$
- C.** $(0, 0), (1, 1), (2, 4), (3, 9)$
- D.** $(1, 1), (2, 3), (3, 6), (4, 10)$

The area of a square is $A = s^2$, where s is the side length.

Is A a linear function of s ? Write yes or no.

Answer: _____

Unit rate problems: unit pricing and constant speed

MAIN	Grade 6 main question	6.RP.A.3.b
BACKWARD 1	Dividing money amounts	5.NBT.B.7
BACKWARD 2	Multiplying by an amount for each one	4.NBT.B.5
BACKWARD 3	Comparing decimals	5.NBT.A.3.b
BACKWARD 4	Hours and minutes	4.MD.A.1
FORWARD 1	Equations for proportional relationships	7.RP.A.2.c
FORWARD 2	Compare functions represented in different ways	8.F.A.2

SET 4 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

4 notebooks cost \$10.

At this rate, how much do 7 notebooks cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 4 • MAIN • Q1/5

It took 7 hours to mow 4 lawns.

At that rate, how many lawns could be mowed in 35 hours?

Lawns: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 4 • MAIN • Q2/5

Which is the better buy?

- A.** 6 bottles of water for \$4.50
- B.** 10 bottles of water for \$7.00
- C.** Both cost the same per bottle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 4 • MAIN • Q3/5

A runner runs 12 kilometers in 60 minutes at a constant speed.

How far does the runner go in 25 minutes?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 4 • MAIN • Q4/5

A machine makes 45 bottles every 3 minutes.

At that rate, how many minutes does it take to make 300 bottles?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 4 • MAIN • Q5/5

SET 4 • 6.RP.A.3.b

BACKWARD 1

Dividing money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Find the quotient.

$$\cancel{\$}4.50 \div 6$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 4 • BACKWARD 1 • Q1/5

Find the quotient.

$$\$10.00 \div 4$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 4 • BACKWARD 1 • Q2/5

$$\$7.00 \div 10 = ?$$

- A.** \$0.70
- B.** \$7.00
- C.** \$0.07
- D.** \$70.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 4 • BACKWARD 1 • Q3/5

Find the quotient.

$$\$3.60 \div 3$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 4 • BACKWARD 1 • Q4/5

True or false?

$$\text{\$}8.40 \div 4 = \text{\$}2.10$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 6.RP.A.3.b

BACKWARD 2

Multiplying by an amount for each one • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Each ticket costs \$15.

How much do 7 tickets cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 4 • BACKWARD 2 • Q1/5

A bus holds 48 people.

How many people can 6 buses hold?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 4 • BACKWARD 2 • Q2/5

Which expression gives the cost of 9 books that cost \$12 each?

A. 9×12

B. $12 + 9$

C. $12 - 9$

D. $12 \div 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 4 • BACKWARD 2 • Q3/5

Each box holds 125 pencils.

How many pencils are in 4 boxes?

Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 4 • BACKWARD 2 • Q4/5

True or false?

A cat sleeps 14 hours each day. In 5 days it sleeps 70 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 6.RP.A.3.b

BACKWARD 3

Comparing decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

Which amount is less?

A. \$0.75

B. \$0.70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 4 • BACKWARD 3 • Q1/5

True or false?

$$0.8 > 0.75$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 4 • BACKWARD 3 • Q2/5

Write $>$, $<$, or $=$ to compare.

2.35 2.4

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 4 • BACKWARD 3 • Q3/5

Which number is greatest?

- A.** 0.6
- B.** 0.59
- C.** 0.506
- D.** 0.065

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 4 • BACKWARD 3 • Q4/5

True or false?

$$1.25 < 1.205$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 4 • BACKWARD 3 • Q5/5

SET 4 • 6.RP.A.3.b

BACKWARD 4

Hours and minutes • Grade 4 • Measurement & Data • 4.MD.A.1

How many minutes are in 3 hours?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 4 • BACKWARD 4 • Q1/5

True or false?

There are 60 seconds in 1 hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 4 • BACKWARD 4 • Q2/5

Complete.

4 hours = ____ minutes

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 4 • BACKWARD 4 • Q3/5

Which is longer?

- A.** 90 minutes
- B.** 2 hours
- C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 4 • BACKWARD 4 • Q4/5

How many minutes are in $\frac{1}{2}$ hour?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 4 • BACKWARD 4 • Q5/5

SET 4 • 6.RP.A.3.b

FORWARD 1

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Apples cost \$2.50 per pound.

Write an equation for the cost c of p pounds of apples.

Equation: _____

A car travels at 55 miles per hour.

Which equation relates the distance d and the time t ?

A. $d = 55t$

B. $t = 55d$

C. $d = t + 55$

D. $d = \frac{t}{55}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 4 • FORWARD 1 • Q2/5

The table shows a proportional relationship.
Write an equation for earnings e after h hours.

Hours (h)	2	3	5
Earnings (e)	36	54	90

Equation: _____

A printer prints 12 pages per minute.

Write an equation relating the pages p and the minutes m .

Equation: _____

True or false?

5 pounds of rice cost \$8. The equation $c = 1.6p$ gives the cost c of p pounds of rice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 4 • FORWARD 1 • Q5/5

SET 4 • 6.RP.A.3.b

FORWARD 2

Compare functions represented in different ways • Grade 8 • Functions •
8.F.A.2

Plan A costs $y = 12x$ dollars for x months. Plan B is shown in the table.

Which plan has the greater rate of change?

Months (x)	2	4	6
Cost (y)	26	52	78

A. Plan A

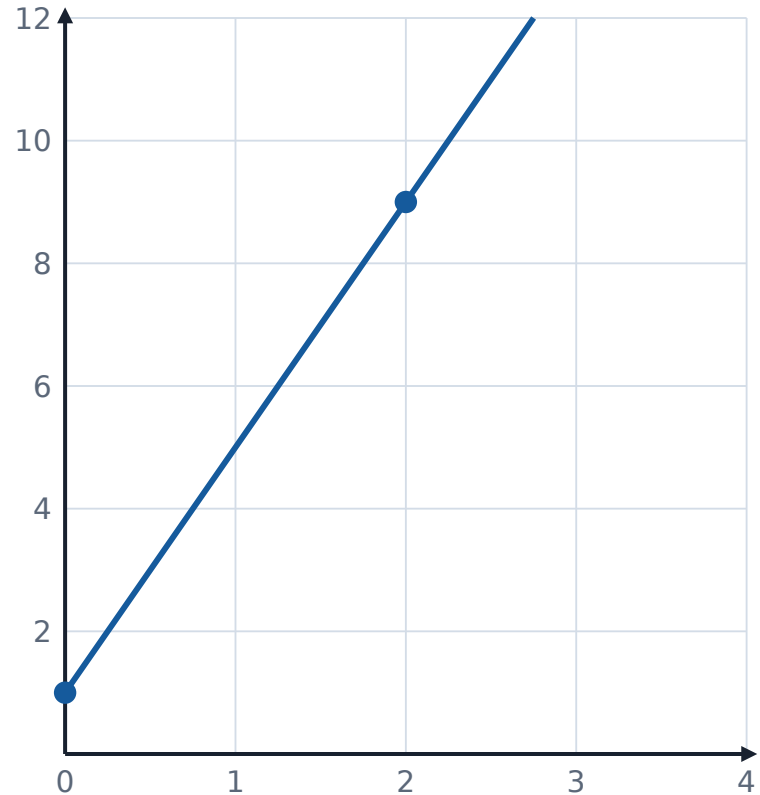
B. Plan B

C. They have the same rate of change.

Function A is $y = 3x + 4$. Function B is shown in the graph.

Which function has the greater rate of change?

- A.** Function A
- B.** Function B
- C.** They have the same rate of change.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.A.2 • Set 4 • FORWARD 2 • Q2/5

Car A travels 150 miles in 3 hours. Car B travels according to $d = 48t$.

How much faster is Car A, in miles per hour?

Miles per hour: _____

Function A is shown in the table. Function B is $y = -3x + 2$. Which function has the greater y-intercept?

x	0	1	2
y	5	3	1

A. Function A

B. Function B

C. They have the same y-intercept.

Store A: 3 pounds of grapes cost \$7.50.

Store B: $c = 2.4p$, where p is pounds.

Which store has the lower price per pound, and by how much?

Answer: _____

Percent of a quantity as a rate per 100

MAIN	Grade 6 main question	6.RP.A.3.c
BACKWARD 1	Hundredths as decimals	4.NF.C.6
BACKWARD 2	Tenths as hundredths	4.NF.C.5
BACKWARD 3	Fraction of a whole number	5.NF.B.4.a
BACKWARD 4	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Multi-step percent problems	7.RP.A.3
FORWARD 2	Build a linear function from a description	8.F.B.4

SET 5 • 6.RP.A.3.c

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 30% of 80?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 5 • MAIN • Q1/5

A jacket costs \$60. The sales tax is 8%.

How much is the tax?

Tax: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 5 • MAIN • Q2/5

A class has 25 students. 40% of the students walk to school.

How many students walk to school?

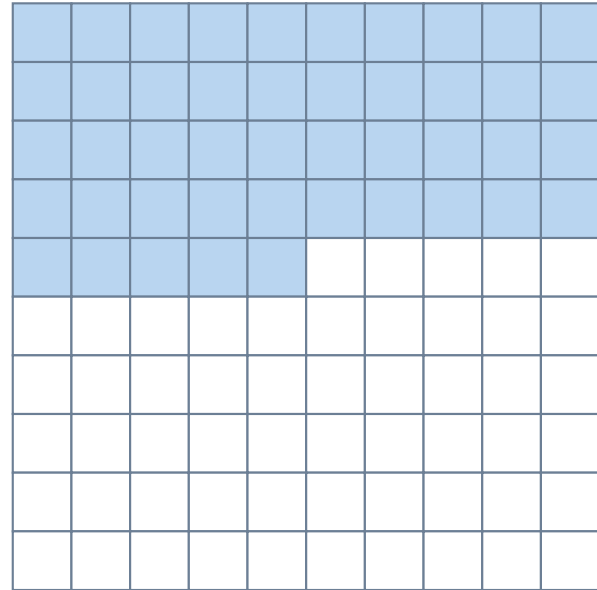
Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 5 • MAIN • Q3/5

What percent of the grid is shaded?

- A.** 45%
- B.** 55%
- C.** 4.5%
- D.** 0.45%



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 5 • MAIN • Q4/5

A school has 450 students. 12% of them play in the band.

How many students play in the band?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 5 • MAIN • Q5/5

SET 5 • 6.RP.A.3.c

BACKWARD 1

Hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{45}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 5 • BACKWARD 1 • Q1/5

True or false?

$$0.07 = \frac{7}{100}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 5 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{30}{100}$?

A. 0.30

B. 3.0

C. 0.03

D. 30.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 5 • BACKWARD 1 • Q3/5

Write 0.62 as a fraction with a denominator of 100.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 5 • BACKWARD 1 • Q4/5

True or false?

$$\frac{8}{100} = 0.8$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 6.RP.A.3.c

BACKWARD 2

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Find the missing number.

$$\frac{3}{10} = \frac{?}{100}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 5 • BACKWARD 2 • Q1/5

True or false?

$$\frac{4}{10} + \frac{5}{100} = \frac{45}{100}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 5 • BACKWARD 2 • Q2/5

Which fraction is equal to $\frac{7}{10}$?

A. $\frac{70}{100}$

B. $\frac{7}{100}$

C. $\frac{17}{100}$

D. $\frac{10}{70}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 5 • BACKWARD 2 • Q3/5

Add.

$$\frac{6}{10} + \frac{25}{100}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 5 • BACKWARD 2 • Q4/5

True or false?

$$\frac{9}{10} = \frac{90}{100}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 6.RP.A.3.c

BACKWARD 3

Fraction of a whole number • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{3}{10} \times 80$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 5 • BACKWARD 3 • Q1/5

Find $\frac{1}{4}$ of 60.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 5 • BACKWARD 3 • Q2/5

True or false?

$$\frac{2}{5} \times 25 = 10$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 5 • BACKWARD 3 • Q3/5

$$\frac{3}{4} \times 20 = ?$$

A. 15

B. 12

C. 16

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 5 • BACKWARD 3 • Q4/5

Find $\frac{7}{10}$ of 50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 5 • BACKWARD 3 • Q5/5

SET 5 • 6.RP.A.3.c

BACKWARD 4

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$0.3 \times 80$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 5 • BACKWARD 4 • Q1/5

Multiply.

$$0.08 \times 60$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 5 • BACKWARD 4 • Q2/5

True or false?

$$0.12 \times 450 = 54$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 5 • BACKWARD 4 • Q3/5

$$0.4 \times 25 = ?$$

A. 10

B. 1

C. 100

D. 0.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 5 • BACKWARD 4 • Q4/5

Multiply.

$$0.25 \times 36$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 5 • BACKWARD 4 • Q5/5

SET 5 • 6.RP.A.3.c

FORWARD 1

Multi-step percent problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A meal costs \$40. You leave a 15% tip.

What is the total cost of the meal with the tip?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 5 • FORWARD 1 • Q1/5

A pair of shoes costs \$80. They are on sale for 25% off.
What is the sale price?

Sale price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 5 • FORWARD 1 • Q2/5

A town's population grew from 2,000 to 2,300.

What is the percent increase?

Percent increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 5 • FORWARD 1 • Q3/5

A \$50 game is marked up 20%. Then a 5% sales tax is added to the new price.

What is the final cost?

A. \$63.00

B. \$62.50

C. \$60.00

D. \$62.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 5 • FORWARD 1 • Q4/5

A salesperson earns a 6% commission on sales. She sells \$3,500 in products.

How much commission does she earn?

Commission: _____

SET 5 • 6.RP.A.3.c

FORWARD 2

Build a linear function from a description • Grade 8 • Functions • 8.F.B.4

A salesperson earns \$300 per week plus 5% of her sales S .

Write a function for her weekly earnings E .

$E =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 2 • Q1/5

A gym charges a \$25 sign-up fee plus \$30 per month. Which function gives the total cost C after m months?

A. $C = 30m + 25$

B. $C = 25m + 30$

C. $C = 55m$

D. $C = 30m - 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 2 • Q2/5

A savings account starts with \$150. Each week, \$12 is added.

What are the initial value and the rate of change?

Initial value: _____

Rate of change: _____

A candle is 20 cm tall. It burns 1.5 cm per hour.
Write a function for its height h after t hours.

$h =$ _____

An agent earns \$500 per month plus 3% of her sales r .
Write a function for her monthly earnings E .

$E =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 2 • Q5/5

Find the whole, given a part and the percent

MAIN	Grade 6 main question	6.RP.A.3.c
BACKWARD 1	Benchmark percents as fractions	4.NF.A.1
BACKWARD 2	Finding an unknown factor	3.OA.A.4
BACKWARD 3	Dividing a whole number by a unit fraction	5.NF.B.7.b
FORWARD 1	Percent problems: find the original amount	7.RP.A.3
FORWARD 2	Linear equations with decimal coefficients	8.EE.C.7.b

SET 6 • 6.RP.A.3.c

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

12 is 25% of what number?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 6 • MAIN • Q1/5

A student answered 18 questions correctly. This was 90% of the questions on the test.

How many questions were on the test?

Questions: _____

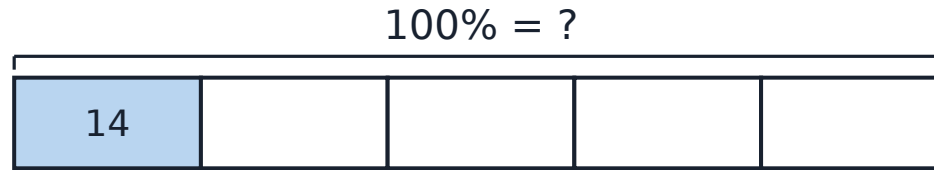
Maria has saved \$45. This is 30% of the cost of a bike.
How much does the bike cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 6 • MAIN • Q3/5

The tape diagram shows that 20% of a number is 14.
What is the number?



A. 70

B. 28

C. 34

D. 280

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 6 • MAIN • Q4/5

In a survey, 36 students chose pizza. This was 40% of the students surveyed.

How many students were surveyed?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 6 • MAIN • Q5/5

SET 6 • 6.RP.A.3.c

BACKWARD 1

Benchmark percents as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.A.1

Find the missing number.

$$\frac{25}{100} = \frac{1}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 1 • Q1/5

Find the missing number.

$$\frac{50}{100} = \frac{?}{2}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 1 • Q2/5

True or false?

$$\frac{20}{100} = \frac{1}{5}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 1 • Q3/5

Which fraction is equivalent to $\frac{75}{100}$?

A. $\frac{3}{4}$

B. $\frac{7}{10}$

C. $\frac{1}{75}$

D. $\frac{4}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 1 • Q4/5

Find the missing number.

$$\frac{10}{100} = \frac{1}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 6.RP.A.3.c

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$\square \times 4 = 36$$

$$\square = \underline{\hspace{10cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 6 • BACKWARD 2 • Q1/5

What number goes in the box?

$$5 \times \square = 45$$

$$\square = \underline{\hspace{10em}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 6 • BACKWARD 2 • Q2/5

True or false?

If $6 \times \square = 42$, then $\square = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 6 • BACKWARD 2 • Q3/5

$$\square \times 8 = 56$$

What number goes in the box?

- A.** 7
- B.** 6
- C.** 8
- D.** 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 6 • BACKWARD 2 • Q4/5

What number goes in the box?

$$9 \times \square = 63$$

$$\square = \underline{\hspace{10cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 6.RP.A.3.c

BACKWARD 3

Dividing a whole number by a unit fraction • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b

Divide.

$$12 \div \frac{1}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 6 • BACKWARD 3 • Q1/5

Divide.

$$5 \div \frac{1}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 6 • BACKWARD 3 • Q2/5

True or false?

$$6 \div \frac{1}{2} = 3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 6 • BACKWARD 3 • Q3/5

How many fifths are in 4?

A. 20

B. 9

C. $\frac{4}{5}$

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 6 • BACKWARD 3 • Q4/5

Divide.

$$8 \div \frac{1}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 6 • BACKWARD 3 • Q5/5

SET 6 • 6.RP.A.3.c

FORWARD 1

Percent problems: find the original amount • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

After a 20% discount, a jacket costs \$48.

What was the original price?

Original price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 6 • FORWARD 1 • Q1/5

A price increased by 10%. The new price is \$55.

What was the original price?

Original price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 6 • FORWARD 1 • Q2/5

The sales tax on an item was \$3.60. The tax rate is 6%.
What was the price before tax?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 6 • FORWARD 1 • Q3/5

A town's population grew by 25% to 1,500 people.
What was the population before the increase?

- A.** 1,200
- B.** 1,125
- C.** 1,875
- D.** 1,475

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 6 • FORWARD 1 • Q4/5

A store marks up the price of a bike by 40%. The selling price is \$210.

How much did the store pay for the bike?

Cost: _____

SET 6 • 6.RP.A.3.c

FORWARD 2

Linear equations with decimal coefficients • Grade 8 • Expressions & Equations
• 8.EE.C.7.b

Solve for x.

$$x - 0.2x = 48$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 6 • FORWARD 2 • Q1/5

Solve for x.

$$0.4x + 12 = 0.1x + 30$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 6 • FORWARD 2 • Q2/5

Solve for x.

$$0.25(x + 40) = 30$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 6 • FORWARD 2 • Q3/5

Which value of n solves the equation?

$$n + 0.15n = 92$$

- A.** 80
- B.** 78.2
- C.** 105.8
- D.** 92.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 6 • FORWARD 2 • Q4/5

Solve for y .

$$2.5y - 7 = 1.5y + 3$$

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 6 • FORWARD 2 • Q5/5

Convert measurement units using ratio reasoning

MAIN	Grade 6 main question	6.RP.A.3.d
BACKWARD 1	Measurement equivalents	4.MD.A.1
BACKWARD 2	Multiplying and dividing by powers of 10	5.NBT.A.2
BACKWARD 3	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Scale drawings	7.G.A.1
FORWARD 2	Dilations on the coordinate plane	8.G.A.3

SET 7 • 6.RP.A.3.d

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.d

There are 3 feet in 1 yard.

How many feet are in 14 yards?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 7 • MAIN • Q1/5

There are 12 inches in 1 foot.

How many inches are in 4.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 7 • MAIN • Q2/5

There are 16 ounces in 1 pound.

How many pounds are in 56 ounces?

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 7 • MAIN • Q3/5

A recipe needs 3 quarts of milk. There are 4 cups in 1 quart.

How many cups of milk are needed?

A. 12 cups

B. 7 cups

C. $\frac{3}{4}$ cup

D. $1\frac{1}{3}$ cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 7 • MAIN • Q4/5

1 inch = 2.54 centimeters.

A book is 10 inches tall. How many centimeters tall is it?

Centimeters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 7 • MAIN • Q5/5

SET 7 • 6.RP.A.3.d

BACKWARD 1

Measurement equivalents • Grade 4 • Measurement & Data • 4.MD.A.1

How many ounces are in 1 pound?

Ounces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 7 • BACKWARD 1 • Q1/5

True or false?

There are 12 inches in 1 yard.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 7 • BACKWARD 1 • Q2/5

How many cups are in 1 quart?

- A.** 4
- B.** 2
- C.** 8
- D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 7 • BACKWARD 1 • Q3/5

How many grams are in 1 kilogram?

Grams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 7 • BACKWARD 1 • Q4/5

True or false?

1 meter = 100 centimeters

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 6.RP.A.3.d

BACKWARD 2

Multiplying and dividing by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$3.4 \times 1,000$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 7 • BACKWARD 2 • Q1/5

Multiply.

$$2.5 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 7 • BACKWARD 2 • Q2/5

True or false?

$$0.6 \times 1,000 = 60$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 7 • BACKWARD 2 • Q3/5

$$450 \div 100 = ?$$

- A.** 4.5
- B.** 45
- C.** 0.45
- D.** 45,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 7 • BACKWARD 2 • Q4/5

Multiply.

$$7.25 \times 10$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 6.RP.A.3.d

BACKWARD 3

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$4.5 \times 12$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 3 • Q1/5

Multiply.

$$3.5 \times 16$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 3 • Q2/5

True or false?

$$1.5 \times 4 = 6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 3 • Q3/5

$$2.25 \times 4 = ?$$

- A.** 9
- B.** 8.1
- C.** 90
- D.** 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 3 • Q4/5

Multiply.

$$6 \times 0.75$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • 6.RP.A.3.d

FORWARD 1

Scale drawings • Grade 7 • Geometry • 7.G.A.1

A map scale is 1 inch : 20 miles. Two towns are 3.5 inches apart on the map.

What is the actual distance between the towns?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.1 • Set 7 • FORWARD 1 • Q1/5

A floor plan uses the scale 1 cm : 2 m. A room is 4.5 cm long on the plan.

How long is the actual room?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.1 • Set 7 • FORWARD 1 • Q2/5

A drawing of a car uses the scale 1 inch : 4 feet. The real car is 14 feet long.

How long is the car in the drawing?

A. 3.5 inches

B. 56 inches

C. 10 inches

D. 18 inches

A garden is 3 cm by 5 cm in a scale drawing. The scale is 1 cm : 2 m.

What is the actual area of the garden?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.1 • Set 7 • FORWARD 1 • Q4/5

On a map, 2 cm represents 15 km.

What actual distance does 7 cm represent?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.1 • Set 7 • FORWARD 1 • Q5/5

SET 7 • 6.RP.A.3.d

FORWARD 2

Dilations on the coordinate plane • Grade 8 • Geometry • 8.G.A.3

Point $A(2, 3)$ is dilated by a scale factor of 4 with the center of dilation at the origin.

What are the coordinates of A' ?

$A' =$ _____

Triangle PQR has vertices P(4, 2), Q(6, 8), and R(2, 6). It is dilated by a scale factor of $\frac{1}{2}$ with the center at the origin.

What are the coordinates of Q'?

A. (3, 4)

B. (12, 16)

C. (5, 7)

D. (6, 4)

A rectangle is dilated with the center at the origin. The vertex $(3, 2)$ maps to $(7.5, 5)$.

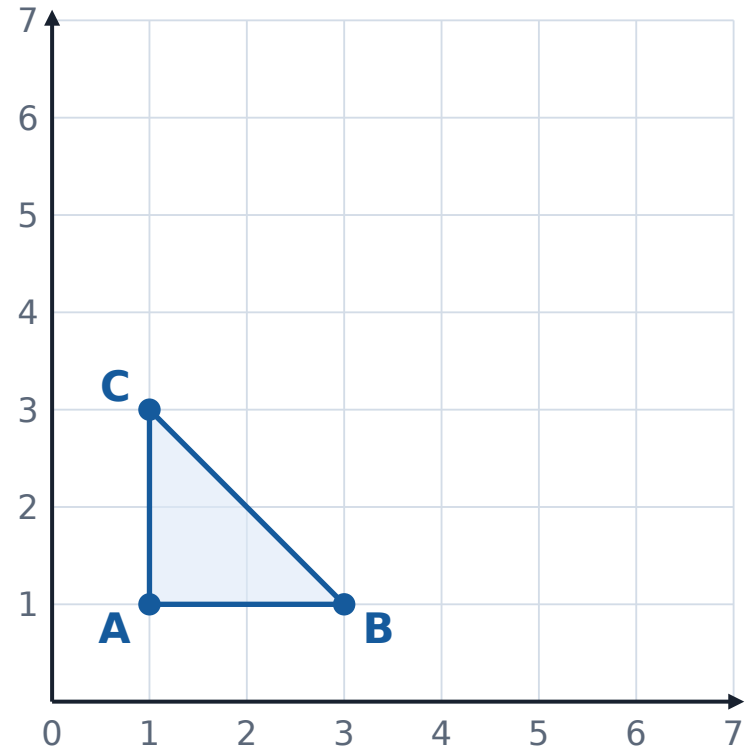
What is the scale factor?

Scale factor: _____

The triangle is dilated by a scale factor of 2 with the center at the origin.

What are the coordinates of C' ?

$C' =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.A.3 • Set 7 • FORWARD 2 • Q4/5

True or false?

A dilation with a scale factor of 3 and the center at the origin maps $(-2, 5)$ to $(-6, 15)$.

A. True

B. False

Divide fractions by fractions

MAIN	Grade 6 main question	6.NS.A.1
BACKWARD 1	Multiplying fractions	5.NF.B.4.a
BACKWARD 2	Whole number $\div$ unit fraction	5.NF.B.7.b
BACKWARD 3	Unit fraction $\div$ whole number	5.NF.B.7.a
BACKWARD 4	Equivalent fractions	4.NF.A.1
BACKWARD 5	Mixed numbers as fractions	4.NF.B.3.b
FORWARD 1	Multiply and divide rational numbers	7.NS.A.2.c
FORWARD 2	Equations with fraction coefficients	8.EE.C.7.b

SET 8 • 6.NS.A.1

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{2}{3} \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 8 • MAIN • Q1/5

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 8 • MAIN • Q2/5

Divide.

$$1\frac{1}{2} \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 8 • MAIN • Q3/5

Which is equal to $\frac{4}{5} \div \frac{2}{3}$?

A. $1\frac{1}{5}$

B. $\frac{8}{15}$

C. $\frac{5}{6}$

D. $\frac{2}{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 8 • MAIN • Q4/5

Divide.

$$3\frac{1}{3} \div \frac{5}{6}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 8 • MAIN • Q5/5

SET 8 • 6.NS.A.1

BACKWARD 1

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{4}{3}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 8 • BACKWARD 1 • Q1/5

True or false?

$$\frac{3}{4} \times \frac{2}{5} = \frac{6}{20}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 8 • BACKWARD 1 • Q2/5

$$\frac{1}{2} \times \frac{5}{6} = ?$$

A. $\frac{5}{12}$

B. $\frac{6}{8}$

C. $\frac{5}{8}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 8 • BACKWARD 1 • Q3/5

Multiply.

$$\frac{4}{5} \times \frac{3}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 8 • BACKWARD 1 • Q4/5

Multiply.

$$\frac{5}{6} \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 6.NS.A.1

BACKWARD 2

Whole number $\div$ unit fraction • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$3 \div \frac{1}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 8 • BACKWARD 2 • Q1/5

True or false?

$$4 \div \frac{1}{2} = 8$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 8 • BACKWARD 2 • Q2/5

How many thirds are in 2?

A. 6

B. $\frac{2}{3}$

C. 5

D. $\frac{1}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 8 • BACKWARD 2 • Q3/5

Divide.

$$6 \div \frac{1}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 8 • BACKWARD 2 • Q4/5

Divide.

$$2 \div \frac{1}{8}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 6.NS.A.1

BACKWARD 3

Unit fraction $\div$ whole number • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{3} \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 8 • BACKWARD 3 • Q1/5

True or false?

$$\frac{1}{2} \div 3 = \frac{1}{6}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 8 • BACKWARD 3 • Q2/5

$$\frac{1}{4} \div 2 = ?$$

A. $\frac{1}{8}$

B. $\frac{1}{2}$

C. 8

D. $\frac{2}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 8 • BACKWARD 3 • Q3/5

Divide.

$$\frac{1}{5} \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 8 • BACKWARD 3 • Q4/5

True or false?

$$\frac{1}{6} \div 2 = \frac{2}{6}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 8 • BACKWARD 3 • Q5/5

SET 8 • 6.NS.A.1

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

$\frac{8}{12}$ is equivalent to $\frac{2}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 8 • BACKWARD 4 • Q1/5

Find the missing number.

$$\frac{10}{15} = \frac{?}{3}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 8 • BACKWARD 4 • Q2/5

Which fraction is equivalent to $\frac{9}{12}$?

A. $\frac{3}{4}$

B. $\frac{2}{3}$

C. $\frac{9}{4}$

D. $\frac{12}{9}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 8 • BACKWARD 4 • Q3/5

Find the missing number.

$$\frac{6}{8} = \frac{?}{4}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 8 • BACKWARD 4 • Q4/5

True or false?

$\frac{4}{6}$ is equivalent to $\frac{6}{8}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 8 • BACKWARD 4 • Q5/5

SET 8 • 6.NS.A.1

BACKWARD 5

Mixed numbers as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $1\frac{1}{2}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 8 • BACKWARD 5 • Q1/5

Write $3\frac{1}{3}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 8 • BACKWARD 5 • Q2/5

True or false?

$$2\frac{1}{4} = \frac{9}{4}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 8 • BACKWARD 5 • Q3/5

Which fraction is equal to $1\frac{2}{3}$?

A. $\frac{5}{3}$

B. $\frac{3}{3}$

C. $\frac{12}{3}$

D. $\frac{7}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 8 • BACKWARD 5 • Q4/5

Write $\frac{11}{4}$ as a mixed number.

Mixed number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 8 • BACKWARD 5 • Q5/5

SET 8 • 6.NS.A.1

FORWARD 1

Multiply and divide rational numbers • Grade 7 • The Number System •
7.NS.A.2.c

Find the value.

$$\left(-\frac{2}{3}\right) \div \frac{4}{5}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.c • Set 8 • FORWARD 1 • Q1/5

Find the value.

$$-\frac{3}{4} \times \left(-\frac{8}{9}\right)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.c • Set 8 • FORWARD 1 • Q2/5

$$\left(-1\frac{1}{2}\right) \div \left(-\frac{3}{8}\right) = ?$$

A. 4

B. -4

C. $\frac{9}{16}$

D. $-\frac{9}{16}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.c • Set 8 • FORWARD 1 • Q3/5

Find the value.

$$\frac{5}{6} \div \left(-\frac{5}{12}\right)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.c • Set 8 • FORWARD 1 • Q4/5

True or false?

$$\left(-\frac{1}{2}\right) \times \frac{4}{5} = -\frac{2}{5}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.c • Set 8 • FORWARD 1 • Q5/5

SET 8 • 6.NS.A.1

FORWARD 2

Equations with fraction coefficients • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$\frac{3}{4}(x - 2) = \frac{1}{2}x + 5$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 8 • FORWARD 2 • Q1/5

Solve for x.

$$\frac{2}{3}x + 4 = \frac{1}{6}x + 7$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 8 • FORWARD 2 • Q2/5

Solve for x.

$$\frac{x}{2} + \frac{x}{3} = 10$$

A. 12

B. 60

C. 5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 8 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{1}{2}(4x + 6) = 3x - 5$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 8 • FORWARD 2 • Q4/5

Solve for y.

$$\frac{5}{8}y - 1 = \frac{3}{8}y + 2$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 8 • FORWARD 2 • Q5/5

Interpret and solve word problems with fraction division

MAIN	Grade 6 main question	6.NS.A.1
BACKWARD 1	Division as "how many groups?"	3.OA.A.2
BACKWARD 2	Unit fractions build a fraction	3.NF.A.1
BACKWARD 3	Area with fractional side lengths	5.NF.B.4.b
BACKWARD 4	Word problems: dividing with unit fractions	5.NF.B.7.c
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Solve linear equations from contexts	8.EE.C.7.b

SET 9 • 6.NS.A.1

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{4}$ –cup servings are in $\frac{2}{3}$ cup of yogurt?

Servings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 9 • MAIN • Q1/5

A path is $\frac{3}{4}$ mile long. It is divided into sections that are each $\frac{1}{8}$ mile long.

How many sections are there?

Sections: _____

A rectangle has an area of $\frac{1}{2}$ square mile. Its width is $\frac{3}{4}$ mile.

What is its length?

Length: _____

Which question can be answered by $\frac{3}{4} \div \frac{2}{5}$?

A. How many $\frac{2}{5}$ s are in $\frac{3}{4}$?

B. What is $\frac{2}{5}$ of $\frac{3}{4}$?

C. How many $\frac{3}{4}$ s are in $\frac{2}{5}$?

D. What is $\frac{3}{4}$ more than $\frac{2}{5}$?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 9 • MAIN • Q4/5

Ana has $\frac{5}{6}$ pound of trail mix. She fills bags that each hold $\frac{5}{12}$ pound.

How many bags can she fill?

Bags: _____

SET 9 • 6.NS.A.1

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 4 are in 20?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 1 • Q1/5

True or false?

$12 \div 3$ can mean "How many groups of 3 are in 12?"

A. True

B. False

Tom has 18 cookies. He puts 6 cookies in each bag.

Which equation finds the number of bags?

A. $18 \div 6 = \square$

B. $18 \times 6 = \square$

C. $18 - 6 = \square$

D. $6 \div 18 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 1 • Q3/5

24 students are split into teams of 8.

How many teams are there?

Teams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 1 • Q4/5

How many 5s are in 35?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 6.NS.A.1

BACKWARD 2

Unit fractions build a fraction • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make $\frac{3}{4}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q1/5

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q2/5

True or false?

$\frac{5}{6}$ is 5 parts of size $\frac{1}{6}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q3/5

$\frac{3}{8}$ is made of how many parts of size $\frac{1}{8}$?

A. 3

B. 8

C. 5

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q4/5

How many $\frac{1}{3}$ s make $\frac{2}{3}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q5/5

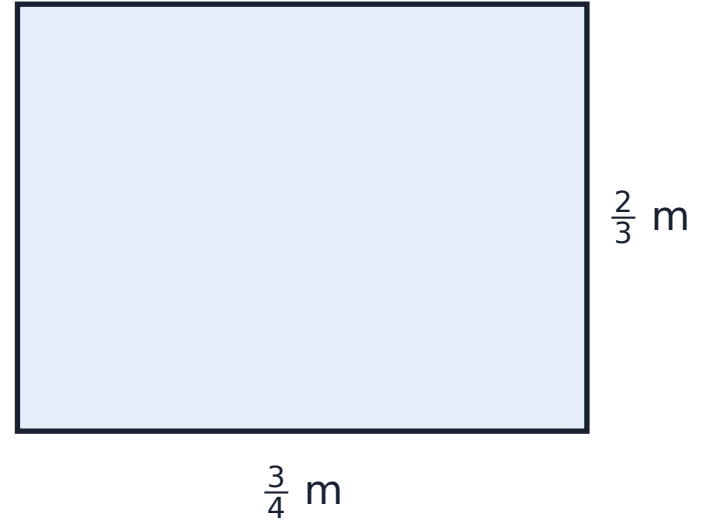
SET 9 • 6.NS.A.1

BACKWARD 3

Area with fractional side lengths • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

What is the area of the rectangle?

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 9 • BACKWARD 3 • Q1/5

True or false?

A rectangle that is $\frac{1}{2}$ foot by $\frac{1}{3}$ foot has an area of $\frac{1}{6}$ square foot.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 9 • BACKWARD 3 • Q2/5

A rectangle is $\frac{2}{5}$ unit by $\frac{1}{2}$ unit.

What is its area?

A. $\frac{1}{5}$ square unit

B. $\frac{3}{7}$ square unit

C. $\frac{2}{7}$ square unit

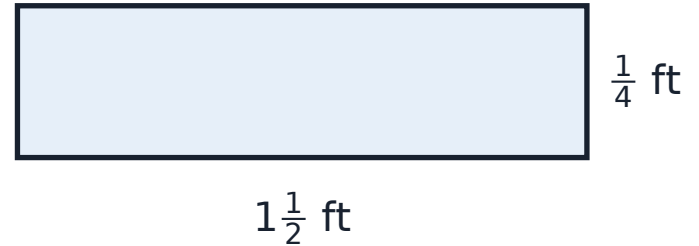
D. $\frac{9}{10}$ square unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 9 • BACKWARD 3 • Q3/5

What is the area of the rectangle?

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 9 • BACKWARD 3 • Q4/5

A garden is $\frac{5}{6}$ yard long and $\frac{3}{5}$ yard wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 9 • BACKWARD 3 • Q5/5

SET 9 • 6.NS.A.1

BACKWARD 4

Word problems: dividing with unit fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{3}$ –cup servings are in 2 cups of raisins?

Servings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 9 • BACKWARD 4 • Q1/5

A 4-pound bag of rice is divided into $\frac{1}{2}$ -pound portions.

How many portions are there?

Portions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 9 • BACKWARD 4 • Q2/5

Which expression finds how many $\frac{1}{5}$ -mile laps are in 3 miles?

A. $3 \div \frac{1}{5}$

B. $\frac{1}{5} \div 3$

C. $3 \times \frac{1}{5}$

D. $3 - \frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 9 • BACKWARD 4 • Q3/5

3 people share $\frac{1}{2}$ pound of chocolate equally.

How much chocolate does each person get?

Each person: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 9 • BACKWARD 4 • Q4/5

True or false?

There are 10 half-hours in 5 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 9 • BACKWARD 4 • Q5/5

SET 9 • 6.NS.A.1

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver descends $\frac{3}{4}$ foot each second.

How many seconds does it take to descend $5\frac{1}{4}$ feet?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • FORWARD 1 • Q1/5

The temperature was 4°F . It dropped $2\frac{1}{2}^{\circ}\text{F}$ each hour for 3 hours.

What was the final temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • FORWARD 1 • Q2/5

A stock's value changed by $-\$1\frac{1}{4}$ each day for 4 days.

What was the total change?

- A.** $-\$5$
- B.** $\$5$
- C.** $-\$4\frac{1}{4}$
- D.** $-\$1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • FORWARD 1 • Q3/5

A recipe needs $\frac{2}{3}$ cup of oil per batch. Jo has $3\frac{1}{3}$ cups of oil.

How many batches can Jo make?

Batches: _____

A hiker descends 1,250 feet in $\frac{5}{6}$ hour at a constant rate.

What is the change in elevation per hour?

Change per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 6.NS.A.1

FORWARD 2

Solve linear equations from contexts • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$\frac{1}{2}(x + 6) = \frac{1}{3}(x + 12)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 2 • Q1/5

Lee's age plus $\frac{1}{3}$ of his age is 24.

Write and solve an equation to find Lee's age.

Age: _____

Tank A has 40 gallons and drains $\frac{3}{4}$ gallon per minute.

Tank B has 10 gallons and fills at $1\frac{1}{4}$ gallons per minute.

After how many minutes will the tanks hold the same amount?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{2}{5}x - 3 = \frac{1}{5}x + 1$$

A. 20

B. 10

C. 4

D. -10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 2 • Q4/5

Solve for n.

$$\frac{3}{4}n + \frac{1}{2} = \frac{1}{4}n + 4$$

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 2 • Q5/5

Divide multi-digit numbers using the standard algorithm

MAIN	Grade 6 main question	6.NS.B.2
BACKWARD 1	Multiplying multi-digit numbers	5.NBT.B.5
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 3	Estimating quotients	5.NBT.B.6
BACKWARD 4	Multi-digit subtraction	4.NBT.B.4
FORWARD 1	Long division: fractions to decimals	7.NS.A.2.d
FORWARD 2	Repeating decimals and rational numbers	8.NS.A.1

SET 10 • 6.NS.B.2

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide.

$$4,508 \div 23$$

Quotient: _____

Use the standard algorithm to divide.

$$7,236 \div 36$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.2 • Set 10 • MAIN • Q2/5

Use the standard algorithm to divide.

$$9,975 \div 75$$

Quotient: _____

A school sets up 1,680 chairs in rows of 48 chairs.

How many rows are there?

- A.** 35
- B.** 30
- C.** 350
- D.** 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.2 • Set 10 • MAIN • Q4/5

Use the standard algorithm to divide.

$$25,494 \div 42$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.2 • Set 10 • MAIN • Q5/5

SET 10 • 6.NS.B.2

BACKWARD 1

Multiplying multi-digit numbers • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Multiply.

$$46 \times 23$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 10 • BACKWARD 1 • Q1/5

Multiply.

$$196 \times 23$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 10 • BACKWARD 1 • Q2/5

True or false?

$$35 \times 48 = 1,680$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 10 • BACKWARD 1 • Q3/5

$$133 \times 75 = ?$$

- A.** 9,975
- B.** 9,875
- C.** 1,596
- D.** 9,775

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 10 • BACKWARD 1 • Q4/5

Multiply.

$$607 \times 42$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 6.NS.B.2

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$852 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 10 • BACKWARD 2 • Q1/5

Divide.

$$1,256 \div 8$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 10 • BACKWARD 2 • Q2/5

True or false?

$$936 \div 3 = 312$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 10 • BACKWARD 2 • Q3/5

$$714 \div 7 = ?$$

- A.** 102
- B.** 12
- C.** 120
- D.** 1,002

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 10 • BACKWARD 2 • Q4/5

Divide.

$$2,345 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 6.NS.B.2

BACKWARD 3

Estimating quotients • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

Which is the best estimate for $4,508 \div 23$?

- A.** 200
- B.** 20
- C.** 2,000
- D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 10 • BACKWARD 3 • Q1/5

Which is the best estimate for $7,236 \div 36$?

- A.** 200
- B.** 20
- C.** 2,000
- D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 10 • BACKWARD 3 • Q2/5

True or false?

9,975 $\div$ 75 is between 100 and 200.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 10 • BACKWARD 3 • Q3/5

Estimate $6,120 \div 29$ by rounding 29 to 30.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 10 • BACKWARD 3 • Q4/5

How many digits are in the quotient of $1,680 \div 48$?

A. 2

B. 1

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • 6.NS.B.2

BACKWARD 4

Multi-digit subtraction • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Subtract.

$$5,012 - 4,698$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 10 • BACKWARD 4 • Q1/5

Subtract.

$$3,000 - 1,847$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 10 • BACKWARD 4 • Q2/5

True or false?

$$7,236 - 7,200 = 36$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 10 • BACKWARD 4 • Q3/5

$$9,975 - 9,000 = ?$$

- A.** 975
- B.** 9,075
- C.** 1,975
- D.** 75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 10 • BACKWARD 4 • Q4/5

Subtract.

$$6,203 - 2,875$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 10 • BACKWARD 4 • Q5/5

SET 10 • 6.NS.B.2

FORWARD 1

Long division: fractions to decimals • Grade 7 • The Number System •
7.NS.A.2.d

Use long division to write $\frac{5}{11}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.d • Set 10 • FORWARD 1 • Q1/5

Use long division to write $\frac{3}{8}$ as a decimal.

Decimal: _____

Which fraction has a repeating decimal?

A. $\frac{2}{3}$

B. $\frac{3}{4}$

C. $\frac{7}{8}$

D. $\frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.d • Set 10 • FORWARD 1 • Q3/5

True or false?

The decimal form of $\frac{7}{20}$ terminates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.d • Set 10 • FORWARD 1 • Q4/5

Use long division to write $\frac{5}{6}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.d • Set 10 • FORWARD 1 • Q5/5

SET 10 • 6.NS.B.2

FORWARD 2

Repeating decimals and rational numbers • Grade 8 • The Number System •
8.NS.A.1

Write $0.272727\ldots$ as a fraction in simplest form.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.1 • Set 10 • FORWARD 2 • Q1/5

Write 0.444... as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.1 • Set 10 • FORWARD 2 • Q2/5

Which number is irrational?

A. $\sqrt{7}$

B. $0.333\ldots$

C. $\frac{5}{8}$

D. $\sqrt{16}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.1 • Set 10 • FORWARD 2 • Q3/5

True or false?

Every repeating decimal is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.1 • Set 10 • FORWARD 2 • Q4/5

Write 1.1666... as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.1 • Set 10 • FORWARD 2 • Q5/5

Add and subtract multi-digit decimals

MAIN	Grade 6 main question	6.NS.B.3
BACKWARD 1	Place value of decimal digits	5.NBT.A.3.a
BACKWARD 2	Equivalent decimals	5.NBT.A.3.b
BACKWARD 3	Lining up decimal points	5.NBT.B.7
BACKWARD 4	Money in decimal notation	4.NF.C.6
BACKWARD 5	Multi-digit addition and subtraction	4.NBT.B.4
FORWARD 1	Add and subtract rational numbers	7.NS.A.1.d
FORWARD 2	Linear equations with decimal coefficients	8.EE.C.7.b

SET 11 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Find the sum.

$$315.4 + 28.75 + 6.309$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • MAIN • Q1/5

Find the difference.

$$52.3 - 18.47$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • MAIN • Q2/5

Jon put this example on the board.

What mistake was made?

What should the sum be?

$$\begin{array}{r} 315.2 \\ 24.8 \\ +7.936 \\ \hline 11.336 \end{array}$$

Mistake: _____

Correct sum: _____

A runner's times were 12.45 seconds and 11.9 seconds.
How much faster was the second time?

- A.** 0.55 second
- B.** 1.45 seconds
- C.** 0.45 second
- D.** 11.26 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • MAIN • Q4/5

Sam has \$50. He buys a book for \$12.89 and a pen for \$3.75.

How much money does he have left?

Money left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • MAIN • Q5/5

SET 11 • 6.NS.B.3

BACKWARD 1

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 7.358, what place value does the digit 5 have?

- A.** Tenths
- B.** Hundredths
- C.** Thousandths
- D.** Ones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 11 • BACKWARD 1 • Q1/5

In 42.906, which digit is in the tenths place?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 11 • BACKWARD 1 • Q2/5

True or false?

In 3.47, the digit 7 is in the hundredths place.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 11 • BACKWARD 1 • Q3/5

What is the value of the 8 in 0.584?

- A.** 0.08
- B.** 0.8
- C.** 0.008
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 11 • BACKWARD 1 • Q4/5

Write $5 + 0.3 + 0.07$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 6.NS.B.3

BACKWARD 2

Equivalent decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

True or false?

2.4 and 2.40 have the same value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 11 • BACKWARD 2 • Q1/5

True or false?

5.6 and 5.06 have the same value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 11 • BACKWARD 2 • Q2/5

Which decimal is equal to 7.5?

- A.** 7.50
- B.** 7.05
- C.** 7.005
- D.** 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 11 • BACKWARD 2 • Q3/5

Write 24.5 with a zero in the hundredths place.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 11 • BACKWARD 2 • Q4/5

True or false?

$$0.300 = 0.3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 6.NS.B.3

BACKWARD 3

Lining up decimal points • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Which setup lines up the decimal points for $36.2 + 4.58$?

A.

$$\begin{array}{r} 36.2 \\ +4.58 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 36.20 \\ + 4.58 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 3 • Q1/5

Which setup lines up the decimal points for $5.07 + 12.3$?

A.

$$\begin{array}{r} 5.07 \\ +12.30 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 5.07 \\ +12.3 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 3 • Q2/5

True or false?

The decimal points are lined up correctly.

- A. True
- B. False

$$\begin{array}{r} 8.4 \\ +2.75 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 3 • Q3/5

Which setup lines up the decimal points for $15.6 - 3.42$?

A.

$$\begin{array}{r} 15.60 \\ - 3.42 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 15.6 \\ - 3.42 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 3 • Q4/5

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 9.30 \\ + 0.45 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 3 • Q5/5

SET 11 • 6.NS.B.3

BACKWARD 4

Money in decimal notation • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

A pen costs 2 dollars and 9 cents.

Write the price using a dollar sign and a decimal point.

Price: _____

A snack costs 75 cents.

Write the price using a dollar sign and a decimal point.

Price: _____

True or false?

Six dollars and forty cents is written as \$6.04.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 4 • Q3/5

Which shows 12 dollars and 30 cents?

A. \$12.30

B. \$12.03

C. \$1.230

D. \$123.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 4 • Q4/5

A toy costs 8 dollars and 50 cents.

Write the price using a dollar sign and a decimal point.

Price: _____

SET 11 • 6.NS.B.3

BACKWARD 5

Multi-digit addition and subtraction • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Subtract.

$$5,230 - 1,847$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 11 • BACKWARD 5 • Q1/5

Add.

$$4,605 + 2,798$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 11 • BACKWARD 5 • Q2/5

True or false?

$$6,000 - 2,458 = 3,542$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 11 • BACKWARD 5 • Q3/5

$$8,014 - 3,276 = ?$$

A. 4,738

B. 5,262

C. 4,838

D. 4,748

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 11 • BACKWARD 5 • Q4/5

Add.

$$1,975 + 3,489 + 612$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 11 • BACKWARD 5 • Q5/5

SET 11 • 6.NS.B.3

FORWARD 1

Add and subtract rational numbers • Grade 7 • The Number System •
7.NS.A.1.d

Find the value.

$$-18.6 + 4.35 =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.d • Set 11 • FORWARD 1 • Q1/5

Find the value.

$$7.2 - 12.85 =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.d • Set 11 • FORWARD 1 • Q2/5

Find the value.

$$-3.4 - (-9.15) =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.d • Set 11 • FORWARD 1 • Q3/5

Find the value.

$$-6.25 + (-2.8) = ?$$

A. -9.05

B. -3.45

C. 3.45

D. 9.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.d • Set 11 • FORWARD 1 • Q4/5

Find the value.

$$-15.75 + 20.3 =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.d • Set 11 • FORWARD 1 • Q5/5

SET 11 • 6.NS.B.3

FORWARD 2

Linear equations with decimal coefficients • Grade 8 • Expressions & Equations
• 8.EE.C.7.b

Solve for x.

$$-1.5x + 6.4 = 0.5x - 1.6$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q1/5

Solve for x.

$$3.6x - 2.5 = 1.6x + 4.5$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q2/5

Solve for x.

$$0.8x + 7.2 = -1.2x + 1.2$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q3/5

Solve for x.

$$-4.5x - 3.25 = -2.5x + 0.75$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q4/5

Solve for x.

$$2.4(x + 1.5) = 1.4x + 6.6$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q5/5

Multiply multi-digit decimals

- MAIN** Grade 6 main question
- BACKWARD 1** Multiplying whole numbers
- BACKWARD 2** Powers of 10 and the decimal point
- BACKWARD 3** Multiplying tenths and hundredths
- BACKWARD 4** Rounding decimals
- FORWARD 1** Multiply rational numbers
- FORWARD 2** Multiply numbers in scientific notation

6.NS.B.3

5.NBT.B.5

5.NBT.A.2

5.NBT.B.7

5.NBT.A.4

7.NS.A.2.c

8.EE.A.4

SET 12 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$3.6 \times 0.45$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 12 • MAIN • Q1/5

Multiply.

$$12.5 \times 3.8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 12 • MAIN • Q2/5

Gas costs \$3.40 per gallon.

How much do 12.5 gallons of gas cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 12 • MAIN • Q3/5

$$0.07 \times 0.6 = ?$$

- A.** 0.042
- B.** 0.42
- C.** 4.2
- D.** 0.0042

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 12 • MAIN • Q4/5

A rectangle is 4.25 meters long and 2.4 meters wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 12 • MAIN • Q5/5

SET 12 • 6.NS.B.3

BACKWARD 1

Multiplying whole numbers • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Multiply.

$$36 \times 45$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 12 • BACKWARD 1 • Q1/5

Multiply.

$$125 \times 38$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 12 • BACKWARD 1 • Q2/5

True or false?

$$425 \times 24 = 10,200$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 12 • BACKWARD 1 • Q3/5

$$34 \times 125 = ?$$

A. 4,250

B. 4,150

C. 425

D. 3,750

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 12 • BACKWARD 1 • Q4/5

Multiply.

$$208 \times 15$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 6.NS.B.3

BACKWARD 2

Powers of 10 and the decimal point • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$0.45 \times 10$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

Multiply.

$$3.6 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

True or false?

$$24 \div 10 = 2.4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

$$1,620 \div 1,000 = ?$$

- A.** 1.62
- B.** 16.2
- C.** 0.162
- D.** 162

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

Multiply.

$$47.5 \times 10$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 6.NS.B.3

BACKWARD 3

Multiplying tenths and hundredths • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

$$0.3 \times 0.2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q1/5

Multiply.

$$0.5 \times 0.8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q2/5

True or false?

$$0.4 \times 0.4 = 1.6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q3/5

$$0.2 \times 0.03 = ?$$

A. 0.006

B. 0.06

C. 0.6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q4/5

Multiply.

$$1.2 \times 0.3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • 6.NS.B.3

BACKWARD 4

Rounding decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 3.62 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 12 • BACKWARD 4 • Q1/5

Round 12.48 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 12 • BACKWARD 4 • Q2/5

True or false?

4.25 rounded to the nearest whole number is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 12 • BACKWARD 4 • Q3/5

Round 0.456 to the nearest hundredth.

A. 0.46

B. 0.45

C. 0.5

D. 0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 12 • BACKWARD 4 • Q4/5

Round 8.75 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 12 • BACKWARD 4 • Q5/5

SET 12 • 6.NS.B.3

FORWARD 1

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the value.

$$(-3.6)(0.45)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.c • Set 12 • FORWARD 1 • Q1/5

Find the value.

$$(-2.5)(-4.2)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.c • Set 12 • FORWARD 1 • Q2/5

$$-0.8 \times 1.5 = ?$$

- A.** -1.2
- B.** 1.2
- C.** -0.12
- D.** 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.c • Set 12 • FORWARD 1 • Q3/5

Find the value.

$$(-1.2)(-0.5)(-3)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.c • Set 12 • FORWARD 1 • Q4/5

True or false?

$$(-0.6)(0.7) = 0.42$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.c • Set 12 • FORWARD 1 • Q5/5

SET 12 • 6.NS.B.3

FORWARD 2

Multiply numbers in scientific notation • Grade 8 • Expressions & Equations •
8.EE.A.4

Multiply. Write the answer in scientific notation.

$$(3.6 \times 10^4)(4.5 \times 10^{-2})$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q1/5

Multiply. Write the answer in scientific notation.

$$(2.5 \times 10^3)(4 \times 10^5)$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q2/5

$$(1.2 \times 10^6)(3 \times 10^{-4}) = ?$$

- A.** 3.6×10^2
- B.** 3.6×10^{10}
- C.** 4.2×10^2
- D.** 3.6×10^{-2}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q3/5

A light-year is about 9.5×10^{12} km.

About how many kilometers are in 4.2 light-years? Write the answer in scientific notation.

Kilometers: _____

True or false?

$$(5 \times 10^3)(6 \times 10^2) = 3 \times 10^6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q5/5

Divide multi-digit decimals

MAIN

Grade 6 main question

6.NS.B.3

BACKWARD 1

Dividing by a two-digit number

5.NBT.B.6

BACKWARD 2

Using powers of 10 to shift the decimal point

5.NBT.A.2

BACKWARD 3

Dividing tenths and hundredths

5.NBT.B.7

FORWARD 1

Divide rational numbers

7.NS.A.2.b

FORWARD 2

Compare quantities with powers of 10

8.EE.A.3

SET 13 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$15.75 \div 0.35$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 13 • MAIN • Q1/5

Divide.

$$8.64 \div 2.4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 13 • MAIN • Q2/5

A rope is 12.6 meters long. It is cut into pieces that are each 0.45 meter long.

How many pieces are there?

Pieces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 13 • MAIN • Q3/5

$$0.72 \div 0.08 = ?$$

A. 9

B. 0.9

C. 90

D. 0.09

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 13 • MAIN • Q4/5

A car traveled 283.5 miles on 13.5 gallons of gas.
How many miles did it travel per gallon?

Miles per gallon: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.3 • Set 13 • MAIN • Q5/5

SET 13 • 6.NS.B.3

BACKWARD 1

Dividing by a two-digit number • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$1,575 \div 35$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 13 • BACKWARD 1 • Q1/5

Divide.

$$864 \div 24$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 13 • BACKWARD 1 • Q2/5

True or false?

$$1,260 \div 45 = 28$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 13 • BACKWARD 1 • Q3/5

$$672 \div 32 = ?$$

- A.** 21
- B.** 12
- C.** 210
- D.** 22

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 13 • BACKWARD 1 • Q4/5

Divide.

$$936 \div 18$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 6.NS.B.3

BACKWARD 2

Using powers of 10 to shift the decimal point • Grade 5 • Number & Operations
in Base Ten • 5.NBT.A.2

Which has the same quotient as $15.75 \div 0.35$?

A. $1,575 \div 35$

B. $157.5 \div 35$

C. $1,575 \div 3.5$

D. $15.75 \div 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 13 • BACKWARD 2 • Q1/5

True or false?

$8.64 \div 2.4$ has the same quotient as $86.4 \div 24$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 13 • BACKWARD 2 • Q2/5

Multiply.

$$0.35 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 13 • BACKWARD 2 • Q3/5

Multiply.

$$15.75 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 13 • BACKWARD 2 • Q4/5

$$0.08 \times \underline{\quad} = 8$$

A. 100

B. 10

C. 1,000

D. 0.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 6.NS.B.3

BACKWARD 3

Dividing tenths and hundredths • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

Divide.

$$2.4 \div 0.6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 3 • Q1/5

Divide.

$$3.5 \div 0.5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 3 • Q2/5

True or false?

$$1.2 \div 0.3 = 0.4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 3 • Q3/5

$$0.9 \div 0.3 = ?$$

A. 3

B. 0.3

C. 30

D. 0.03

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 3 • Q4/5

Divide.

$$4.8 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 3 • Q5/5

SET 13 • 6.NS.B.3

FORWARD 1

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the value.

$$(-15.75) \div 0.35$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.b • Set 13 • FORWARD 1 • Q1/5

Find the value.

$$-8.64 \div (-2.4)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.b • Set 13 • FORWARD 1 • Q2/5

True or false?

$$-(12 \div 4) = (-12) \div 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.b • Set 13 • FORWARD 1 • Q3/5

$$7.2 \div (-0.9) = ?$$

A. -8

B. 8

C. -0.8

D. 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.b • Set 13 • FORWARD 1 • Q4/5

The temperature fell 13.5°F in 4.5 hours at a constant rate.

What was the change in temperature per hour?

Change per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.b • Set 13 • FORWARD 1 • Q5/5

SET 13 • 6.NS.B.3

FORWARD 2

Compare quantities with powers of 10 • Grade 8 • Expressions & Equations •
8.EE.A.3

The population of a city is about 3×10^6 . The population of a town is about 6×10^4 .

How many times as large is the city's population?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 13 • FORWARD 2 • Q1/5

About how many times as large is 9×10^8 as 3×10^5 ?

- A.** 3,000
- B.** 300
- C.** 30,000
- D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 13 • FORWARD 2 • Q2/5

Estimate 0.00047 as a single digit times an integer power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 13 • FORWARD 2 • Q3/5

An elephant has a mass of about 6×10^3 kg. An ant has a mass of about 3×10^{-6} kg.

How many times as large is the elephant's mass?

Times as large: _____

True or false?

8×10^7 is 20 times as large as 4×10^6 .

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 13 • FORWARD 2 • Q5/5

Greatest common factor and least common multiple

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Factors	4.OA.B.4
BACKWARD 2	Multiples	4.OA.B.4
BACKWARD 3	Prime and composite numbers	4.OA.B.4
FORWARD 1	Add and subtract rational numbers with unlike denominators	7.NS.A.1.d
FORWARD 2	Equations with fractions: clearing denominators	8.EE.C.7.b

SET 14 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 14 • MAIN • Q1/5

Find the least common multiple of 6 and 8.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 14 • MAIN • Q2/5

Hot dogs come in packs of 10. Buns come in packs of 8.

What is the least number of hot dogs you can buy to have the same number of hot dogs and buns?

Hot dogs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 14 • MAIN • Q3/5

What is the greatest common factor of 18 and 45?

- A.** 9
- B.** 3
- C.** 90
- D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 14 • MAIN • Q4/5

A teacher has 42 pencils and 56 erasers. She makes identical kits and uses every item.

What is the greatest number of kits she can make?

Kits: _____

SET 14 • 6.NS.B.4

BACKWARD 1

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factors of 18.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 1 • Q1/5

True or false?

7 is a factor of 42.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 1 • Q2/5

Which number is NOT a factor of 36?

A. 8

B. 9

C. 12

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 1 • Q3/5

True or false?

5 is a factor of 32.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 1 • Q4/5

List all the factors of 24.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 6.NS.B.4

BACKWARD 2

Multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five multiples of 6.

Multiples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 2 • Q1/5

True or false?

36 is a multiple of 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 2 • Q2/5

Which number is a multiple of both 4 and 6?

A. 12

B. 18

C. 8

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 2 • Q3/5

True or false?

45 is a multiple of 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 2 • Q4/5

What is the 4th multiple of 7?

Multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 6.NS.B.4

BACKWARD 3

Prime and composite numbers • Grade 4 • Operations & Algebraic Thinking •
4.OA.B.4

True or false?

13 is a prime number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 3 • Q1/5

Which number is composite?

A. 21

B. 17

C. 11

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 3 • Q2/5

True or false?

1 is a prime number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 3 • Q3/5

Is 29 prime or composite?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 3 • Q4/5

Which number is prime?

A. 31

B. 27

C. 33

D. 39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • 6.NS.B.4

FORWARD 1

Add and subtract rational numbers with unlike denominators • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-\frac{5}{6} + \frac{3}{8}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.d • Set 14 • FORWARD 1 • Q1/5

Find the value.

$$\frac{3}{4} - \left(-\frac{1}{6}\right)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.d • Set 14 • FORWARD 1 • Q2/5

$$-\frac{2}{3} - \frac{1}{4} = ?$$

A. $-\frac{11}{12}$

B. $-\frac{1}{12}$

C. $\frac{5}{12}$

D. $-\frac{3}{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.d • Set 14 • FORWARD 1 • Q3/5

Find the value.

$$-1\frac{1}{2} + \frac{5}{6}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.d • Set 14 • FORWARD 1 • Q4/5

True or false?

$$\frac{3}{10} + \left(-\frac{2}{15}\right) = \frac{1}{6}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.d • Set 14 • FORWARD 1 • Q5/5

SET 14 • 6.NS.B.4

FORWARD 2

Equations with fractions: clearing denominators • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$\frac{x}{6} + \frac{x}{8} = 7$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q1/5

Solve for x.

$$\frac{x}{4} - 1 = \frac{x}{6} + 2$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q2/5

Solve for x.

$$\frac{2x}{3} + \frac{x}{2} = 14$$

A. 12

B. 7

C. 42

D. $\frac{84}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{x}{5} + 3 = \frac{x}{2}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q4/5

Solve for x.

$$\frac{(x+1)}{4} = \frac{(x-2)}{3}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q5/5

Use the distributive property to factor a sum with the GCF

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Factors of a number	4.OA.B.4
BACKWARD 2	Distributive property with whole numbers	3.OA.B.5
BACKWARD 3	Parentheses and order of operations	5.OA.A.1
FORWARD 1	Factor and expand linear expressions	7.EE.A.1
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 15 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write $36 + 8$ as a product of the GCF and a sum of two whole numbers.

Expression: _____

Which expression is equal to $42 + 56$ and uses the greatest common factor?

- A.** $14(3 + 4)$
- B.** $7(6 + 8)$
- C.** $2(21 + 28)$
- D.** $14(3 + 8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 15 • MAIN • Q2/5

Use the greatest common factor to write $27 + 45$ as a product.

Expression: _____

True or false?

$$24 + 60 = 12(2 + 5)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 15 • MAIN • Q4/5

Complete the equation.

$$30 + 75 = 15(\underline{\quad} + \underline{\quad})$$

First blank: _____

Second blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 15 • MAIN • Q5/5

SET 15 • 6.NS.B.4

BACKWARD 1

Factors of a number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

True or false?

4 is a factor of both 36 and 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 15 • BACKWARD 1 • Q1/5

List the factors of 12 and the factors of 18. Which factors do they share?

Shared factors: _____

Which number is a factor of both 27 and 45?

- A.** 9
- B.** 5
- C.** 6
- D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 15 • BACKWARD 1 • Q3/5

List all the factors of 42.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 15 • BACKWARD 1 • Q4/5

True or false?

6 is a factor of 56.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 6.NS.B.4

BACKWARD 2

Distributive property with whole numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$8 \times 7 = 8 \times 5 + 8 \times 2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 15 • BACKWARD 2 • Q1/5

True or false?

$$4 \times (9 + 2) = 4 \times 9 + 2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 15 • BACKWARD 2 • Q2/5

Which is equal to 6×12 ?

A. $6 \times 10 + 6 \times 2$

B. $6 \times 10 + 2$

C. $6 + 10 \times 2$

D. $6 \times 10 \times 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 15 • BACKWARD 2 • Q3/5

Fill in the blank.

$$5 \times 13 = 5 \times 10 + 5 \times \underline{\quad}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 15 • BACKWARD 2 • Q4/5

Fill in the blank.

$$3 \times (4 + 5) = 3 \times 4 + 3 \times \underline{\hspace{1cm}}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 6.NS.B.4

BACKWARD 3

Parentheses and order of operations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$4 \times (9 + 2)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 15 • BACKWARD 3 • Q1/5

Evaluate.

$$14 \times (3 + 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 15 • BACKWARD 3 • Q2/5

True or false?

$$9 \times (3 + 5) = 32$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 15 • BACKWARD 3 • Q3/5

$$12 \times (2 + 5) = ?$$

A. 84

B. 29

C. 17

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 15 • BACKWARD 3 • Q4/5

Evaluate.

$$(20 - 5) \times 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • 6.NS.B.4

FORWARD 1

Factor and expand linear expressions • Grade 7 • Expressions & Equations •
7.EE.A.1

Factor.

$$12x - 18$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 15 • FORWARD 1 • Q1/5

Expand.

$$-3(2x - 5)$$

Expanded form: _____

Which expression is equivalent to $8y + 20$?

- A.** $4(2y + 5)$
- B.** $8(y + 20)$
- C.** $2(4y + 20)$
- D.** $4(2y + 20)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 15 • FORWARD 1 • Q3/5

Factor.

$$15a + 10b - 5$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 15 • FORWARD 1 • Q4/5

True or false?

$$\frac{1}{2}(6x - 4) = 3x - 2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 6.NS.B.4

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

How many solutions does the equation have?

$$4(x + 3) = 4x + 12$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 15 • FORWARD 2 • Q1/5

How many solutions does the equation have?

$$2(3x - 1) = 6x + 5$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 15 • FORWARD 2 • Q2/5

How many solutions does the equation have?

$$3(x + 2) = 2x + 9$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 15 • FORWARD 2 • Q3/5

What number goes in the box so that the equation has infinitely many solutions?

$$5(x - 2) = 5x - \square$$

$$\square = \underline{\hspace{10cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 15 • FORWARD 2 • Q4/5

Which equation has no solution?

A. $2(x + 4) = 2x + 5$

B. $2(x + 4) = 2x + 8$

C. $2(x + 4) = x + 8$

D. $2(x + 4) = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 15 • FORWARD 2 • Q5/5

Positive and negative numbers in real-world contexts

MAIN	Grade 6 main question	6.NS.C.5
BACKWARD 1	Whole numbers as distances from 0	2.MD.B.6
BACKWARD 2	Reading a vertical scale	4.MD.A.2
BACKWARD 3	Increase and decrease situations	2.OA.A.1
FORWARD 1	Adding integers as moving on a number line	7.NS.A.1.b
FORWARD 2	Negative rate of change in a linear function	8.F.B.4

SET 16 • 6.NS.C.5

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.5

A diver is 30 feet below sea level.

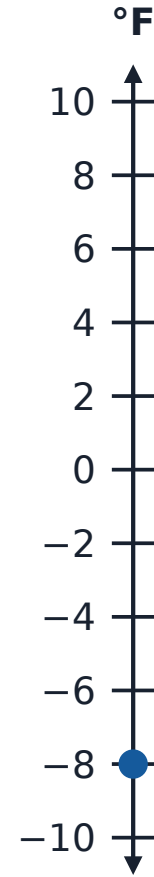
Write an integer to represent the diver's position.

Integer: _____

The thermometer shows the temperature in °F.

What is the temperature?

Temperature: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • MAIN • Q2/5

A bank account shows a withdrawal of \$45.
Which number represents this change?

- A.** -45
- B.** 45
- C.** 0
- D.** -4.5

True or false?

When integers describe elevation, 0 represents sea level.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 16 • MAIN • Q4/5

A hawk flies 120 feet above sea level. A submarine is 120 feet below sea level.

Write an integer for each position.

Hawk: _____

Submarine: _____

SET 16 • 6.NS.C.5

BACKWARD 1

Whole numbers as distances from 0 • Grade 2 • Measurement & Data •
2.MD.B.6

What number does point A represent?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 16 • BACKWARD 1 • Q1/5

How many units from 0 is the number 6?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 16 • BACKWARD 1 • Q2/5

True or false?

On a number line, 9 is farther from 0 than 4 is.

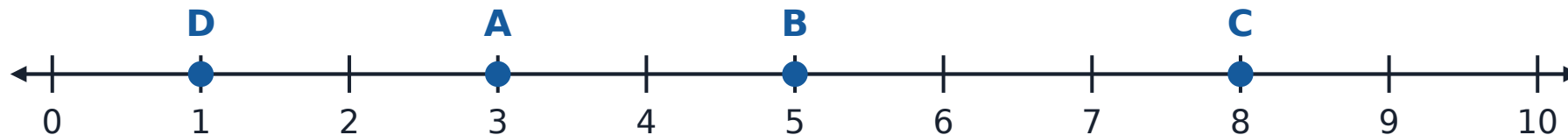
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 16 • BACKWARD 1 • Q3/5

Which point is 3 units from 0?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 16 • BACKWARD 1 • Q4/5

What number does point B represent?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 16 • BACKWARD 1 • Q5/5

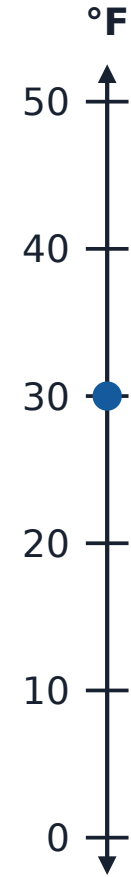
SET 16 • 6.NS.C.5

BACKWARD 2

Reading a vertical scale • Grade 4 • Measurement & Data • 4.MD.A.2

What temperature does the thermometer show?

Temperature: _____

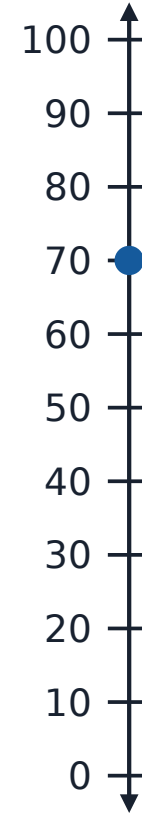


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 16 • BACKWARD 2 • Q1/5

What number does the point show?

Number: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 16 • BACKWARD 2 • Q2/5

True or false?

The point is at 40.

A. True

B. False

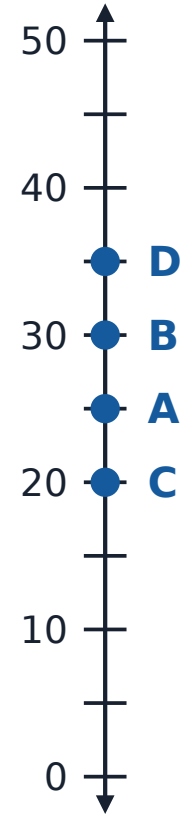


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 16 • BACKWARD 2 • Q3/5

Which point shows 25?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

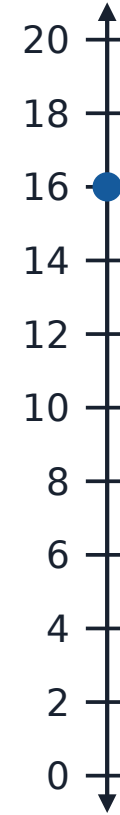


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 16 • BACKWARD 2 • Q4/5

What number does the point show?

Number: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 6.NS.C.5

BACKWARD 3

Increase and decrease situations • Grade 2 • Operations & Algebraic Thinking •
2.OA.A.1

Kim had \$20. She spent \$8.

How much money does she have now?

Money: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 16 • BACKWARD 3 • Q1/5

A plant was 12 cm tall. It grew 5 cm.

How tall is the plant now?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 16 • BACKWARD 3 • Q2/5

Which situation describes a decrease?

- A.** A balloon loses 6 feet of height.
- B.** A plant grows 4 inches.
- C.** Tom earns \$10.
- D.** A team gains 5 yards.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 16 • BACKWARD 3 • Q3/5

True or false?

"The temperature fell 7 degrees" describes a decrease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 16 • BACKWARD 3 • Q4/5

Which word describes an increase in money?

- A.** Deposit
- B.** Withdrawal
- C.** Loss
- D.** Drop

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • 6.NS.C.5

FORWARD 1

Adding integers as moving on a number line • Grade 7 • The Number System •
7.NS.A.1.b

The temperature is -4°F . It rises 9°F .

What is the new temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.b • Set 16 • FORWARD 1 • Q1/5

A submarine is at -120 feet. It rises 45 feet.
What is its new position?

Position: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

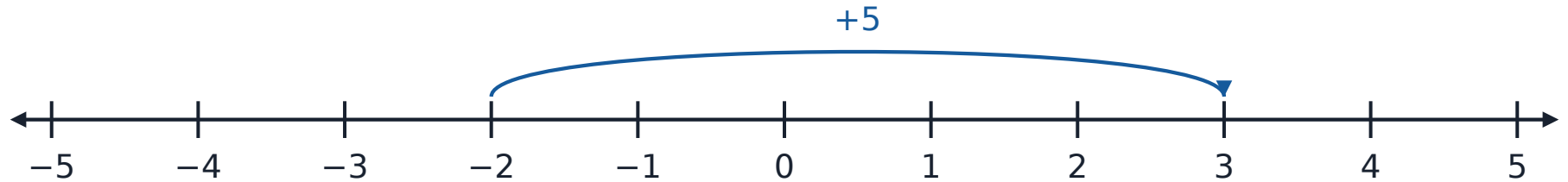
Grade 7 • The Number System • 7.NS.A.1.b • Set 16 • FORWARD 1 • Q2/5

On a number line, where is $6 + (-4)$ compared to 6?

- A.** 4 units to the left of 6
- B.** 4 units to the right of 6
- C.** 6 units to the left of 4
- D.** At 0

The number line shows $-2 + 5$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.b • Set 16 • FORWARD 1 • Q4/5

True or false?

A \$30 debt plus a \$30 deposit makes a balance of \$0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.b • Set 16 • FORWARD 1 • Q5/5

SET 16 • 6.NS.C.5

FORWARD 2

Negative rate of change in a linear function • Grade 8 • Functions • 8.F.B.4

A tank holds 60 gallons. It drains 3 gallons per minute.
Write a function for the amount A after t minutes.

$A =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 2 • Q1/5

An airplane at 12,000 feet descends 800 feet per minute.

What is the rate of change of its altitude?

- A.** –800 feet per minute
- B.** 800 feet per minute
- C.** 12,000 feet per minute
- D.** –12,000 feet per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 2 • Q2/5

The table shows a linear function.

What is the rate of change in degrees per hour?

Hours	0	2	4
Temperature (°F)	5	1	-3

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 2 • Q3/5

A diver starts at -10 feet and descends 4 feet per second.

Write a function for the diver's position d after t seconds.

$d =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 2 • Q4/5

True or false?

A line through $(0, 8)$ and $(4, 0)$ has a rate of change of -2 .

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 2 • Q5/5

Opposite signs as opposite sides of 0; the opposite of the opposite

MAIN	Grade 6 main question	6.NS.C.6.a
BACKWARD 1	Distance from 0 on a number line	2.MD.B.6
BACKWARD 2	Positive and negative numbers in context	6.NS.C.5
FORWARD 1	Opposite quantities combine to make 0	7.NS.A.1.a
FORWARD 2	Solve equations using inverse terms	8.EE.C.7.b

SET 17 • 6.NS.C.6.a

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of -7 ?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 17 • MAIN • Q1/5

Simplify.

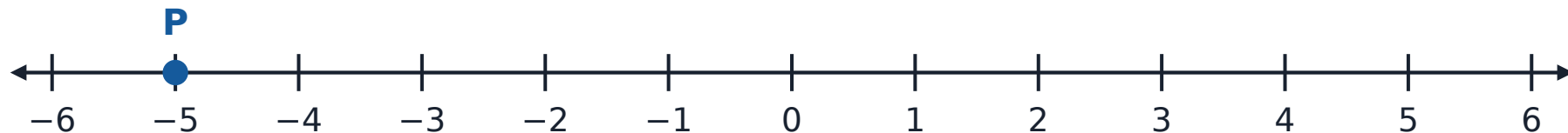
$$-(-12)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 17 • MAIN • Q2/5

Which number is the opposite of the number at point P?



A. 5

B. -5

C. 0

D. $\frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 17 • MAIN • Q3/5

True or false?

0 is its own opposite.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 17 • MAIN • Q4/5

What is the opposite of the opposite of 9?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 17 • MAIN • Q5/5

SET 17 • 6.NS.C.6.a

BACKWARD 1

Distance from 0 on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

How far is point A from 0?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 17 • BACKWARD 1 • Q1/5

How many units from 0 is the number 8?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 17 • BACKWARD 1 • Q2/5

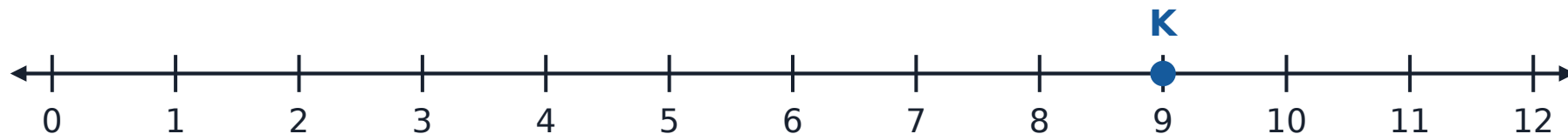
Which number is 2 units from 0?

- A.** 2
- B.** 0
- C.** 4
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 17 • BACKWARD 1 • Q3/5

What number does point K represent?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 17 • BACKWARD 1 • Q4/5

True or false?

The distance from 0 to 7 is 7 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 6.NS.C.6.a

BACKWARD 2

Positive and negative numbers in context • Grade 6 • The Number System •
6.NS.C.5

True or false?

–5 can represent 5 degrees below zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 17 • BACKWARD 2 • Q1/5

Write an integer for a gain of 12 yards.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 17 • BACKWARD 2 • Q2/5

Which integer represents 20 feet below sea level?

A. -20

B. 20

C. 0

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 17 • BACKWARD 2 • Q3/5

Write an integer for a loss of \$7.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 17 • BACKWARD 2 • Q4/5

True or false?

A deposit of \$15 is written as -15 .

A. True

B. False

SET 17 • 6.NS.C.6.a

FORWARD 1

Opposite quantities combine to make 0 • Grade 7 • The Number System •
7.NS.A.1.a

A hydrogen atom has 1 proton (charge $+1$) and 1 electron (charge -1).

What is its total charge?

Total charge: _____

True or false?

$$-8 + 8 = 0$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.a • Set 17 • FORWARD 1 • Q2/5

Which sum is equal to 0?

A. $-3.5 + 3.5$

B. $-3.5 + (-3.5)$

C. $3.5 + 3.5$

D. $-3.5 + 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.a • Set 17 • FORWARD 1 • Q3/5

You earn \$25 and then spend some money. The total change in your money is \$0.

How much did you spend?

Amount spent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.a • Set 17 • FORWARD 1 • Q4/5

What number added to 14.2 gives 0?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.a • Set 17 • FORWARD 1 • Q5/5

SET 17 • 6.NS.C.6.a

FORWARD 2

Solve equations using inverse terms • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$5x - 7 = 2x + 8$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q1/5

Solve for x.

$$-3x + 4 = 3x - 8$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q2/5

Solve for x.

$$6 - 2x = 4x - 12$$

A. 3

B. -3

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q3/5

Solve for x.

$$7x + 3 = -2x + 21$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q4/5

Solve for x.

$$4(x - 1) = 2x + 10$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q5/5

Signs of coordinates in the quadrants; reflections across the axes

MAIN	Grade 6 main question	6.NS.C.6.b
BACKWARD 1	Axes, origin, and ordered pairs	5.G.A.1
BACKWARD 2	Plotting points in the first quadrant	5.G.A.2
BACKWARD 3	Opposite numbers	6.NS.C.6.a
BACKWARD 4	Lines of symmetry	4.G.A.3
FORWARD 1	Interpret points on the graph of a proportional relationship	7.RP.A.2.d
FORWARD 2	Transformations with coordinates	8.G.A.3

SET 18 • 6.NS.C.6.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.b

In which quadrant is the point $(-4, 7)$?

- A.** Quadrant II
- B.** Quadrant I
- C.** Quadrant III
- D.** Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.b • Set 18 • MAIN • Q1/5

The point $(3, -5)$ is reflected across the x -axis.
What are the coordinates of its image?

Image: _____

The point $(-2, 6)$ is reflected across the y -axis.
What are the coordinates of its image?

Image: _____

How are the points (a, b) and $(a, -b)$ related when b is not 0?

- A.** They are reflections of each other across the x-axis.
- B.** They are reflections of each other across the y-axis.
- C.** They are the same point.
- D.** They are always in the same quadrant.

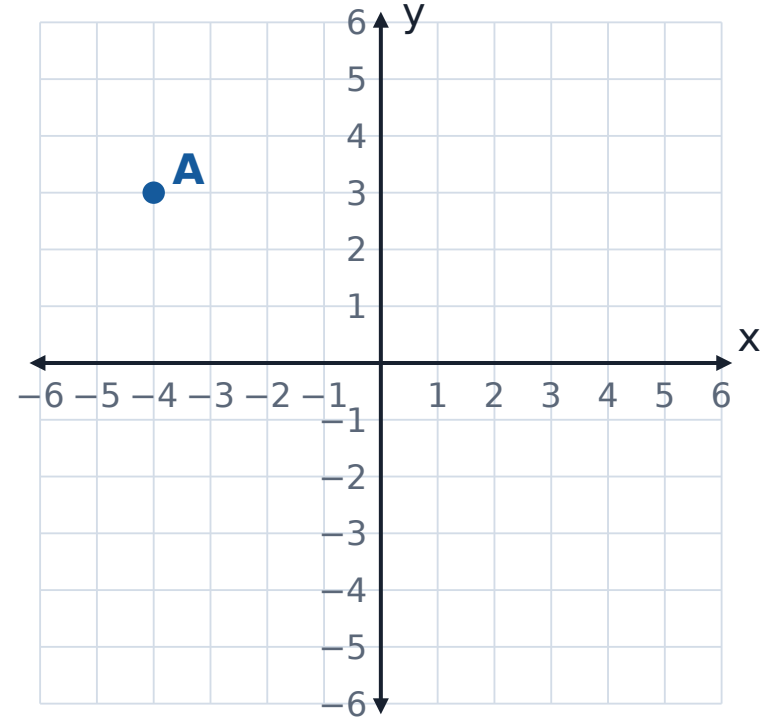
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.b • Set 18 • MAIN • Q4/5

Point A is reflected across the y-axis.

What are the coordinates of its image?

Image: _____



SET 18 • 6.NS.C.6.b

BACKWARD 1

Axes, origin, and ordered pairs • Grade 5 • Geometry • 5.G.A.1

True or false?

The origin is the point $(0, 0)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.1 • Set 18 • BACKWARD 1 • Q1/5

In the ordered pair (5, 2), which number tells how far to move along the x-axis?

A. 5

B. 2

C. 7

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.1 • Set 18 • BACKWARD 1 • Q2/5

What is the name of the point where the x-axis and the y-axis meet?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.1 • Set 18 • BACKWARD 1 • Q3/5

True or false?

In the ordered pair $(3, 8)$, the y -coordinate is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.1 • Set 18 • BACKWARD 1 • Q4/5

Which axis is horizontal?

A. The x-axis

B. The y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.1 • Set 18 • BACKWARD 1 • Q5/5

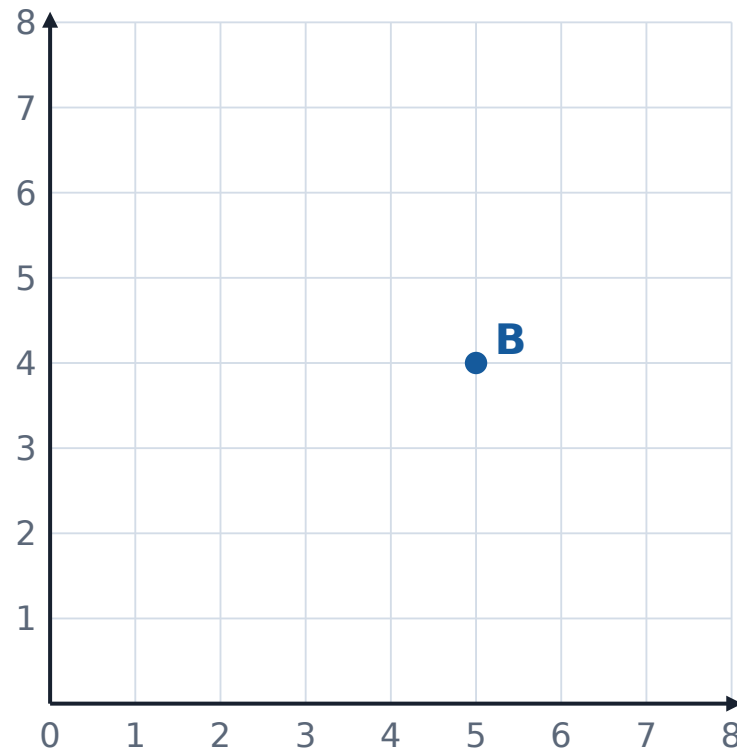
SET 18 • 6.NS.C.6.b

BACKWARD 2

Plotting points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point B?

B = _____

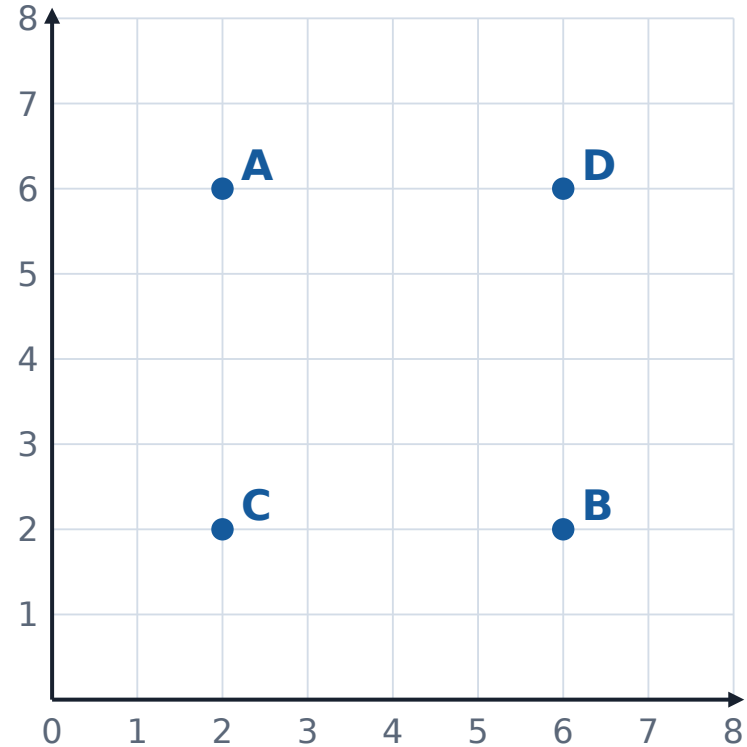


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 18 • BACKWARD 2 • Q1/5

Which point is located at (2, 6)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

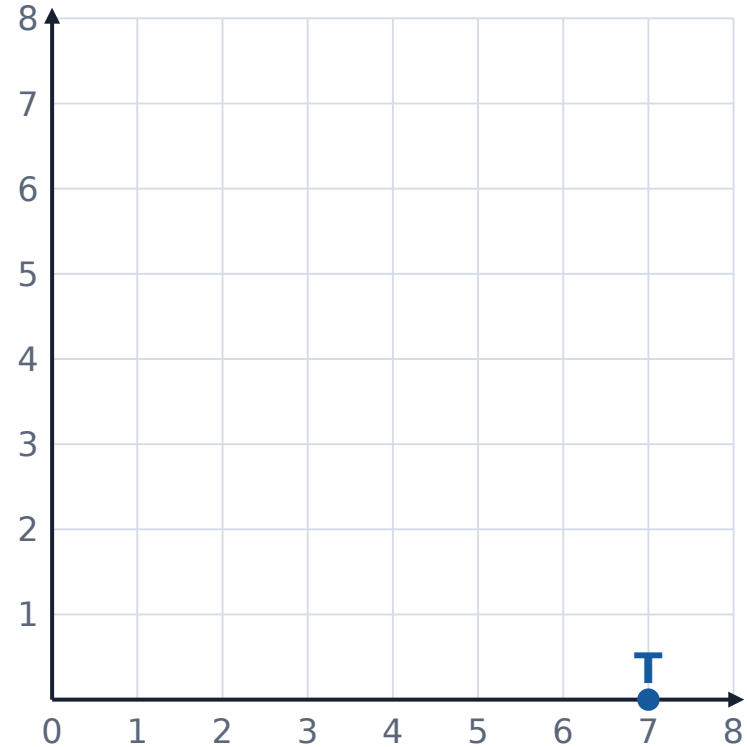


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 18 • BACKWARD 2 • Q2/5

What are the coordinates of point T?

T = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

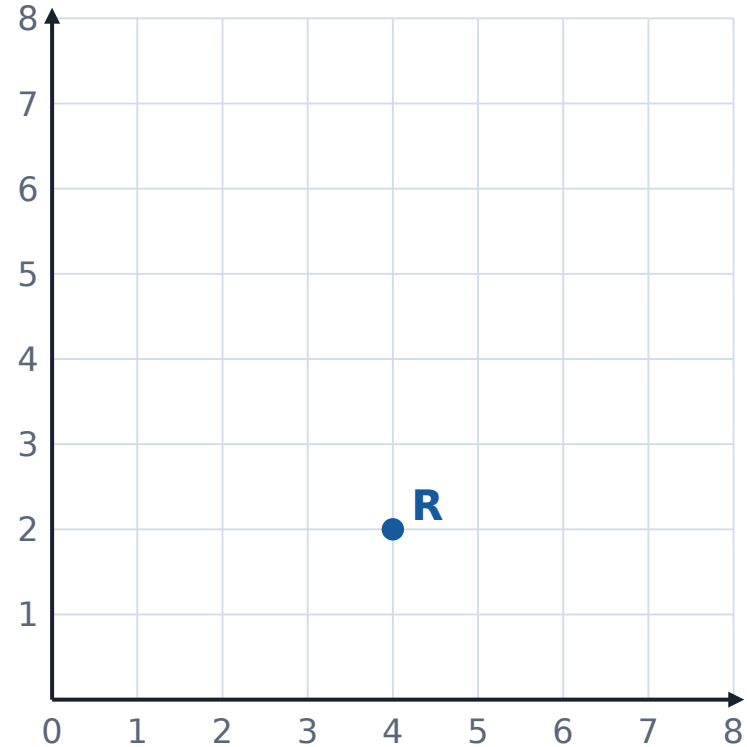
Grade 5 • Geometry • 5.G.A.2 • Set 18 • BACKWARD 2 • Q3/5

True or false?

Point R is located at (2, 4).

A. True

B. False

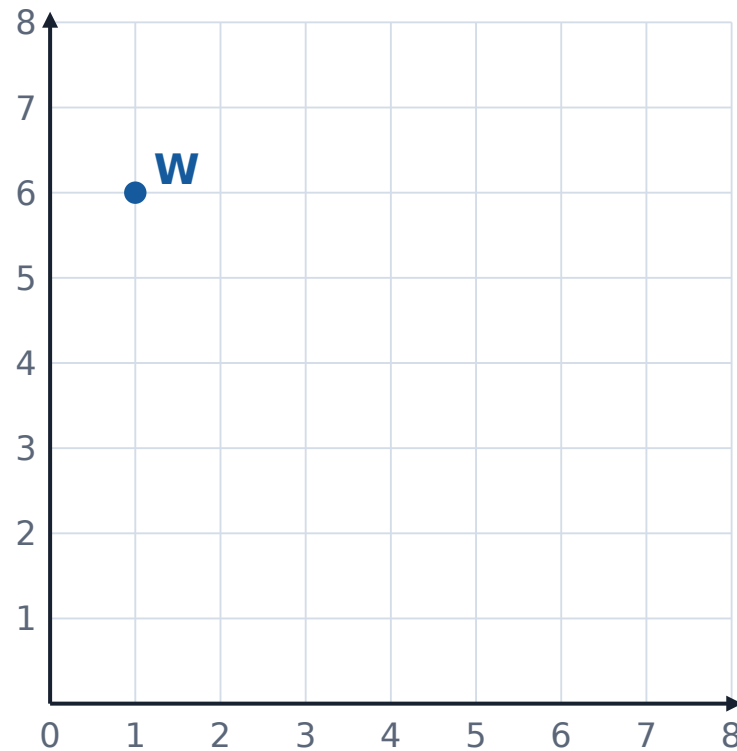


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 18 • BACKWARD 2 • Q4/5

What are the coordinates of point W?

W = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 6.NS.C.6.b

BACKWARD 3

Opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 6?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 18 • BACKWARD 3 • Q1/5

True or false?

The opposite of -9 is 9 .

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 18 • BACKWARD 3 • Q2/5

Which pair of numbers are opposites?

- A.** -4 and 4
- B.** -4 and -4
- C.** 4 and $\frac{1}{4}$
- D.** 0 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 18 • BACKWARD 3 • Q3/5

What is the opposite of -2.5 ?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 18 • BACKWARD 3 • Q4/5

True or false?

$$-(-3) = -3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 18 • BACKWARD 3 • Q5/5

SET 18 • 6.NS.C.6.b

BACKWARD 4

Lines of symmetry • Grade 4 • Geometry • 4.G.A.3

True or false?

A line of symmetry divides a figure into two matching halves.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.3 • Set 18 • BACKWARD 4 • Q1/5

How many lines of symmetry does a square have?

A. 4

B. 2

C. 1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.3 • Set 18 • BACKWARD 4 • Q2/5

True or false?

The dashed line is a line of symmetry of the rectangle.

A. True

B. False



Which letter has a vertical line of symmetry?

A. A

B. F

C. J

D. R

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.3 • Set 18 • BACKWARD 4 • Q4/5

How many lines of symmetry does a rectangle that is not a square have?

Lines: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.3 • Set 18 • BACKWARD 4 • Q5/5

SET 18 • 6.NS.C.6.b

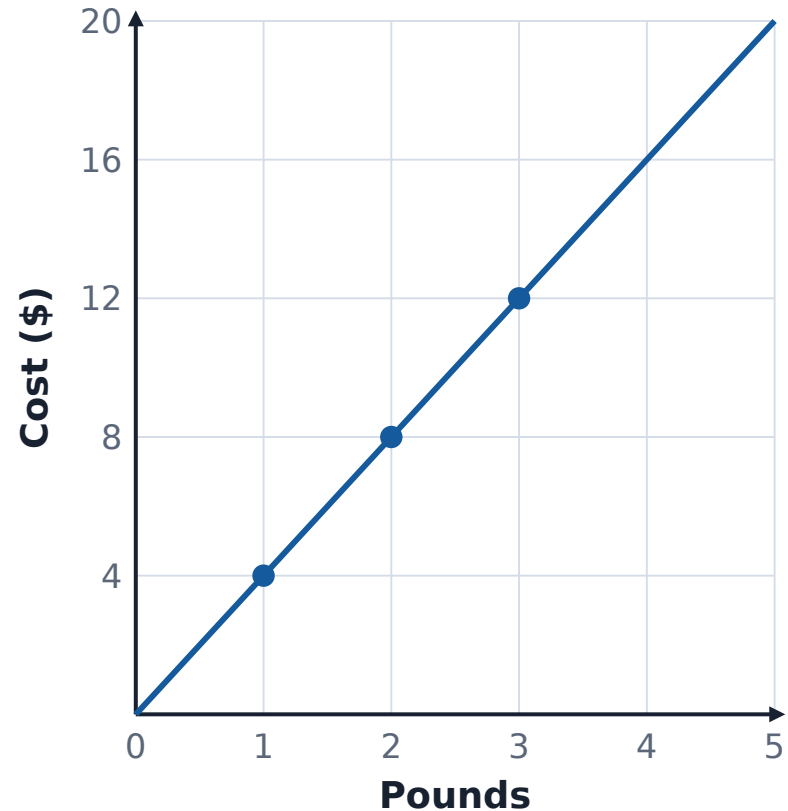
FORWARD 1

Interpret points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of apples.

What does the point (2, 8) represent?

Meaning: _____



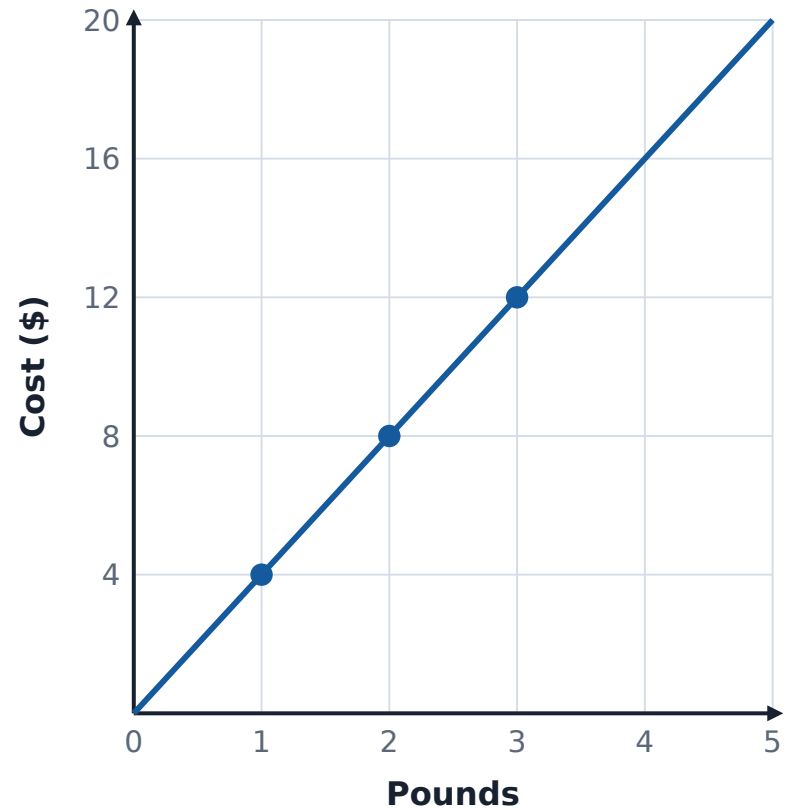
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 18 • FORWARD 1 • Q1/5

The graph shows the cost of apples.

What does the point (1, 4) represent?

- A.** 1 pound costs \$4.
- B.** 4 pounds cost \$1.
- C.** The cost starts at \$4.
- D.** \$1 buys 4 pounds.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 18 • FORWARD 1 • Q2/5

True or false?

On the graph of a proportional relationship between pounds and cost, the point $(0, 0)$ means 0 pounds cost \$0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 18 • FORWARD 1 • Q3/5

The graph of a proportional relationship passes through $(1, r)$ and $(5, 30)$.

What is r ?

$r =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 18 • FORWARD 1 • Q4/5

A graph shows the distance y in miles after x hours. It passes through $(3, 150)$.

What does this point mean?

- A.** In 3 hours, the distance is 150 miles.
- B.** In 150 hours, the distance is 3 miles.
- C.** The speed is 3 miles per hour.
- D.** The speed is 150 miles per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 18 • FORWARD 1 • Q5/5

SET 18 • 6.NS.C.6.b

FORWARD 2

Transformations with coordinates • Grade 8 • Geometry • 8.G.A.3

A triangle has a vertex at $(2, -3)$. The triangle is reflected across the y -axis.

What are the coordinates of the image of that vertex?

Image: _____

Which transformation is described by $(x, y) \rightarrow (x, -y)$?

- A.** A reflection across the x-axis
- B.** A reflection across the y-axis
- C.** A rotation of 180°
- D.** A translation

The point $(4, 1)$ is rotated 180° about the origin.
What are the coordinates of its image?

Image: _____

The point $(-3, 5)$ is translated 4 units right and 2 units down.

What are the coordinates of its image?

Image: _____

True or false?

Rotating $(2, 5)$ 90° counterclockwise about the origin gives $(-5, 2)$.

A. True

B. False

Find and position rational numbers on a number line

MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Fractions on a number line	3.NF.A.2
BACKWARD 2	Decimals on a number line	4.NF.C.6
BACKWARD 3	Opposites on a number line	6.NS.C.6.a
FORWARD 1	Adding rational numbers on a number line	7.NS.A.1.b
FORWARD 2	Approximate irrational numbers on a number line	8.NS.A.2

SET 19 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 19 • MAIN • Q1/5

Which point is located at $-2\frac{1}{4}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 19 • MAIN • Q2/5

Write the number at point Q as a mixed number.



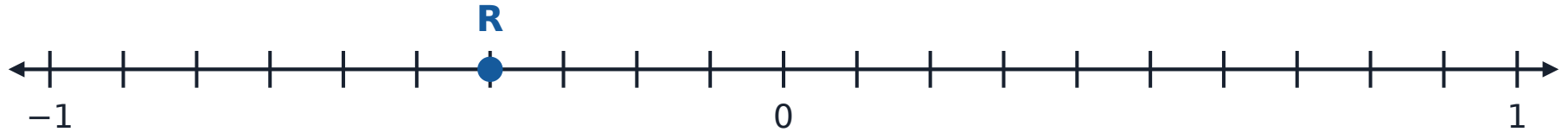
Q = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 19 • MAIN • Q3/5

True or false?

Point R is located at -0.6 .



A. True

B. False

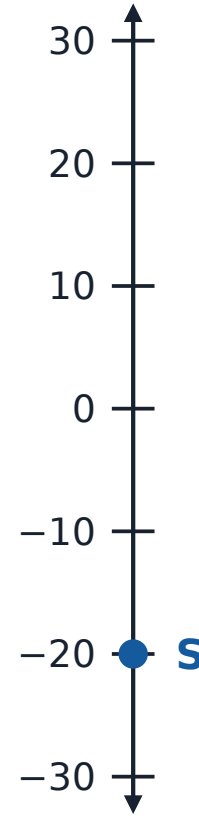
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 19 • MAIN • Q4/5

The number line shows
elevations in meters.

What elevation does point S
show?

S = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 19 • MAIN • Q5/5

SET 19 • 6.NS.C.6.c

BACKWARD 1

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction does point A represent?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 19 • BACKWARD 1 • Q1/5

What fraction does point B represent?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 19 • BACKWARD 1 • Q2/5

True or false?

Point C is located at $\frac{5}{6}$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 19 • BACKWARD 1 • Q3/5

The space from 0 to 1 is split into equal parts.
How many equal parts are there?



A. 8

B. 7

C. 9

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 19 • BACKWARD 1 • Q4/5

Write the number at point D as a fraction.



D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 6.NS.C.6.c

BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

What decimal does point A represent?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 2 • Q1/5

What decimal does point B represent?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 2 • Q2/5

True or false?

Point C is located at 0.3.



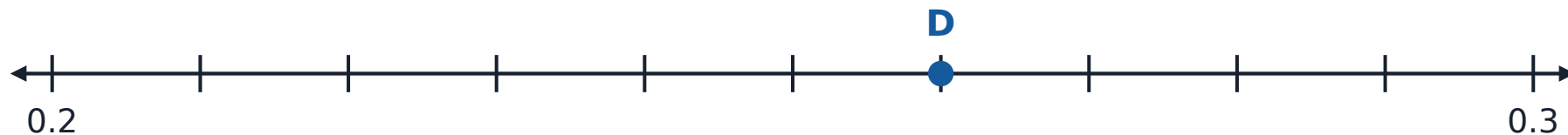
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 2 • Q3/5

What decimal does point D represent?



- A.** 0.26
- C.** 0.206

- B.** 2.6
- D.** 0.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 2 • Q4/5

What decimal does point E represent?



E = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 2 • Q5/5

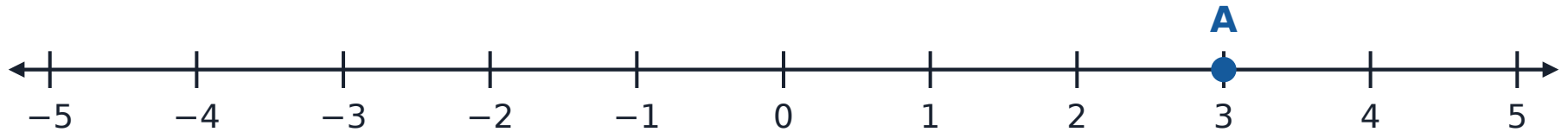
SET 19 • 6.NS.C.6.c

BACKWARD 3

Opposites on a number line • Grade 6 • The Number System • 6.NS.C.6.a

Point B (not shown) is the opposite of point A.

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 19 • BACKWARD 3 • Q1/5

True or false?

–2.5 and 2.5 are the same distance from 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 19 • BACKWARD 3 • Q2/5

What is the opposite of $1\frac{3}{4}$?

A. $-1\frac{3}{4}$

B. $1\frac{3}{4}$

C. $-\frac{4}{7}$

D. $\frac{4}{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 19 • BACKWARD 3 • Q3/5

What is the opposite of -0.8 ?

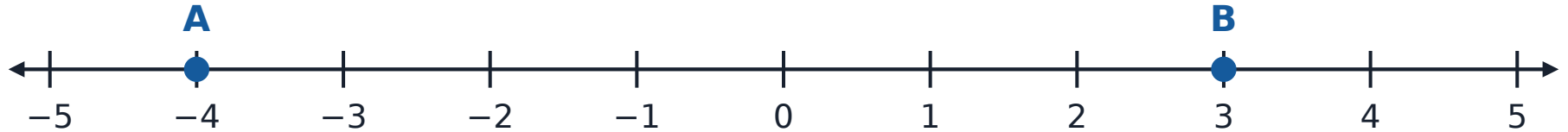
Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 19 • BACKWARD 3 • Q4/5

True or false?

Points A and B represent opposite numbers.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 19 • BACKWARD 3 • Q5/5

SET 19 • 6.NS.C.6.c

FORWARD 1

Adding rational numbers on a number line • Grade 7 • The Number System •
7.NS.A.1.b

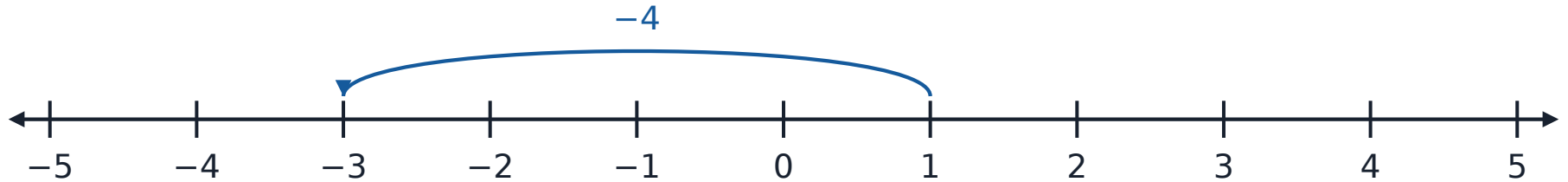
Start at -2.5 on a number line and move 4 units to the right.

Where do you end?

Number: _____

The number line shows $1 + (-4)$.

What is the sum?

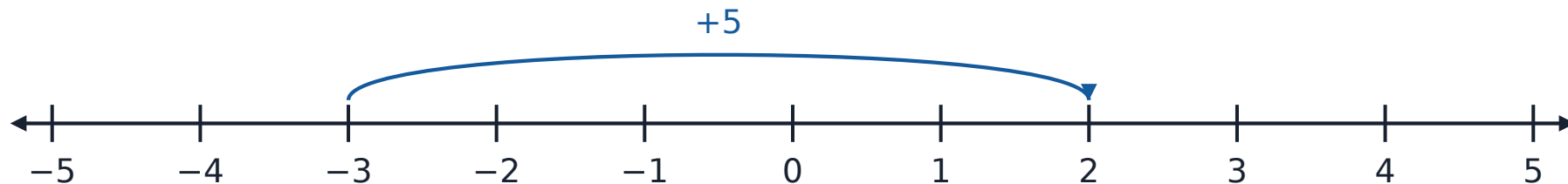


Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.b • Set 19 • FORWARD 1 • Q2/5

Which expression does the number line show?



A. $-3 + 5$

B. $5 + 3$

C. $-3 - 5$

D. $2 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.b • Set 19 • FORWARD 1 • Q3/5

Find the sum.

$$-1\frac{1}{2} + 3\frac{1}{4}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.b • Set 19 • FORWARD 1 • Q4/5

True or false?

On a number line, $-3 + 7$ is 7 units to the right of -3 .

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.b • Set 19 • FORWARD 1 • Q5/5

SET 19 • 6.NS.C.6.c

FORWARD 2

Approximate irrational numbers on a number line • Grade 8 • The Number System • 8.NS.A.2

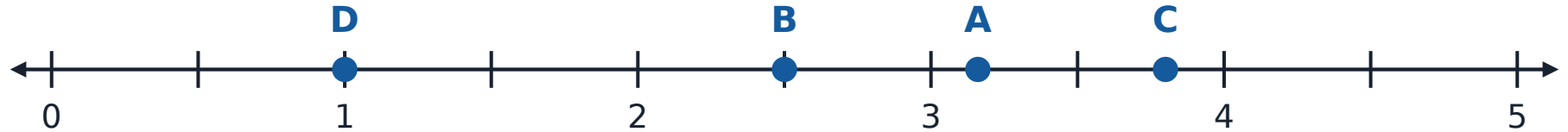
Between which two consecutive whole numbers is $\sqrt{20}$?

Between: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.2 • Set 19 • FORWARD 2 • Q1/5

Which point best shows $\sqrt{10}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.2 • Set 19 • FORWARD 2 • Q2/5

Estimate $\sqrt{50}$ to the nearest tenth.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.2 • Set 19 • FORWARD 2 • Q3/5

Which list is ordered from least to greatest?

A. $\sqrt{5}$, 2.5, $\sqrt{8}$

B. 2.5, $\sqrt{5}$, $\sqrt{8}$

C. $\sqrt{8}$, $\sqrt{5}$, 2.5

D. $\sqrt{5}$, $\sqrt{8}$, 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.2 • Set 19 • FORWARD 2 • Q4/5

True or false?

π is between 3.1 and 3.2 on the number line.

A. True

B. False

Find and position pairs of numbers on the coordinate plane

MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Ordered pairs and the axes	5.G.A.1
BACKWARD 2	Reading points in the first quadrant	5.G.A.2
BACKWARD 3	Quadrants and signs of coordinates	6.NS.C.6.b
FORWARD 1	Proportional relationships on a graph	7.RP.A.2.a
FORWARD 2	Functions as sets of ordered pairs	8.F.A.1

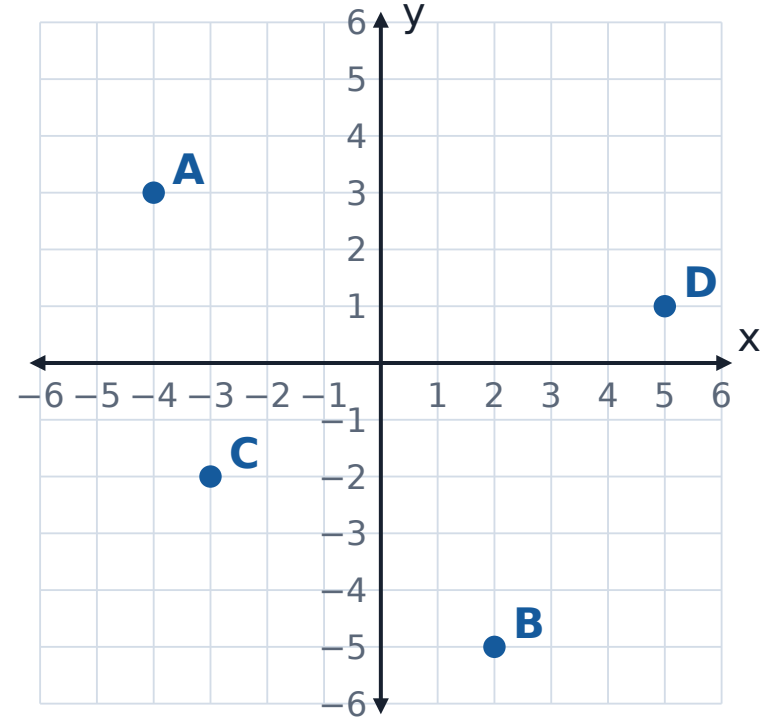
SET 20 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?

A = _____

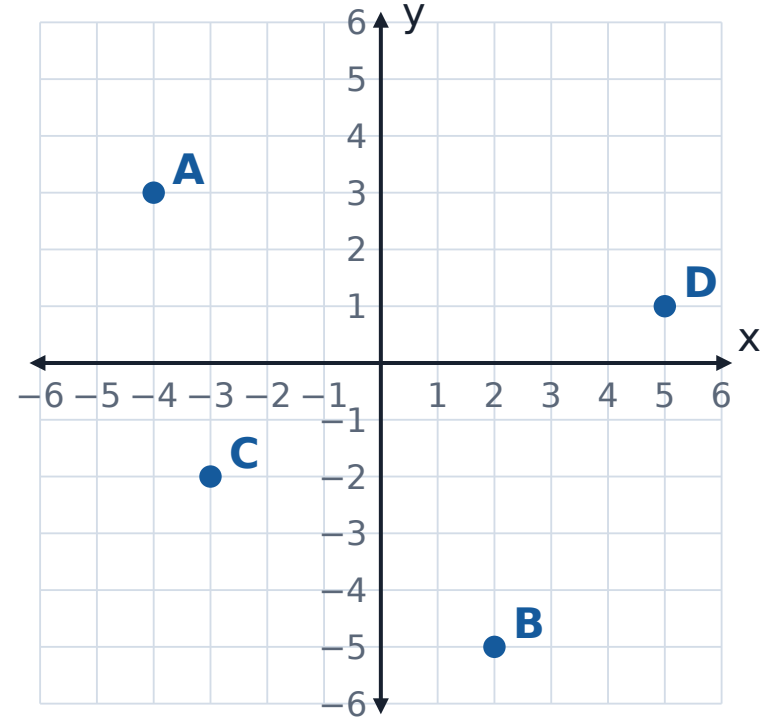


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 20 • MAIN • Q1/5

Which point is located at $(-3, -2)$?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

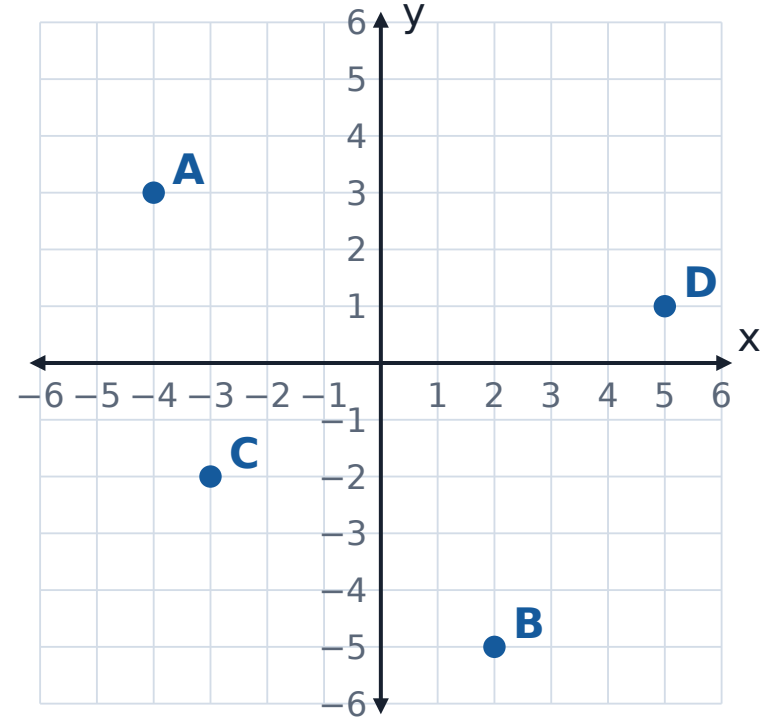


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 20 • MAIN • Q2/5

What are the coordinates of point B?

B = _____

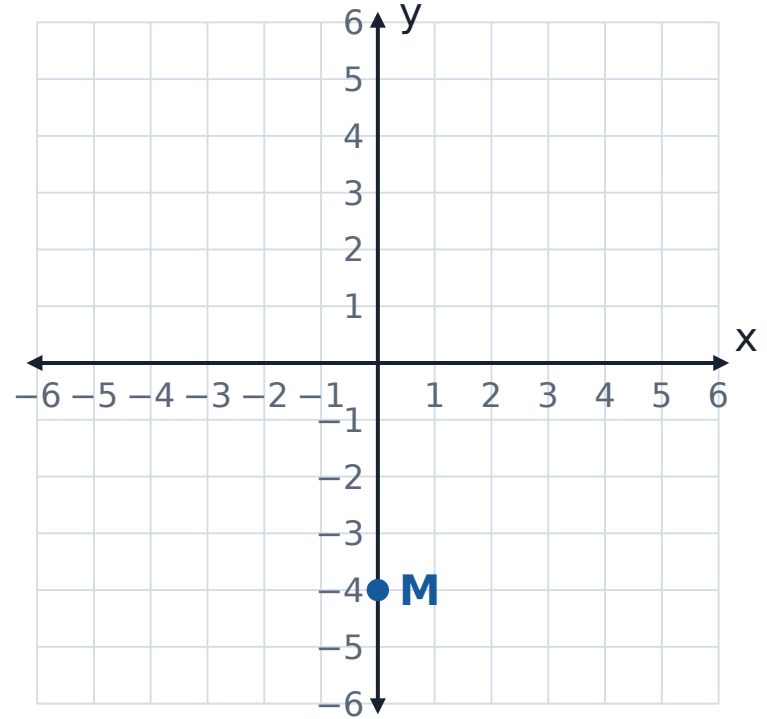


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 20 • MAIN • Q3/5

Which ordered pair names point M?

- A.** $(0, -4)$
- B.** $(-4, 0)$
- C.** $(4, 0)$
- D.** $(0, 4)$

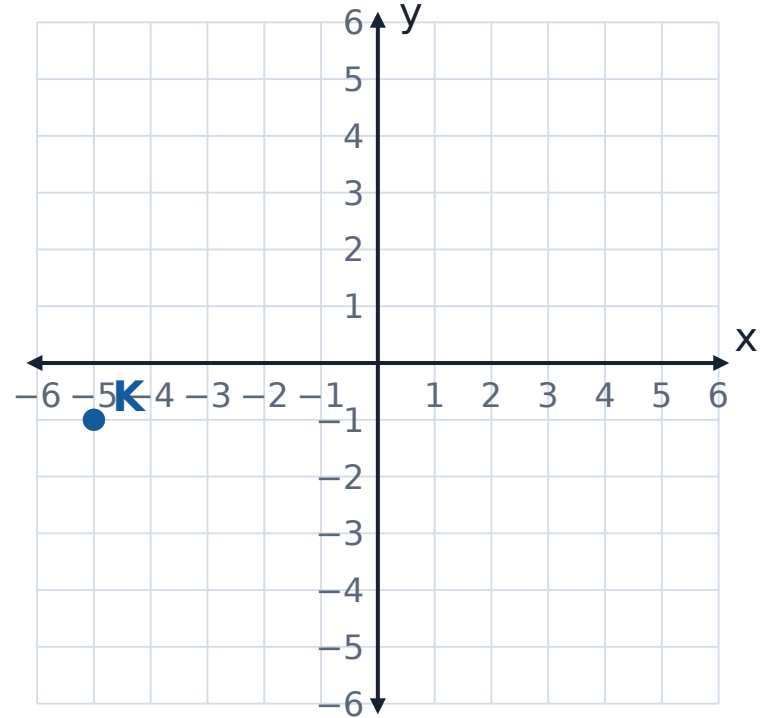


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 20 • MAIN • Q4/5

What are the coordinates of point K?

K = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 20 • MAIN • Q5/5

SET 20 • 6.NS.C.6.c

BACKWARD 1

Ordered pairs and the axes • Grade 5 • Geometry • 5.G.A.1

True or false?

The x-coordinate tells how far to move from the origin along the x-axis.

A. True

B. False

In the ordered pair $(7, 3)$, what is the y-coordinate?

A. 3

B. 7

C. 10

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.1 • Set 20 • BACKWARD 1 • Q2/5

What are the coordinates of the origin?

Origin: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.1 • Set 20 • BACKWARD 1 • Q3/5

True or false?

$(2, 5)$ and $(5, 2)$ name the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.1 • Set 20 • BACKWARD 1 • Q4/5

To plot (6, 1) starting from the origin, what do you do first?

- A.** Move 6 units right
- B.** Move 6 units up
- C.** Move 1 unit right
- D.** Move 1 unit up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.1 • Set 20 • BACKWARD 1 • Q5/5

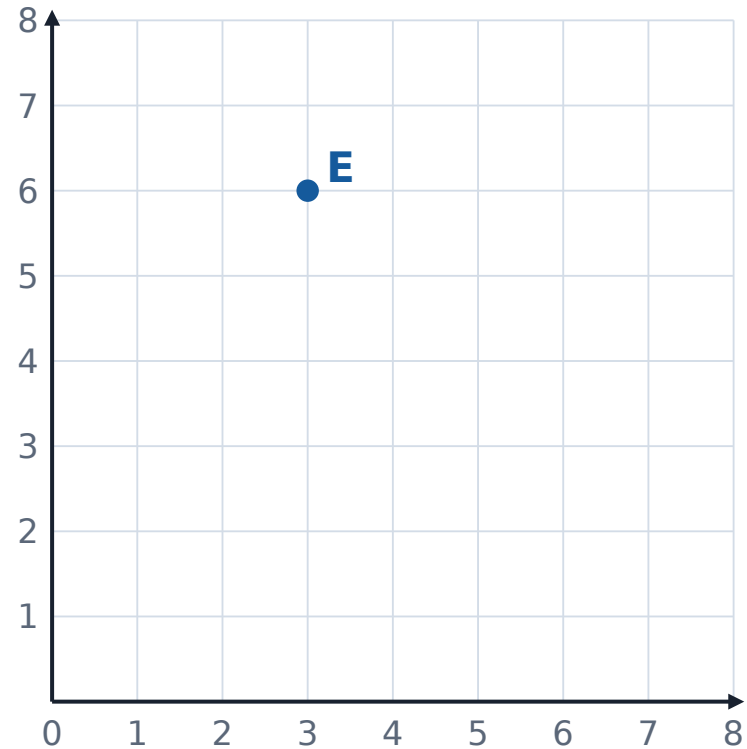
SET 20 • 6.NS.C.6.c

BACKWARD 2

Reading points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point E?

E = _____

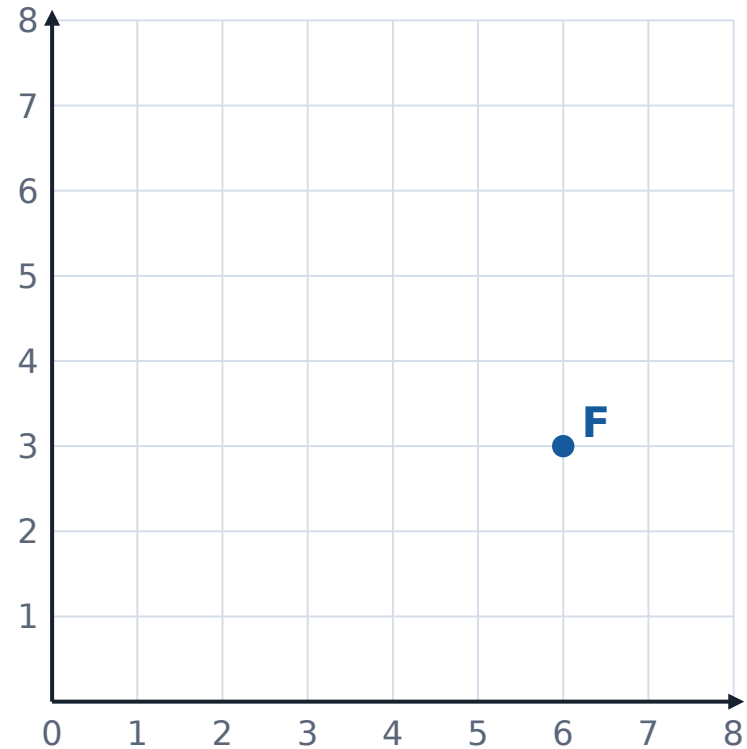


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 20 • BACKWARD 2 • Q1/5

What are the coordinates of point F?

F = _____

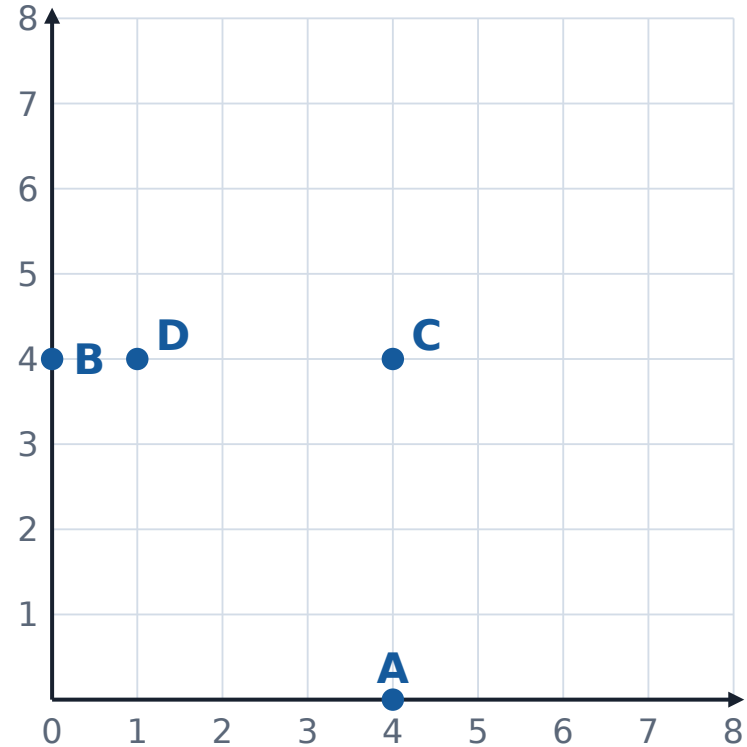


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 20 • BACKWARD 2 • Q2/5

Which point is located at (4, 0)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

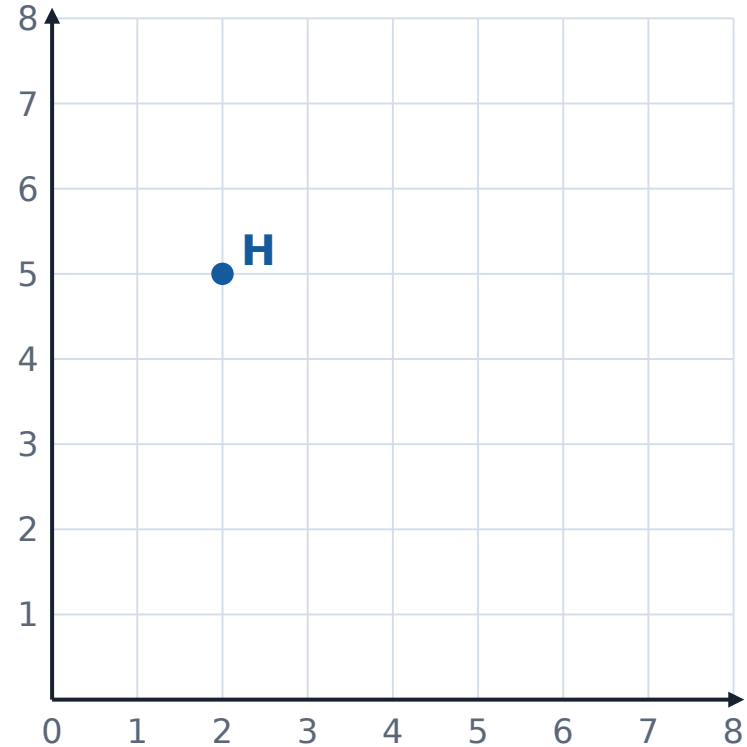
Grade 5 • Geometry • 5.G.A.2 • Set 20 • BACKWARD 2 • Q3/5

True or false?

Point H is located at (5, 2).

A. True

B. False

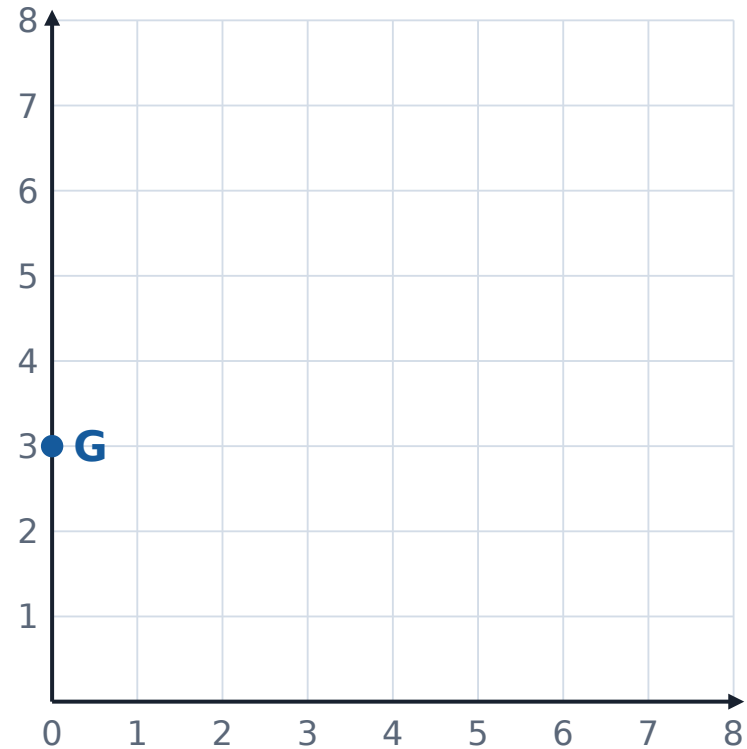


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 20 • BACKWARD 2 • Q4/5

What are the coordinates of point G?

G = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 6.NS.C.6.c

BACKWARD 3

Quadrants and signs of coordinates • Grade 6 • The Number System •
6.NS.C.6.b

In which quadrant are both coordinates negative?

- A.** Quadrant III
- B.** Quadrant I
- C.** Quadrant II
- D.** Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.b • Set 20 • BACKWARD 3 • Q1/5

In which quadrant is $(5, -2)$?

- A.** Quadrant IV
- B.** Quadrant I
- C.** Quadrant II
- D.** Quadrant III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.b • Set 20 • BACKWARD 3 • Q2/5

True or false?

$(-3, 4)$ is in Quadrant II.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.b • Set 20 • BACKWARD 3 • Q3/5

A point is in Quadrant IV.

Write "positive" or "negative" for each coordinate.

x-coordinate: _____

y-coordinate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.b • Set 20 • BACKWARD 3 • Q4/5

True or false?

The point $(0, -6)$ is in Quadrant III.

A. True

B. False

SET 20 • 6.NS.C.6.c

FORWARD 1

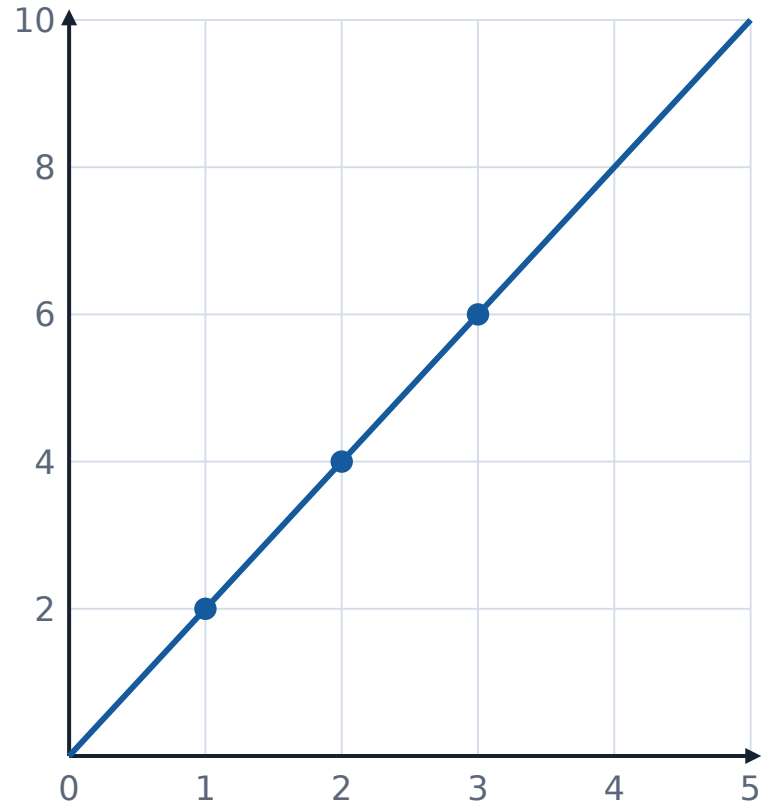
Proportional relationships on a graph • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The graph shows a proportional relationship.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 20 • FORWARD 1 • Q1/5

The points (1, 3), (2, 5), and (3, 7) are graphed.

Do they show a proportional relationship?

- A.** No, because the line through them does not pass through the origin.
- B.** Yes, because they form a straight line.
- C.** Yes, because y increases by 2.
- D.** No, because they are not on a line.

The table shows x-values 2, 4, 6 and y-values 5, 10, 15.
Would the graph be a straight line through the origin?
Write yes or no.

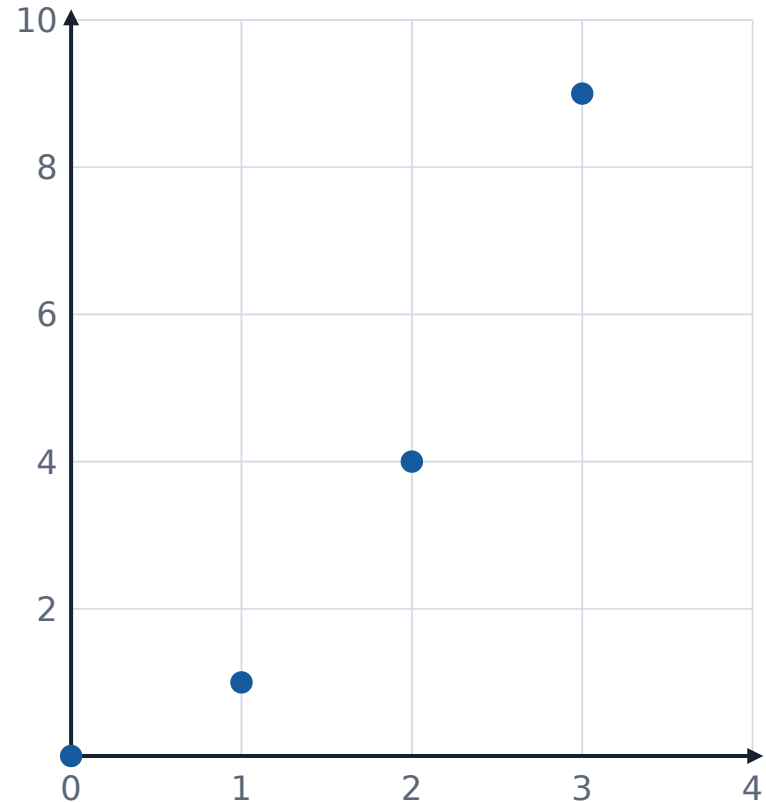
Answer: _____

True or false?

The graph shows a proportional relationship.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 20 • FORWARD 1 • Q4/5

Which set of points lies on a straight line through the origin?

A. (2, 3), (4, 6), (6, 9)

B. (1, 3), (2, 4), (3, 5)

C. (0, 2), (1, 4), (2, 6)

D. (1, 1), (2, 4), (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 20 • FORWARD 1 • Q5/5

SET 20 • 6.NS.C.6.c

FORWARD 2

Functions as sets of ordered pairs • Grade 8 • Functions • 8.F.A.1

True or false?

The set of ordered pairs $(1, 2)$, $(2, 4)$, $(3, 6)$, $(4, 8)$ represents a function.

A. True

B. False

Which set of ordered pairs does NOT represent a function?

- A.** $(2, 5), (2, 7), (3, 9)$
- B.** $(1, 4), (2, 4), (3, 4)$
- C.** $(0, 0), (1, 1), (2, 2)$
- D.** $(-1, 3), (0, 5), (1, 7)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.A.1 • Set 20 • FORWARD 2 • Q2/5

Does the set $(4, 1), (5, 2), (4, 3)$ represent a function?
Write yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

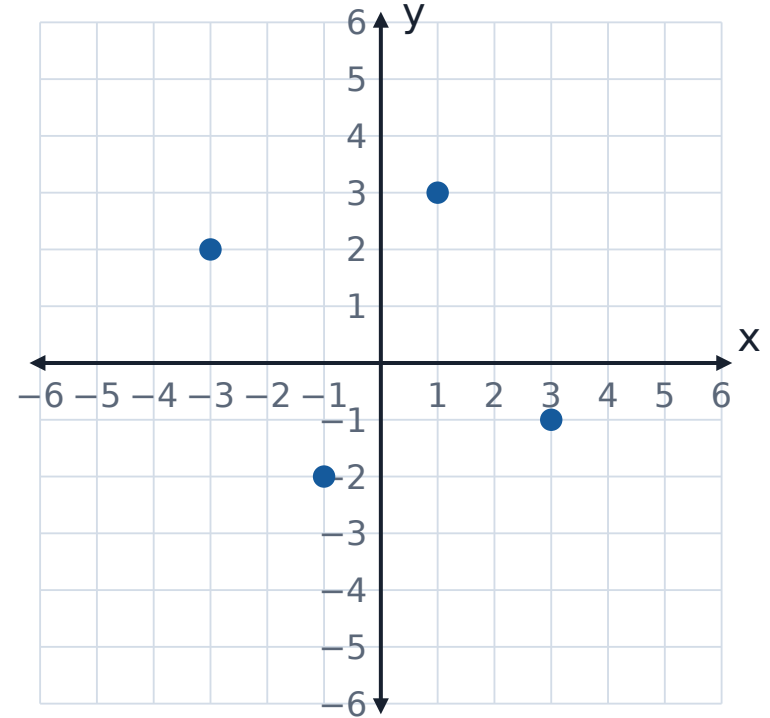
Grade 8 • Functions • 8.F.A.1 • Set 20 • FORWARD 2 • Q3/5

True or false?

The graph represents a function.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.A.1 • Set 20 • FORWARD 2 • Q4/5

A function assigns to each input ____.

- A.** exactly one output
- B.** at least two outputs
- C.** the same output as the input
- D.** no output

Interpret inequalities as relative position on a number line

MAIN	Grade 6 main question
BACKWARD 1	Comparison symbols
BACKWARD 2	Position on a number line
BACKWARD 3	Negative numbers on a number line
FORWARD 1	Solve and graph inequalities
FORWARD 2	Compare and order irrational numbers

6.NS.C.7.a

2.NBT.A.4

2.MD.B.6

6.NS.C.6.c

7.EE.B.4.b

8.NS.A.2

SET 21 • 6.NS.C.7.a

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.a

$$-3 > -7$$

Which statement explains this inequality?

- A.** -3 is to the right of -7 on a number line.
- B.** -3 is to the left of -7 on a number line.
- C.** 3 is less than 7 .
- D.** -3 is farther from 0 than -7 .

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 21 • MAIN • Q1/5

True or false?

The number line shows that $-6 > -1$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 21 • MAIN • Q2/5

Write an inequality that compares -4 and -9 .

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 21 • MAIN • Q3/5

Which statement is true?

A. $-5 < -2$

B. $-5 > -2$

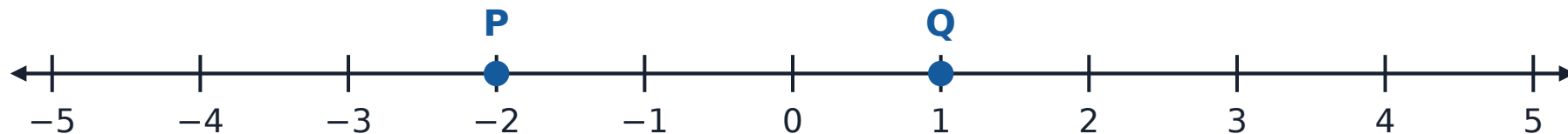
C. $-2 < -5$

D. $-5 > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 21 • MAIN • Q4/5

Write an inequality using the numbers at points P and Q.



Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 21 • MAIN • Q5/5

SET 21 • 6.NS.C.7.a

BACKWARD 1

Comparison symbols • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $<$, or $=$ to compare.

452 ____ 425

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 21 • BACKWARD 1 • Q1/5

True or false?

$$67 < 76$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 21 • BACKWARD 1 • Q2/5

Which statement is true?

A. $309 > 290$

B. $309 < 290$

C. $309 = 290$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 21 • BACKWARD 1 • Q3/5

Write $>$, $<$, or $=$ to compare.

138 183

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 21 • BACKWARD 1 • Q4/5

True or false?

500 > 505

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 6.NS.C.7.a

BACKWARD 2

Position on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

True or false?

On a number line, 8 is to the right of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 21 • BACKWARD 2 • Q1/5

Which number is farthest to the left on a number line?

A. 2

B. 7

C. 5

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 21 • BACKWARD 2 • Q2/5

Which point is farther to the left, A or B?



Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 21 • BACKWARD 2 • Q3/5

True or false?

Numbers get greater as you move to the left on a number line.

A. True

B. False

Which number is to the right of 14 on a number line?

A. 19

B. 11

C. 9

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 6.NS.C.7.a

BACKWARD 3

Negative numbers on a number line • Grade 6 • The Number System •
6.NS.C.6.c

What number is at point A?

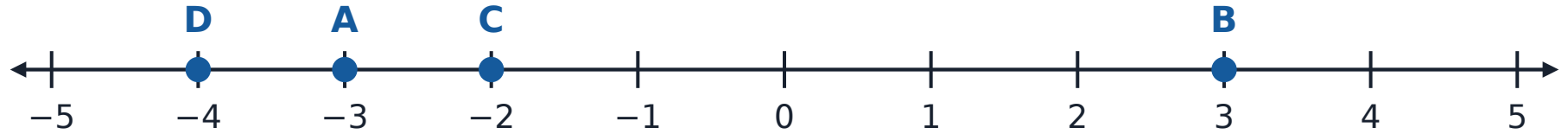


A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 21 • BACKWARD 3 • Q1/5

Which point is at -3 ?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 21 • BACKWARD 3 • Q2/5

True or false?

-4 is to the left of -1 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 21 • BACKWARD 3 • Q3/5

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 21 • BACKWARD 3 • Q4/5

Which point is at -5 ?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

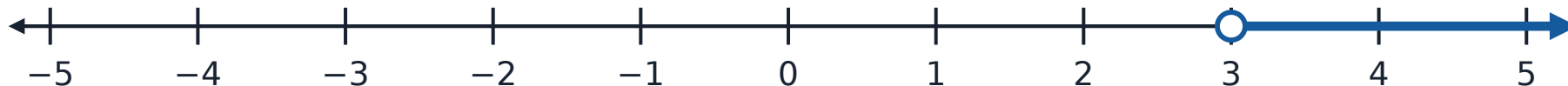
Grade 6 • The Number System • 6.NS.C.6.c • Set 21 • BACKWARD 3 • Q5/5

SET 21 • 6.NS.C.7.a

FORWARD 1

Solve and graph inequalities • Grade 7 • Expressions & Equations • 7.EE.B.4.b

Which inequality does the graph show?



A. $x > 3$

B. $x \geq 3$

C. $x < 3$

D. $x \leq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 21 • FORWARD 1 • Q1/5

Solve the inequality.

$$2x + 5 > 11$$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 21 • FORWARD 1 • Q2/5

Solve the inequality.

$$-3x + 4 \leq 16$$

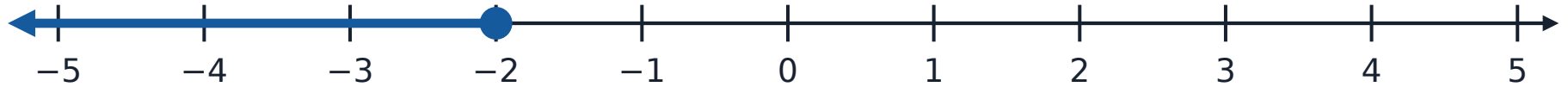
Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 21 • FORWARD 1 • Q3/5

True or false?

The graph shows the solutions of $4x - 1 \leq -9$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 21 • FORWARD 1 • Q4/5

Solve the inequality.

$$5x - 2 < 18$$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 21 • FORWARD 1 • Q5/5

SET 21 • 6.NS.C.7.a

FORWARD 2

Compare and order irrational numbers • Grade 8 • The Number System •
8.NS.A.2

Which is greater, $\sqrt{50}$ or 7.2?

Greater: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.2 • Set 21 • FORWARD 2 • Q1/5

Which number is between 3 and 4?

A. $\sqrt{12}$

B. $\sqrt{8}$

C. $\sqrt{17}$

D. $\sqrt{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.2 • Set 21 • FORWARD 2 • Q2/5

True or false?

$$\sqrt{30} < 5.5$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.2 • Set 21 • FORWARD 2 • Q3/5

Order from least to greatest.

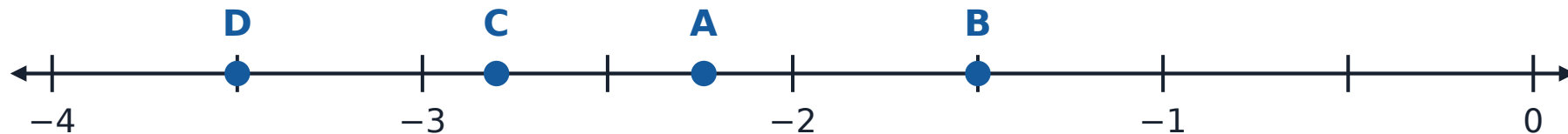
π , $\sqrt{8}$, 3.1

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.2 • Set 21 • FORWARD 2 • Q4/5

Which point best shows $-\sqrt{5}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • The Number System • 8.NS.A.2 • Set 21 • FORWARD 2 • Q5/5

Write, interpret, and explain order statements in real-world contexts

MAIN	Grade 6 main question	6.NS.C.7.b
BACKWARD 1	Negative numbers in context	6.NS.C.5
BACKWARD 2	Inequality symbols	1.NBT.B.3
BACKWARD 3	Order as position on a number line	6.NS.C.7.a
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Solve real-world problems with two linear equations	8.EE.C.8.c

SET 22 • 6.NS.C.7.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.b

A temperature of -3°C is warmer than a temperature of -7°C .

Write an inequality to show this.

Inequality: _____

Account A has a balance of $-\$25$. Account B has a balance of $-\$40$.

Which statement is true?

- A.** $-25 > -40$, so Account A has the greater balance.
- B.** $-40 > -25$, so Account B has the greater balance.
- C.** $-25 < -40$, so Account A owes more.
- D.** The balances are equal.

A fish swims at -12 feet. A turtle swims at -5 feet.

Write an inequality to compare the positions. Which animal is deeper?

Inequality: _____

Deeper animal: _____

True or false?

$-15 < -10$ means that -15°F is colder than -10°F .

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.b • Set 22 • MAIN • Q4/5

The lowest point in Death Valley is -86 meters. A beach is at 0 meters.

Which statement is true?

A. $-86 < 0$, so Death Valley's lowest point is lower.

B. $-86 > 0$, so Death Valley's lowest point is higher.

C. $0 < -86$, so the beach is lower.

D. $-86 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.b • Set 22 • MAIN • Q5/5

SET 22 • 6.NS.C.7.b

BACKWARD 1

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 12 degrees below zero.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 22 • BACKWARD 1 • Q1/5

True or false?

A debt of \$30 can be written as -30 .

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 22 • BACKWARD 1 • Q2/5

Which situation could be represented by -6 ?

- A.** A loss of 6 yards
- B.** A gain of 6 yards
- C.** 6 feet above sea level
- D.** A deposit of \$6

Write an integer for 50 feet below sea level.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 22 • BACKWARD 1 • Q4/5

True or false?

A temperature of 4 degrees above zero is written as -4 .

A. True

B. False

SET 22 • 6.NS.C.7.b

BACKWARD 2

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

$$42 \text{ ___ } 24$$

A. $>$

B. $<$

C. $=$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 22 • BACKWARD 2 • Q1/5

True or false?

$$15 > 51$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 22 • BACKWARD 2 • Q2/5

Write $>$, $<$, or $=$ to compare.

73 37

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 22 • BACKWARD 2 • Q3/5

Which symbol means "is less than"?

A. $<$

B. $>$

C. $=$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 22 • BACKWARD 2 • Q4/5

True or false?

$$89 < 98$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 6.NS.C.7.b

BACKWARD 3

Order as position on a number line • Grade 6 • The Number System •
6.NS.C.7.a

True or false?

$-2 > -6$ because -2 is to the right of -6 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 22 • BACKWARD 3 • Q1/5

Which number is greatest?

A. -1

B. -4

C. -9

D. -7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 22 • BACKWARD 3 • Q2/5

–5 is to the left of 3 on a number line.

Write an inequality to show this.

Inequality: _____

True or false?

-8 is to the right of -3 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 22 • BACKWARD 3 • Q4/5

Which number is to the left of -4 on a number line?

A. -6

B. -2

C. 0

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 22 • BACKWARD 3 • Q5/5

SET 22 • 6.NS.C.7.b

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

At 6 a.m., the temperature was -7°F . By noon, it was 12°F .

How many degrees did the temperature rise?

Degrees: _____

A submarine is at -250 feet. It rises 95 feet and then dives 130 feet.

What is its final position?

Position: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 1 • Q2/5

An account balance is $-\$42$. Then $\$65$ is deposited.

What is the new balance?

- A.** $\$23$
- B.** $-\$23$
- C.** $\$107$
- D.** $-\$107$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 1 • Q3/5

A mountain peak has an elevation of 1,250 m. An ocean trench has an elevation of $-3,400$ m.

What is the difference in elevation?

Difference: _____

True or false?

A temperature change from -3°C to -11°C is a change of -8°C .

A. True

B. False

SET 22 • 6.NS.C.7.b

FORWARD 2

Solve real-world problems with two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.c

Plan A costs \$20 plus \$5 per month. Plan B costs \$35 plus \$2 per month.

After how many months do the plans cost the same?

Months: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • FORWARD 2 • Q1/5

Tank A has 100 gallons and drains 6 gallons per minute.
Tank B has 40 gallons and fills 4 gallons per minute.

When will the tanks hold the same amount?

- A.** After 6 minutes
- B.** After 10 minutes
- C.** After 14 minutes
- D.** After 4 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • FORWARD 2 • Q2/5

Sara has \$50 and saves \$8 per week. Tom has \$90 and saves \$3 per week.

After how many weeks will they have the same amount?

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • FORWARD 2 • Q3/5

The sum of two numbers is 24. Their difference is 6.
What are the two numbers?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • FORWARD 2 • Q4/5

Adult tickets cost \$8 and child tickets cost \$5. A group buys 10 tickets for \$65.

How many adult tickets did they buy?

A. 5

B. 3

C. 7

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • FORWARD 2 • Q5/5

Absolute value as distance from 0

MAIN	Grade 6 main question	6.NS.C.7.c
BACKWARD 1	Distance on a number line	2.MD.B.6
BACKWARD 2	Opposite numbers	6.NS.C.6.a
BACKWARD 3	Negative numbers in context	6.NS.C.5
FORWARD 1	Distance between rational numbers as $ p - q $	7.NS.A.1.c
FORWARD 2	Square roots and cube roots as solutions	8.EE.A.2

SET 23 • 6.NS.C.7.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-18|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 23 • MAIN • Q1/5

An account balance is $-\$30$.

What is the absolute value of the balance, and what does it describe?

Absolute value: _____

It describes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 23 • MAIN • Q2/5

Which number has the greatest absolute value?

A. -12

B. 8

C. -5

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 23 • MAIN • Q3/5

A diver is at an elevation of -42 feet.

How far is the diver from sea level?

Distance: _____

True or false?

$$|-6| = |6|$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

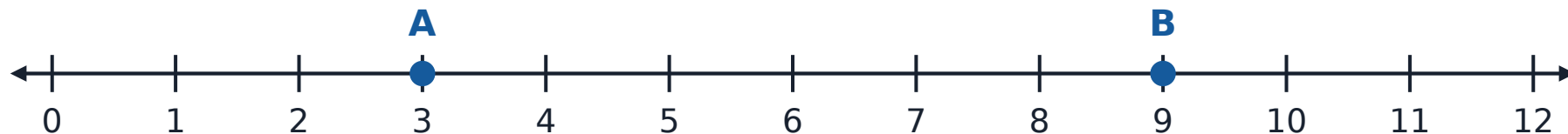
Grade 6 • The Number System • 6.NS.C.7.c • Set 23 • MAIN • Q5/5

SET 23 • 6.NS.C.7.c

BACKWARD 1

Distance on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

How many units apart are points A and B?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 1 • Q1/5

True or false?

The distance from 0 to 5 is the same as the distance from 5 to 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 1 • Q2/5

How many units from 0 is the number 11?

A. 11

B. 0

C. 1

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 1 • Q3/5

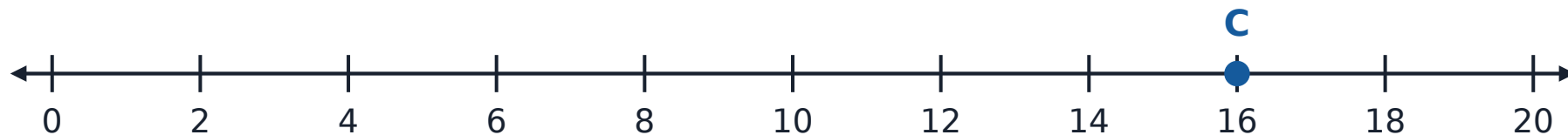
What is the distance from 4 to 10 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 1 • Q4/5

How many units is point C from 0?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 6.NS.C.7.c

BACKWARD 2

Opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 15?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 23 • BACKWARD 2 • Q1/5

True or false?

Opposite numbers are the same distance from 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 23 • BACKWARD 2 • Q2/5

Which number is the opposite of $-\frac{2}{3}$?

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. $\frac{3}{2}$

D. $-\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 23 • BACKWARD 2 • Q3/5

What is the opposite of 0?

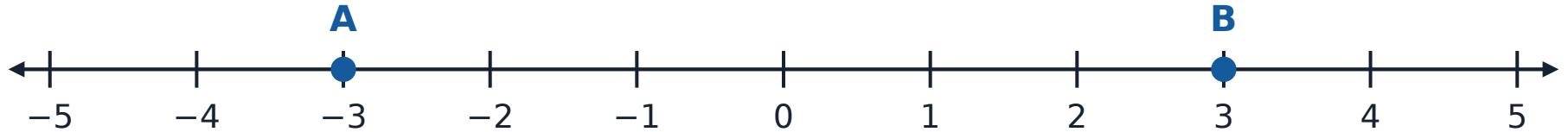
Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 23 • BACKWARD 2 • Q4/5

True or false?

Points A and B represent opposite numbers.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.a • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 6.NS.C.7.c

BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

True or false?

–30 can represent owing \$30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 23 • BACKWARD 3 • Q1/5

Write an integer for 42 feet below sea level.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 23 • BACKWARD 3 • Q2/5

Which situation is represented by $+25$?

- A.** A deposit of \$25
- B.** A debt of \$25
- C.** 25 feet below sea level
- D.** A loss of 25 yards

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 23 • BACKWARD 3 • Q3/5

Write an integer for a debt of \$18.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 23 • BACKWARD 3 • Q4/5

True or false?

A temperature of 0°F means there is no temperature.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • 6.NS.C.7.c

FORWARD 1

Distance between rational numbers as $|p - q|$ • Grade 7 • The Number System
• 7.NS.A.1.c

Find the distance between -4 and 7 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.c • Set 23 • FORWARD 1 • Q1/5

Find the value.

$$|-3.5 - 2|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.c • Set 23 • FORWARD 1 • Q2/5

Which expression gives the distance between -9 and -2 ?

A. $|-9 - (-2)|$

B. $-9 - 2$

C. $|-9 + (-2)|$

D. $-9 + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.c • Set 23 • FORWARD 1 • Q3/5

True or false?

The distance between -6 and 6 is 0 .

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.c • Set 23 • FORWARD 1 • Q4/5

A bird is 25 feet above sea level. A fish is 12 feet below sea level.

How far apart are they?

Distance: _____

SET 23 • 6.NS.C.7.c

FORWARD 2

Square roots and cube roots as solutions • Grade 8 • Expressions & Equations •
8.EE.A.2

Solve.

$$x^2 = 49$$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 23 • FORWARD 2 • Q1/5

Solve.

$$x^3 = 64$$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 23 • FORWARD 2 • Q2/5

What are the solutions of $x^2 = 81$?

A. $x = 9$ or $x = -9$

B. $x = 9$ only

C. $x = 40.5$

D. $x = -9$ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 23 • FORWARD 2 • Q3/5

Solve. Write the answer using a square root symbol.

$$x^2 = 20$$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 23 • FORWARD 2 • Q4/5

True or false?

Both 6 and -6 are solutions of $x^2 = 36$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 23 • FORWARD 2 • Q5/5

Distinguish comparisons of absolute value from statements about order

MAIN	Grade 6 main question	6.NS.C.7.d
BACKWARD 1	Absolute value	6.NS.C.7.c
BACKWARD 2	Ordering negative numbers	6.NS.C.7.a
BACKWARD 3	Negative numbers in context	6.NS.C.5
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Compare magnitudes with powers of 10	8.EE.A.3

SET 24 • 6.NS.C.7.d

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.d

True or false?

An account balance less than -30 dollars represents a debt greater than 30 dollars.

- A. True
- B. False

Which account balance shows the greatest debt?

A. $-\$55$

B. $-\$20$

C. $-\$38$

D. $\$10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.d • Set 24 • MAIN • Q2/5

True or false?

$-12 < -5$, so $|-12| < |-5|$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.d • Set 24 • MAIN • Q3/5

Two temperatures are -15°F and -9°F .

Which temperature is colder? Which has the greater absolute value?

Colder: _____

Greater absolute value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.d • Set 24 • MAIN • Q4/5

A submarine is at -300 feet. A diver is at -45 feet.

Which statement is true?

- A.** The submarine is deeper because $|-300| > |-45|$.
- B.** The diver is deeper because $-45 > -300$.
- C.** The submarine is deeper because $-300 > -45$.
- D.** They are at the same depth.

SET 24 • 6.NS.C.7.d

BACKWARD 1

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-25|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 24 • BACKWARD 1 • Q1/5

Find the value.

$|14|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 24 • BACKWARD 1 • Q2/5

True or false?

$$|-3| = -3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 24 • BACKWARD 1 • Q3/5

Which number has an absolute value of 7?

A. -7

B. $\frac{1}{7}$

C. 0

D. $-\frac{1}{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 24 • BACKWARD 1 • Q4/5

Find the value.

$$|-2.5|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 6.NS.C.7.d

BACKWARD 2

Ordering negative numbers • Grade 6 • The Number System • 6.NS.C.7.a

Which number is least?

A. -20

B. -5

C. 0

D. -12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 24 • BACKWARD 2 • Q1/5

True or false?

$$-9 > -4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 24 • BACKWARD 2 • Q2/5

Order from least to greatest.

$-3, -8, 1$

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 24 • BACKWARD 2 • Q3/5

True or false?

$$-1 > -100$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 24 • BACKWARD 2 • Q4/5

Which number is greater?

A. -6

B. -16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.a • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 6.NS.C.7.d

BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for a debt of \$40.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • BACKWARD 3 • Q1/5

What could -8 represent?

- A.** 8 feet below sea level
- B.** 8 feet above sea level
- C.** A gain of 8 yards
- D.** A deposit of \$8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • BACKWARD 3 • Q2/5

True or false?

An account balance of $-\$10$ means you owe $\$10$.

- A. True
- B. False

Write an integer for 15 degrees below zero.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • BACKWARD 3 • Q4/5

True or false?

A 20-foot descent can be written as +20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • BACKWARD 3 • Q5/5

SET 24 • 6.NS.C.7.d

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A company lost \$2.5 million in one year and earned a profit of \$1.8 million the next year.

What is the total for the two years?

Total: _____

The temperature was -6°F . It dropped 11°F .
What is the new temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 24 • FORWARD 1 • Q2/5

A diver at -35.5 feet rises 12.25 feet.

What is the diver's new position?

- A.** -23.25 feet
- B.** -47.75 feet
- C.** 23.25 feet
- D.** -22.75 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 24 • FORWARD 1 • Q3/5

A hiker goes from an elevation of 1,200 feet to an elevation of -80 feet.

What is the change in elevation?

Change: _____

True or false?

If the temperature drops from -4°C to -9°C , it gets 5°C colder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 6.NS.C.7.d

FORWARD 2

Compare magnitudes with powers of 10 • Grade 8 • Expressions & Equations •
8.EE.A.3

One company's debt is 2×10^6 dollars. Another company's debt is 4×10^4 dollars.

How many times as large is the first debt?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 24 • FORWARD 2 • Q1/5

The deepest ocean point is about -1×10^4 meters. A lake bottom is about -5×10^1 meters.

About how many times as deep is the ocean point?

A. 200

B. 20

C. 2,000

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 24 • FORWARD 2 • Q2/5

Estimate 7,980,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 24 • FORWARD 2 • Q3/5

True or false?

6×10^9 is 300 times as large as 2×10^7 .

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 24 • FORWARD 2 • Q4/5

Estimate 0.0000031 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 24 • FORWARD 2 • Q5/5

Distances between points on the coordinate plane

MAIN	Grade 6 main question	6.NS.C.8
BACKWARD 1	Absolute value	6.NS.C.7.c
BACKWARD 2	Naming points on the coordinate plane	6.NS.C.6.c
BACKWARD 3	Distance between whole numbers	2.MD.B.6
BACKWARD 4	Perimeter of rectangles	3.MD.D.8
FORWARD 1	Distance between rational numbers	7.NS.A.1.c
FORWARD 2	Distance between points using the Pythagorean Theorem	8.G.B.8

SET 25 • 6.NS.C.8

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-3, 4)$ and $(5, 4)$.

Distance: _____

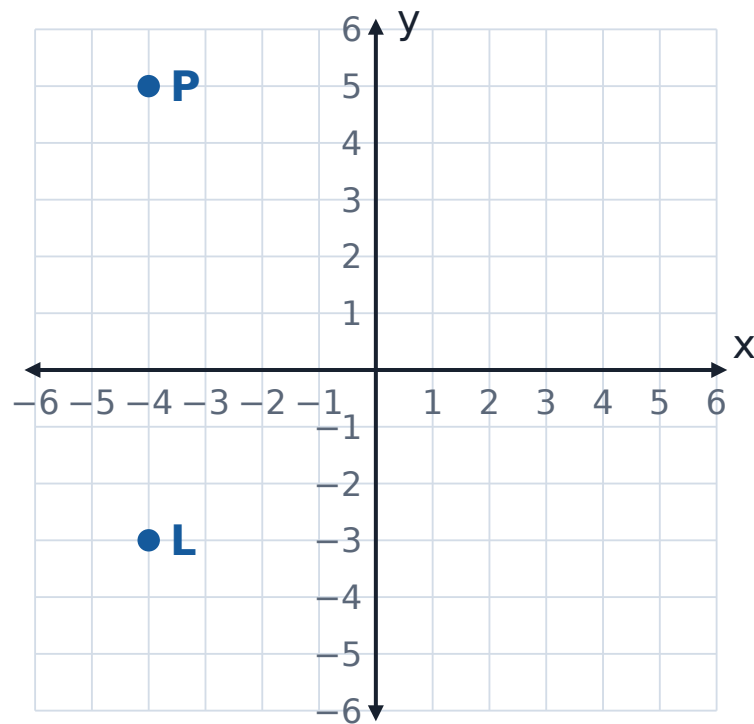
Find the distance between $(2, -6)$ and $(2, 5)$.

Distance: _____

On the map, each unit is 1 block.

How many blocks is it from the library L to the park P?

Blocks: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.8 • Set 25 • MAIN • Q3/5

What is the distance between $(-7, -2)$ and $(-1, -2)$?

- A.** 6
- B.** 8
- C.** -6
- D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.8 • Set 25 • MAIN • Q4/5

A rectangle has vertices at $(-2, 3)$, $(4, 3)$, $(4, -1)$, and $(-2, -1)$.

What is its perimeter?

Perimeter: _____

SET 25 • 6.NS.C.8

BACKWARD 1

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-7|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 25 • BACKWARD 1 • Q1/5

True or false?

$$|-3| + |5| = 8$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 25 • BACKWARD 1 • Q2/5

$$|-4| + |6| = ?$$

A. 10

B. 2

C. -10

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 25 • BACKWARD 1 • Q3/5

Find the value.

$$|-9| + |2|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 25 • BACKWARD 1 • Q4/5

True or false?

The distance from -5 to 0 is $|-5|$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 25 • BACKWARD 1 • Q5/5

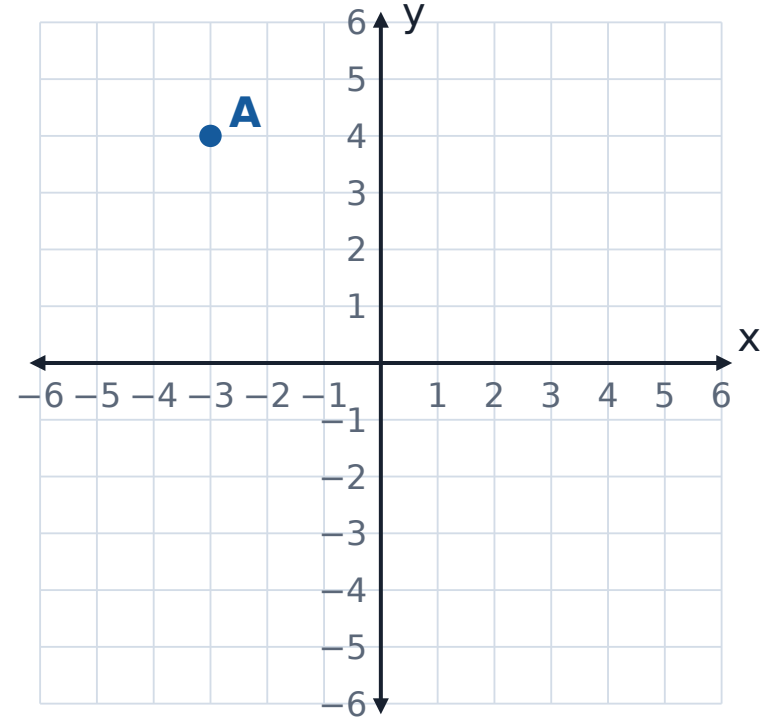
SET 25 • 6.NS.C.8

BACKWARD 2

Naming points on the coordinate plane • Grade 6 • The Number System •
6.NS.C.6.c

What are the coordinates of point A?

A = _____

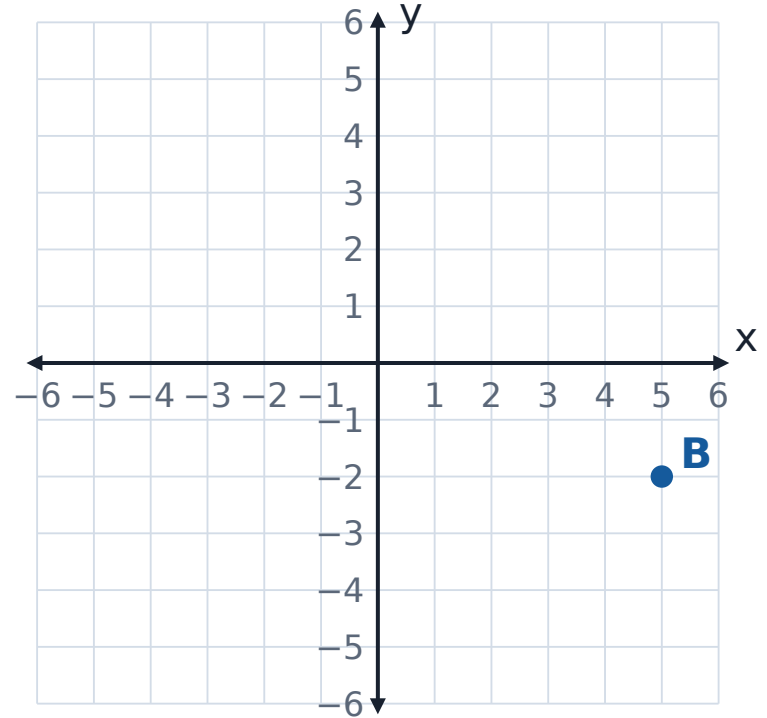


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 25 • BACKWARD 2 • Q1/5

What are the coordinates of point B?

B = _____

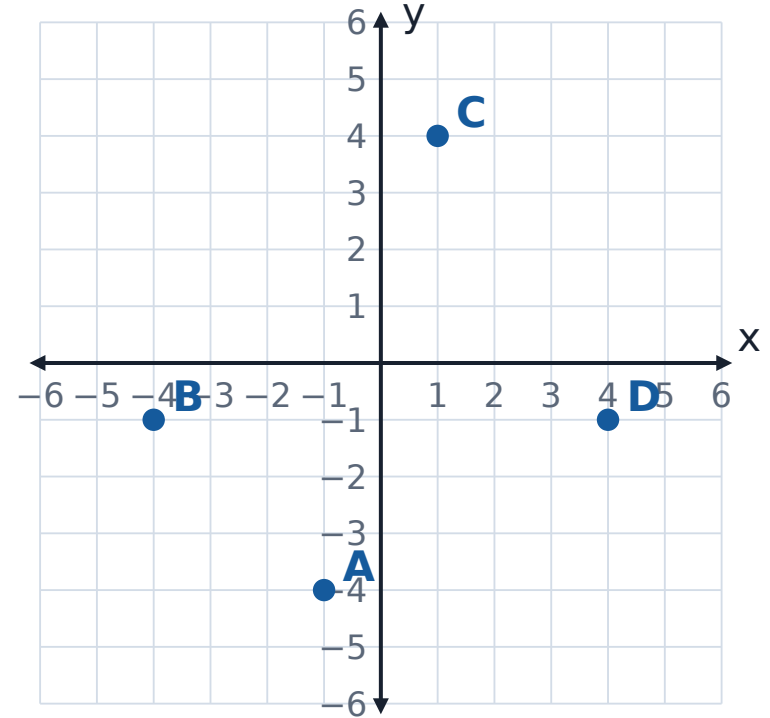


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 25 • BACKWARD 2 • Q2/5

Which point is at $(-1, -4)$?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

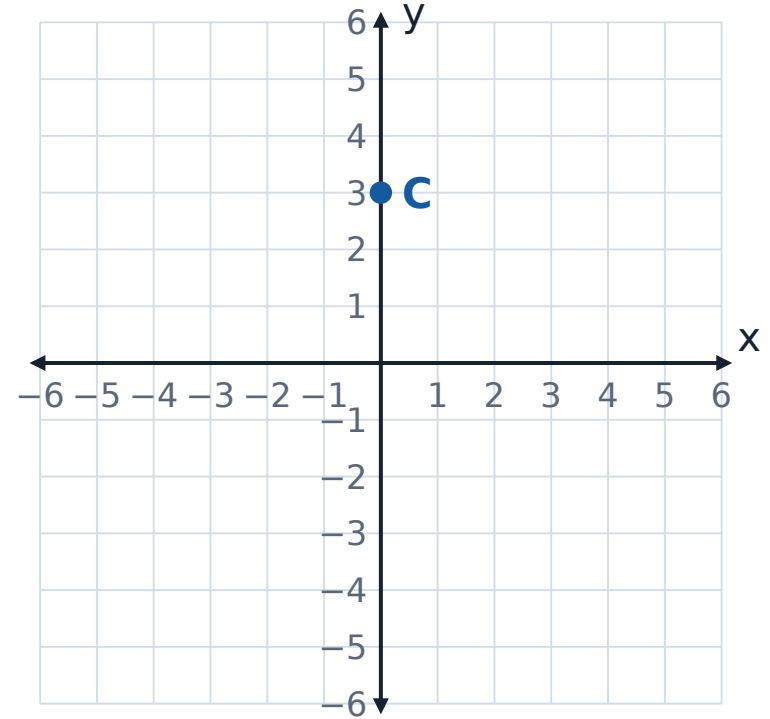
Grade 6 • The Number System • 6.NS.C.6.c • Set 25 • BACKWARD 2 • Q3/5

True or false?

Point C is located at $(3, 0)$.

A. True

B. False

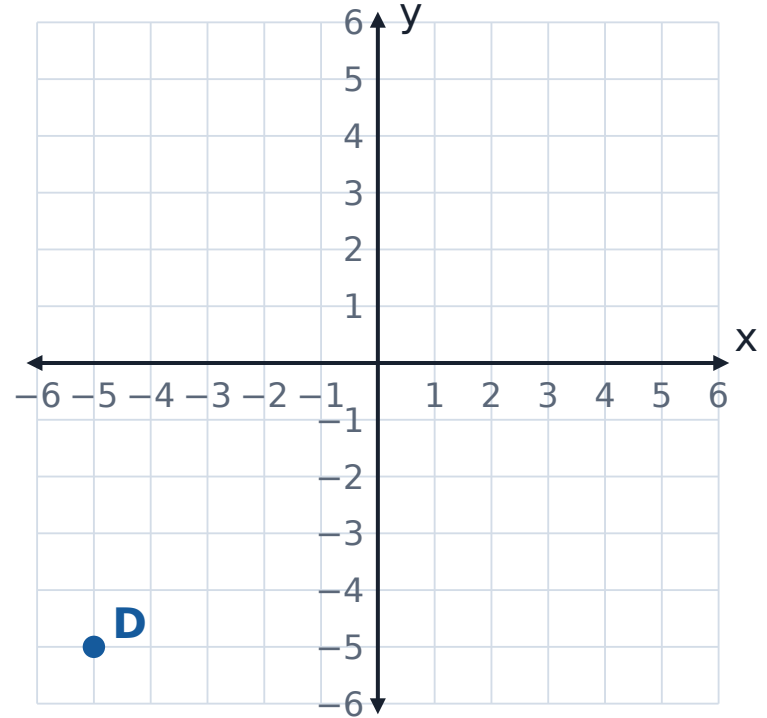


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 25 • BACKWARD 2 • Q4/5

What are the coordinates of point D?

D = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 6.NS.C.8

BACKWARD 3

Distance between whole numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 3 to 11 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 3 • Q1/5

How many units apart are points A and B?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 3 • Q2/5

True or false?

The distance from 12 to 4 is 8 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 3 • Q3/5

What is the distance from 5 to 14?

- A.** 9
- B.** 19
- C.** 8
- D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 3 • Q4/5

What is the distance from 0 to 13 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • 6.NS.C.8

BACKWARD 4

Perimeter of rectangles • Grade 3 • Measurement & Data • 3.MD.D.8

What is the perimeter of the rectangle?

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 25 • BACKWARD 4 • Q1/5

A rectangle is 9 cm long and 3 cm wide.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 25 • BACKWARD 4 • Q2/5

True or false?

A square with 5-meter sides has a perimeter of 25 meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 25 • BACKWARD 4 • Q3/5

A rectangle is 8 inches long and 2 inches wide.

What is its perimeter?

- A.** 20 inches
- B.** 16 inches
- C.** 10 inches
- D.** 18 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 25 • BACKWARD 4 • Q4/5

A rectangle has a perimeter of 18 feet and a length of 5 feet.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 25 • BACKWARD 4 • Q5/5

SET 25 • 6.NS.C.8

FORWARD 1

Distance between rational numbers • Grade 7 • The Number System •
7.NS.A.1.c

Find the distance between -2.5 and 4.5 on a number line.

Distance: _____

Find the value.

$$|-8.4 - 3.6|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.c • Set 25 • FORWARD 1 • Q2/5

What is the distance between $-3\frac{1}{2}$ and $2\frac{1}{4}$ on a number line?

A. $5\frac{3}{4}$

B. $1\frac{1}{4}$

C. $-5\frac{3}{4}$

D. $6\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.c • Set 25 • FORWARD 1 • Q3/5

True or false?

The distance between -7 and -2 is $|-7 - (-2)|$, which is 5.

A. True

B. False

Find the distance between -4 and 2.5 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.1.c • Set 25 • FORWARD 1 • Q5/5

SET 25 • 6.NS.C.8

FORWARD 2

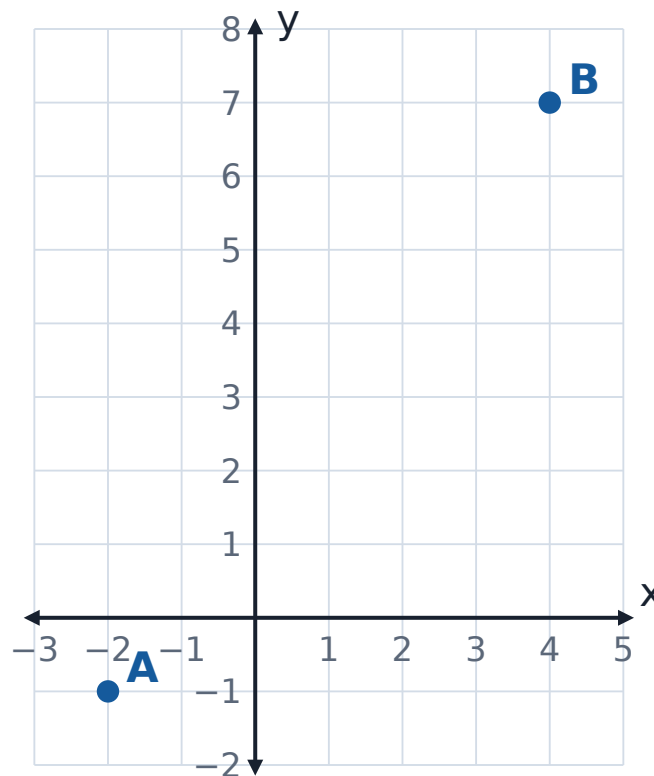
Distance between points using the Pythagorean Theorem • Grade 8 •
Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

Find the distance between points A and B.

Distance: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.8 • Set 25 • FORWARD 2 • Q2/5

What is the distance between $(0, 0)$ and $(5, 12)$?

- A.** 13
- B.** 17
- C.** 7
- D.** $\sqrt{119}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.8 • Set 25 • FORWARD 2 • Q3/5

Find the distance between $(-3, 5)$ and $(1, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.8 • Set 25 • FORWARD 2 • Q4/5

True or false?

The distance between $(2, 1)$ and $(5, 5)$ is 5.

A. True

B. False

Write and evaluate numerical expressions with whole-number exponents

MAIN	Grade 6 main question	6.EE.A.1
BACKWARD 1	Powers of 10 with exponents	5.NBT.A.2
BACKWARD 2	Order of operations with parentheses	5.OA.A.1
BACKWARD 3	Repeated multiplication	3.OA.C.7
FORWARD 1	Products of signed numbers	7.NS.A.2.a
FORWARD 2	Properties of integer exponents	8.EE.A.1

SET 26 • 6.EE.A.1

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2^4 + 3^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 26 • MAIN • Q1/5

Evaluate.

$$3 \times 2^3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 26 • MAIN • Q2/5

Evaluate.

$$(5 - 2)^3 \div 9$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 26 • MAIN • Q3/5

Which expression is equal to 4^3 ?

A. $4 \times 4 \times 4$

B. 4×3

C. $3 \times 3 \times 3 \times 3$

D. $4 + 4 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 26 • MAIN • Q4/5

Evaluate.

$$\left(\frac{2}{3}\right)^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 26 • MAIN • Q5/5

SET 26 • 6.EE.A.1

BACKWARD 1

Powers of 10 with exponents • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Write 10^3 as a whole number.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 26 • BACKWARD 1 • Q1/5

True or false?

$$10^4 = 10,000$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 26 • BACKWARD 1 • Q2/5

Which is equal to 100,000?

A. 10^5

B. 10^4

C. 10^6

D. 5^{10}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 26 • BACKWARD 1 • Q3/5

Write 1,000,000 as a power of 10.

Power of 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 26 • BACKWARD 1 • Q4/5

True or false?

$$10^2 = 20$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 6.EE.A.1

BACKWARD 2

Order of operations with parentheses • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$(8 + 4) \div 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 26 • BACKWARD 2 • Q1/5

Evaluate.

$$20 - (3 \times 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 26 • BACKWARD 2 • Q2/5

True or false?

$$2 \times (3 + 5) = 11$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 26 • BACKWARD 2 • Q3/5

Evaluate.

$$18 \div (2 + 4)$$

A. 3

B. 13

C. 6

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 26 • BACKWARD 2 • Q4/5

Evaluate.

$$(9 - 3) \times 5$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 6.EE.A.1

BACKWARD 3

Repeated multiplication • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$2 \times 2 \times 2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 26 • BACKWARD 3 • Q1/5

Multiply.

$$5 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 26 • BACKWARD 3 • Q2/5

True or false?

$$3 \times 3 \times 3 = 9$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 26 • BACKWARD 3 • Q3/5

$$4 \times 4 \times 4 = ?$$

A. 64

B. 12

C. 16

D. 44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 26 • BACKWARD 3 • Q4/5

Multiply.

$$6 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • 6.EE.A.1

FORWARD 1

Products of signed numbers • Grade 7 • The Number System • 7.NS.A.2.a

Evaluate.

$$(-3)^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.a • Set 26 • FORWARD 1 • Q1/5

Evaluate.

$$(-2)^3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.a • Set 26 • FORWARD 1 • Q2/5

True or false?

$$-3^2 = (-3)^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.a • Set 26 • FORWARD 1 • Q3/5

$$(-1)^5 = ?$$

A. -1

B. 1

C. -5

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.a • Set 26 • FORWARD 1 • Q4/5

Evaluate.

$$\left(-\frac{1}{2}\right)^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.2.a • Set 26 • FORWARD 1 • Q5/5

SET 26 • 6.EE.A.1

FORWARD 2

Properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$$3^2 \times 3^{-5}$$

Single power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 26 • FORWARD 2 • Q1/5

Evaluate.

$$2^{-3}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 26 • FORWARD 2 • Q2/5

$$(5^2)^3 = ?$$

A. 5^6

B. 5^5

C. 5^8

D. 5^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 26 • FORWARD 2 • Q3/5

Evaluate.

$$\frac{4^5}{4^3}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 26 • FORWARD 2 • Q4/5

True or false?

$$7^0 = 1$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 26 • FORWARD 2 • Q5/5

Write expressions in which letters stand for numbers

MAIN	Grade 6 main question	6.EE.A.2.a
BACKWARD 1	Words for operations	5.OA.A.2
BACKWARD 2	A symbol for an unknown number	3.OA.A.4
BACKWARD 3	Order matters in subtraction and division	3.OA.B.5
FORWARD 1	Write and solve equations of the form $px + q = r$	7.EE.B.4.a
FORWARD 2	Construct a function from a description	8.F.B.4

SET 27 • 6.EE.A.2.a

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.a

Write an expression for "subtract y from 5."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 27 • MAIN • Q1/5

Write an expression for "5 more than twice a number n ."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 27 • MAIN • Q2/5

Which expression means "the quotient of x and 4"?

A. $\frac{x}{4}$

B. $\frac{4}{x}$

C. $x - 4$

D. $4x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 27 • MAIN • Q3/5

Write an expression for "3 less than the product of 7 and a number m ."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 27 • MAIN • Q4/5

True or false?

"8 less than k" can be written as $8 - k$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 27 • MAIN • Q5/5

SET 27 • 6.EE.A.2.a

BACKWARD 1

Words for operations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Which expression shows "add 8 and 7, then multiply by 2"?

A. $2 \times (8 + 7)$

B. $2 \times 8 + 7$

C. $8 + 7 \times 2$

D. $(2 + 8) \times 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 27 • BACKWARD 1 • Q1/5

Write an expression for "subtract 4 from 10, then divide by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 27 • BACKWARD 1 • Q2/5

True or false?

"The product of 6 and 9" means 6×9 .

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 27 • BACKWARD 1 • Q3/5

Which word describes the answer to a subtraction problem?

- A.** Difference
- B.** Sum
- C.** Product
- D.** Quotient

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 27 • BACKWARD 1 • Q4/5

Write an expression for "the sum of 12 and 5, multiplied by 3."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 6.EE.A.2.a

BACKWARD 2

A symbol for an unknown number • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.4

What number goes in the box?

$$\square + 7 = 12$$

$$\square = \underline{\hspace{10em}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 27 • BACKWARD 2 • Q1/5

True or false?

In $8 \times \square = 48$, the box stands for 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 27 • BACKWARD 2 • Q2/5

$$\square - 5 = 9$$

What number goes in the box?

A. 14

B. 4

C. 45

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 27 • BACKWARD 2 • Q3/5

What number goes in the box?

$$20 \div \square = 4$$

$$\square = \underline{\hspace{10cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 27 • BACKWARD 2 • Q4/5

What number goes in the box?

$$\square \times 3 = 27$$

$$\square = \underline{\hspace{10cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • 6.EE.A.2.a

BACKWARD 3

Order matters in subtraction and division • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$7 - 3 = 3 - 7$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 27 • BACKWARD 3 • Q1/5

True or false?

$$6 \times 4 = 4 \times 6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 27 • BACKWARD 3 • Q2/5

True or false?

$$12 \div 3 = 3 \div 12$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 27 • BACKWARD 3 • Q3/5

Which statement is true?

A. $5 + 9 = 9 + 5$

B. $9 - 5 = 5 - 9$

C. $10 \div 2 = 2 \div 10$

D. $8 - 2 = 2 - 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 27 • BACKWARD 3 • Q4/5

True or false?

$$3 + 11 = 11 + 3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • 6.EE.A.2.a

FORWARD 1

Write and solve equations of the form $px + q = r$ • Grade 7 • Expressions & Equations • 7.EE.B.4.a

The perimeter of a rectangle is 54 cm. Its length is 6 cm.

Write and solve an equation to find its width.

Width: _____

Jo has \$40. She buys 3 shirts that cost the same amount and has \$4 left.

How much does each shirt cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 27 • FORWARD 1 • Q2/5

Which equation represents "5 more than 3 times a number is 26"?

A. $3n + 5 = 26$

B. $5n + 3 = 26$

C. $3(n + 5) = 26$

D. $3n - 5 = 26$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 27 • FORWARD 1 • Q3/5

Twice a number minus 7 is 15.

Write and solve an equation to find the number.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 27 • FORWARD 1 • Q4/5

A phone plan costs \$30 per month plus \$0.10 per text.
One month's bill is \$42.

How many texts were sent?

Texts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 27 • FORWARD 1 • Q5/5

SET 27 • 6.EE.A.2.a

FORWARD 2

Construct a function from a description • Grade 8 • Functions • 8.F.B.4

A pool has 500 gallons of water. It is filled at 25 gallons per minute.

Write a function for the gallons G after t minutes.

$G =$ _____

Tickets cost \$4 each plus a \$6 fee for the order.

Which function gives the total cost y for x tickets?

A. $y = 4x + 6$

B. $y = 6x + 4$

C. $y = 10x$

D. $y = 4x - 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 27 • FORWARD 2 • Q2/5

The table shows a linear function.

Write an equation for the function.

x	0	1	2
y	7	10	13

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 27 • FORWARD 2 • Q3/5

A line passes through $(0, -2)$ and $(3, 4)$.

Write an equation for the line.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 27 • FORWARD 2 • Q4/5

True or false?

The function $y = 15x + 20$ has an initial value of 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 27 • FORWARD 2 • Q5/5

Identify parts of an expression: term, sum, product, factor, quotient, coefficient

MAIN	Grade 6 main question	6.EE.A.2.b
BACKWARD 1	Sum, difference, product, and quotient	5.OA.A.2
BACKWARD 2	Factors	4.OA.B.4
BACKWARD 3	Interpreting expressions without evaluating	5.OA.A.2
FORWARD 1	Combine like terms with rational coefficients	7.EE.A.1
FORWARD 2	Interpret the parts of a linear function	8.F.B.4

SET 28 • 6.EE.A.2.b

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.b

In the expression $2(8 + 7)$, how can the 2 be described?

- A.** A factor
- B.** A sum
- C.** A quotient
- D.** A term of $8 + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 28 • MAIN • Q1/5

How many terms are in the expression $3x + 5y - 2$?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 28 • MAIN • Q2/5

What is the coefficient of y in $4x + 9y$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 28 • MAIN • Q3/5

True or false?

In $6(a + b)$, the quantity $(a + b)$ is a single factor and also a sum of two terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 28 • MAIN • Q4/5

In the expression $7m + 3$, which part is the constant term?

- A.** 3
- B.** 7
- C.** m
- D.** $7m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 28 • MAIN • Q5/5

SET 28 • 6.EE.A.2.b

BACKWARD 1

Sum, difference, product, and quotient • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$8 + 7$ is a sum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 1 • Q1/5

What is 6×9 called?

- A.** A product
- B.** A sum
- C.** A difference
- D.** A quotient

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 1 • Q2/5

Is $24 \div 6$ a sum, difference, product, or quotient?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 1 • Q3/5

True or false?

15 – 4 is called a product.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 1 • Q4/5

Which expression is a quotient?

A. $20 \div 5$

B. $20 - 5$

C. 20×5

D. $20 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 6.EE.A.2.b

BACKWARD 2

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

True or false?

In $4 \times 7 = 28$, the numbers 4 and 7 are factors of 28.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 28 • BACKWARD 2 • Q1/5

Name the two factors in 5×9 .

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 28 • BACKWARD 2 • Q2/5

Which number is NOT a factor of 20?

- A.** 6
- B.** 4
- C.** 5
- D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 28 • BACKWARD 2 • Q3/5

True or false?

3 is a factor of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 28 • BACKWARD 2 • Q4/5

Write one factor pair of 18.

Factor pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 6.EE.A.2.b

BACKWARD 3

Interpreting expressions without evaluating • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$3 \times (18,932 + 921)$ is three times as large as $18,932 + 921$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 3 • Q1/5

How does $5 \times (26 + 4)$ compare to $26 + 4$?

- A.** It is 5 times as large.
- B.** It is 5 more.
- C.** It is 5 less.
- D.** It is the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 3 • Q2/5

True or false?

$(9 + 4) \times 2$ is 2 more than $9 + 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 3 • Q3/5

Which expression is 4 times as large as $125 - 25$?

A. $4 \times (125 - 25)$

B. $4 + (125 - 25)$

C. $125 - 25 \times 4$

D. $(125 - 25) \div 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 3 • Q4/5

Without calculating, how many times as large is
 $10 \times (63 + 7)$ as $63 + 7$?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 3 • Q5/5

SET 28 • 6.EE.A.2.b

FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$$4x + 3 - 2x + 5$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 28 • FORWARD 1 • Q1/5

Simplify.

$$0.5y + 1.25y$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 28 • FORWARD 1 • Q2/5

Which pair shows like terms?

A. $3a$ and $-7a$

B. $3a$ and $3b$

C. $4x$ and 4

D. x^2 and x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 28 • FORWARD 1 • Q3/5

Simplify.

$$-3(x - 2) + 5x$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 28 • FORWARD 1 • Q4/5

True or false?

$$\frac{1}{2}x + \frac{1}{4}x = \frac{3}{4}x$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 28 • FORWARD 1 • Q5/5

SET 28 • 6.EE.A.2.b

FORWARD 2

Interpret the parts of a linear function • Grade 8 • Functions • 8.F.B.4

The cost to bowl is $C = 4.5n + 12$, where n is the number of games.

What does 12 represent?

Meaning: _____

In $y = -3x + 7$, what is the rate of change?

A. -3

B. 7

C. x

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 28 • FORWARD 2 • Q2/5

The gas left in a car's tank is $G = 14 - 0.04m$, where m is miles driven.

What is the initial value?

Initial value: _____

A car's distance from home is $d = 60t + 25$, where t is hours.

What does 60 represent?

- A.** The speed in miles per hour
- B.** The starting distance
- C.** The total time
- D.** The total distance

True or false?

In $y = 8x$, the initial value is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 28 • FORWARD 2 • Q5/5

Evaluate expressions and formulas for given values

MAIN	Grade 6 main question	6.EE.A.2.c
BACKWARD 1	Whole-number exponents	6.EE.A.1
BACKWARD 2	Order of operations with grouping symbols	5.OA.A.1
BACKWARD 3	A letter stands for a number	3.OA.D.8
BACKWARD 4	Multiplying fractions	5.NF.B.4.a
FORWARD 1	Evaluate expressions with rational numbers	7.NS.A.3
FORWARD 2	Volume formulas for cylinders, cones, and spheres	8.G.C.9

SET 29 • 6.EE.A.2.c

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $3x^2 + 2$ when $x = 4$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 29 • MAIN • Q1/5

The volume of a cube is $V = s^3$.

Find V when $s = \frac{1}{2}$ inch.

$V =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 29 • MAIN • Q2/5

The surface area of a cube is $A = 6s^2$.

Find A when $s = 3$ cm.

$A =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 29 • MAIN • Q3/5

Evaluate $2(a + b)$ when $a = 3.5$ and $b = 4$.

A. 15

B. 11.5

C. 9.5

D. 7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 29 • MAIN • Q4/5

Evaluate $\frac{x}{4} + 5y$ when $x = 12$ and $y = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 29 • MAIN • Q5/5

SET 29 • 6.EE.A.2.c

BACKWARD 1

Whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$4^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 29 • BACKWARD 1 • Q1/5

Evaluate.

$$\left(\frac{1}{2}\right)^3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 29 • BACKWARD 1 • Q2/5

True or false?

$$3^2 = 6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 29 • BACKWARD 1 • Q3/5

$$2^5 = ?$$

A. 32

B. 10

C. 25

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 29 • BACKWARD 1 • Q4/5

Evaluate.

$$5^2 + 1$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 6.EE.A.2.c

BACKWARD 2

Order of operations with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$3 \times (16 + 2)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 29 • BACKWARD 2 • Q1/5

Evaluate.

$$2 \times (6 + 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 29 • BACKWARD 2 • Q2/5

True or false?

$$(3 + 4) \times 2 = 11$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 29 • BACKWARD 2 • Q3/5

Evaluate.

$$(12 \div 4) + 10$$

A. 13

B. 1

C. 22

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 29 • BACKWARD 2 • Q4/5

Evaluate.

$$[2 \times (5 + 3)] - 4$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 6.EE.A.2.c

BACKWARD 3

A letter stands for a number • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

If $n = 6$, what is $n + 9$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 29 • BACKWARD 3 • Q1/5

True or false?

If $a = 4$, then $a \times 5 = 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 29 • BACKWARD 3 • Q2/5

If $k = 7$, what is $3 \times k$?

A. 21

B. 37

C. 10

D. 73

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 29 • BACKWARD 3 • Q3/5

If $m = 20$, what is $m - 8$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 29 • BACKWARD 3 • Q4/5

If $t = 3$, what is $4 \times t + 2$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • 6.EE.A.2.c

BACKWARD 4

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 29 • BACKWARD 4 • Q1/5

Multiply.

$$6 \times \frac{1}{3}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 29 • BACKWARD 4 • Q2/5

True or false?

$$\frac{2}{3} \times \frac{2}{3} = \frac{4}{6}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 29 • BACKWARD 4 • Q3/5

$$\frac{3}{4} \times 8 = ?$$

A. 6

B. $\frac{3}{32}$

C. 24

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 29 • BACKWARD 4 • Q4/5

Multiply.

$$\frac{1}{2} \times 12$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 29 • BACKWARD 4 • Q5/5

SET 29 • 6.EE.A.2.c

FORWARD 1

Evaluate expressions with rational numbers • Grade 7 • The Number System •
7.NS.A.3

Use $C = \frac{5}{9}(F - 32)$ to find C when $F = -4$.

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 29 • FORWARD 1 • Q1/5

Evaluate $-2x + 5$ when $x = -3.5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 29 • FORWARD 1 • Q2/5

Evaluate $a^2 - b$ when $a = -3$ and $b = 4$.

A. 5

B. -13

C. 13

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 29 • FORWARD 1 • Q3/5

Evaluate $\frac{(x+y)}{2}$ when $x = -6.5$ and $y = 2.5$.

Value: _____

True or false?

When $x = -2$, the value of $x^3 + 1$ is -7 .

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • The Number System • 7.NS.A.3 • Set 29 • FORWARD 1 • Q5/5

SET 29 • 6.EE.A.2.c

FORWARD 2

Volume formulas for cylinders, cones, and spheres • Grade 8 • Geometry •
8.G.C.9

A cylinder has a radius of 3 cm and a height of 10 cm.
Find its volume. Use 3.14 for π .

Volume: _____

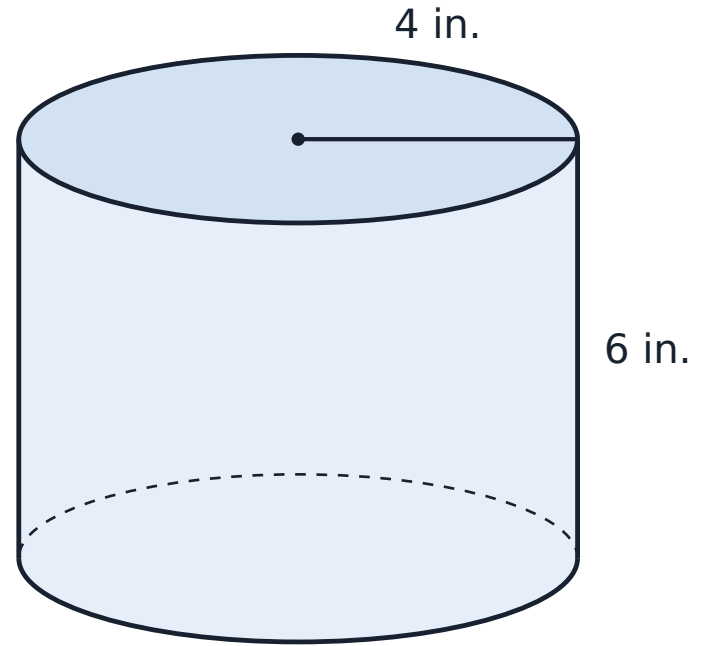
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 29 • FORWARD 2 • Q1/5

Find the volume of the cylinder.

Use 3.14 for π .

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 29 • FORWARD 2 • Q2/5

A cone has a radius of 3 units and a height of 4 units.

What is its volume in terms of π ?

A. 12π

B. 36π

C. 4π

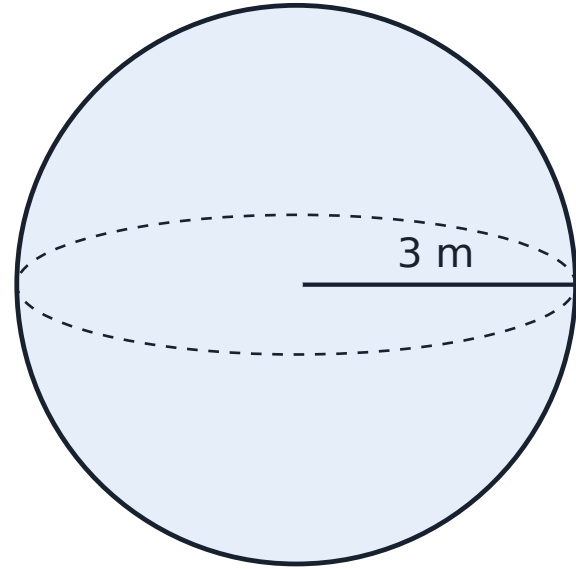
D. 48π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 29 • FORWARD 2 • Q3/5

Find the volume of the sphere
in terms of π .

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 29 • FORWARD 2 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 5 has a volume of 20π .

A. True

B. False

Apply properties of operations to generate equivalent expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Distributive property with numbers	3.OA.B.5
BACKWARD 2	Greatest common factor	6.NS.B.4
BACKWARD 3	Repeated addition as multiplication	3.OA.A.1
FORWARD 1	Expand and factor with rational coefficients	7.EE.A.1
FORWARD 2	Solve equations by expanding and collecting like terms	8.EE.C.7.b

SET 30 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(2 + x)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 30 • MAIN • Q1/5

Factor using the greatest common factor.

$$24x + 18y$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 30 • MAIN • Q2/5

Simplify.

$$y + y + y$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 30 • MAIN • Q3/5

Which expression is equivalent to $6(4x + 3y)$?

A. $24x + 18y$

B. $24x + 3y$

C. $10x + 9y$

D. $4x + 18y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 30 • MAIN • Q4/5

Expand.

$$5(3a - 2)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 30 • MAIN • Q5/5

SET 30 • 6.EE.A.3

BACKWARD 1

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$6 \times (4 + 3) = 6 \times 4 + 6 \times 3$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 30 • BACKWARD 1 • Q1/5

Fill in the blank.

$$5 \times (2 + 7) = 5 \times 2 + 5 \times \underline{\quad}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 30 • BACKWARD 1 • Q2/5

Which is equal to 3×14 ?

A. $3 \times 10 + 3 \times 4$

B. $3 \times 10 + 4$

C. $3 + 10 \times 4$

D. $30 + 4 + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 30 • BACKWARD 1 • Q3/5

True or false?

$$7 \times 12 = 7 \times 10 + 2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 30 • BACKWARD 1 • Q4/5

Fill in the blank.

$$8 \times (5 + 1) = 8 \times 5 + \underline{\hspace{2cm}}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 6.EE.A.3

BACKWARD 2

Greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 18.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 30 • BACKWARD 2 • Q1/5

Find the greatest common factor of 15 and 25.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 30 • BACKWARD 2 • Q2/5

True or false?

The greatest common factor of 12 and 20 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 30 • BACKWARD 2 • Q3/5

What is the greatest common factor of 16 and 40?

- A.** 8
- B.** 4
- C.** 2
- D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 30 • BACKWARD 2 • Q4/5

Find the greatest common factor of 9 and 21.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.B.4 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 6.EE.A.3

BACKWARD 3

Repeated addition as multiplication • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$$4 + 4 + 4 = 3 \times 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 30 • BACKWARD 3 • Q1/5

Write $7 + 7 + 7 + 7$ as a multiplication expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 30 • BACKWARD 3 • Q2/5

What does 5×6 mean?

- A.** 5 groups of 6
- B.** 5 more than 6
- C.** 6 minus 5
- D.** The number 56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 30 • BACKWARD 3 • Q3/5

True or false?

$$2 + 2 + 2 + 2 + 2 = 2 \times 2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 30 • BACKWARD 3 • Q4/5

Write $9 + 9$ as a multiplication expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • 6.EE.A.3

FORWARD 1

Expand and factor with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$$\frac{1}{2}(8x - 6)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 30 • FORWARD 1 • Q1/5

Factor.

$$2.5x + 10$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 30 • FORWARD 1 • Q2/5

$$-4(2y - 3) = ?$$

A. $-8y + 12$

B. $-8y - 12$

C. $8y - 12$

D. $-8y - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 30 • FORWARD 1 • Q3/5

Simplify.

$$3(x + 4) - 2(x - 1)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 30 • FORWARD 1 • Q4/5

True or false?

$$0.2(5a + 10) = a + 2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 30 • FORWARD 1 • Q5/5

SET 30 • 6.EE.A.3

FORWARD 2

Solve equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$3(x + 2) = 5x - 4$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 30 • FORWARD 2 • Q1/5

Solve for x.

$$2(4x - 1) - 3x = 18$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 30 • FORWARD 2 • Q2/5

Solve for x.

$$-2(x - 5) = 3(x + 5)$$

A. -1

B. 1

C. 5

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 30 • FORWARD 2 • Q3/5

Solve for x.

$$6(x - 1) = 4(x + 2)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 30 • FORWARD 2 • Q4/5

Solve for x.

$$5(2x + 1) = 3(3x + 4)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 30 • FORWARD 2 • Q5/5

Identify when two expressions are equivalent

MAIN	Grade 6 main question	6.EE.A.4
BACKWARD 1	Commutative and associative properties	3.OA.B.5
BACKWARD 2	Evaluate an expression for a given value	6.EE.A.2.c
BACKWARD 3	Equal groups	3.OA.A.1
FORWARD 1	Rewrite expressions to show how quantities are related	7.EE.A.2
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 31 • 6.EE.A.4

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.4

True or false?

$y + y + y$ and $3y$ are equivalent expressions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 31 • MAIN • Q1/5

Which expression is equivalent to $2(x + 4)$?

A. $2x + 8$

B. $2x + 4$

C. $x + 8$

D. $2x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 31 • MAIN • Q2/5

True or false?

$3(x + 2)$ and $3x + 2$ are equivalent expressions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 31 • MAIN • Q3/5

Nia substitutes $x = 5$ into $4x + 2x$ and into $6x$. Both equal 30.

Which statement is true?

- A.** This supports that the expressions are equivalent, since they can be rewritten as the same expression.
- B.** This proves the expressions are equal only when $x = 5$.
- C.** The expressions are not equivalent.
- D.** Only $6x$ equals 30.

Write an expression without parentheses that is equivalent to $5(a + 3)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 31 • MAIN • Q5/5

SET 31 • 6.EE.A.4

BACKWARD 1

Commutative and associative properties • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$8 + 5 = 5 + 8$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 31 • BACKWARD 1 • Q1/5

True or false?

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 31 • BACKWARD 1 • Q2/5

Which equation shows the commutative property of multiplication?

A. $6 \times 7 = 7 \times 6$

B. $6 \times 7 = 42$

C. $6 \times (7 + 1) = 6 \times 7 + 6$

D. $6 + 0 = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 31 • BACKWARD 1 • Q3/5

True or false?

$$9 - 4 = 4 - 9$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 31 • BACKWARD 1 • Q4/5

Fill in the blank.

$$3 \times 8 = 8 \times \underline{\quad}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • 6.EE.A.4

BACKWARD 2

Evaluate an expression for a given value • Grade 6 • Expressions & Equations •
6.EE.A.2.c

Evaluate $4x + 2x$ when $x = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 31 • BACKWARD 2 • Q1/5

Evaluate $3(x + 2)$ when $x = 5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 31 • BACKWARD 2 • Q2/5

True or false?

When $x = 2$, the value of $3x + 2$ is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 31 • BACKWARD 2 • Q3/5

Evaluate $2(x + 4)$ when $x = 1$.

A. 10

B. 6

C. 9

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 31 • BACKWARD 2 • Q4/5

Evaluate $5a + 15$ when $a = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 6.EE.A.4

BACKWARD 3

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$$6 + 6 + 6 + 6 = 4 \times 6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 31 • BACKWARD 3 • Q1/5

Which expression means 3 groups of 8?

A. 3×8

B. $3 + 8$

C. $8 - 3$

D. 38

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 31 • BACKWARD 3 • Q2/5

Write $5 + 5 + 5$ as a multiplication expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 31 • BACKWARD 3 • Q3/5

True or false?

$$7 \times 2 = 7 + 7$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 31 • BACKWARD 3 • Q4/5

How many groups of 4 are in $4 + 4 + 4 + 4 + 4$?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 31 • BACKWARD 3 • Q5/5

SET 31 • 6.EE.A.4

FORWARD 1

Rewrite expressions to show how quantities are related • Grade 7 •
Expressions & Equations • 7.EE.A.2

A price p increased by 5% can be written as $p + 0.05p$.

Which expression is equivalent?

A. $1.05p$

B. $0.05p$

C. $5.05p$

D. $1.5p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 31 • FORWARD 1 • Q1/5

True or false?

$$a - 0.2a = 0.8a$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 31 • FORWARD 1 • Q2/5

Write an expression equivalent to $x + 0.15x$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 31 • FORWARD 1 • Q3/5

Which expression shows the price p after a 30% discount?

- A.** $0.7p$
- B.** $0.3p$
- C.** $p - 30$
- D.** $1.3p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 31 • FORWARD 1 • Q4/5

The perimeter of a square is $s + s + s + s$.

Write an equivalent expression with one term.

Expression: _____

SET 31 • 6.EE.A.4

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

True or false?

The equation $2(x + 3) = 2x + 6$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 31 • FORWARD 2 • Q1/5

How many solutions does the equation have?

$$5x - 3 = 5x + 1$$

- A.** No solution
- B.** One solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 31 • FORWARD 2 • Q2/5

How many solutions does the equation have?

$$3(2x + 4) = 6x + 12$$

- A.** No solution
- B.** One solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 31 • FORWARD 2 • Q3/5

Write a number in the box so the equation has no solution.

$$4x + 7 = 4x + \square$$

$$\square = \underline{\hspace{10cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 31 • FORWARD 2 • Q4/5

Which equation has exactly one solution?

A. $3x + 1 = 2x + 1$

B. $3x + 1 = 3x + 1$

C. $3x + 1 = 3x + 2$

D. $3(x + 1) = 3x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 31 • FORWARD 2 • Q5/5

Use substitution to decide whether a number makes an equation or inequality true

MAIN	Grade 6 main question	6.EE.B.5
BACKWARD 1	Evaluate an expression	6.EE.A.2.c
BACKWARD 2	Comparing numbers	2.NBT.A.4
BACKWARD 3	Meaning of the equal sign	1.OA.D.7
FORWARD 1	Solve equations of the form $px + q = r$ and $p(x + q) = r$	7.EE.B.4.a
FORWARD 2	Solutions of a system of two linear equations	8.EE.C.8.a

SET 32 • 6.EE.B.5

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.5

Which value from the set $\{2, 4, 6, 8\}$ makes $3x + 1 = 19$ true?

A. 6

B. 2

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • MAIN • Q1/5

True or false?

$x = 5$ is a solution of $2x - 3 > 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • MAIN • Q2/5

Which numbers from the set $\{1, 3, 5, 7\}$ make $x + 4 < 10$ true?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • MAIN • Q3/5

Is $y = 2.5$ a solution of $4y = 10$?

A. Yes

B. No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • MAIN • Q4/5

Which value from the set $\{12, 15, 18\}$ makes $\frac{n}{3} = 6$ true?

$n =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • MAIN • Q5/5

SET 32 • 6.EE.B.5

BACKWARD 1

Evaluate an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $3x + 1$ when $x = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 32 • BACKWARD 1 • Q1/5

Evaluate $2x - 3$ when $x = 5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 32 • BACKWARD 1 • Q2/5

True or false?

When $y = 4$, the value of $5y$ is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 32 • BACKWARD 1 • Q3/5

Evaluate $x + 4$ when $x = 7$.

A. 11

B. 3

C. 28

D. 47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 32 • BACKWARD 1 • Q4/5

Evaluate $\frac{n}{3}$ when $n = 15$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 6.EE.B.5

BACKWARD 2

Comparing numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $<$, or $=$ to compare.

19 21

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 32 • BACKWARD 2 • Q1/5

True or false?

$$7 > 8$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 32 • BACKWARD 2 • Q2/5

Which statement is true?

A. $9 > 6$

B. $11 < 10$

C. $6 > 9$

D. $10 < 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 32 • BACKWARD 2 • Q3/5

True or false?

$$24 = 24$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 32 • BACKWARD 2 • Q4/5

Write $>$, $<$, or $=$ to compare.

105 150

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 6.EE.B.5

BACKWARD 3

Meaning of the equal sign • Grade 1 • Operations & Algebraic Thinking •
1.OA.D.7

True or false?

$$6 = 6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 32 • BACKWARD 3 • Q1/5

True or false?

$$7 = 8 - 1$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 32 • BACKWARD 3 • Q2/5

True or false?

$$5 + 2 = 2 + 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 32 • BACKWARD 3 • Q3/5

Which equation is true?

A. $4 + 3 = 5 + 2$

B. $4 + 3 = 8$

C. $4 + 3 = 5 + 3$

D. $7 = 4 - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 32 • BACKWARD 3 • Q4/5

True or false?

$$9 - 1 = 8$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 32 • BACKWARD 3 • Q5/5

SET 32 • 6.EE.B.5

FORWARD 1

Solve equations of the form $px + q = r$ and $p(x + q) = r$ • Grade 7 •
Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 7 = 22$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • FORWARD 1 • Q1/5

Solve for x.

$$4(x - 2) = 20$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • FORWARD 1 • Q2/5

Solve for x.

$$0.5x + 6 = 10$$

A. 8

B. 2

C. 32

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • FORWARD 1 • Q3/5

Solve for x.

$$\frac{2}{3}x - 4 = 10$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • FORWARD 1 • Q4/5

A movie ticket costs \$12. Popcorn costs p dollars. Two tickets and one popcorn cost \$31.

What is p ?

$p =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • FORWARD 1 • Q5/5

SET 32 • 6.EE.B.5

FORWARD 2

Solutions of a system of two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.a

True or false?

$(2, 3)$ is a solution of both $y = x + 1$ and $y = 2x - 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 32 • FORWARD 2 • Q1/5

Which point is a solution of both $y = 2x$ and $y = x + 3$?

A. (3, 6)

B. (1, 2)

C. (0, 3)

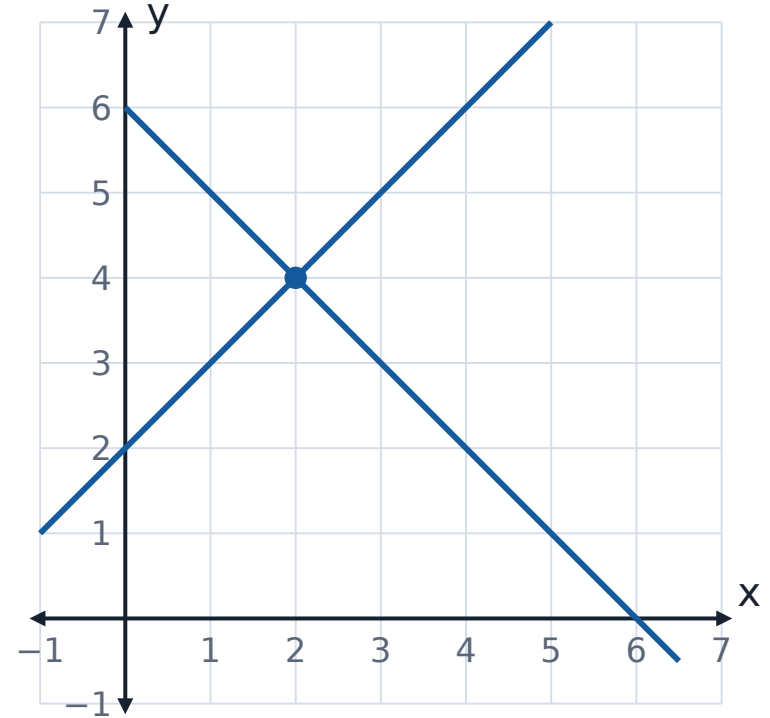
D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 32 • FORWARD 2 • Q2/5

What is the solution of the system shown by the graph?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 32 • FORWARD 2 • Q3/5

True or false?

(1, 4) is a solution of both $y = 3x + 1$ and $y = -x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 32 • FORWARD 2 • Q4/5

Is $(0, 2)$ a solution of both $y = 2x + 2$ and $y = x - 2$?

A. No

B. Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 32 • FORWARD 2 • Q5/5

Use variables to write expressions for real-world problems

MAIN	Grade 6 main question	6.EE.B.6
BACKWARD 1	Word problems with an unknown number	3.OA.A.3
BACKWARD 2	Multiplicative comparison word problems	4.OA.A.2
BACKWARD 3	Number patterns that follow a rule	4.OA.C.5
FORWARD 1	Write and solve equations for word problems	7.EE.B.4.a
FORWARD 2	Model a linear relationship with a function	8.F.B.4

SET 33 • 6.EE.B.6

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.6

Tickets cost \$8 each.

Write an expression for the cost of t tickets.

Expression: _____

Kim is k years old. Sam is 3 years older than Kim.
Write an expression for Sam's age.

Expression: _____

A pizza is cut into s slices. 4 friends share the slices equally.

Which expression shows how many slices each friend gets?

A. $\frac{s}{4}$

B. $4s$

C. $s - 4$

D. $s + 4$

The length of a rectangle is 5 inches more than its width w .

Write an expression for the length.

Expression: _____

A gym charges \$20 to join plus \$15 per month.

Write an expression for the total cost after m months.

Expression: _____

SET 33 • 6.EE.B.6

BACKWARD 1

Word problems with an unknown number • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 6 bags with the same number of apples.
There are 42 apples in all.

How many apples are in each bag?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 33 • BACKWARD 1 • Q1/5

There are 5 boxes with $\square$ pencils in each box. There are 35 pencils in all.

Which equation matches?

A. $5 \times \square = 35$

B. $5 + \square = 35$

C. $35 \times 5 = \square$

D. $\square - 5 = 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 33 • BACKWARD 1 • Q2/5

True or false?

If $4 \times \square = 28$, then $\square = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 33 • BACKWARD 1 • Q3/5

There are 8 rows with $\square$ chairs in each row. There are 56 chairs in all.

What number is $\square$?

$\square =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 33 • BACKWARD 1 • Q4/5

Jan split 45 stickers equally among 9 friends.
How many stickers did each friend get?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 6.EE.B.6

BACKWARD 2

Multiplicative comparison word problems • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A red ribbon is 7 inches long. A blue ribbon is 3 times as long.

How long is the blue ribbon?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 2 • Q1/5

Ben has 6 books. Ali has 4 times as many books as Ben.
Which equation finds b , the number of books Ali has?

A. $b = 4 \times 6$

B. $b = 4 + 6$

C. $b = 6 \div 4$

D. $b = 6 - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 2 • Q2/5

True or false?

A cat weighs 5 pounds. A dog weighs 8 times as much. The dog weighs 40 pounds.

A. True

B. False

A bush is 4 feet tall. A tree is 5 times as tall.

How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 2 • Q4/5

Sara ran 24 miles. That is 3 times as far as Tom ran.
How far did Tom run?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 6.EE.B.6

BACKWARD 3

Number patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking
• 4.OA.C.5

The rule is "add 4." The pattern starts at 3.
Write the first five numbers.

Numbers: _____

The rule is "multiply by 2." The pattern starts at 1.
What is the 5th number?

- A.** 16
- B.** 10
- C.** 32
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 33 • BACKWARD 3 • Q2/5

True or false?

The rule "add 5," starting at 5, gives 5, 10, 15, 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 33 • BACKWARD 3 • Q3/5

What is the rule for this pattern?

7, 11, 15, 19, ...

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 33 • BACKWARD 3 • Q4/5

The rule is "add 6." The pattern starts at 2.
What is the 4th number?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • 6.EE.B.6

FORWARD 1

Write and solve equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Sam has \$60. He spends \$8 each week.

Write and solve an equation to find when he will have \$20 left.

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • FORWARD 1 • Q1/5

A rectangle has a perimeter of 50 cm and a length of 15 cm.

Which equation can be used to find its width w ?

A. $2(15 + w) = 50$

B. $15 + w = 50$

C. $2(15w) = 50$

D. $15 + 2w = 50$

A taxi charges \$3 plus \$2 per mile. The fare was \$17.
How many miles was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • FORWARD 1 • Q3/5

You buy 4 notebooks and a \$6 backpack. You spend \$18 in all.

How much does each notebook cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • FORWARD 1 • Q4/5

True or false?

"Five less than twice a number is 17" can be written as $2n - 5 = 17$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • FORWARD 1 • Q5/5

SET 33 • 6.EE.B.6

FORWARD 2

Model a linear relationship with a function • Grade 8 • Functions • 8.F.B.4

A plant is 4 cm tall. It grows 1.5 cm each week.
Write a function for its height h after w weeks.

$h =$ _____

Which situation matches $y = 2x + 10$, where y is dollars in savings after x weeks?

- A.** Start with \$10 and save \$2 each week.
- B.** Start with \$2 and save \$10 each week.
- C.** Save \$12 each week.
- D.** Start with \$10 and spend \$2 each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 33 • FORWARD 2 • Q2/5

The table shows savings over time.

Write a function for the savings y after x weeks.

Weeks (x)	0	1	2
Savings (y)	30	45	60

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 33 • FORWARD 2 • Q3/5

A rental truck costs \$50 plus \$0.25 per mile.

Write a function for the cost C of driving m miles.

$C =$ _____

True or false?

A line through $(0, 5)$ and $(2, 9)$ has an initial value of 5 and a rate of change of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.B.4 • Set 33 • FORWARD 2 • Q5/5

Solve one-step equations of the form

$x + p = q$ and $px = q$

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Subtraction as an unknown-addend problem	1.OA.B.4
BACKWARD 2	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 3	Decimal operations	5.NBT.B.7
BACKWARD 4	Dividing by a fraction	6.NS.A.1
FORWARD 1	Solve two-step equations	7.EE.B.4.a
FORWARD 2	Solve equations with variables on both sides	8.EE.C.7.b

SET 34 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 7.5 = 12$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 34 • MAIN • Q1/5

Solve for x.

$$4x = 30$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 34 • MAIN • Q2/5

Solve for y.

$$\frac{2}{3}y = 12$$

$$y = \underline{\hspace{10cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 34 • MAIN • Q3/5

Solve for n.

$$n - \frac{3}{4} = 5\frac{1}{2}$$

A. $6\frac{1}{4}$

B. $4\frac{3}{4}$

C. $6\frac{3}{4}$

D. $5\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 34 • MAIN • Q4/5

A pack of 6 pens costs \$4.50.

Write and solve an equation to find the cost p of one pen.

$p =$ _____

SET 34 • 6.EE.B.7

BACKWARD 1

Subtraction as an unknown-addend problem • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4

True or false?

$12 - 7 = \square$ can be solved by thinking
 $7 + \square = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 34 • BACKWARD 1 • Q1/5

What number goes in the box?

$$7 + \square = 15$$

$$\square = \underline{\hspace{10em}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 34 • BACKWARD 1 • Q2/5

To find $16 - 9$, think $9 + \square = 16$.

What number goes in the box?

- A.** 7
- B.** 25
- C.** 8
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 34 • BACKWARD 1 • Q3/5

What number goes in the box?

$$\square + 8 = 13$$

$$\square = \underline{\hspace{10em}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 34 • BACKWARD 1 • Q4/5

True or false?

To solve $10 - 4$, you can think $4 + 6 = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 6.EE.B.7

BACKWARD 2

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $32 \div 8$ by thinking $8 \times \square = 32$.

$\square =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 34 • BACKWARD 2 • Q1/5

True or false?

$56 \div 7 = 8$ because $7 \times 8 = 56$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 34 • BACKWARD 2 • Q2/5

Which fact helps you find $45 \div 5$?

A. $5 \times 9 = 45$

B. $5 + 40 = 45$

C. $45 \times 5 = 225$

D. $9 - 5 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 34 • BACKWARD 2 • Q3/5

What number goes in the box?

$$\square \times 6 = 42$$

$$\square = \underline{\hspace{10em}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 34 • BACKWARD 2 • Q4/5

True or false?

$24 \div 4 = 5$ because $4 \times 5 = 24$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • 6.EE.B.7

BACKWARD 3

Decimal operations • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Subtract.

$$12 - 7.5$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 34 • BACKWARD 3 • Q1/5

Divide.

$$30 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 34 • BACKWARD 3 • Q2/5

True or false?

$$8.2 - 3.7 = 4.5$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 34 • BACKWARD 3 • Q3/5

$$4.50 \div 6 = ?$$

- A.** 0.75
- B.** 0.70
- C.** 1.33
- D.** 7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 34 • BACKWARD 3 • Q4/5

Add.

$$6.25 + 2.8$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • 6.EE.B.7

BACKWARD 4

Dividing by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$12 \div \frac{2}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 34 • BACKWARD 4 • Q1/5

Divide.

$$\frac{3}{4} \div \frac{1}{2}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 34 • BACKWARD 4 • Q2/5

True or false?

$$5 \div \frac{5}{8} = 8$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 34 • BACKWARD 4 • Q3/5

$$\frac{9}{10} \div \frac{3}{5} = ?$$

A. $1\frac{1}{2}$

B. $\frac{27}{50}$

C. $\frac{2}{3}$

D. $1\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 34 • BACKWARD 4 • Q4/5

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.A.1 • Set 34 • BACKWARD 4 • Q5/5

SET 34 • 6.EE.B.7

FORWARD 1

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 8 = 29$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 34 • FORWARD 1 • Q1/5

Solve for x.

$$-2x + 5 = 17$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 34 • FORWARD 1 • Q2/5

Solve for x.

$$\frac{x}{4} - 3 = 2$$

A. 20

B. -4

C. 5

D. $\frac{5}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 34 • FORWARD 1 • Q3/5

Solve for x.

$$0.6x + 1.5 = 4.5$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 34 • FORWARD 1 • Q4/5

Solve for x.

$$5(x + 2) = 45$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 34 • FORWARD 1 • Q5/5

SET 34 • 6.EE.B.7

FORWARD 2

Solve equations with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$7x - 4 = 3x + 12$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 2 • Q1/5

Solve for x.

$$-5x + 9 = x - 3$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 2 • Q2/5

Solve for x.

$$2.5x + 4 = 1.5x + 9$$

A. 5

B. 13

C. -5

D. 6.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{1}{2}x + 6 = 2x - 3$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 2 • Q4/5

Solve for x.

$$3(2x - 5) = 4x + 1$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 2 • Q5/5

Write inequalities $x > c$ or $x < c$ and graph their solutions

MAIN	Grade 6 main question	6.EE.B.8
BACKWARD 1	Inequality symbols	1.NBT.B.3
BACKWARD 2	Locating numbers on a number line	6.NS.C.6.c
BACKWARD 3	Meaning of "at least," "at most," "more than," and "fewer than"	6.NS.C.7.b
FORWARD 1	Write, solve, and graph two-step inequalities	7.EE.B.4.b
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 35 • 6.EE.B.8

MAIN

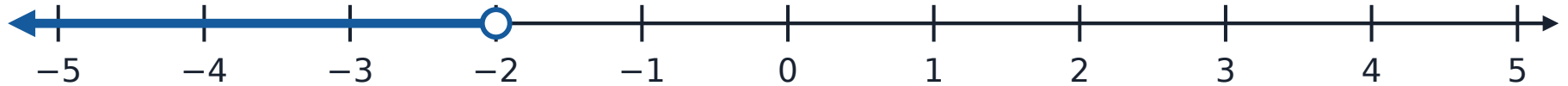
Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8

Riders must be at least 48 inches tall.

Write an inequality for the height h of a rider.

Inequality: _____

Which inequality does the graph show?



A. $x < -2$

B. $x \leq -2$

C. $x > -2$

D. $x \geq -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • MAIN • Q2/5

True or false?

The inequality $x > 5$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • MAIN • Q3/5

A bag can hold at most 20 pounds.

Which inequality describes the weight w the bag can hold?

A. $w \leq 20$

B. $w < 20$

C. $w \geq 20$

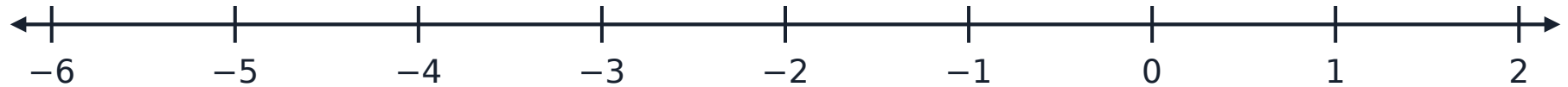
D. $w > 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • MAIN • Q4/5

The temperature must stay below -3°C .

Write an inequality for the temperature t , then graph it.



Inequality: _____

SET 35 • 6.EE.B.8

BACKWARD 1

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

58 85

A. <

B. >

C. =

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 35 • BACKWARD 1 • Q1/5

True or false?

$$30 > 29$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 35 • BACKWARD 1 • Q2/5

Write $>$, $<$, or $=$ to compare.

64 46

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 35 • BACKWARD 1 • Q3/5

Which symbol means "is greater than"?

A. $>$

B. $<$

C. $=$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 35 • BACKWARD 1 • Q4/5

True or false?

$$17 < 12$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 6.EE.B.8

BACKWARD 2

Locating numbers on a number line • Grade 6 • The Number System •
6.NS.C.6.c

What number is at point A?

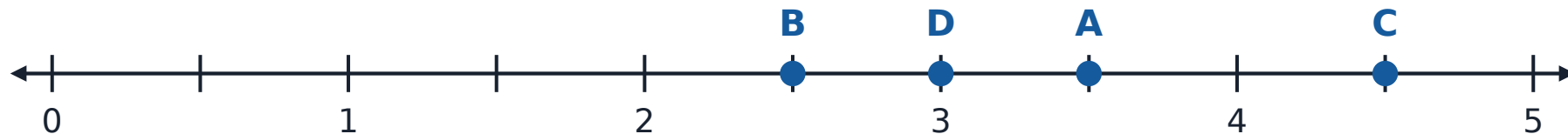


A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 35 • BACKWARD 2 • Q1/5

Which point is at 3.5?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 35 • BACKWARD 2 • Q2/5

True or false?

Point B is at -1.5 .



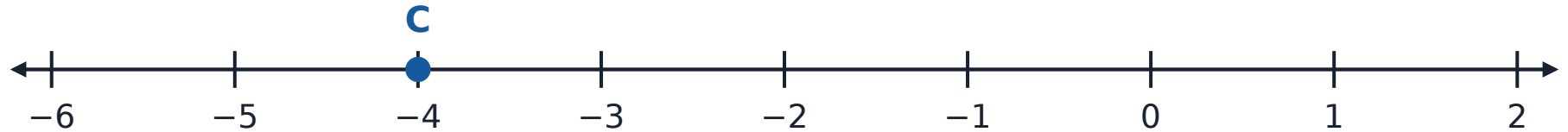
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 35 • BACKWARD 2 • Q3/5

What number is at point C?

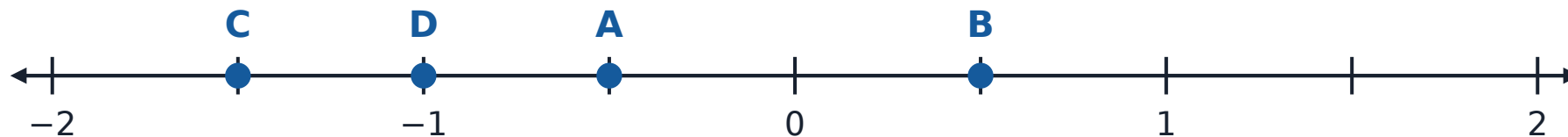


C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 35 • BACKWARD 2 • Q4/5

Which point is at $-\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 6.EE.B.8

BACKWARD 3

Meaning of "at least," "at most," "more than," and "fewer than" • Grade 6 • The Number System • 6.NS.C.7.b

True or false?

"At least 10" includes the number 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.b • Set 35 • BACKWARD 3 • Q1/5

What does "at most 6" mean?

- A.** 6 or less
- B.** More than 6
- C.** 6 or more
- D.** Less than 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.b • Set 35 • BACKWARD 3 • Q2/5

True or false?

"Fewer than 8" includes the number 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.b • Set 35 • BACKWARD 3 • Q3/5

Which number is NOT "at least 12"?

A. 11

B. 12

C. 13

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.b • Set 35 • BACKWARD 3 • Q4/5

True or false?

9 is "more than 7."

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.b • Set 35 • BACKWARD 3 • Q5/5

SET 35 • 6.EE.B.8

FORWARD 1

Write, solve, and graph two-step inequalities • Grade 7 • Expressions & Equations • 7.EE.B.4.b

You have \$50. Each ride costs \$6.

Write and solve an inequality to find how many rides r you can afford.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • FORWARD 1 • Q1/5

Solve the inequality.

$$3x - 4 > 11$$

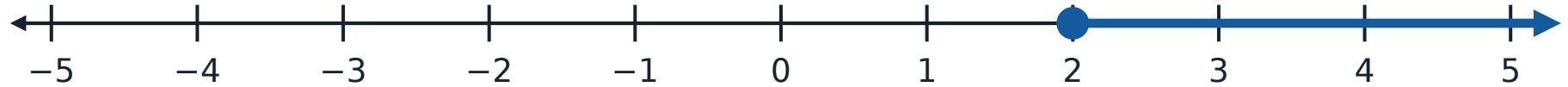
Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • FORWARD 1 • Q2/5

True or false?

The graph shows the solutions of $2x + 3 \geq 7$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • FORWARD 1 • Q3/5

Solve the inequality.

$$-2x < 8$$

A. $x > -4$

B. $x < -4$

C. $x > 4$

D. $x < 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • FORWARD 1 • Q4/5

A salesperson earns \$80 plus \$12 for each sale.

Write and solve an inequality for the number of sales s needed to earn more than \$200.

Solution: _____

SET 35 • 6.EE.B.8

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

True or false?

$x + 3 = x + 3$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 35 • FORWARD 2 • Q1/5

How many solutions does the equation have?

$$2x + 5 = 2x - 1$$

- A.** No solution
- B.** One solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 35 • FORWARD 2 • Q2/5

Which equation has infinitely many solutions?

A. $4(x - 1) = 4x - 4$

B. $4(x - 1) = 4x - 1$

C. $4x - 1 = x$

D. $4x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 35 • FORWARD 2 • Q3/5

How many solutions does the equation have?

$$3x - 7 = 3x - 7$$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 35 • FORWARD 2 • Q4/5

How many solutions does the equation have?

$$6x - 2 = 4x + 8$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 35 • FORWARD 2 • Q5/5

Dependent and independent variables: tables, graphs, and equations

MAIN	Grade 6 main question	6.EE.C.9
BACKWARD 1	Two related number patterns	5.OA.B.3
BACKWARD 2	Plotting and reading points in the first quadrant	5.G.A.2
BACKWARD 3	Input-output rules	4.OA.C.5
FORWARD 1	Equations for proportional relationships	7.RP.A.2.c
FORWARD 2	Compare functions represented in different ways	8.F.A.2

SET 36 • 6.EE.C.9

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

A car travels at 65 miles per hour. The equation $d = 65t$ gives the distance d after t hours.

Which is the independent variable?

- A.** t (time)
- B.** d (distance)
- C.** 65
- D.** There is none.

Write an equation that shows how y depends on x .

x	1	2	3	4
y	6	12	18	24

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 36 • MAIN • Q2/5

A plant is 5 cm tall. It grows 2 cm each week.

Write an equation for its height h after w weeks.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 36 • MAIN • Q3/5

Which set of ordered pairs matches $y = x + 3$?

A. (1, 4), (2, 5), (3, 6)

B. (1, 3), (2, 6), (3, 9)

C. (4, 1), (5, 2), (6, 3)

D. (0, 0), (1, 3), (2, 6)

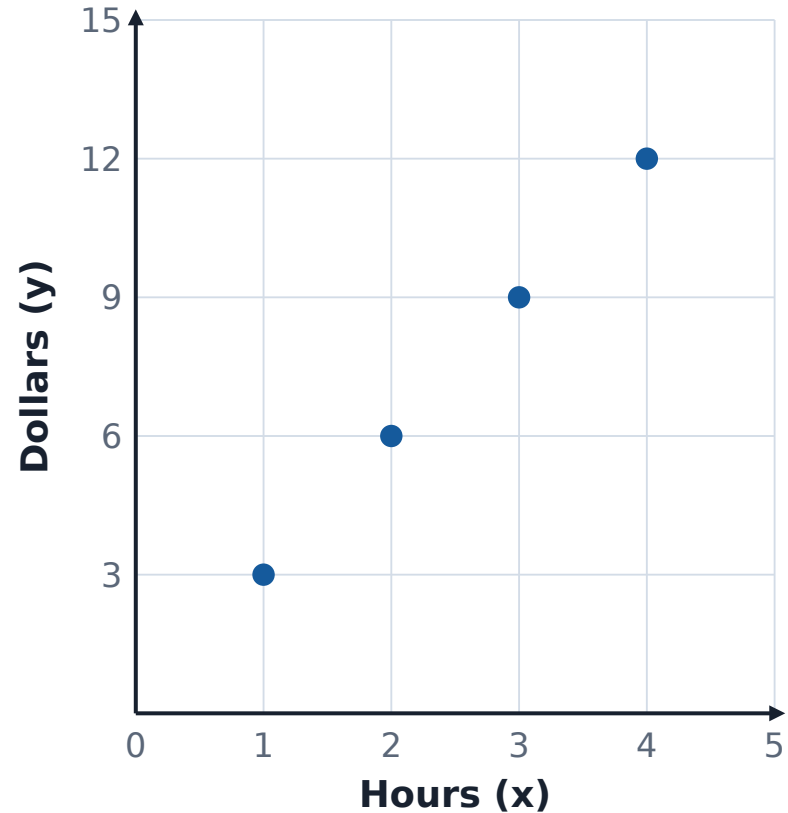
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 36 • MAIN • Q4/5

The graph shows y dollars earned for x hours worked.

Write an equation for the relationship.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 36 • MAIN • Q5/5

SET 36 • 6.EE.C.9

BACKWARD 1

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

True or false?

Pattern A: 0, 2, 4, 6

Pattern B: 0, 6, 12, 18

Each number in Pattern B is 3 times the matching number in Pattern A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 36 • BACKWARD 1 • Q1/5

The rule is "add 5," starting at 0.

What number comes next? 0, 5, 10, ____

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 36 • BACKWARD 1 • Q2/5

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

When Pattern A is 12, what is the matching number in Pattern B?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 36 • BACKWARD 1 • Q3/5

Pattern A starts at 0 and adds 1. Pattern B starts at 0 and adds 4.

Which ordered pairs come from the patterns?

A. (0, 0), (1, 4), (2, 8)

B. (0, 0), (4, 1), (8, 2)

C. (1, 4), (2, 5), (3, 6)

D. (0, 4), (1, 5), (2, 6)

True or false?

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Each number in Pattern B is 5 more than the matching number in Pattern A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 36 • BACKWARD 1 • Q5/5

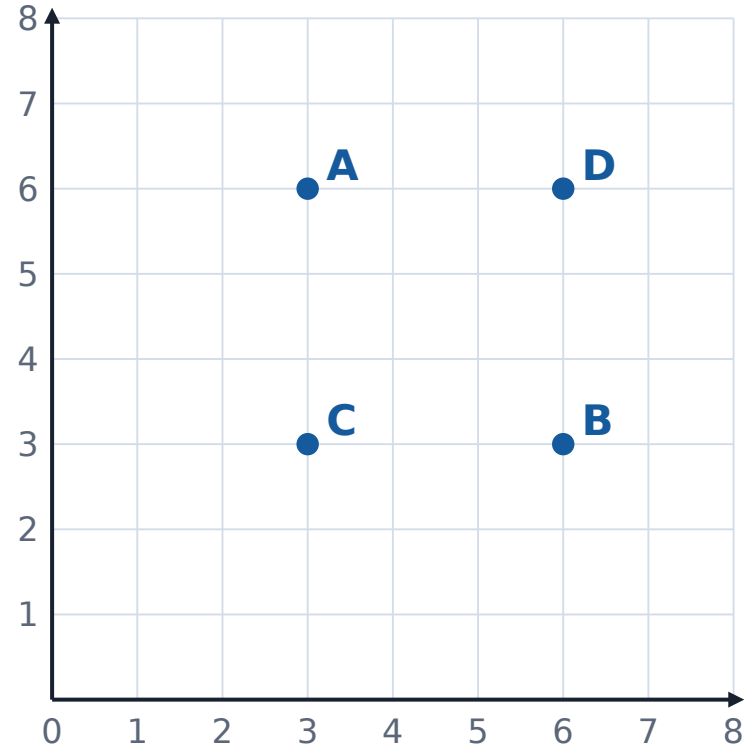
SET 36 • 6.EE.C.9

BACKWARD 2

Plotting and reading points in the first quadrant • Grade 5 • Geometry •
5.G.A.2

Which point is at (3, 6)?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D

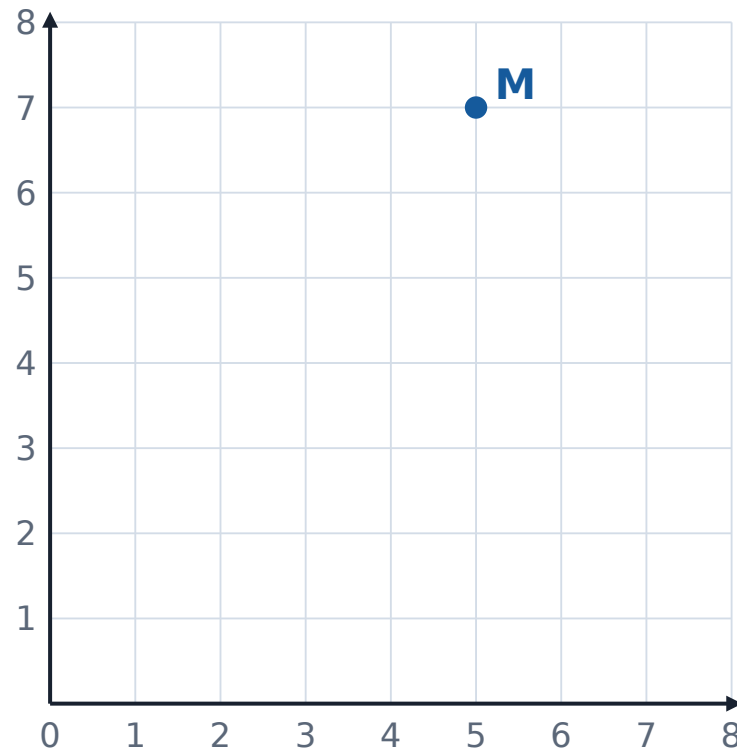


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 36 • BACKWARD 2 • Q1/5

What are the coordinates of point M?

M = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 36 • BACKWARD 2 • Q2/5

True or false?

To plot (2, 8), move 2 units right and 8 units up from the origin.

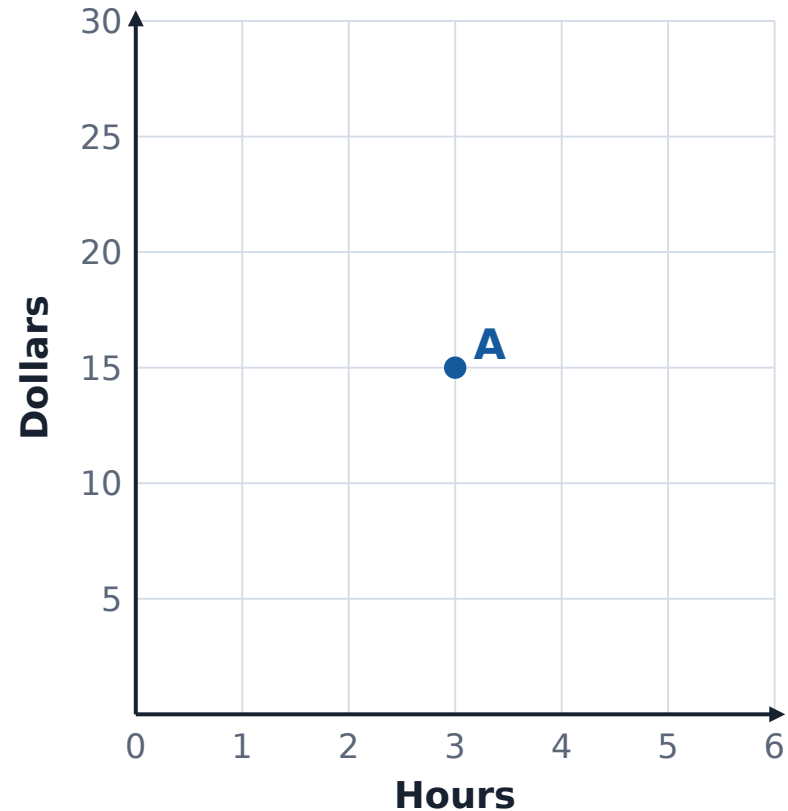
A. True

B. False

The graph shows dollars earned after working some hours.

How many dollars were earned after 3 hours?

Dollars: _____

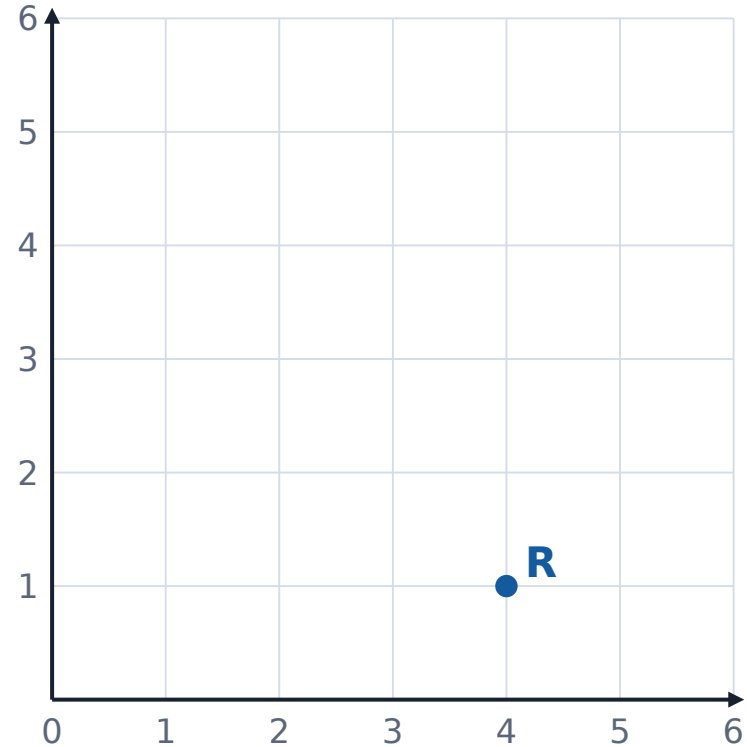


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 36 • BACKWARD 2 • Q4/5

Which ordered pair names point R?

- A.** (4, 1)
- B.** (1, 4)
- C.** (4, 4)
- D.** (1, 1)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 6.EE.C.9

BACKWARD 3

Input-output rules • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

What is the rule?

In	1	2	3	4
Out	5	10	15	20

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 36 • BACKWARD 3 • Q1/5

The rule is "multiply by 3."

What is the output when the input is 7?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 36 • BACKWARD 3 • Q2/5

True or false?

The rule is "add 6." When the input is 4, the output is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 36 • BACKWARD 3 • Q3/5

What is the rule?

In	2	4	6
Out	6	8	10

A. Add 4

B. Multiply by 3

C. Add 2

D. Multiply by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 36 • BACKWARD 3 • Q4/5

The rule is "subtract 2."

What is the output when the input is 15?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 36 • BACKWARD 3 • Q5/5

SET 36 • 6.EE.C.9

FORWARD 1

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Tomatoes cost \$3.20 per pound.

Write an equation for the cost c of p pounds.

Equation: _____

The table shows a proportional relationship.
Write an equation for the cost c of n ounces.

Ounces (n)	4	10
Cost (c)	\$1.20	\$3.00

Equation: _____

Which equation represents a proportional relationship?

A. $y = 4.5x$

B. $y = 4.5x + 1$

C. $y = x + 4.5$

D. $y = 4.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 36 • FORWARD 1 • Q3/5

A car uses 3 gallons of gas to travel 84 miles.

Write an equation for the miles m traveled on g gallons.

Equation: _____

True or false?

$y = \frac{2}{3}x$ represents a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 36 • FORWARD 1 • Q5/5

SET 36 • 6.EE.C.9

FORWARD 2

Compare functions represented in different ways • Grade 8 • Functions •
8.F.A.2

Function A is $y = 2x + 5$. Function B is shown in the table.

Which function has the greater rate of change?

x	0	1	2
y	1	4	7

- A.** Function B **B.** Function A **C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.A.2 • Set 36 • FORWARD 2 • Q1/5

Function A is a line through (0, 3) and (2, 7). Function B is $y = 3x - 1$.

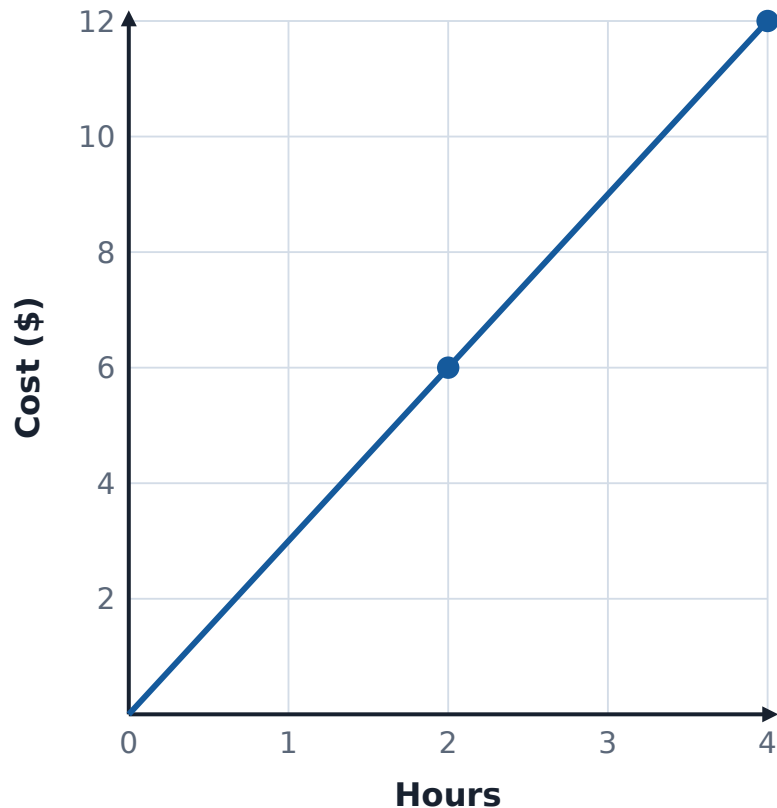
Which function has the greater initial value?

Answer: _____

Plan A costs \$5 plus \$2 per hour.
Plan B is shown in the graph.

Which plan has the greater rate of change?

- A.** Plan B
- B.** Plan A
- C.** They are the same.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.A.2 • Set 36 • FORWARD 2 • Q3/5

Function A is $y = -x + 10$. Function B is shown in the table.

Which function decreases faster?

x	0	2	4
y	10	6	2

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.A.2 • Set 36 • FORWARD 2 • Q4/5

True or false?

$y = 4x + 1$ has a greater rate of change than the function in the table.

x	0	1	2
y	0	5	10

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Functions • 8.F.A.2 • Set 36 • FORWARD 2 • Q5/5

Area of right triangles and other triangles

MAIN

Grade 6 main question

6.G.A.1

BACKWARD 1

Area of rectangles

3.MD.C.7.b

BACKWARD 2

Half of a product

5.NF.B.4.a

BACKWARD 3

Right angles and perpendicular segments

4.G.A.1

FORWARD 1

Area and volume of figures made of triangles

7.G.B.6

FORWARD 2

Use the Pythagorean Theorem in right triangles

8.G.B.7

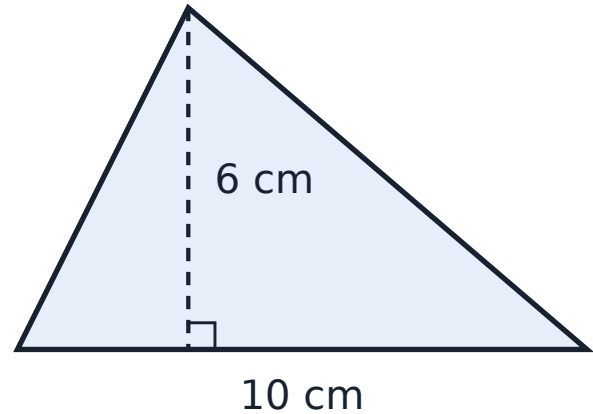
SET 37 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____

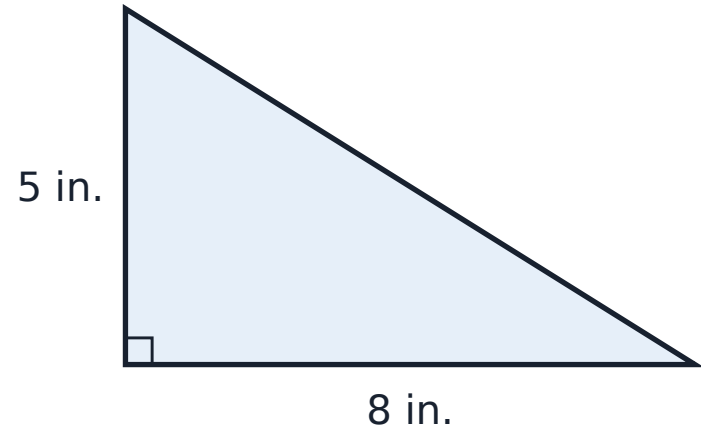


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 37 • MAIN • Q1/5

Find the area of the right triangle.

Area: _____

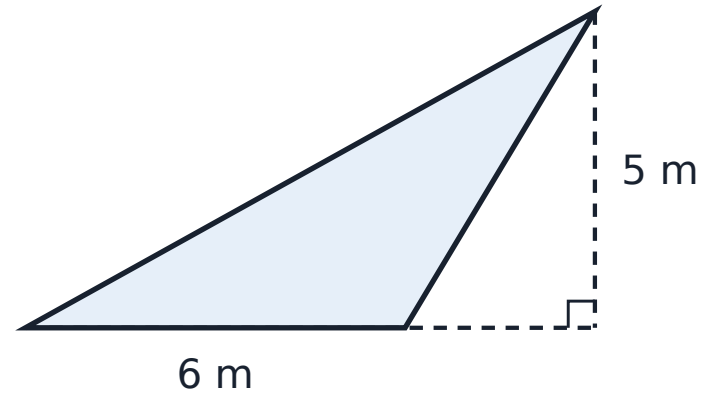


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 37 • MAIN • Q2/5

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 37 • MAIN • Q3/5

A triangle has a base of 12 feet and a height of 7 feet.

What is its area?

- A.** 42 square feet
- B.** 84 square feet
- C.** 19 square feet
- D.** 38 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 37 • MAIN • Q4/5

A triangular sail has an area of 36 square meters. Its base is 8 meters.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 37 • MAIN • Q5/5

SET 37 • 6.G.A.1

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 37 • BACKWARD 1 • Q1/5

A rectangle is 9 feet long and 6 feet wide.
What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 37 • BACKWARD 1 • Q2/5

True or false?

A rectangle that is 5 cm long and 3 cm wide has an area of 15 square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 37 • BACKWARD 1 • Q3/5

A rectangle is 12 m long and 4 m wide.

What is its area?

- A.** 48 square meters
- B.** 32 square meters
- C.** 16 square meters
- D.** 24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 37 • BACKWARD 1 • Q4/5

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 6.G.A.1

BACKWARD 2

Half of a product • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times 10 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 37 • BACKWARD 2 • Q1/5

Multiply.

$$\frac{1}{2} \times 14$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 37 • BACKWARD 2 • Q2/5

True or false?

$$\frac{1}{2} \times 8 \times 5 = 20$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 37 • BACKWARD 2 • Q3/5

$$\frac{1}{2} \times 12 \times 7 = ?$$

A. 42

B. 84

C. 19

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 37 • BACKWARD 2 • Q4/5

Multiply.

$$\frac{1}{2} \times 9 \times 4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 6.G.A.1

BACKWARD 3

Right angles and perpendicular segments • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.1 • Set 37 • BACKWARD 3 • Q1/5

Which angle measure is a right angle?

- A.** 90°
- B.** 45°
- C.** 180°
- D.** 100°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

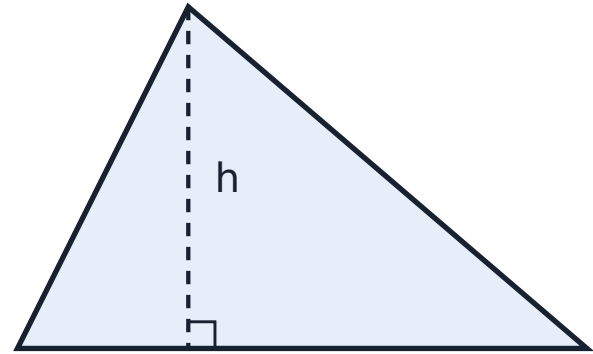
Grade 4 • Geometry • 4.G.A.1 • Set 37 • BACKWARD 3 • Q2/5

True or false?

The dashed segment is perpendicular to the base of the triangle.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.1 • Set 37 • BACKWARD 3 • Q3/5

How many degrees are in a right angle?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.1 • Set 37 • BACKWARD 3 • Q4/5

True or false?

A square has 4 right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.1 • Set 37 • BACKWARD 3 • Q5/5

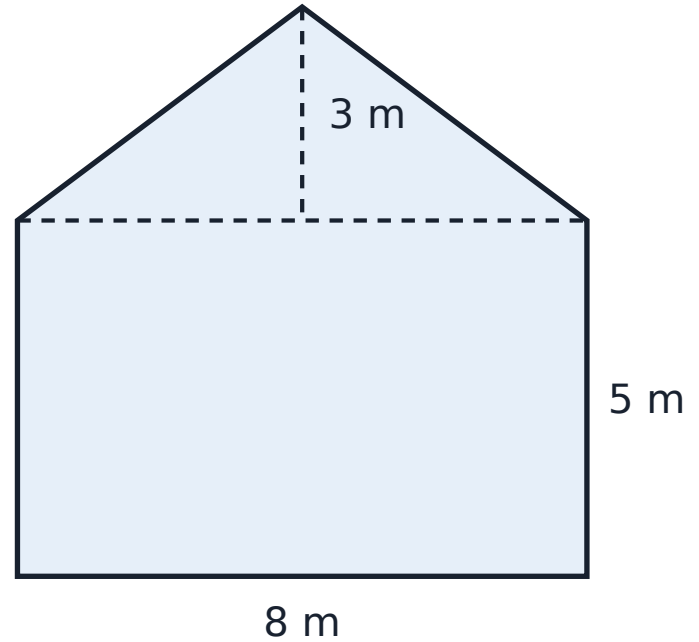
SET 37 • 6.G.A.1

FORWARD 1

Area and volume of figures made of triangles • Grade 7 • Geometry • 7.G.B.6

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.6 • Set 37 • FORWARD 1 • Q1/5

A kite is made of two triangles. Each triangle has a base of 10 inches and a height of 4 inches.

What is the area of the kite?

Area: _____

A triangle with a base of 4 cm and a height of 3 cm is cut out of a rectangle that is 12 cm by 6 cm.

What is the area of the part that is left?

- A.** 66 square cm
- B.** 72 square cm
- C.** 78 square cm
- D.** 60 square cm

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

True or false?

A figure made of a 6-by-4 rectangle and a triangle with a base of 3 and a height of 4 has an area of 30 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.6 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 6.G.A.1

FORWARD 2

Use the Pythagorean Theorem in right triangles • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs of 6 cm and 8 cm.
What is the length of the hypotenuse?

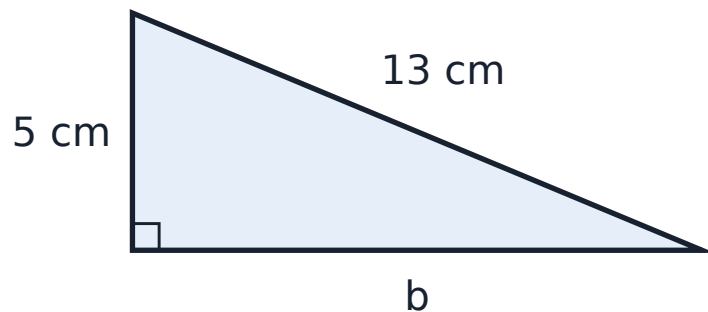
Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.7 • Set 37 • FORWARD 2 • Q1/5

Find the length b .

$b =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.7 • Set 37 • FORWARD 2 • Q2/5

A 10-foot ladder leans against a wall. The top reaches 8 feet up the wall.

How far is the bottom of the ladder from the wall?

A. 6 feet

B. 2 feet

C. 18 feet

D. 12.8 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.7 • Set 37 • FORWARD 2 • Q3/5

An isosceles triangle has a base of 16 inches and two sides of 10 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.7 • Set 37 • FORWARD 2 • Q4/5

A right triangle has legs of 9 m and 12 m.
What is the length of the hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.7 • Set 37 • FORWARD 2 • Q5/5

Area of parallelograms and trapezoids

MAIN	Grade 6 main question	6.G.A.1
BACKWARD 1	Area of rectangles	3.MD.C.7.b
BACKWARD 2	Area by splitting into rectangles	3.MD.C.7.d
BACKWARD 3	Classifying quadrilaterals	5.G.B.4
BACKWARD 4	Height is perpendicular to the base	4.G.A.1
FORWARD 1	Actual areas from scale drawings	7.G.A.1
FORWARD 2	Similar figures and dilations	8.G.A.4

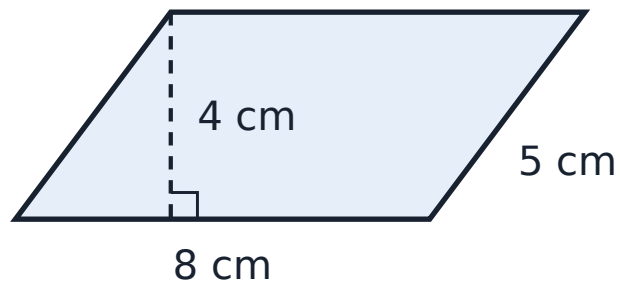
SET 38 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

Find the area of the
parallelogram.

Area: _____

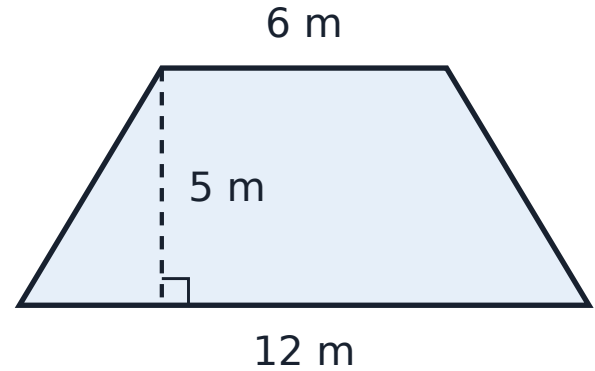


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 38 • MAIN • Q1/5

Find the area of the trapezoid.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 38 • MAIN • Q2/5

A parallelogram has a base of 9 feet and a height of 5 feet.

What is its area?

A. 45 square feet

B. 22.5 square feet

C. 14 square feet

D. 28 square feet

A trapezoid has bases of 7 inches and 11 inches. Its height is 4 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 38 • MAIN • Q4/5

A parallelogram has an area of 72 square centimeters and a base of 12 centimeters.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 38 • MAIN • Q5/5

SET 38 • 6.G.A.1

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 38 • BACKWARD 1 • Q1/5

A rectangle is 11 cm long and 5 cm wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 38 • BACKWARD 1 • Q2/5

True or false?

A square with 6-meter sides has an area of 24 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 38 • BACKWARD 1 • Q3/5

A rectangle is 15 inches long and 2 inches wide.

What is its area?

- A.** 30 square inches
- B.** 17 square inches
- C.** 34 square inches
- D.** 60 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 38 • BACKWARD 1 • Q4/5

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 38 • BACKWARD 1 • Q5/5

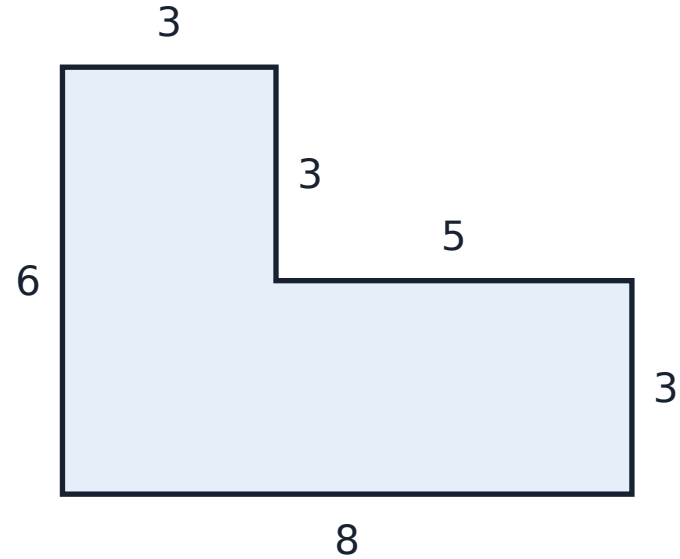
SET 38 • 6.G.A.1

BACKWARD 2

Area by splitting into rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

Find the area of the figure.

Area: _____

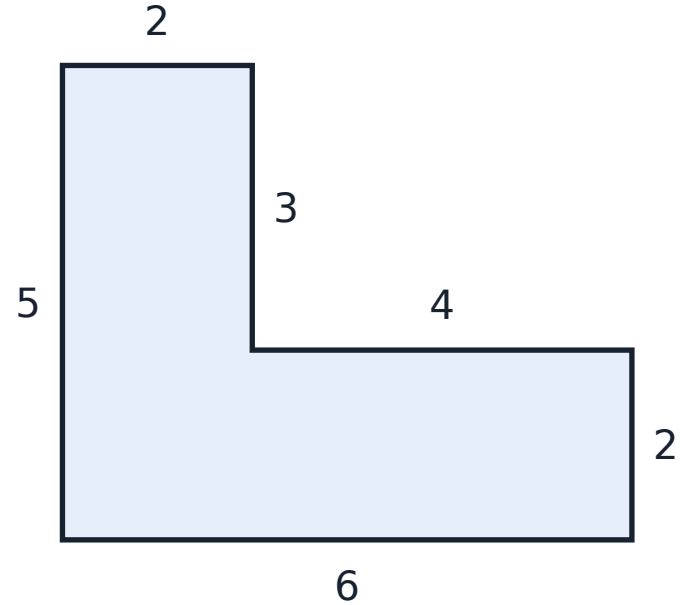


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 38 • BACKWARD 2 • Q1/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 38 • BACKWARD 2 • Q2/5

True or false?

The area of an L-shaped figure can be found by adding the areas of two rectangles.

- A.** True
- B.** False

A figure is made of a 4-by-5 rectangle and a 2-by-3 rectangle that do not overlap.

What is its area?

A. 26 square units

B. 14 square units

C. 20 square units

D. 120 square units

A figure is made of a 10-by-2 rectangle and a 3-by-3 square that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 6.G.A.1

BACKWARD 3

Classifying quadrilaterals • Grade 5 • Geometry • 5.G.B.4

True or false?

Every rectangle is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.B.4 • Set 38 • BACKWARD 3 • Q1/5

True or false?

Every parallelogram is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.B.4 • Set 38 • BACKWARD 3 • Q2/5

Which quadrilateral always has four right angles?

- A.** Rectangle
- B.** Rhombus
- C.** Trapezoid
- D.** Parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.B.4 • Set 38 • BACKWARD 3 • Q3/5

How many pairs of parallel sides does a parallelogram have?

A. 2

B. 1

C. 0

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.B.4 • Set 38 • BACKWARD 3 • Q4/5

True or false?

A square is a rhombus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.B.4 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • 6.G.A.1

BACKWARD 4

Height is perpendicular to the base • Grade 4 • Geometry • 4.G.A.1

True or false?

The height of a parallelogram is perpendicular to its base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

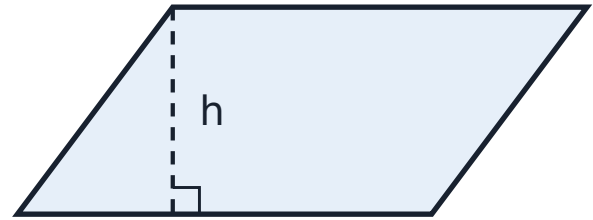
Grade 4 • Geometry • 4.G.A.1 • Set 38 • BACKWARD 4 • Q1/5

True or false?

The dashed segment shows the height of the parallelogram.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.1 • Set 38 • BACKWARD 4 • Q2/5

Which statement describes perpendicular lines?

- A.** They meet at right angles.
- B.** They never meet.
- C.** They meet at a 45° angle.
- D.** They are always the same length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.1 • Set 38 • BACKWARD 4 • Q3/5

True or false?

The slanted side of a parallelogram is always its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.1 • Set 38 • BACKWARD 4 • Q4/5

What is the measure of the angle between a base and its height?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Geometry • 4.G.A.1 • Set 38 • BACKWARD 4 • Q5/5

SET 38 • 6.G.A.1

FORWARD 1

Actual areas from scale drawings • Grade 7 • Geometry • 7.G.A.1

A scale drawing uses 1 cm : 3 m. A rectangle on the drawing is 4 cm by 2 cm.

What is the actual area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.1 • Set 38 • FORWARD 1 • Q1/5

A scale drawing uses 1 in. : 5 ft. A parallelogram on the drawing has a base of 3 in. and a height of 2 in.

What is the actual area?

Area: _____

A drawing uses 1 cm : 10 m. A trapezoid on the drawing has bases of 2 cm and 4 cm and a height of 3 cm.

What is the actual area?

- A.** 900 square meters
- B.** 90 square meters
- C.** 9 square meters
- D.** 9,000 square meters

A park is 400 m by 300 m. A scale drawing uses 1 cm : 100 m.

What are the dimensions of the park in the drawing?

Dimensions: _____

True or false?

A scale drawing uses 1 in. : 2 ft. A 3 in. by 4 in. rectangle on the drawing represents 24 square feet.

A. True

B. False

SET 38 • 6.G.A.1

FORWARD 2

Similar figures and dilations • Grade 8 • Geometry • 8.G.A.4

Rectangle A is 3 by 5. Rectangle B is 9 by 15. Rectangle B is a dilation of Rectangle A.

What is the scale factor?

Scale factor: _____

Which rectangle is similar to a 4-by-6 rectangle?

- A.** 6 by 9
- B.** 5 by 7
- C.** 8 by 10
- D.** 4 by 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.A.4 • Set 38 • FORWARD 2 • Q2/5

True or false?

Two parallelograms have matching angles that are equal. One has sides 2 and 5, and the other has sides 6 and 15. The parallelograms are similar.

A. True

B. False

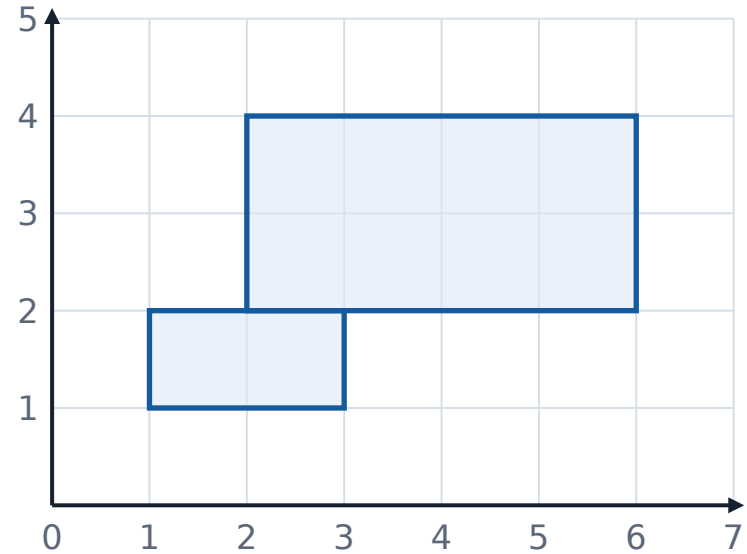
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.A.4 • Set 38 • FORWARD 2 • Q3/5

The smaller rectangle is dilated with the center at the origin to form the larger rectangle.

What is the scale factor?

Scale factor: _____



Triangle PQR is similar to triangle STU. The scale factor from PQR to STU is $\frac{1}{2}$.

PQ = 10. What is ST?

ST = _____

Area of polygons by composing and decomposing

MAIN	Grade 6 main question	6.G.A.1
BACKWARD 1	Area of rectilinear figures	3.MD.C.7.d
BACKWARD 2	Missing side lengths of rectangles	4.MD.A.3
BACKWARD 3	Area of triangles	6.G.A.1
FORWARD 1	Area of composite figures in real-world problems	7.G.B.6
FORWARD 2	Volume of composite solids	8.G.C.9

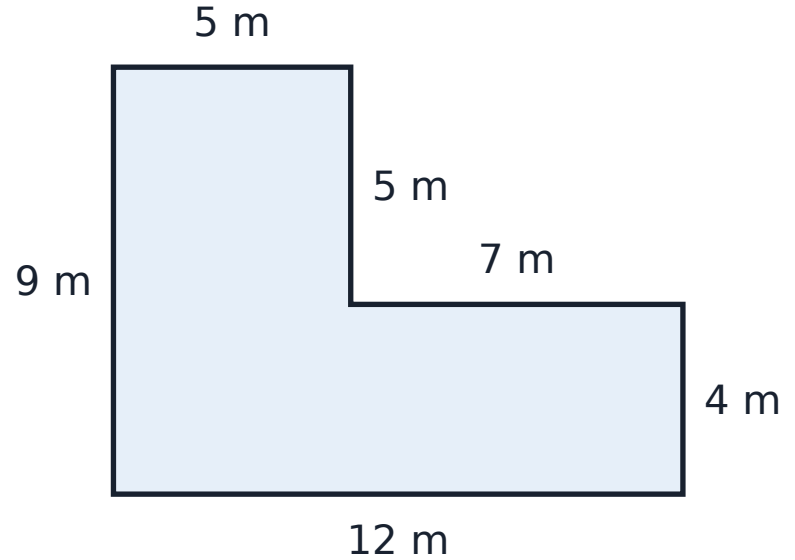
SET 39 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

Find the area of the figure.

Area: _____

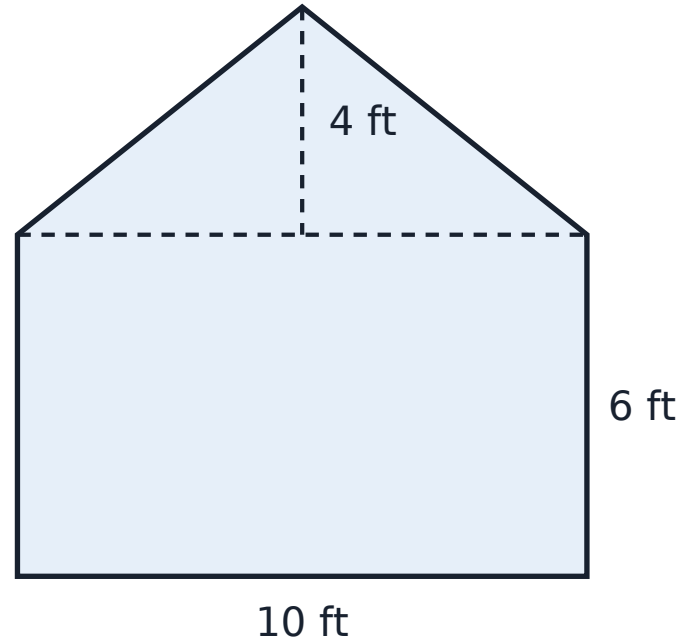


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 39 • MAIN • Q1/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 39 • MAIN • Q2/5

A rectangular yard is 20 m by 15 m. A triangular garden in the yard has a base of 6 m and a height of 5 m.

What is the area of the yard that is NOT garden?

Area: _____

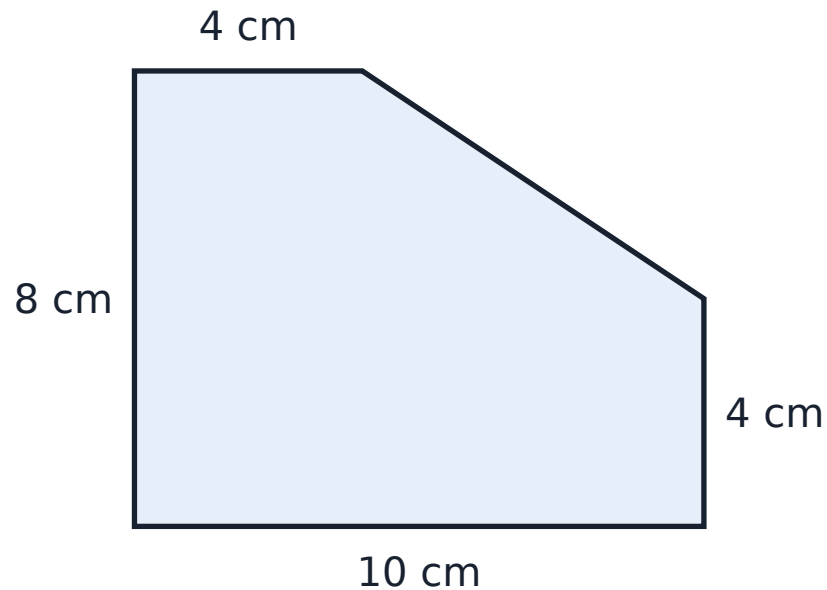
A polygon is split into a rectangle that is 8 by 3 and two triangles. Each triangle has a base of 2 and a height of 3.

What is the area of the polygon?

- A.** 30 square units
- B.** 27 square units
- C.** 36 square units
- D.** 24 square units

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 39 • MAIN • Q5/5

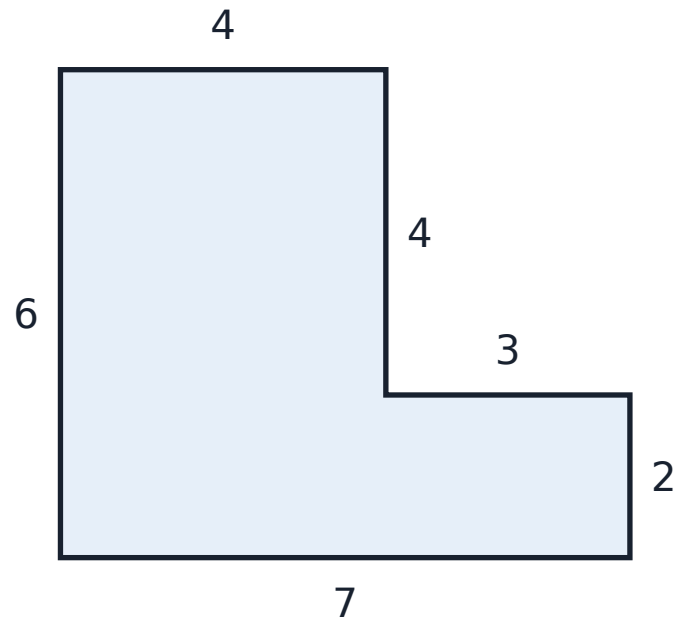
SET 39 • 6.G.A.1

BACKWARD 1

Area of rectilinear figures • Grade 3 • Measurement & Data • 3.MD.C.7.d

Find the area of the figure.

Area: _____

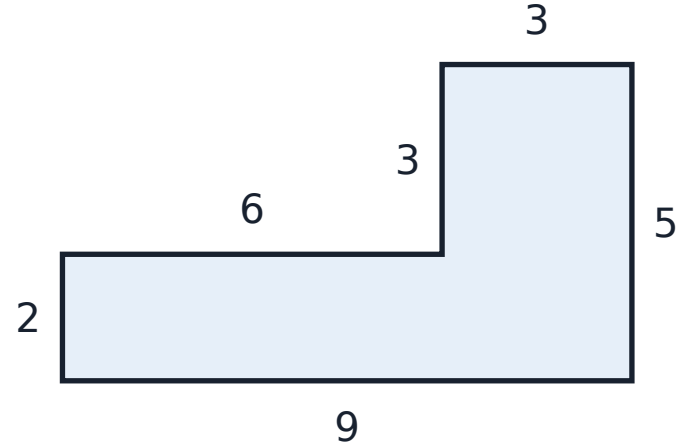


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 39 • BACKWARD 1 • Q1/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 39 • BACKWARD 1 • Q2/5

True or false?

A figure made of a 5-by-2 rectangle and a 3-by-4 rectangle has an area of 22 square units.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 39 • BACKWARD 1 • Q3/5

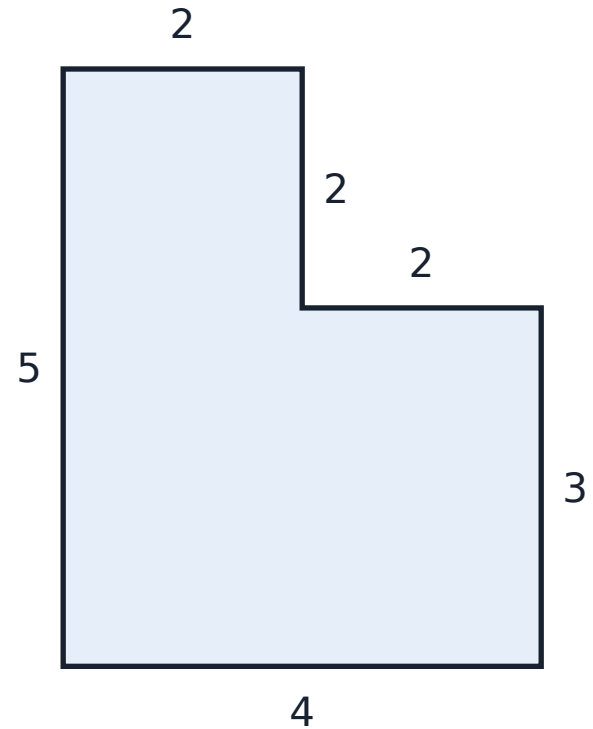
Which expression gives the area of the figure?

A. $4 \times 3 + 2 \times 2$

B. 4×5

C. $4 + 3 + 2 + 2$

D. $4 \times 3 \times 2$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 39 • BACKWARD 1 • Q4/5

A figure is made of two rectangles: 6 by 3 and 2 by 4.
What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 6.G.A.1

BACKWARD 2

Missing side lengths of rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has an area of 48 square feet and a width of 6 feet.

What is its length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 39 • BACKWARD 2 • Q1/5

A rectangle has a perimeter of 30 cm and a length of 9 cm.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 39 • BACKWARD 2 • Q2/5

True or false?

A rectangle with an area of 36 square cm and a length of 9 cm has a width of 4 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 39 • BACKWARD 2 • Q3/5

A rectangle has an area of 56 square inches and a width of 7 inches.

What is its length?

A. 8 inches

B. 49 inches

C. 63 inches

D. 9 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 39 • BACKWARD 2 • Q4/5

The rectangle has an area of 60 square meters.

What is the missing side length?



Length: _____

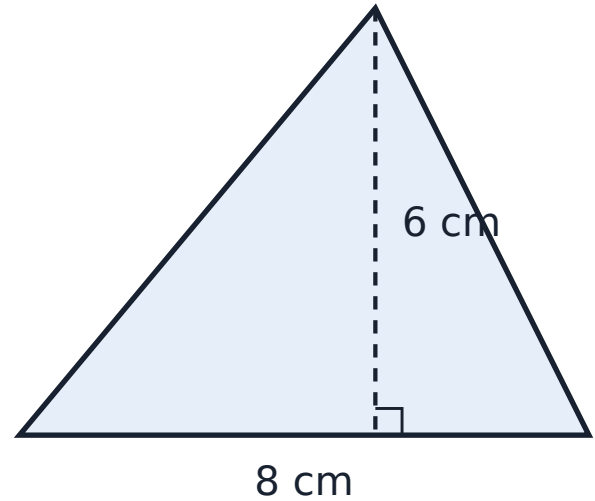
SET 39 • 6.G.A.1

BACKWARD 3

Area of triangles • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 39 • BACKWARD 3 • Q1/5

A triangle has a base of 10 m and a height of 3 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 39 • BACKWARD 3 • Q2/5

True or false?

A right triangle with legs of 4 in. and 6 in. has an area of 24 square inches.

A. True

B. False

A triangle has a base of 5 ft and a height of 8 ft.

What is its area?

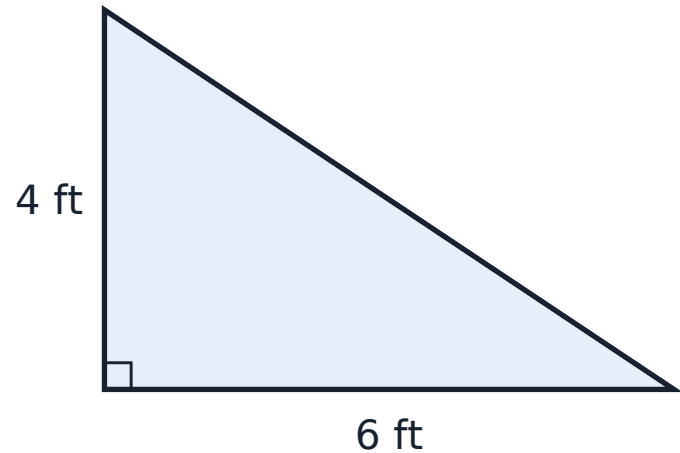
- A.** 20 square feet
- B.** 40 square feet
- C.** 13 square feet
- D.** 26 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 39 • BACKWARD 3 • Q4/5

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 39 • BACKWARD 3 • Q5/5

SET 39 • 6.G.A.1

FORWARD 1

Area of composite figures in real-world problems • Grade 7 • Geometry •
7.G.B.6

A deck is made of a rectangle that is 12 ft by 10 ft and a triangle with a base of 10 ft and a height of 6 ft.

What is the area of the deck?

Area: _____

A trapezoid-shaped garden has bases of 8 m and 14 m and a height of 5 m. It has a square pond with 2 m sides.

What is the area of the garden without the pond?

Area: _____

A regular hexagon is split into 6 equal triangles. Each triangle has a base of 4 cm and a height of 3.5 cm.

What is the area of the hexagon?

- A.** 42 square cm
- B.** 84 square cm
- C.** 21 square cm
- D.** 24 square cm

An L-shaped prism has a base area of 30 square feet and a height of 4 feet.

What is its volume?

Volume: _____

True or false?

A figure made of a 5-by-5 square and a triangle with a base of 5 and a height of 2 has an area of 30 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.6 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 6.G.A.1

FORWARD 2

Volume of composite solids • Grade 8 • Geometry • 8.G.C.9

A silo is a cylinder with a radius of 3 m and a height of 10 m, topped by a hemisphere with a radius of 3 m.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 39 • FORWARD 2 • Q1/5

A cone has a radius of 3 in. and a height of 4 in. A hemisphere with a radius of 3 in. sits on top.

What is the total volume in terms of π ?

Volume: _____

A cone with a radius of 2 and a height of 3 is cut out of a cylinder with a radius of 2 and a height of 6.

What volume is left?

A. 20π

B. 28π

C. 24π

D. 4π

A cylinder and a cone each have a radius of 1 and a height of 2.

How much more volume does the cylinder have, in terms of π ?

Difference: _____

True or false?

A sphere with a radius of 3 has the same volume as a cylinder with a radius of 3 and a height of 4.

A. True

B. False

Volume of right rectangular prisms with fractional edge lengths

MAIN	Grade 6 main question	6.G.A.2
BACKWARD 1	Volume by counting unit cubes	5.MD.C.4
BACKWARD 2	Volume formula with whole numbers	5.MD.C.5.b
BACKWARD 3	Multiplying mixed numbers	5.NF.B.6
FORWARD 1	Volume of prisms in real-world problems	7.G.B.6
FORWARD 2	Volume of cylinders, cones, and spheres	8.G.C.9

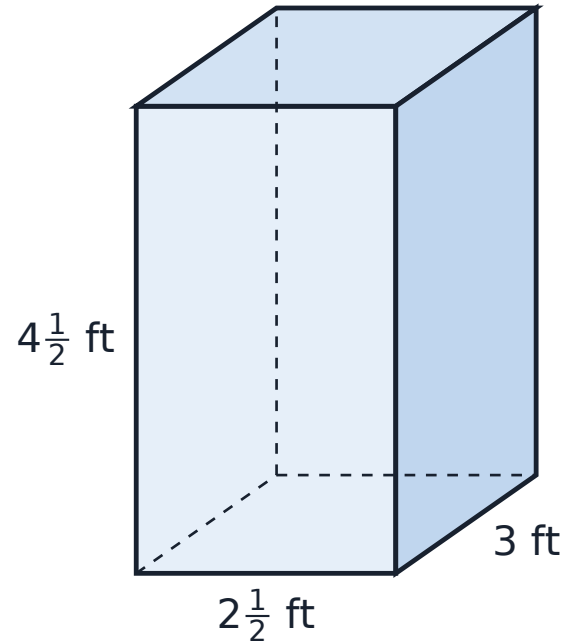
SET 40 • 6.G.A.2

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

Find the volume of the rectangular prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.2 • Set 40 • MAIN • Q1/5

How many cubes with $\frac{1}{2}$ -inch edges fill a box that is 2 in. by $1\frac{1}{2}$ in. by 1 in.?

Cubes: _____

A box is $1\frac{1}{2}$ ft long, 2 ft wide, and $\frac{3}{4}$ ft tall.

What is its volume?

- A.** $2\frac{1}{4}$ cubic feet
- B.** $4\frac{1}{4}$ cubic feet
- C.** 3 cubic feet
- D.** $2\frac{3}{4}$ cubic feet

Use $V = l \times w \times h$ to find the volume of a prism with $l = 5$ cm, $w = 2\frac{1}{2}$ cm, and $h = 1\frac{1}{5}$ cm.

$V =$ _____

True or false?

A prism is packed with 48 cubes that each have $\frac{1}{2}$ -unit edges. Its volume is 6 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.2 • Set 40 • MAIN • Q5/5

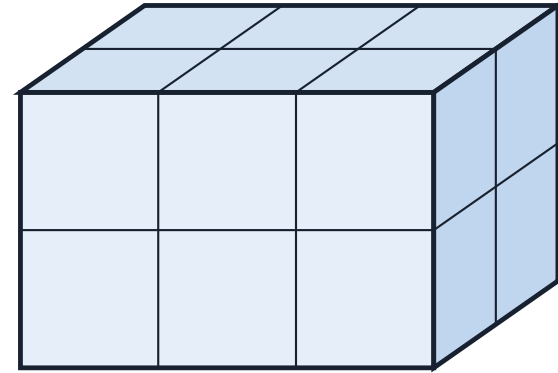
SET 40 • 6.G.A.2

BACKWARD 1

Volume by counting unit cubes • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic unit.

What is the volume of the prism?



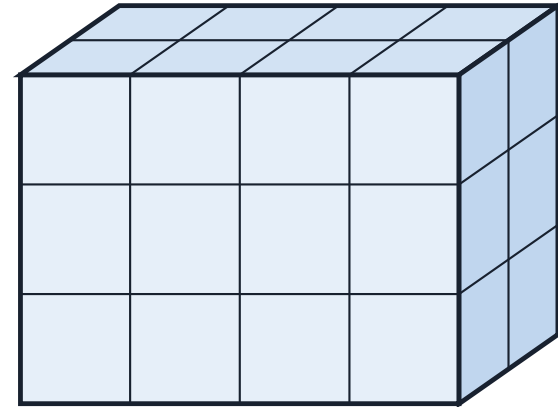
Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 40 • BACKWARD 1 • Q1/5

Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 40 • BACKWARD 1 • Q2/5

True or false?

A prism made of 2 layers with 6 unit cubes in each layer has a volume of 12 cubic units.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 40 • BACKWARD 1 • Q3/5

A box holds 3 layers with 5 unit cubes in each layer.

What is its volume?

- A.** 15 cubic units
- B.** 8 cubic units
- C.** 35 cubic units
- D.** 53 cubic units

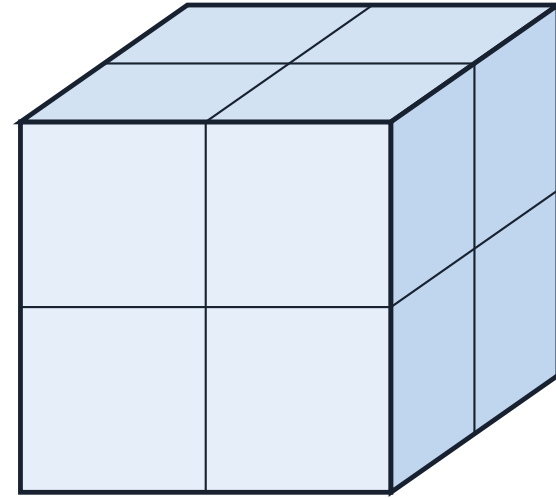
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 40 • BACKWARD 1 • Q4/5

Each cube is 1 cubic unit.

What is the volume of the prism?

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 40 • BACKWARD 1 • Q5/5

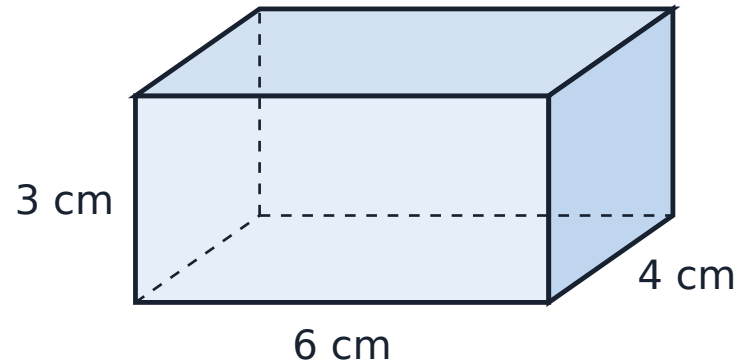
SET 40 • 6.G.A.2

BACKWARD 2

Volume formula with whole numbers • Grade 5 • Measurement & Data •
5.MD.C.5.b

Find the volume of the prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 40 • BACKWARD 2 • Q1/5

A prism has a length of 10 m, a width of 2 m, and a height of 5 m.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 40 • BACKWARD 2 • Q2/5

True or false?

A cube with 3-inch edges has a volume of 27 cubic inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 40 • BACKWARD 2 • Q3/5

A box is 8 in. by 5 in. by 2 in.

What is its volume?

- A.** 80 cubic inches
- B.** 15 cubic inches
- C.** 40 cubic inches
- D.** 160 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 40 • BACKWARD 2 • Q4/5

A prism has a base area of 20 square cm and a height of 6 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 6.G.A.2

BACKWARD 3

Multiplying mixed numbers • Grade 5 • Number & Operations–Fractions •
5.NF.B.6

Multiply.

$$2\frac{1}{2} \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 40 • BACKWARD 3 • Q1/5

Multiply.

$$1\frac{1}{2} \times \frac{3}{4}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 40 • BACKWARD 3 • Q2/5

True or false?

$$2\frac{1}{2} \times 2 = 5$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 40 • BACKWARD 3 • Q3/5

$$3 \times 4\frac{1}{2} = ?$$

A. $13\frac{1}{2}$

B. $12\frac{1}{2}$

C. $7\frac{1}{2}$

D. $12\frac{1}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 40 • BACKWARD 3 • Q4/5

Multiply.

$$1\frac{1}{3} \times 1\frac{1}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • 6.G.A.2

FORWARD 1

Volume of prisms in real-world problems • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 5 in. and a height of 4 in. The prism is 8 in. long.

What is its volume?

Volume: _____

A prism has a base area of $12\frac{1}{2}$ square cm and a height of 4 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.6 • Set 40 • FORWARD 1 • Q2/5

A cube has edges of 2.5 cm.

What is its volume?

- A.** 15.625 cubic cm
- B.** 7.5 cubic cm
- C.** 6.25 cubic cm
- D.** 15 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.6 • Set 40 • FORWARD 1 • Q3/5

An aquarium is 30 in. long, 12 in. wide, and 16 in. tall. It is filled with water to a height of 12 in.

What is the volume of the water?

Volume: _____

True or false?

A triangular prism with a base area of 15 square units and a length of 6 units has a volume of 90 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.6 • Set 40 • FORWARD 1 • Q5/5

SET 40 • 6.G.A.2

FORWARD 2

Volume of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 5 and a height of 4.
What is its volume in terms of π ?

Volume: _____

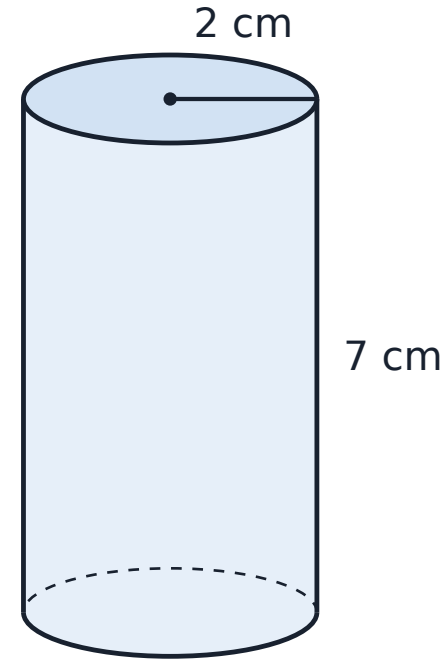
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 2 • Q1/5

Find the volume of the cylinder.

Use 3.14 for π .

Volume: _____



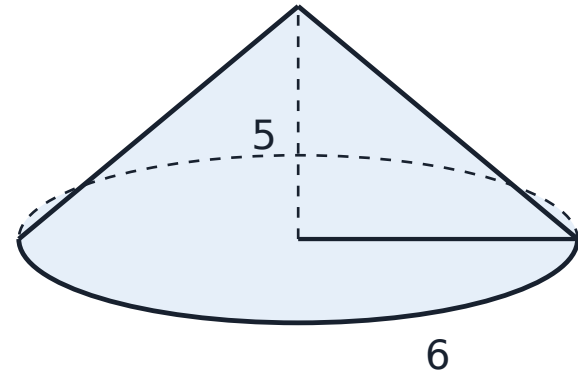
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 2 • Q2/5

A cone has a radius of 6 and a height of 5.

What is its volume?

- A.** 60π **B.** 180π
C. 30π **D.** 150π



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 2 • Q3/5

A sphere has a radius of 6.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 2 • Q4/5

True or false?

A cylinder has a greater volume than a cone with the same radius and height.

A. True

B. False

Polygons in the coordinate plane; side lengths from coordinates

MAIN	Grade 6 main question	6.G.A.3
BACKWARD 1	Naming points in all four quadrants	6.NS.C.6.c
BACKWARD 2	Distance between points with the same x or y	6.NS.C.8
BACKWARD 3	Perimeter of polygons	3.MD.D.8
BACKWARD 4	Properties of rectangles	3.G.A.1
FORWARD 1	Scale drawings on a grid	7.G.A.1
FORWARD 2	Transform polygons using coordinates	8.G.A.3

SET 41 • 6.G.A.3

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.3

A rectangle has vertices at $(-3, 2)$, $(4, 2)$, $(4, -5)$, and $(-3, -5)$.

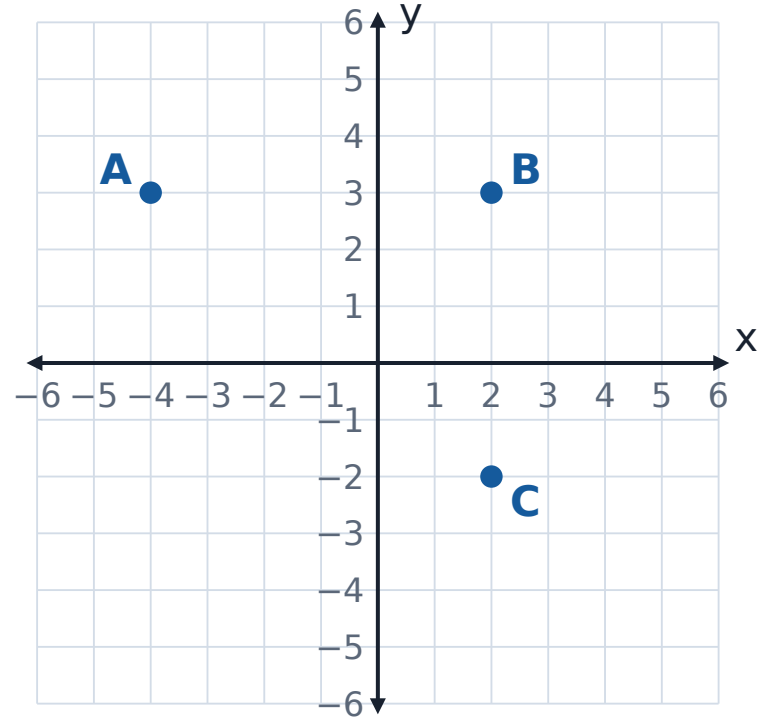
What is its perimeter?

Perimeter: _____

Points A, B, and C are three vertices of rectangle ABCD.

What are the coordinates of point D?

D = _____



One side of a polygon goes from $(-2, -3)$ to $(-2, 5)$.

How long is this side?

Length: _____

A triangle has vertices at $(0, 0)$, $(6, 0)$, and $(0, 4)$.

What is its area?

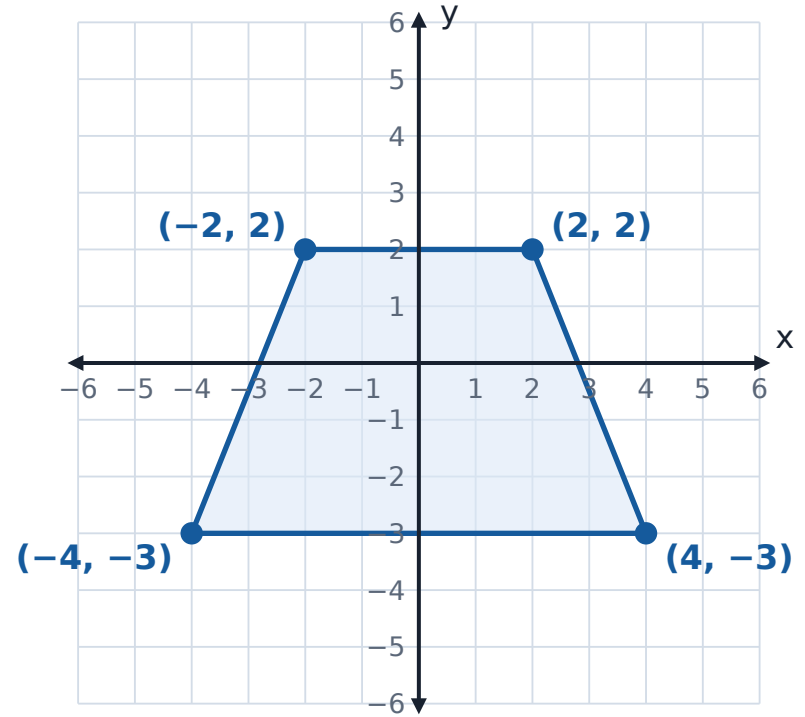
- A.** 12 square units
- B.** 24 square units
- C.** 10 square units
- D.** 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.3 • Set 41 • MAIN • Q4/5

Find the area of the trapezoid.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.3 • Set 41 • MAIN • Q5/5

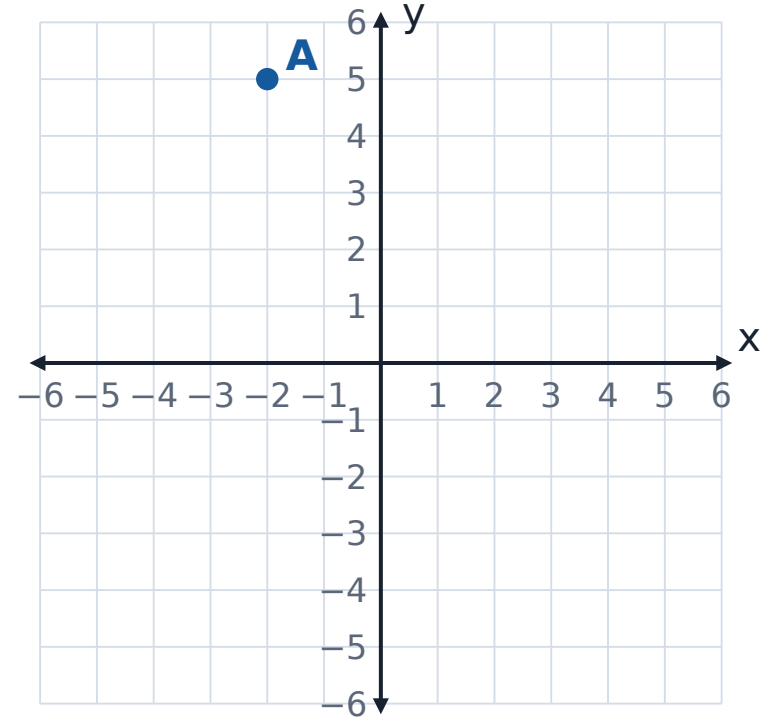
SET 41 • 6.G.A.3

BACKWARD 1

Naming points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

What are the coordinates of point A?

A = _____

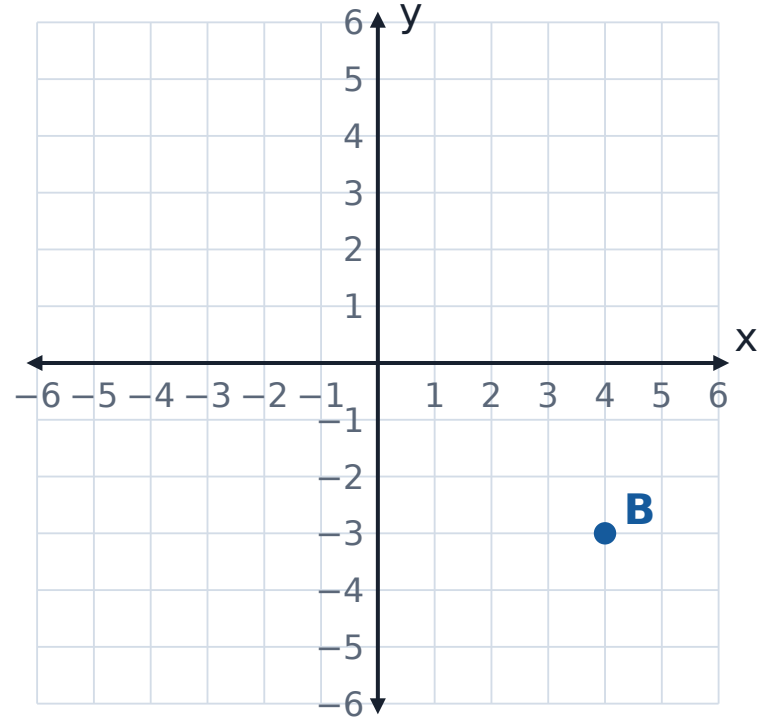


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 41 • BACKWARD 1 • Q1/5

What are the coordinates of point B?

B = _____

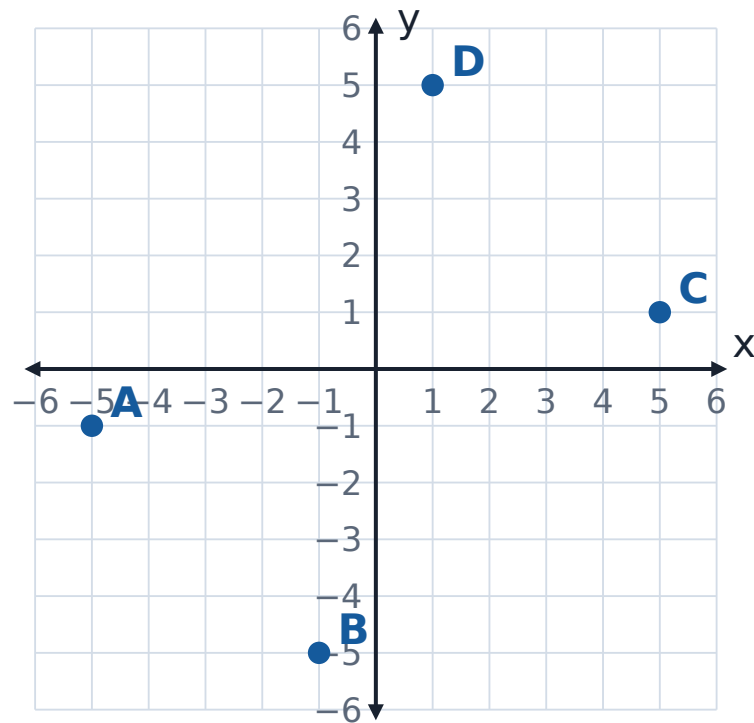


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 41 • BACKWARD 1 • Q2/5

Which point is at $(-5, -1)$?

- A.** Point A
- B.** Point B
- C.** Point C
- D.** Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

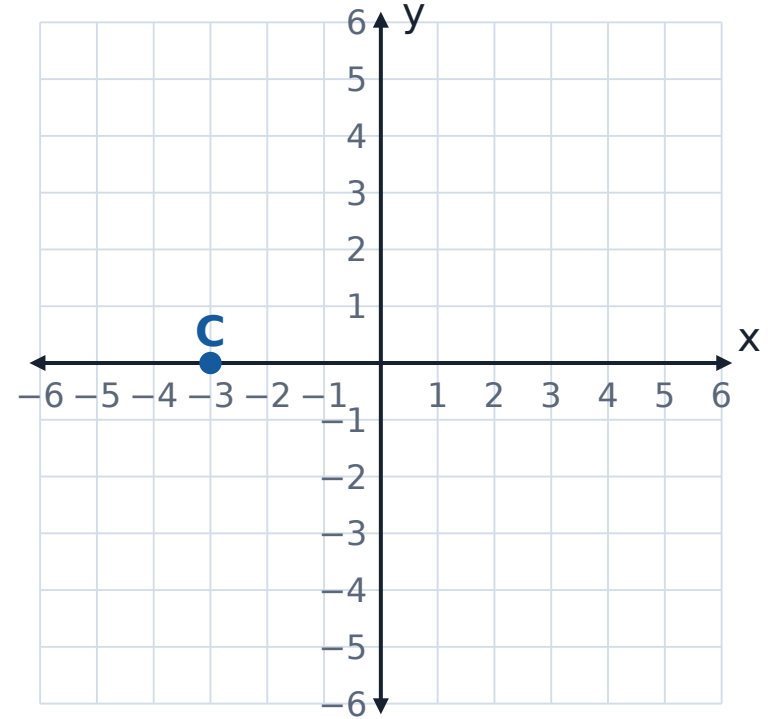
Grade 6 • The Number System • 6.NS.C.6.c • Set 41 • BACKWARD 1 • Q3/5

True or false?

Point C is at $(-3, 0)$.

A. True

B. False

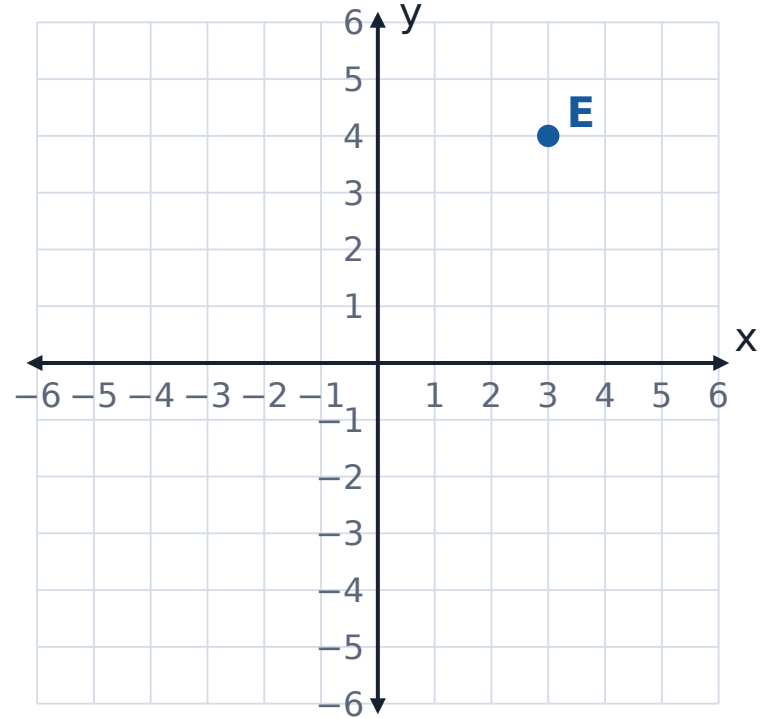


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 41 • BACKWARD 1 • Q4/5

What are the coordinates of point E?

E = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.6.c • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 6.G.A.3

BACKWARD 2

Distance between points with the same x or y • Grade 6 • The Number System
• 6.NS.C.8

Find the distance between $(-3, 2)$ and $(4, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.8 • Set 41 • BACKWARD 2 • Q1/5

Find the distance between $(1, -4)$ and $(1, 6)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.8 • Set 41 • BACKWARD 2 • Q2/5

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.8 • Set 41 • BACKWARD 2 • Q3/5

What is the distance between $(2, -3)$ and $(2, -8)$?

- A.** 5
- B.** 11
- C.** -5
- D.** -11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.8 • Set 41 • BACKWARD 2 • Q4/5

Find the distance between $(-6, -1)$ and $(-2, -1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.8 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • 6.G.A.3

BACKWARD 3

Perimeter of polygons • Grade 3 • Measurement & Data • 3.MD.D.8

Find the perimeter of the rectangle.

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 41 • BACKWARD 3 • Q1/5

A square has sides of 9 inches.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 41 • BACKWARD 3 • Q2/5

True or false?

A rectangle that is 10 m by 4 m has a perimeter of 28 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 41 • BACKWARD 3 • Q3/5

A triangle has sides of 5 cm, 6 cm, and 7 cm.

What is its perimeter?

- A.** 18 cm
- B.** 30 cm
- C.** 11 cm
- D.** 210 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 41 • BACKWARD 3 • Q4/5

A rectangle is 12 ft long and 5 ft wide.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 41 • BACKWARD 3 • Q5/5

SET 41 • 6.G.A.3

BACKWARD 4

Properties of rectangles • Grade 3 • Geometry • 3.G.A.1

True or false?

Opposite sides of a rectangle have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Geometry • 3.G.A.1 • Set 41 • BACKWARD 4 • Q1/5

True or false?

A rectangle has 4 right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Geometry • 3.G.A.1 • Set 41 • BACKWARD 4 • Q2/5

Which shape always has 4 sides of equal length?

- A.** Rhombus
- B.** Rectangle
- C.** Trapezoid
- D.** Parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Geometry • 3.G.A.1 • Set 41 • BACKWARD 4 • Q3/5

True or false?

Every quadrilateral has 4 right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Geometry • 3.G.A.1 • Set 41 • BACKWARD 4 • Q4/5

How many sides does a quadrilateral have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Geometry • 3.G.A.1 • Set 41 • BACKWARD 4 • Q5/5

SET 41 • 6.G.A.3

FORWARD 1

Scale drawings on a grid • Grade 7 • Geometry • 7.G.A.1

A rectangle that is 3 units by 2 units is redrawn with a scale factor of 2.

What are the new dimensions?

Dimensions: _____

On a floor plan, each grid unit is 5 feet. A room has corners at $(1, 1)$, $(5, 1)$, $(5, 4)$, and $(1, 4)$.

What are the actual length and width of the room?

Length and width: _____

A triangle on a grid has a base of 4 units and a height of 3 units. It is enlarged by a scale factor of 3.

What is the area of the new triangle?

- A.** 54 square units
- B.** 18 square units
- C.** 36 square units
- D.** 6 square units

On a map grid, 1 unit represents 2 km. A park is a rectangle 3 units by 5 units.

What is the actual area of the park?

Area: _____

True or false?

When a scale drawing is enlarged by a scale factor of 2, its area doubles.

A. True

B. False

SET 41 • 6.G.A.3

FORWARD 2

Transform polygons using coordinates • Grade 8 • Geometry • 8.G.A.3

A rectangle has vertices $(1, 2)$, $(4, 2)$, $(4, 5)$, and $(1, 5)$.
It is translated 3 units left and 4 units down.

What are the new vertices?

Vertices: _____

A triangle has vertices $(2, 1)$, $(5, 1)$, and $(2, 4)$. It is dilated by a scale factor of 2 with the center at the origin.

What is the image of $(5, 1)$?

A. $(10, 2)$

B. $(7, 3)$

C. $(5, 2)$

D. $(2.5, 0.5)$

A square has vertices $(1, 1)$, $(3, 1)$, $(3, 3)$, and $(1, 3)$. It is reflected across the x -axis.

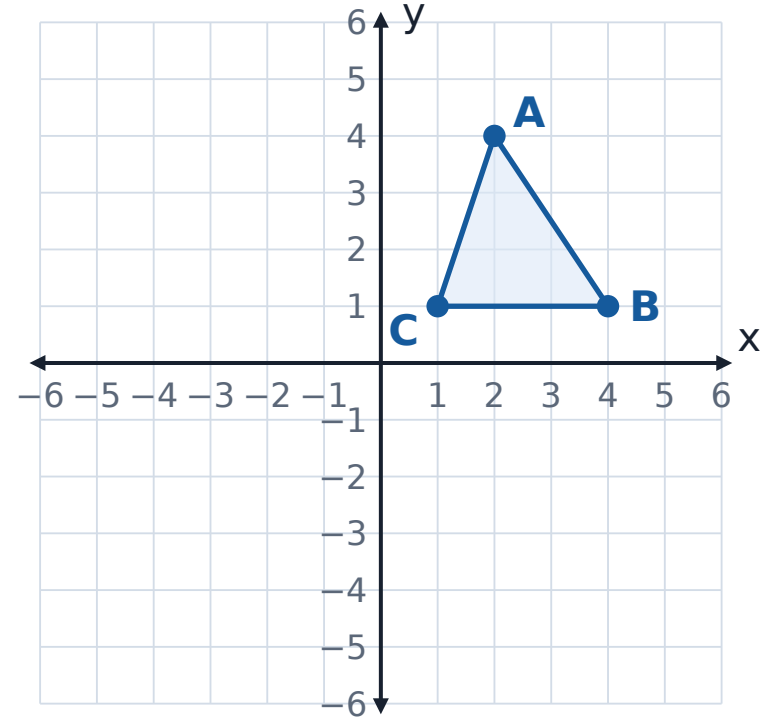
What is the image of $(3, 3)$?

Image: _____

Triangle ABC is rotated 90° clockwise about the origin.

What are the coordinates of A' ?

$A' =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.A.3 • Set 41 • FORWARD 2 • Q4/5

True or false?

Translating a polygon changes its side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.A.3 • Set 41 • FORWARD 2 • Q5/5

Nets of three-dimensional figures and surface area

MAIN	Grade 6 main question	6.G.A.4
BACKWARD 1	Area of rectangles	3.MD.C.7.b
BACKWARD 2	Area of triangles	6.G.A.1
BACKWARD 3	Faces of three-dimensional shapes	2.G.A.1
FORWARD 1	Surface area of prisms	7.G.B.6
FORWARD 2	Pythagorean Theorem in three dimensions	8.G.B.7

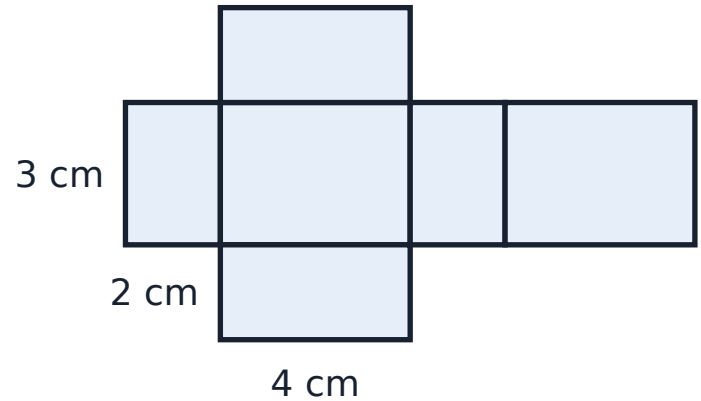
SET 42 • 6.G.A.4

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.4

The net folds into a rectangular prism.

What is the surface area of the prism?



Surface area: _____

A cube has edges of 5 inches.

What is its surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.4 • Set 42 • MAIN • Q2/5

A rectangular prism is 6 ft by 2 ft by 3 ft.

What is its surface area?

- A.** 72 square feet
- B.** 36 square feet
- C.** 66 square feet
- D.** 11 square feet

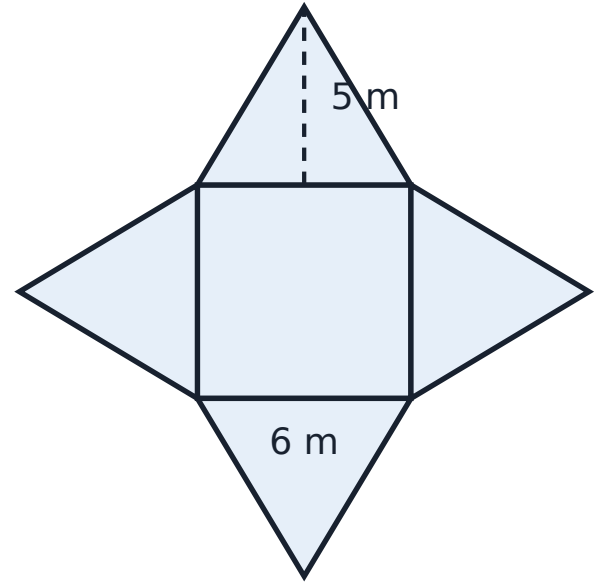
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.4 • Set 42 • MAIN • Q3/5

The net folds into a square pyramid.

What is the surface area of the pyramid?

Surface area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.4 • Set 42 • MAIN • Q4/5

A box with no lid is 10 in. long, 4 in. wide, and 3 in. tall.
How much cardboard is needed to make its 5 faces?

Cardboard: _____

SET 42 • 6.G.A.4

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 42 • BACKWARD 1 • Q1/5

A rectangle is 5 m by 7 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 42 • BACKWARD 1 • Q2/5

True or false?

A rectangle that is 3 units by 6 units has an area of 18 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 42 • BACKWARD 1 • Q3/5

A rectangle is 6 m by 2 m.

What is its area?

- A.** 12 square meters
- B.** 8 square meters
- C.** 16 square meters
- D.** 24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 42 • BACKWARD 1 • Q4/5

A square face has 5-inch edges.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 42 • BACKWARD 1 • Q5/5

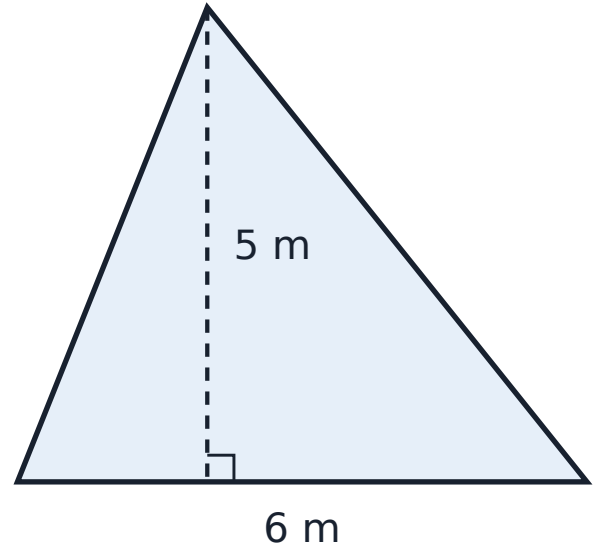
SET 42 • 6.G.A.4

BACKWARD 2

Area of triangles • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 42 • BACKWARD 2 • Q1/5

A triangle has a base of 8 cm and a height of 3 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 42 • BACKWARD 2 • Q2/5

True or false?

A triangle with a base of 10 and a height of 4 has an area of 40 square units.

A. True

B. False

A triangle has a base of 7 in. and a height of 6 in.

What is its area?

- A.** 21 square inches
- B.** 42 square inches
- C.** 13 square inches
- D.** 26 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 42 • BACKWARD 2 • Q4/5

Four triangles each have a base of 6 m and a height of 5 m.

What is their total area?

Total area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.1 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • 6.G.A.4

BACKWARD 3

Faces of three-dimensional shapes • Grade 2 • Geometry • 2.G.A.1

How many faces does a cube have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Geometry • 2.G.A.1 • Set 42 • BACKWARD 3 • Q1/5

How many faces does a rectangular prism have?

- A.** 6
- B.** 4
- C.** 8
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Geometry • 2.G.A.1 • Set 42 • BACKWARD 3 • Q2/5

True or false?

A square pyramid has 5 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Geometry • 2.G.A.1 • Set 42 • BACKWARD 3 • Q3/5

What shape are the faces of a cube?

- A.** Squares
- B.** Triangles
- C.** Circles
- D.** Pentagons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Geometry • 2.G.A.1 • Set 42 • BACKWARD 3 • Q4/5

How many triangular faces does a square pyramid have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Geometry • 2.G.A.1 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • 6.G.A.4

FORWARD 1

Surface area of prisms • Grade 7 • Geometry • 7.G.B.6

A triangular prism has right-triangle bases with sides 3 cm, 4 cm, and 5 cm. The prism is 10 cm long.

What is its surface area?

Surface area: _____

A cube has edges of 2.5 cm.

What is its surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.6 • Set 42 • FORWARD 1 • Q2/5

A rectangular prism is 5 in. by 4 in. by 2.5 in.

What is its surface area?

- A.** 85 square inches
- B.** 50 square inches
- C.** 42.5 square inches
- D.** 65 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.6 • Set 42 • FORWARD 1 • Q3/5

A gift box is 8 in. by 6 in. by 4 in.

How much wrapping paper covers the box exactly?

Paper: _____

True or false?

The surface area of a prism is the sum of the areas of all of its faces.

A. True

B. False

SET 42 • 6.G.A.4

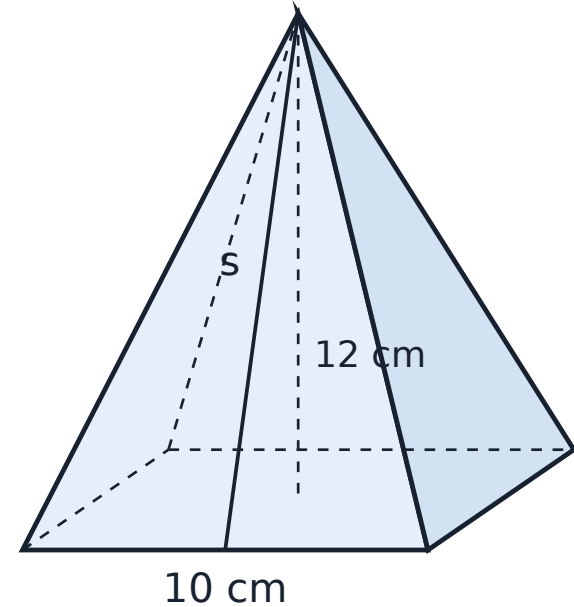
FORWARD 2

Pythagorean Theorem in three dimensions • Grade 8 • Geometry • 8.G.B.7

The square pyramid has a base edge of 10 cm and a height of 12 cm.

What is its slant height s ?

$s =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.7 • Set 42 • FORWARD 2 • Q1/5

A box is 3 in. by 4 in. by 12 in.

How long is the longest diagonal from one corner to the opposite corner?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.7 • Set 42 • FORWARD 2 • Q2/5

A square pyramid has a base edge of 16 cm and a slant height of 10 cm.

What is its height?

A. 6 cm

B. 8 cm

C. 12.8 cm

D. 18.9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.7 • Set 42 • FORWARD 2 • Q3/5

A cone has a radius of 5 in. and a slant height of 13 in.
What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.7 • Set 42 • FORWARD 2 • Q4/5

True or false?

The bottom of a box is 6 in. by 8 in. The diagonal across the bottom is 10 in.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.B.7 • Set 42 • FORWARD 2 • Q5/5

Recognize statistical questions

MAIN Grade 6 main question

BACKWARD 1 Reading a bar graph

BACKWARD 2 Reading a line plot

FORWARD 1 Random samples and representative samples

FORWARD 2 Scatter plots of bivariate data

6.SP.A.1

3.MD.B.3

4.MD.B.4

7.SP.A.1

8.SP.A.1

SET 43 • 6.SP.A.1

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.1

Which question is a statistical question?

- A.** How many hours of sleep do students in my class get each night?
- B.** How many hours did I sleep last night?
- C.** How many days are in a week?
- D.** What time does our school start?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 43 • MAIN • Q1/5

True or false?

"How tall is the flagpole at our school?" is a statistical question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 43 • MAIN • Q2/5

Which question is NOT a statistical question?

- A.** What is the population of our city right now?
- B.** How many pets do the students in grade 6 have?
- C.** How long do students spend on homework each day?
- D.** How much do the dogs at the shelter weigh?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 43 • MAIN • Q3/5

True or false?

"What are the ages of the teachers at our school?" is a statistical question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 43 • MAIN • Q4/5

Rewrite this question so that it is a statistical question.

"How many books did Mia read this year?"

Question: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 43 • MAIN • Q5/5

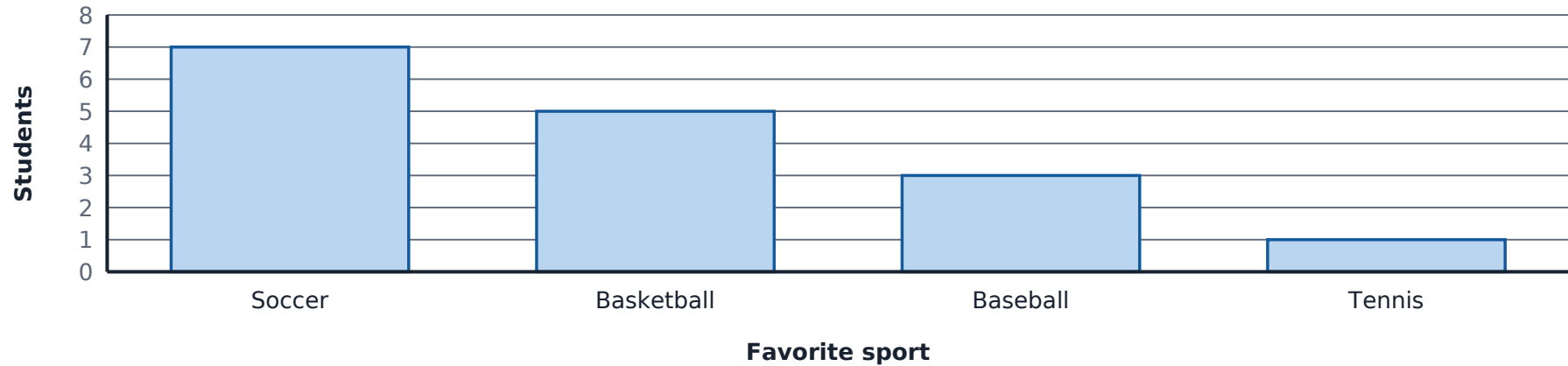
SET 43 • 6.SP.A.1

BACKWARD 1

Reading a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

The graph shows the favorite sports of a class.

How many students chose basketball?



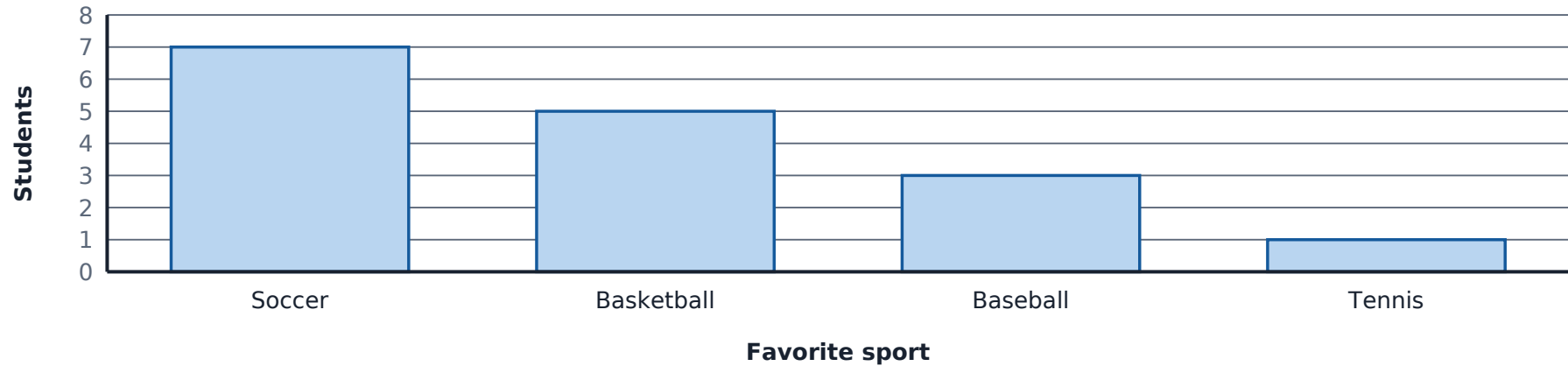
Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 43 • BACKWARD 1 • Q1/5

The graph shows the favorite sports of a class.

How many more students chose soccer than baseball?

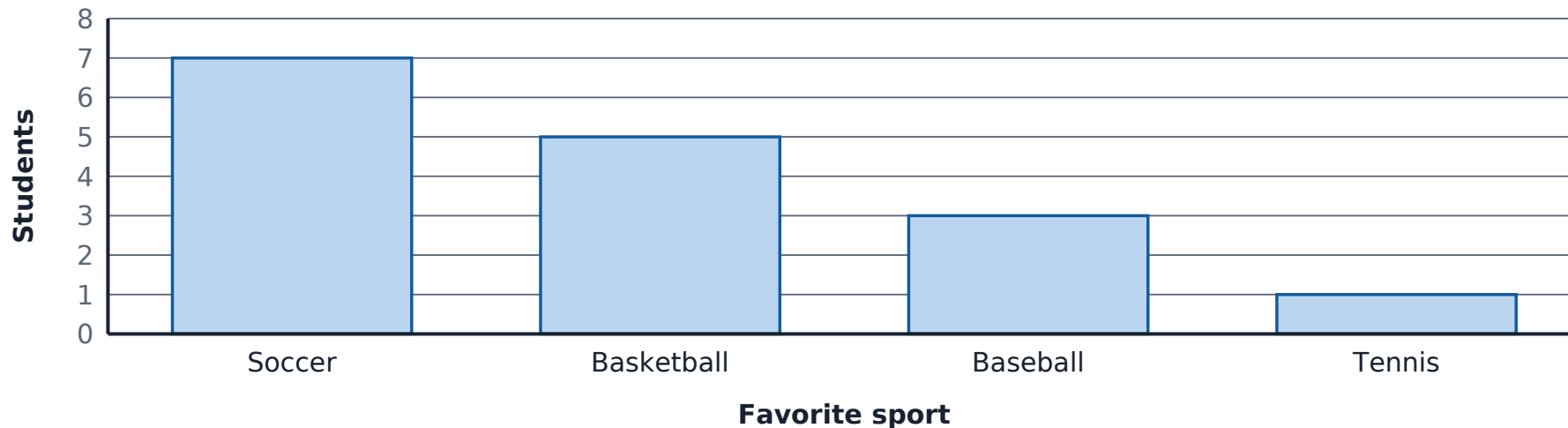


Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 43 • BACKWARD 1 • Q2/5

Which sport did 3 students choose?



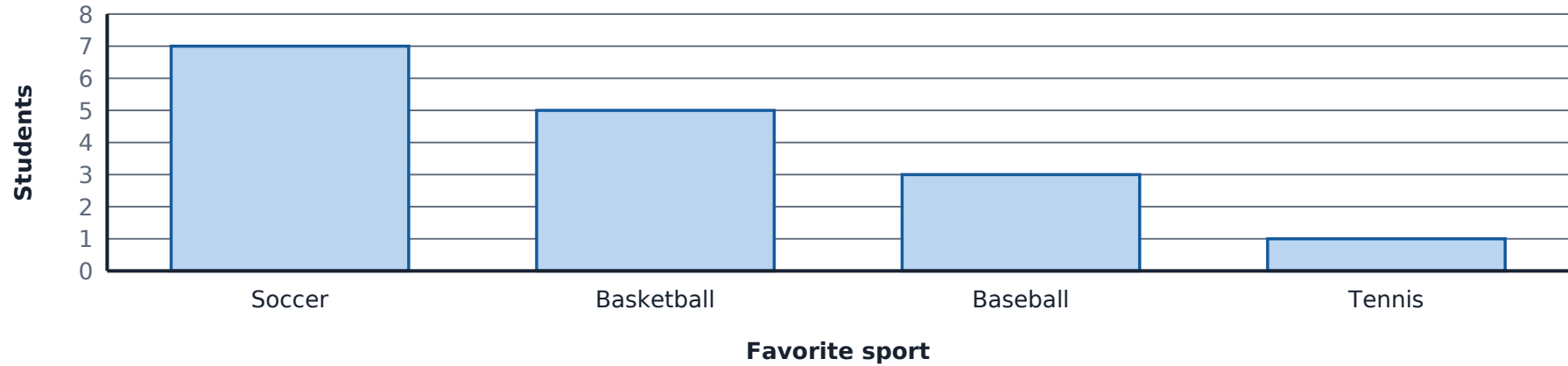
A. Baseball **B.** Soccer **C.** Basketball **D.** Tennis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 43 • BACKWARD 1 • Q3/5

The graph shows the favorite sports of a class.

How many students answered in all?



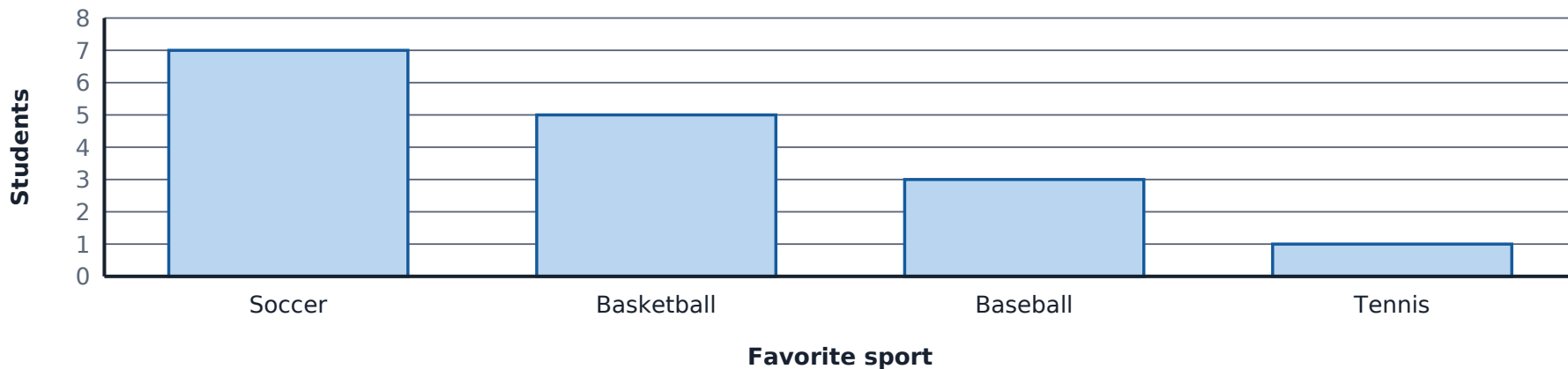
Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 43 • BACKWARD 1 • Q4/5

True or false?

More students chose basketball than soccer.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 43 • BACKWARD 1 • Q5/5

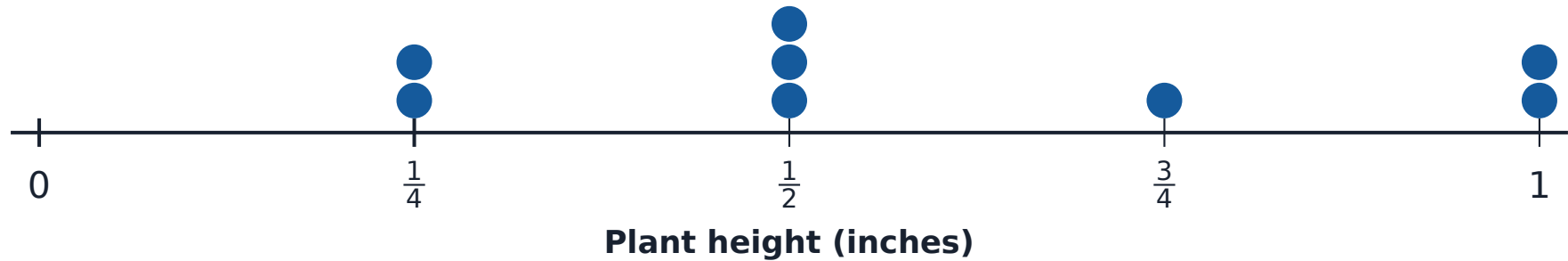
SET 43 • 6.SP.A.1

BACKWARD 2

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows the heights of plants.

How many plants are $\frac{1}{2}$ inch tall?



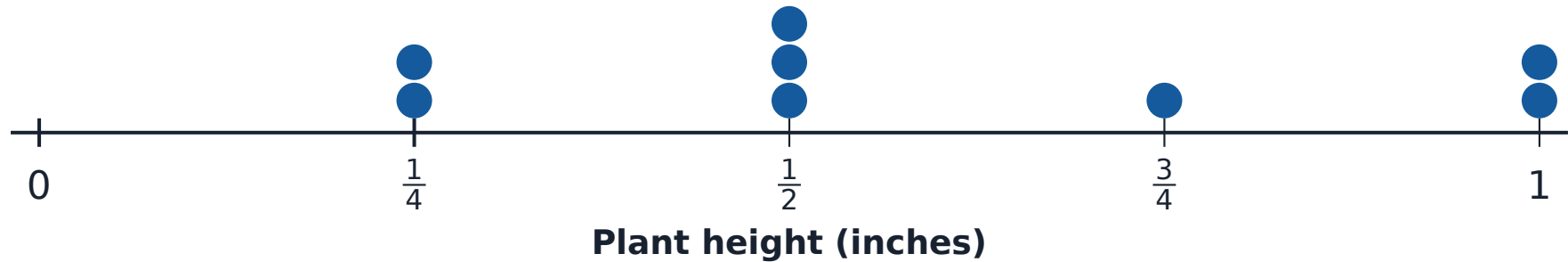
Plants: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 2 • Q1/5

The line plot shows the heights of plants.

How many plants were measured?



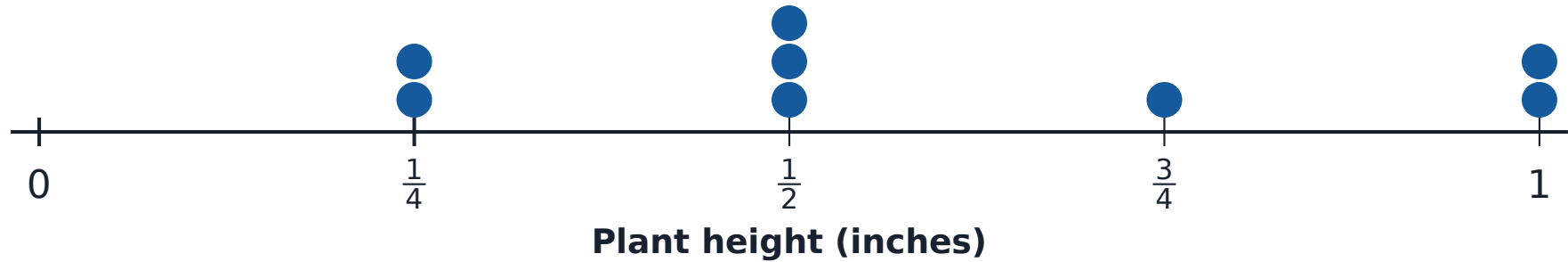
Plants: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 2 • Q2/5

True or false?

The most common plant height is $\frac{3}{4}$ inch.



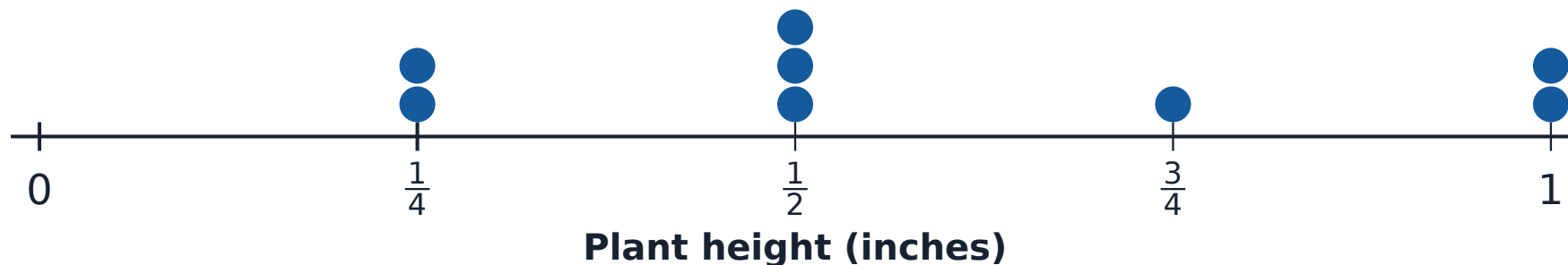
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 2 • Q3/5

Which height appears least often?



A. $\frac{3}{4}$ inch

B. $\frac{1}{4}$ inch

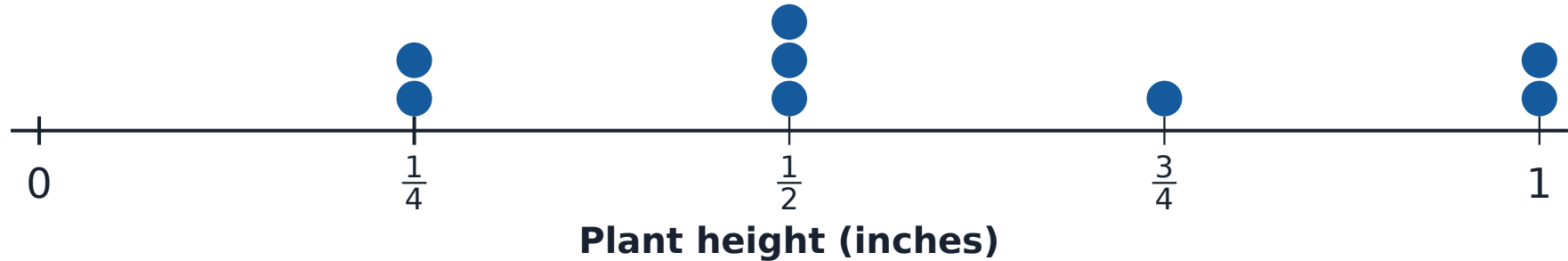
C. $\frac{1}{2}$ inch

D. 1 inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 2 • Q4/5

What is the difference between the tallest and the shortest plant heights?



Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 6.SP.A.1

FORWARD 1

Random samples and representative samples • Grade 7 • Statistics & Probability • 7.SP.A.1

A principal wants to know the favorite lunch of all 600 students.

Which sample is most likely to represent all students?

- A.** 50 students chosen at random from the whole school
- B.** The 30 students in one class
- C.** The first 50 students in line on pizza day
- D.** The students on the soccer team

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 43 • FORWARD 1 • Q1/5

True or false?

A random sample is more likely to represent a population than a sample of your friends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 43 • FORWARD 1 • Q2/5

A reporter asks only basketball players, "What is your favorite sport?"

Why is this sample biased?

Reason: _____

A random sample of 40 students out of 800 shows that 10 walk to school.

About how many of the 800 students walk to school?

A. 200

B. 10

C. 40

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 43 • FORWARD 1 • Q4/5

True or false?

Conclusions about a population from a sample are valid only if the sample represents the population.

- A.** True
- B.** False

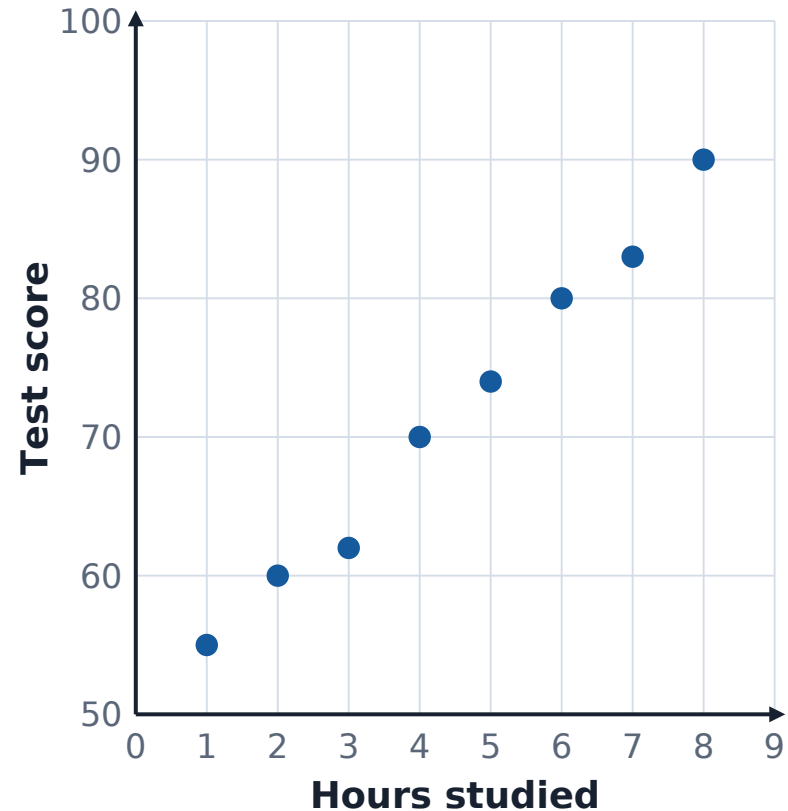
SET 43 • 6.SP.A.1

FORWARD 2

Scatter plots of bivariate data • Grade 8 • Statistics & Probability • 8.SP.A.1

What type of association does the scatter plot show?

- A.** Positive linear
- B.** Negative linear
- C.** No association
- D.** Nonlinear

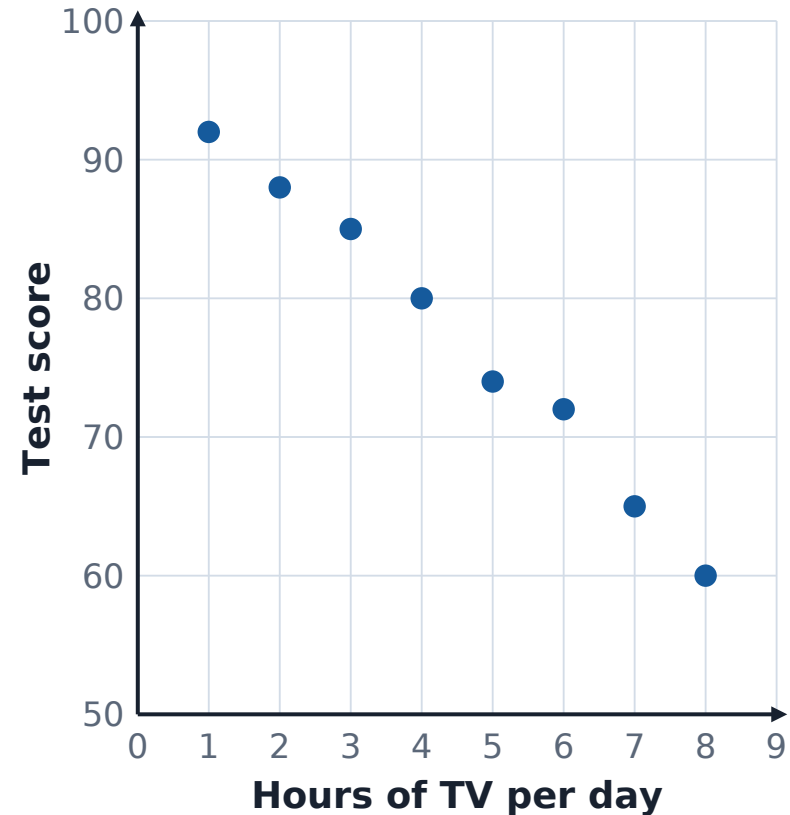


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 43 • FORWARD 2 • Q1/5

What type of association does the scatter plot show?

- A.** Negative linear
- B.** Positive linear
- C.** No association
- D.** Nonlinear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 43 • FORWARD 2 • Q2/5

True or false?

A scatter plot shows data about two variables for each person or object.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 43 • FORWARD 2 • Q3/5

Which question would be best answered with a scatter plot?

- A.** Is there a relationship between height and arm span?
- B.** What is the most common eye color?
- C.** How many students ride the bus?
- D.** What is the median age in the class?

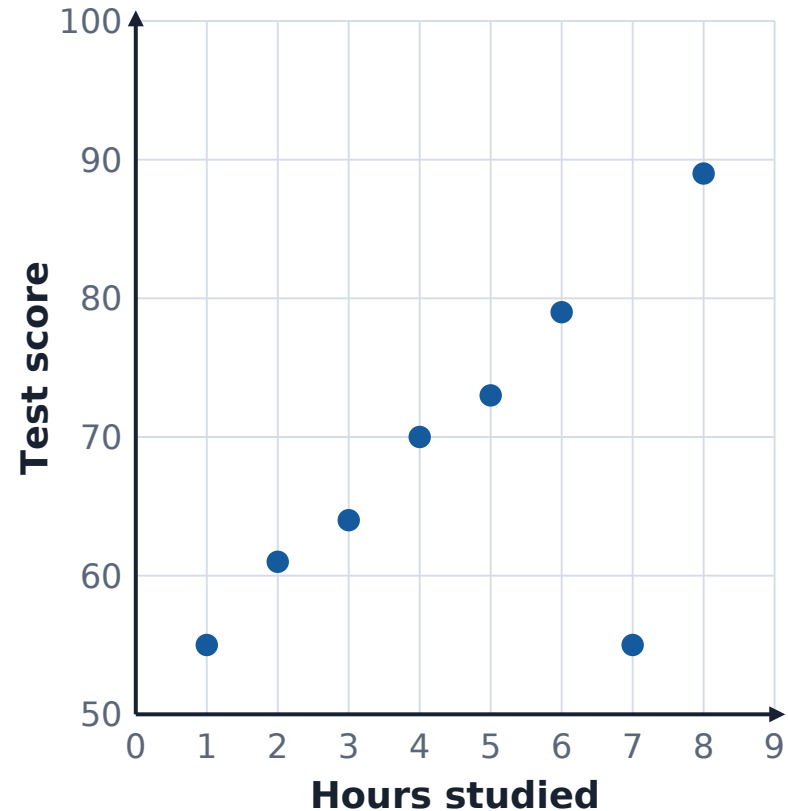
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 43 • FORWARD 2 • Q4/5

Which point is an outlier?

Write its coordinates.

Outlier: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 43 • FORWARD 2 • Q5/5

Describe a distribution by its center, spread, and shape

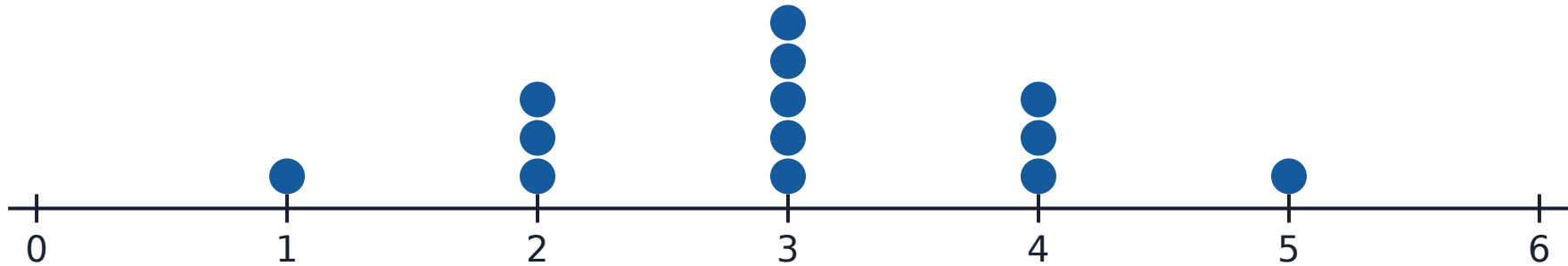
MAIN	Grade 6 main question	6.SP.A.2
BACKWARD 1	Line plots with fractional measurements	5.MD.B.2
BACKWARD 2	Finding a difference	2.NBT.B.5
BACKWARD 3	Ordering numbers	2.NBT.A.4
FORWARD 1	Compare two distributions visually	7.SP.B.3
FORWARD 2	Patterns in scatter plots	8.SP.A.1

SET 44 • 6.SP.A.2

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.2

Which word best describes the shape of the data?



A. Symmetric

B. Skewed right

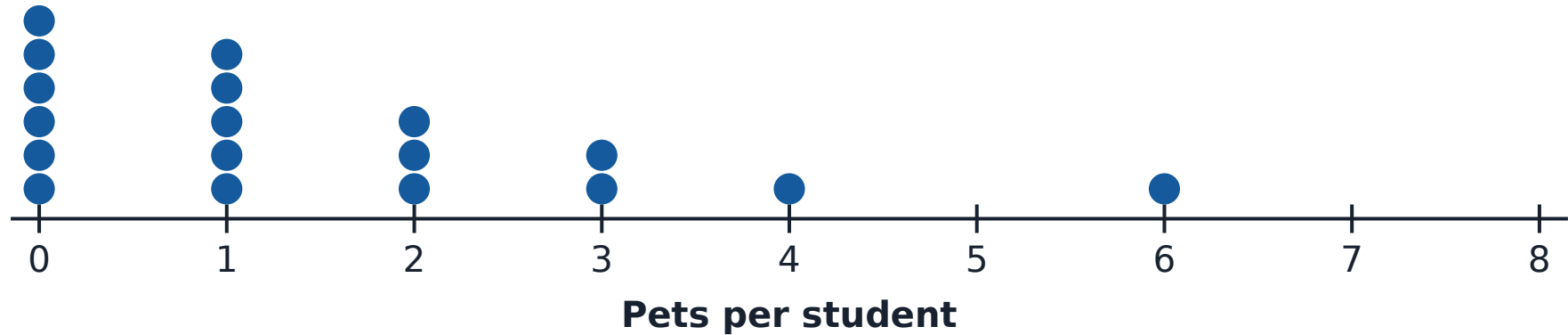
C. Skewed left

D. Uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 44 • MAIN • Q1/5

Which word best describes the shape of the data?



A. Skewed right

B. Symmetric

C. Skewed left

D. Uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 44 • MAIN • Q2/5

What is the range of the data?

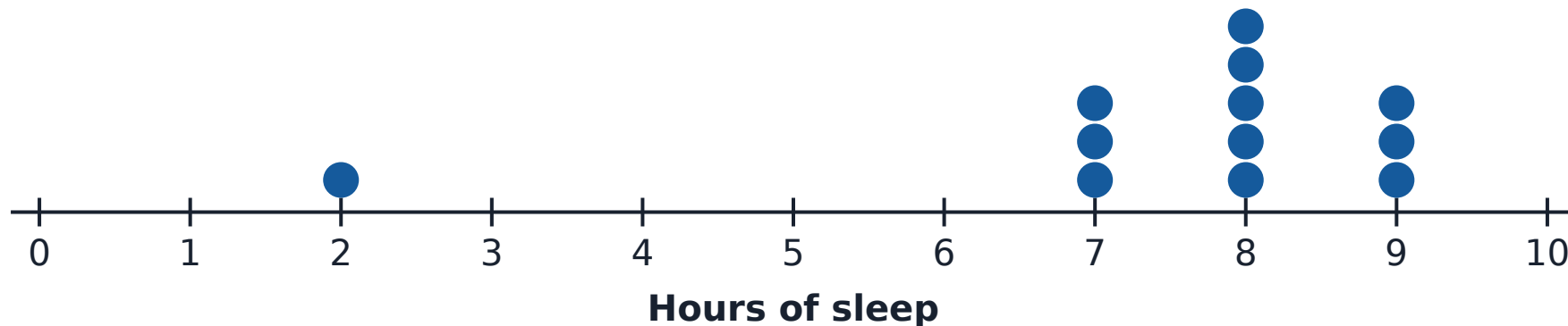


Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 44 • MAIN • Q3/5

Where do most of the data values cluster?



A. Between 7 and 9

B. Between 0 and 2

C. Between 3 and 5

D. At 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 44 • MAIN • Q4/5

True or false?

The distribution of a data set can be described by its center, its spread, and its overall shape.

- A. True
- B. False

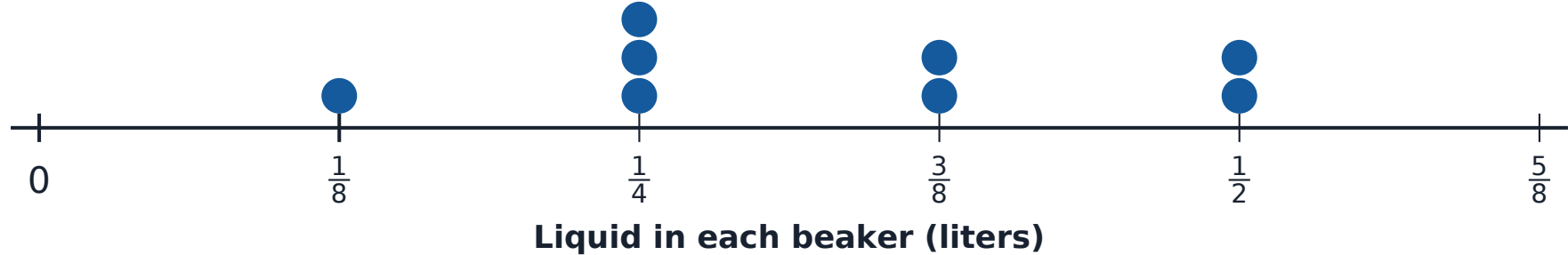
SET 44 • 6.SP.A.2

BACKWARD 1

Line plots with fractional measurements • Grade 5 • Measurement & Data •
5.MD.B.2

The line plot shows the liquid in some beakers.

How many beakers hold $\frac{1}{4}$ liter?



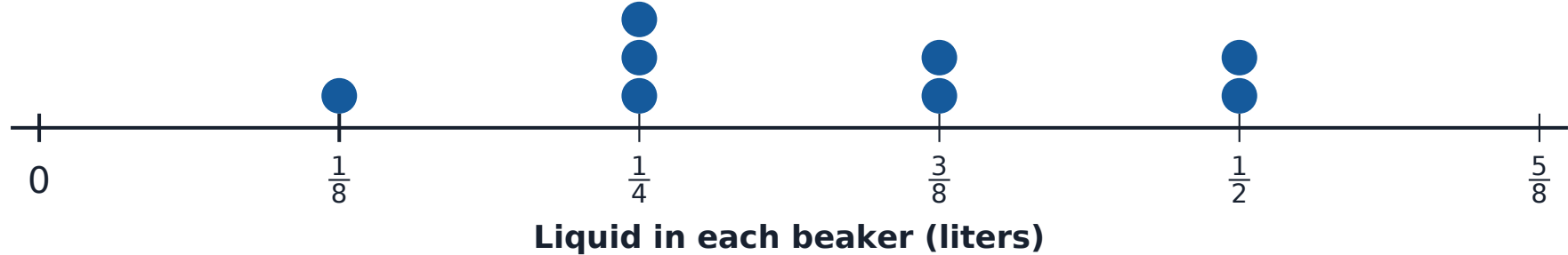
Beakers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • BACKWARD 1 • Q1/5

The line plot shows the liquid in some beakers.

How much liquid is in all of the $\frac{1}{2}$ -liter beakers together?



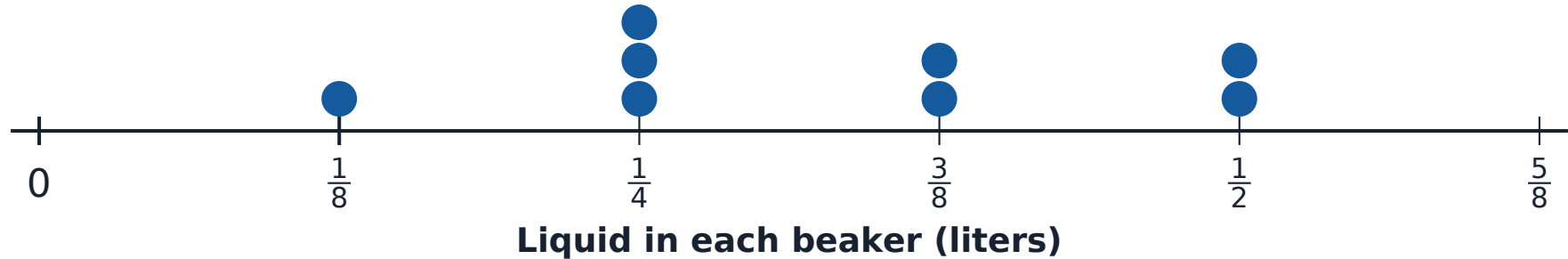
Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • BACKWARD 1 • Q2/5

True or false?

There are 8 beakers in all.



A. True

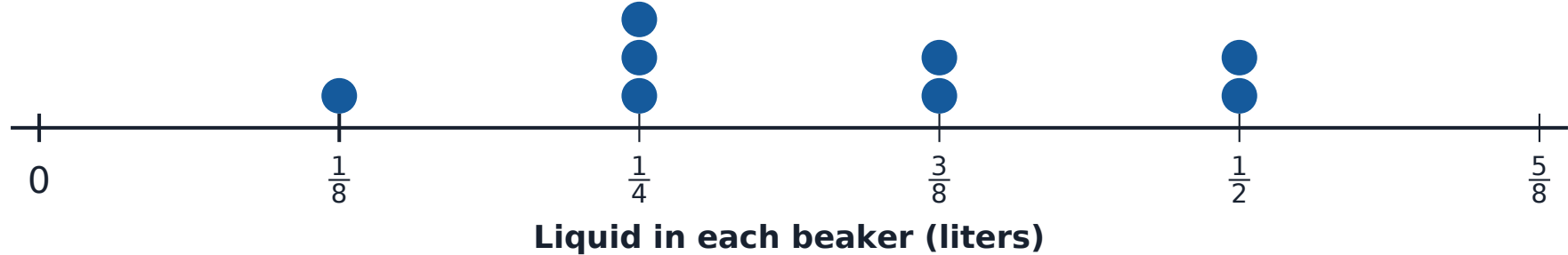
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • BACKWARD 1 • Q3/5

The liquid in the $\frac{3}{8}$ -liter beakers is poured together.

How much liquid is there?



A. $\frac{3}{4}$ liter

B. $\frac{3}{8}$ liter

C. $\frac{3}{16}$ liter

D. 1 liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • BACKWARD 1 • Q4/5

How much more liquid is in a $\frac{1}{2}$ -liter beaker than in the $\frac{1}{8}$ -liter beaker?



Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 6.SP.A.2

BACKWARD 2

Finding a difference • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

The greatest value is 47. The least value is 18.

What is the difference?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 44 • BACKWARD 2 • Q1/5

Subtract.

$$83 - 29$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 44 • BACKWARD 2 • Q2/5

True or false?

$$62 - 35 = 27$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 44 • BACKWARD 2 • Q3/5

$$91 - 46 = ?$$

A. 45

B. 55

C. 47

D. 137

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 44 • BACKWARD 2 • Q4/5

What is the difference between 100 and 64?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 6.SP.A.2

BACKWARD 3

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

42, 24, 36, 18

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 44 • BACKWARD 3 • Q1/5

Which number is greatest?

A. 350

B. 305

C. 299

D. 53

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 44 • BACKWARD 3 • Q2/5

True or false?

These numbers are in order from least to greatest: 12, 21, 19, 30

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 44 • BACKWARD 3 • Q3/5

Order from least to greatest.

105, 150, 115

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 44 • BACKWARD 3 • Q4/5

Which number is least?

- A.** 89
- B.** 98
- C.** 91
- D.** 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

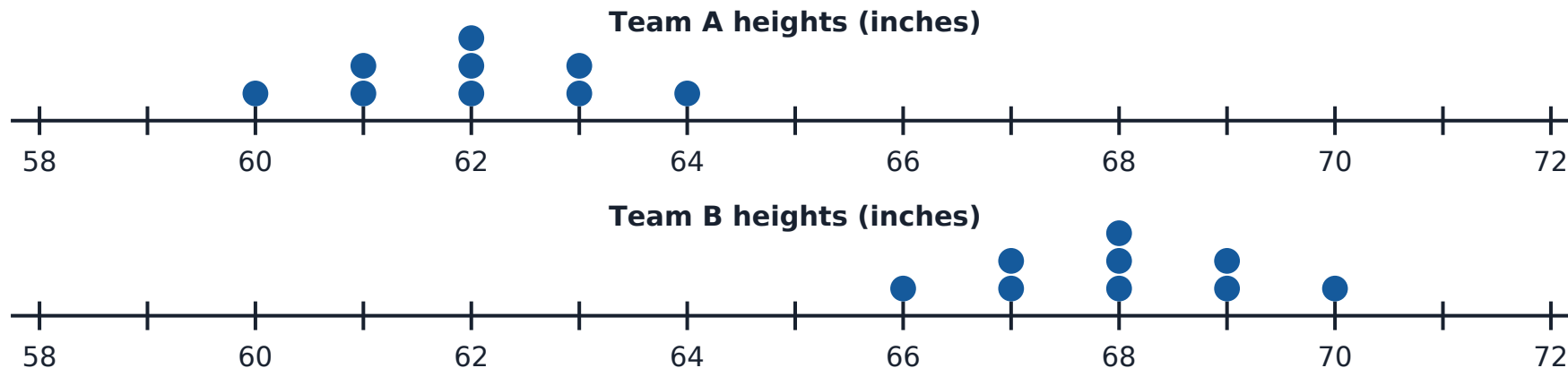
Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • 6.SP.A.2

FORWARD 1

Compare two distributions visually • Grade 7 • Statistics & Probability •
7.SP.B.3

About how many inches greater is the center of Team B's heights than the center of Team A's heights?



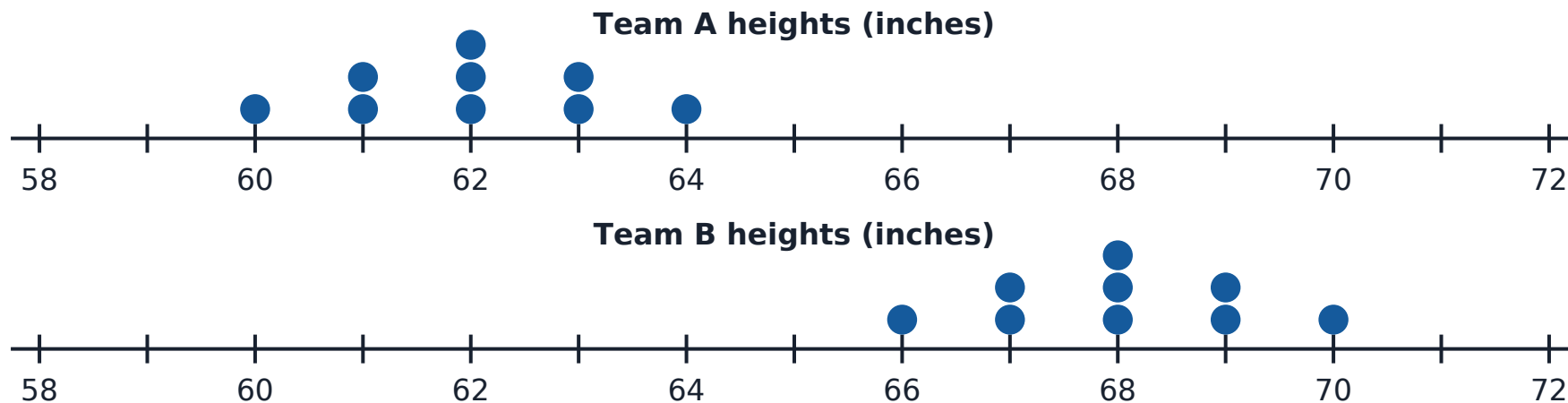
Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 44 • FORWARD 1 • Q1/5

True or false?

The two distributions have about the same spread.



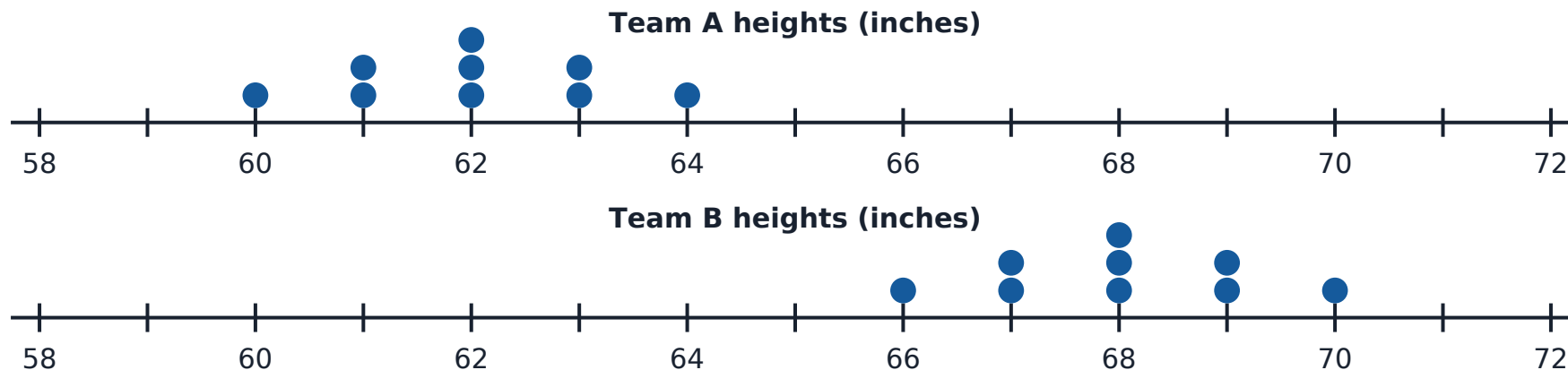
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 44 • FORWARD 1 • Q2/5

Which statement is true?



- A.** Team B's heights are typically greater. **B.** Team A's heights are typically greater.
- C.** The teams have the same center. **D.** The distributions overlap a lot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 44 • FORWARD 1 • Q3/5

Class A's mean score is 60 and Class B's mean score is 66. Each class has a mean absolute deviation of 3.

The difference in means is how many times the MAD?

Times: _____

Two classes have mean scores of 78 and 84. Each has a MAD of 6.

The difference in means is how many times the MAD?

A. 1

B. 2

C. 6

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 44 • FORWARD 1 • Q5/5

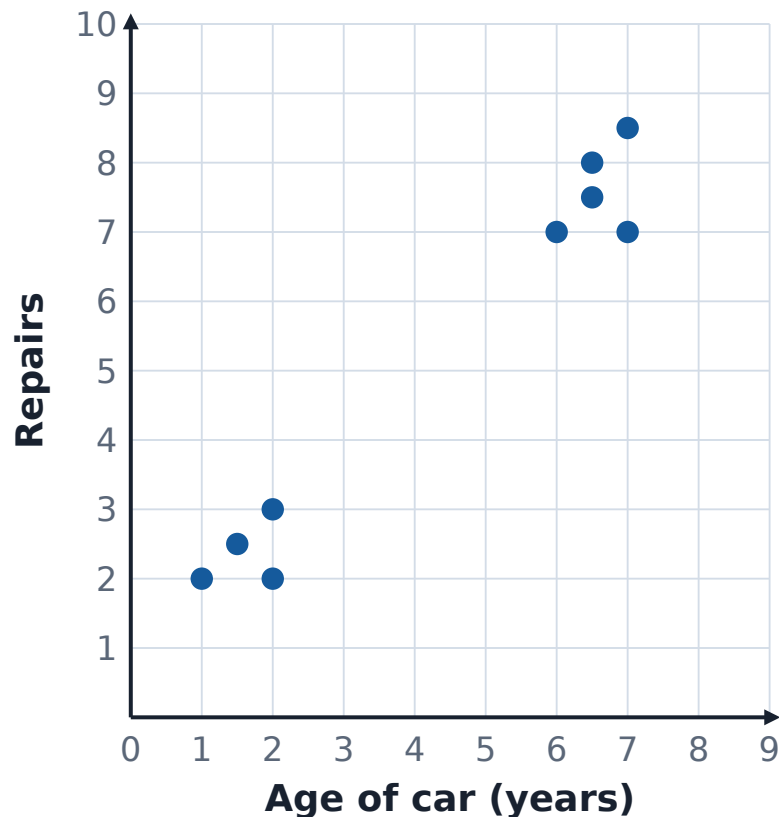
SET 44 • 6.SP.A.2

FORWARD 2

Patterns in scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

Which describes the pattern in the scatter plot?

- A.** Two clusters
- B.** A strong negative association
- C.** No pattern
- D.** A single outlier



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

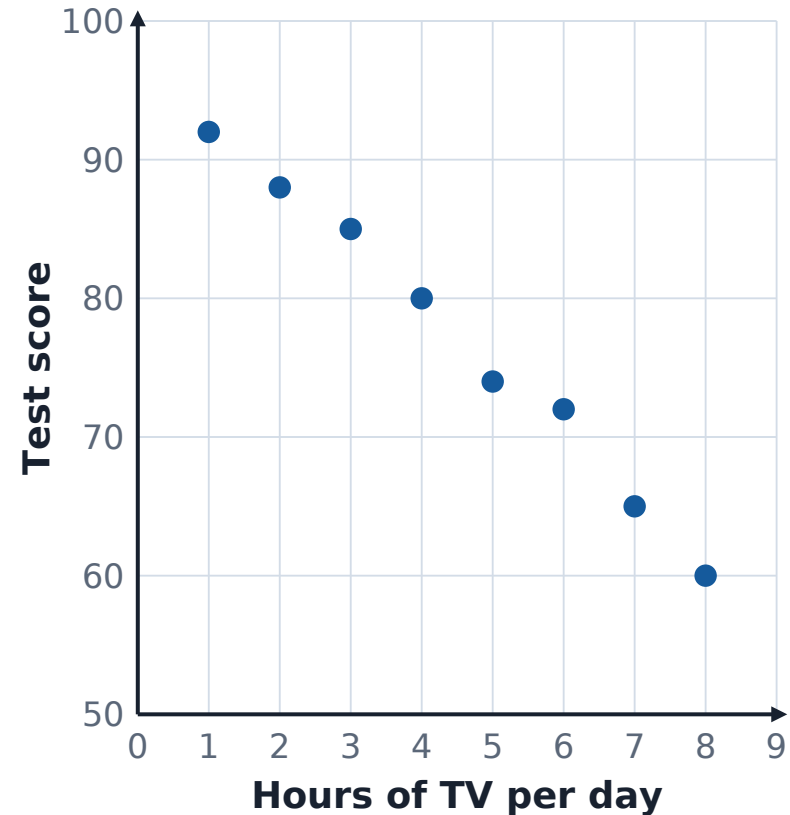
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 44 • FORWARD 2 • Q1/5

True or false?

The scatter plot shows a negative association.

A. True

B. False

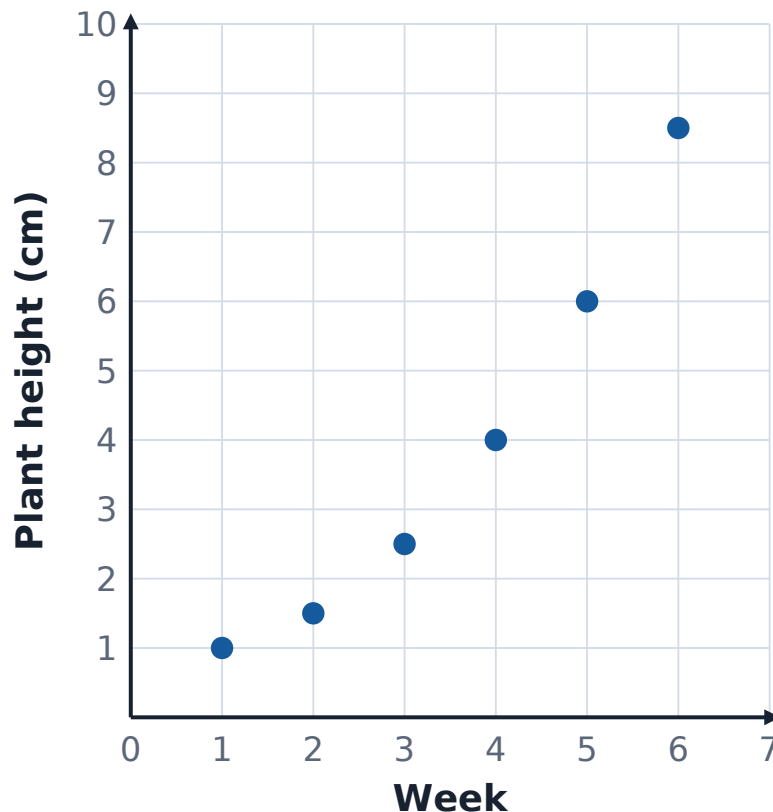


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 44 • FORWARD 2 • Q2/5

Which describes the association in the scatter plot?

- A.** Nonlinear
- B.** Negative linear
- C.** No association
- D.** Constant



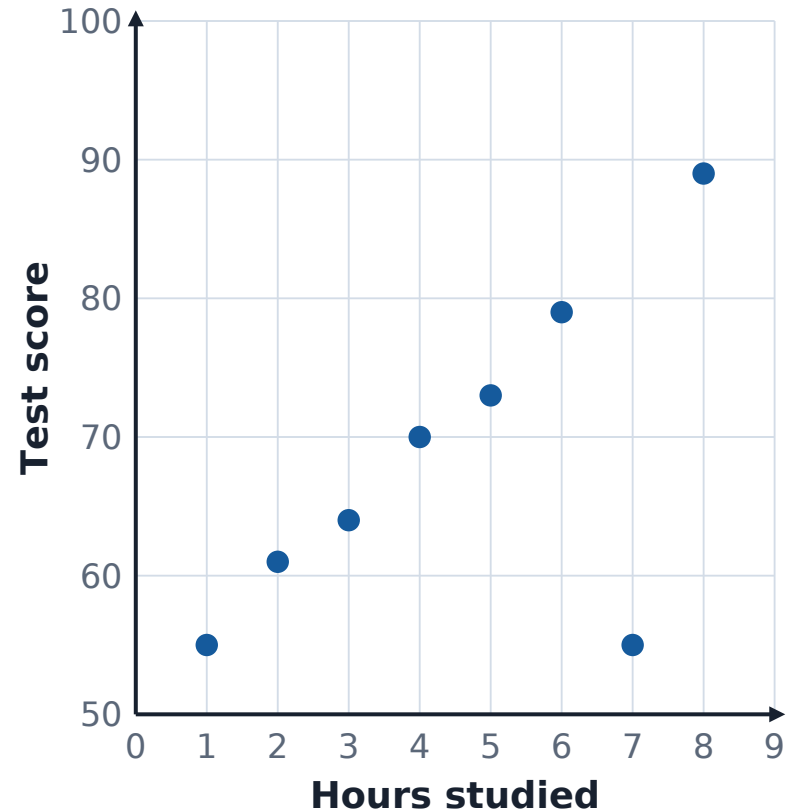
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 44 • FORWARD 2 • Q3/5

Which point is an outlier?

Write its coordinates.

Outlier: _____

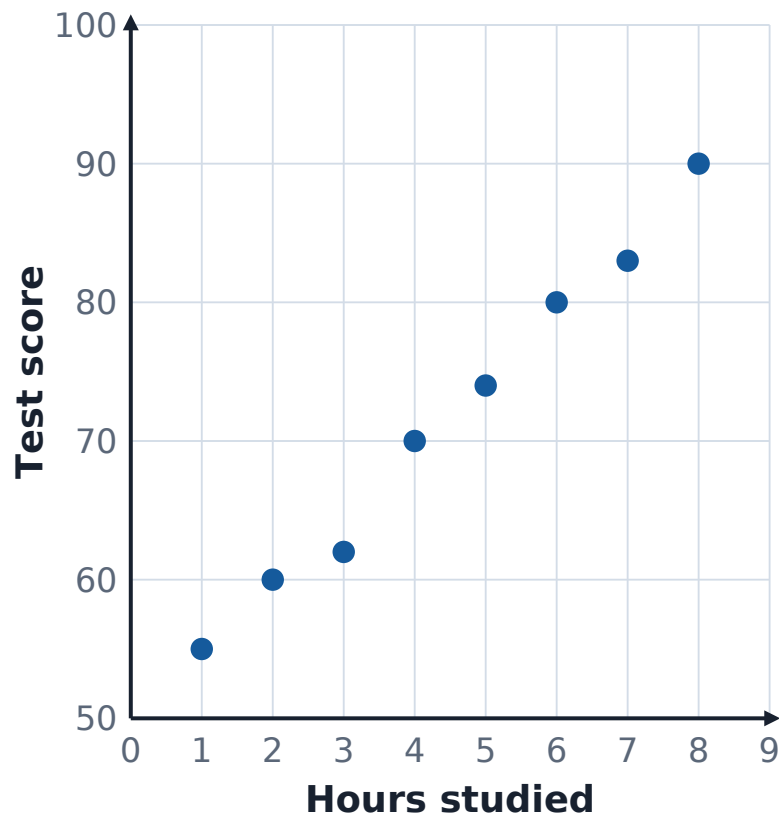


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 44 • FORWARD 2 • Q4/5

In the scatter plot, as x increases, y tends to ____.

- A.** increase
- B.** decrease
- C.** stay the same
- D.** equal x



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 44 • FORWARD 2 • Q5/5

A measure of center vs. a measure of variation

MAIN	Grade 6 main question	6.SP.A.3
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Sharing a total equally	5.MD.B.2
BACKWARD 3	Ordering numbers	2.NBT.A.4
FORWARD 1	Compare populations using center and variability	7.SP.B.4
FORWARD 2	Lines of best fit	8.SP.A.2

SET 45 • 6.SP.A.3

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.3

Which measure describes the center of a data set?

- A.** Median
- B.** Range
- C.** Interquartile range
- D.** Mean absolute deviation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 45 • MAIN • Q1/5

Which measure describes how much the values in a data set vary?

- A.** Interquartile range
- B.** Mean
- C.** Median
- D.** The middle value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 45 • MAIN • Q2/5

True or false?

A measure of center summarizes all of the values in a data set with a single number.

A. True

B. False

True or false?

A measure of variation describes how the values in a data set vary with a single number.

- A.** True
- B.** False

A teacher wants one number that describes a typical test score.

Which measure should she use?

- A.** The mean
- B.** The range
- C.** The interquartile range
- D.** The mean absolute deviation

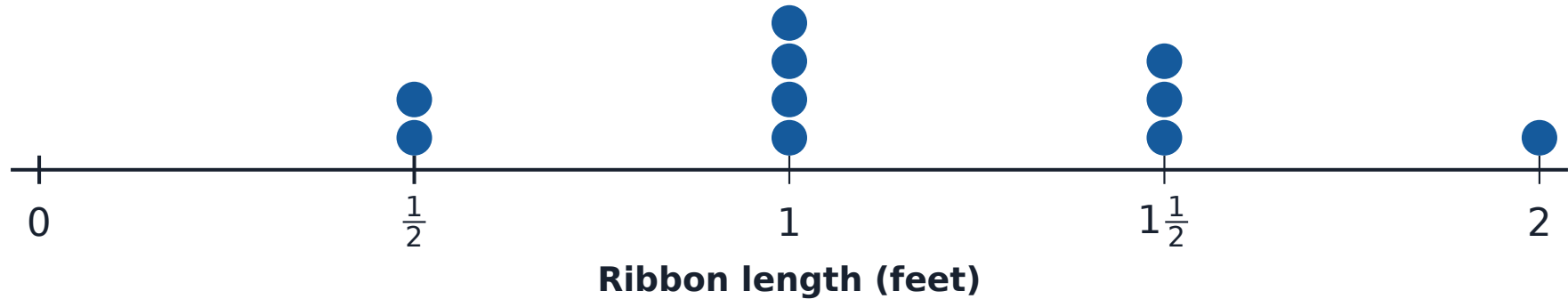
SET 45 • 6.SP.A.3

BACKWARD 1

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows ribbon lengths.

How many ribbons are 1 foot long?



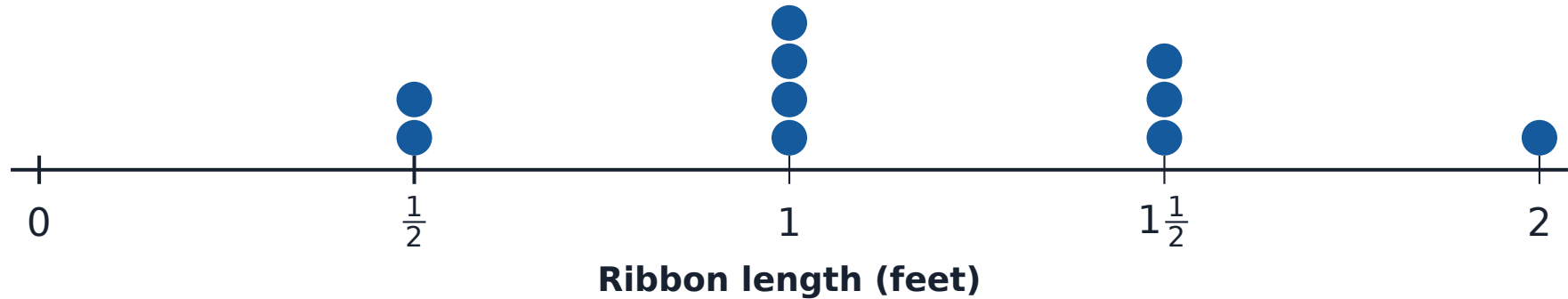
Ribbons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 45 • BACKWARD 1 • Q1/5

The line plot shows ribbon lengths.

How many ribbons are there in all?



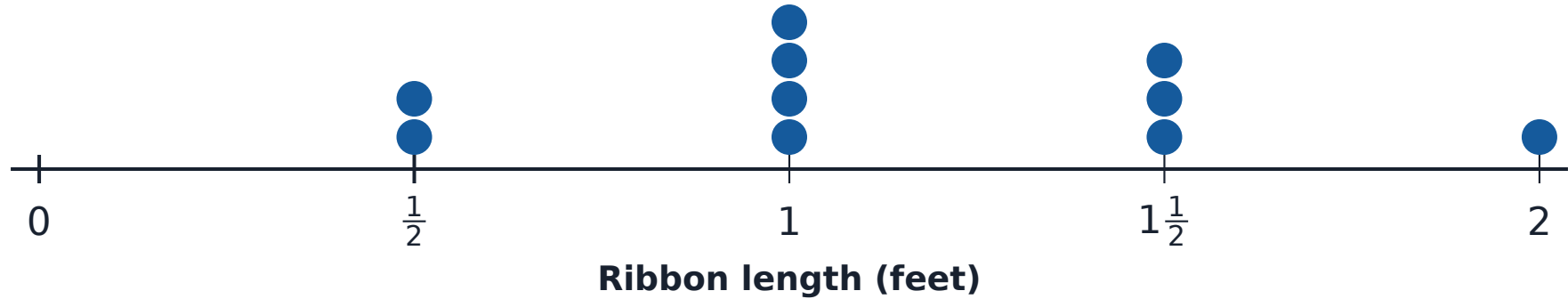
Ribbons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 45 • BACKWARD 1 • Q2/5

True or false?

Two ribbons are $\frac{1}{2}$ foot long.



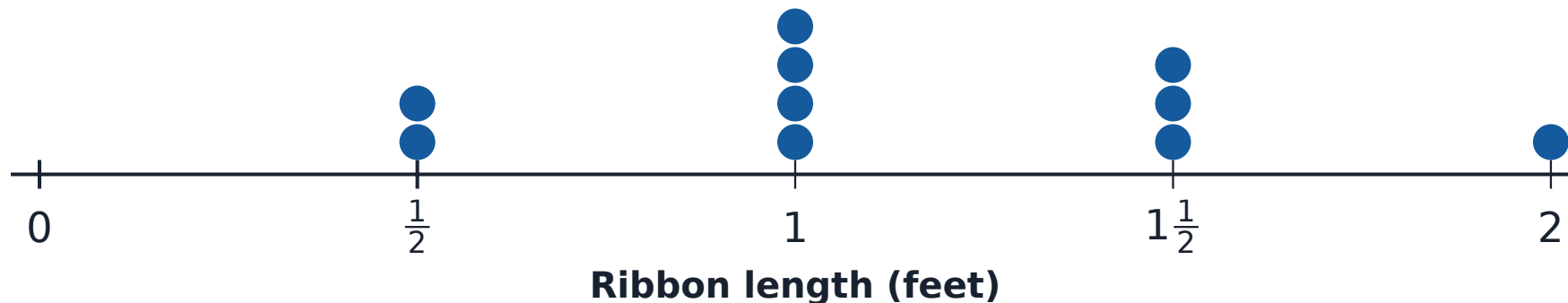
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 45 • BACKWARD 1 • Q3/5

Which ribbon length is most common?



A. 1 foot

B. $\frac{1}{2}$ foot

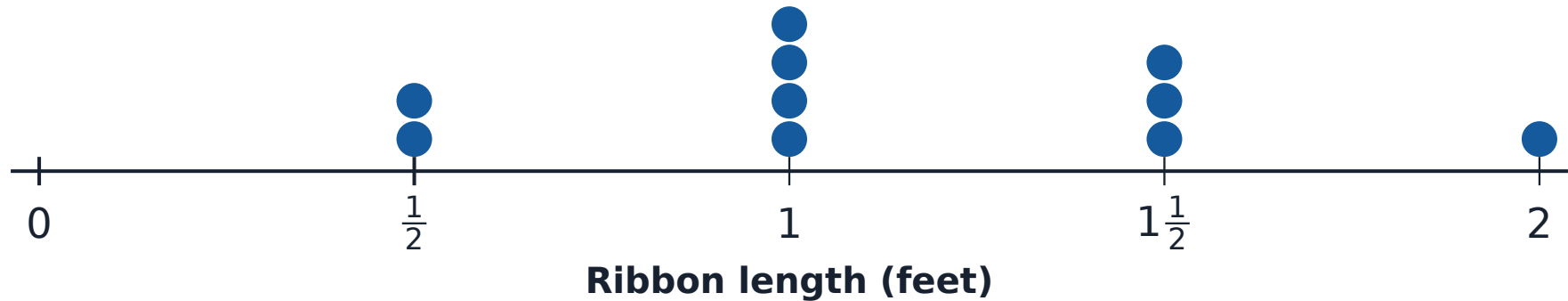
C. $1\frac{1}{2}$ feet

D. 2 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 45 • BACKWARD 1 • Q4/5

What is the total length of the $1\frac{1}{2}$ -foot ribbons?



Total length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 6.SP.A.3

BACKWARD 2

Sharing a total equally • Grade 5 • Measurement & Data • 5.MD.B.2

Three friends have 4, 6, and 8 stickers. They share all of the stickers equally.

How many stickers does each friend get?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 45 • BACKWARD 2 • Q1/5

Four jars hold 2, 3, 5, and 6 liters of water. The water is shared equally among the jars.

How much water is in each jar?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 45 • BACKWARD 2 • Q2/5

True or false?

Cups holding 1, 2, and 3 liters are shared equally.
Each cup then holds 2 liters.

- A.** True
- B.** False

Five friends collected 3, 5, 6, 7, and 9 cans. They share the cans equally.

How many cans does each friend get?

A. 6

B. 5

C. 30

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 45 • BACKWARD 2 • Q4/5

Two bottles hold $\frac{1}{4}$ liter and $\frac{3}{4}$ liter. The juice is shared equally between the bottles.

How much juice is in each bottle?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 6.SP.A.3

BACKWARD 3

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

56, 65, 38, 83

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 45 • BACKWARD 3 • Q1/5

Which number is greatest?

A. 720

B. 702

C. 270

D. 207

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 45 • BACKWARD 3 • Q2/5

True or false?

These numbers are in order from least to greatest: 14, 41, 44, 114

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 45 • BACKWARD 3 • Q3/5

Order from least to greatest.

209, 290, 92

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 45 • BACKWARD 3 • Q4/5

Which number is least?

A. 147

B. 174

C. 417

D. 471

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • 6.SP.A.3

FORWARD 1

Compare populations using center and variability • Grade 7 • Statistics & Probability • 7.SP.B.4

Class A has a median score of 82 and an IQR of 10.

Class B has a median score of 75 and an IQR of 10.

Which statement is true?

A. Class A typically scored higher.

B. Class B typically scored higher.

C. The classes scored the same.

D. Class A's scores vary more.

Random samples: Town A has a mean age of 34 years.
Town B has a mean age of 42 years.

What is the difference in mean ages?

Difference: _____

True or false?

Two samples have the same median. The sample with the greater IQR has more variability.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 45 • FORWARD 1 • Q3/5

Which measure is best for comparing the variability of two samples?

- A.** IQR
- B.** Median
- C.** Mean
- D.** Sample size

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 45 • FORWARD 1 • Q4/5

In random samples, 7th graders sleep a median of 8 hours, and 6th graders sleep a median of 9 hours.

Which group typically sleeps more?

Group: _____

SET 45 • 6.SP.A.3

FORWARD 2

Lines of best fit • Grade 8 • Statistics & Probability • 8.SP.A.2

True or false?

A line of best fit usually has about the same number of points above it as below it.

A. True

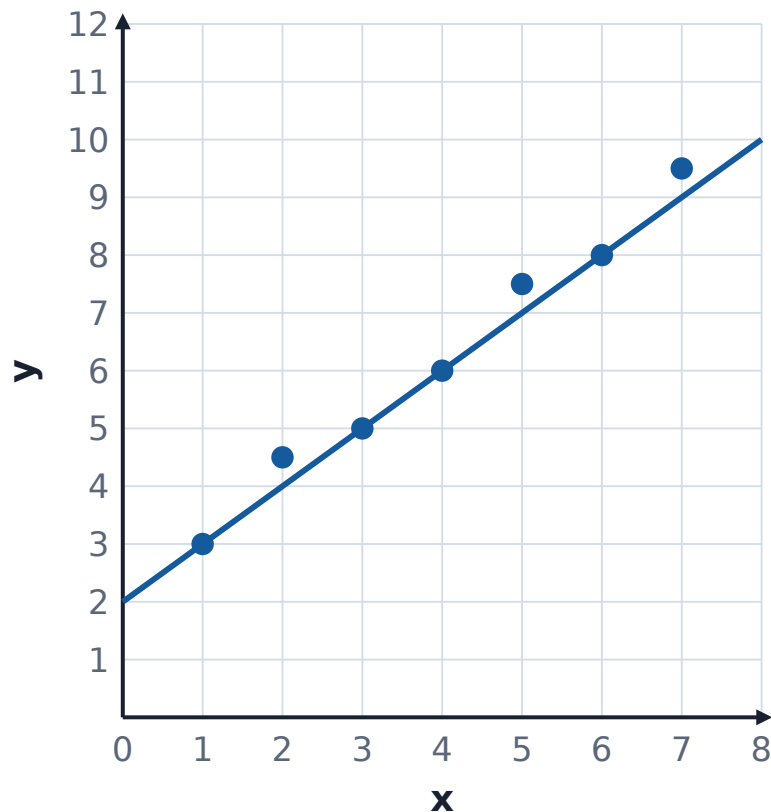
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 45 • FORWARD 2 • Q1/5

Use the line of best fit to
predict y when $x = 6$.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 45 • FORWARD 2 • Q2/5

Which statement about a line of best fit is true?

- A.** It shows the overall trend of the data.
- B.** It must pass through every point.
- C.** It must pass through the origin.
- D.** It is always horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

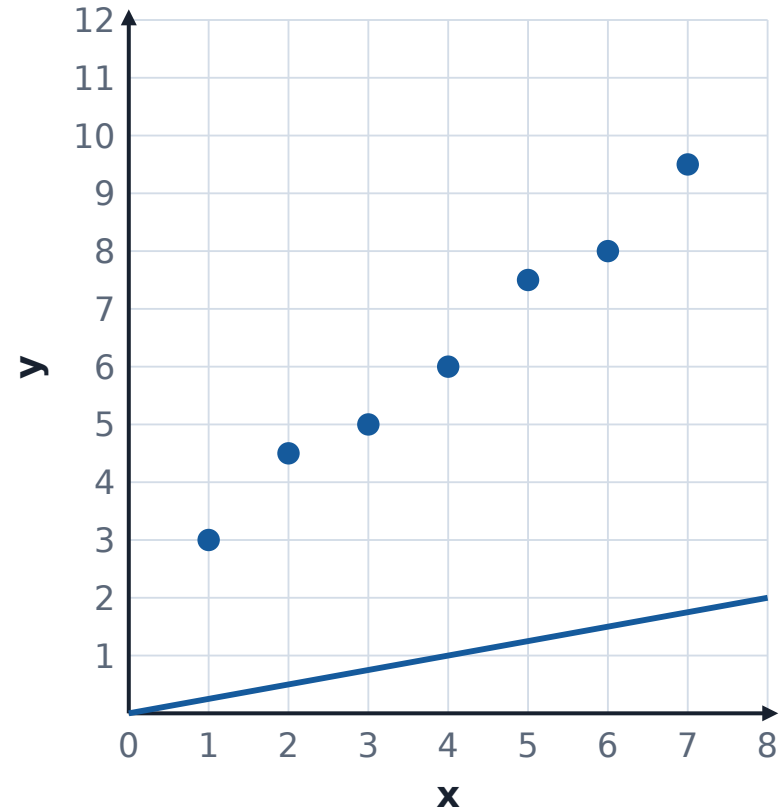
Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 45 • FORWARD 2 • Q3/5

True or false?

The line is a good fit for the data.

A. True

B. False

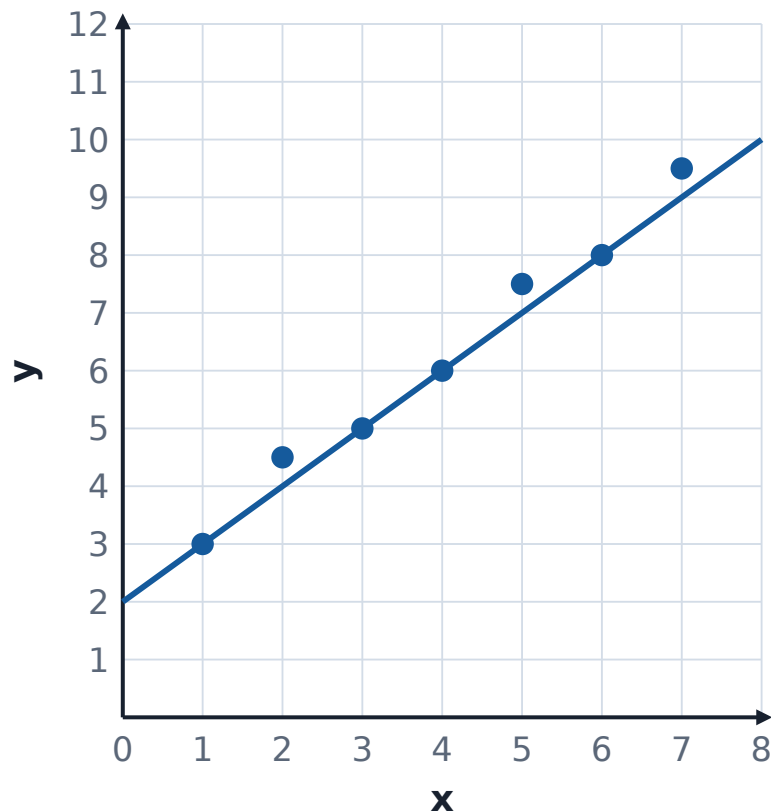


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 45 • FORWARD 2 • Q4/5

Use the line of best fit to
predict y when $x = 2$.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 45 • FORWARD 2 • Q5/5

Display data in dot plots and histograms

MAIN Grade 6 main question

6.SP.B.4

BACKWARD 1 Making a line plot

4.MD.B.4

BACKWARD 2 Scaled bar graphs

3.MD.B.3

BACKWARD 3 Placing numbers in intervals

2.NBT.A.4

FORWARD 1 Compare two dot plots

7.SP.B.3

FORWARD 2 Construct a scatter plot

8.SP.A.1

SET 46 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

The histogram shows how many books students read.

How many students read 10–19 books?

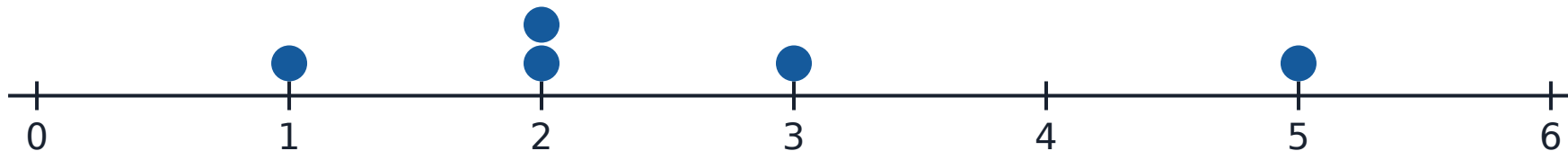


Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 46 • MAIN • Q1/5

Which data set matches the dot plot?



A. 1, 2, 2, 3, 5

B. 1, 2, 3, 5

C. 1, 1, 2, 3, 5

D. 1, 2, 2, 3, 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 46 • MAIN • Q2/5

Data: 2, 4, 4, 5, 7, 7, 7, 9

Make a dot plot of the data on the number line. How many dots will be above 7?



Dots: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 46 • MAIN • Q3/5

Which interval has the most students?



- A.** 10-19 **B.** 0-9 **C.** 20-29 **D.** 30-39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 46 • MAIN • Q4/5

True or false?

6 students read 20–29 books.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 46 • MAIN • Q5/5

SET 46 • 6.SP.B.4

BACKWARD 1

Making a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

Lengths (inches): $\frac{1}{2}$, 1, 1, $1\frac{1}{2}$, 1, $\frac{1}{2}$

In a line plot of these lengths, how many X's go above 1?

X's: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 46 • BACKWARD 1 • Q1/5

True or false?

In a line plot of 2, $2\frac{1}{2}$, $2\frac{1}{2}$, 3, there are two X's above $2\frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 46 • BACKWARD 1 • Q2/5

Which number line would you use to make a line plot of these lengths?

$$\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{1}{2}$$

- A.** A number line from 0 to 1 marked in fourths
- B.** A number line from 0 to 10 marked in ones
- C.** A number line from 0 to 100 marked in tens
- D.** A number line from 1 to 2 marked in halves

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 46 • BACKWARD 1 • Q3/5

Lengths (feet): 3 , $3\frac{1}{2}$, 4 , $3\frac{1}{2}$, $3\frac{1}{2}$

Which length gets the most X's?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 46 • BACKWARD 1 • Q4/5

True or false?

A line plot shows each data value as a mark above a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 46 • BACKWARD 1 • Q5/5

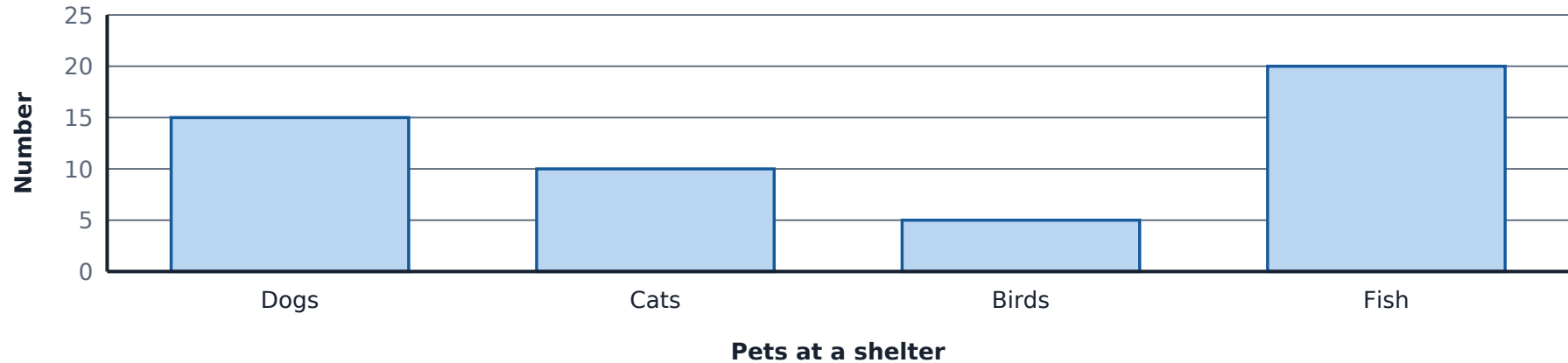
SET 46 • 6.SP.B.4

BACKWARD 2

Scaled bar graphs • Grade 3 • Measurement & Data • 3.MD.B.3

The graph shows the pets at a shelter.

How many cats are there?



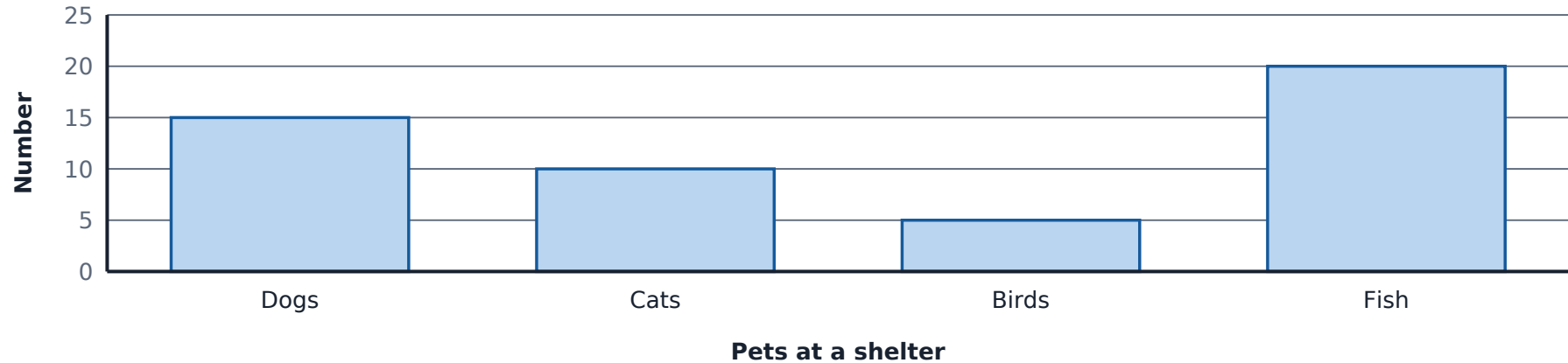
Cats: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 46 • BACKWARD 2 • Q1/5

The graph shows the pets at a shelter.

How many more fish than birds are there?



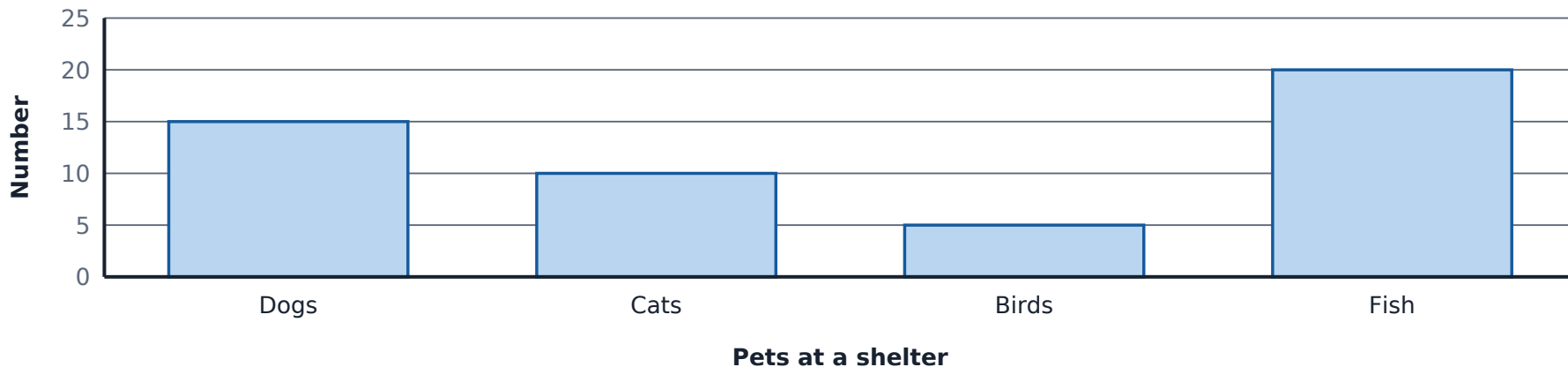
Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 46 • BACKWARD 2 • Q2/5

True or false?

Each grid line on the graph stands for 5 pets.



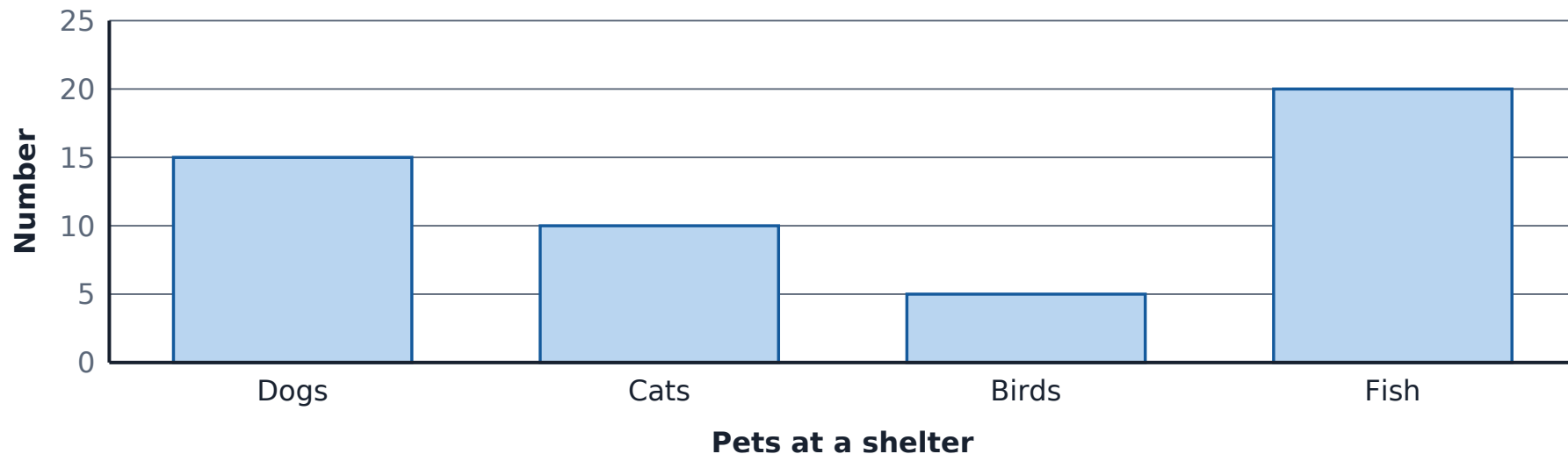
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 46 • BACKWARD 2 • Q3/5

Which type of pet has 15 at the shelter?



A. Dogs

B. Cats

C. Birds

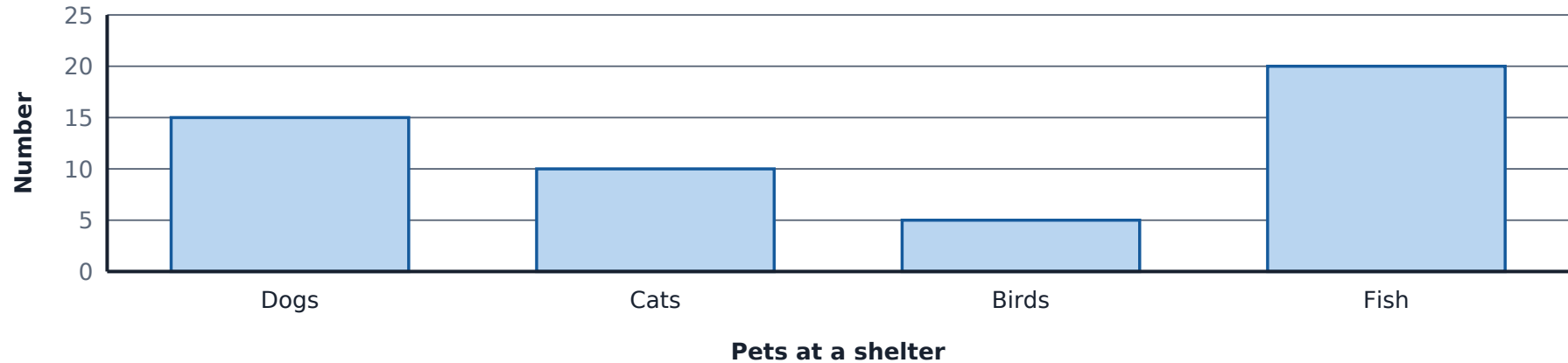
D. Fish

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 46 • BACKWARD 2 • Q4/5

The graph shows the pets at a shelter.

How many dogs and cats are there together?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 6.SP.B.4

BACKWARD 3

Placing numbers in intervals • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

Which interval contains 17?

A. 10–19

B. 0–9

C. 20–29

D. 30–39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 46 • BACKWARD 3 • Q1/5

True or false?

25 is in the interval 20–29.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 46 • BACKWARD 3 • Q2/5

Data: 3, 12, 15, 18, 22

How many of the numbers are in the interval 10–19?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 46 • BACKWARD 3 • Q3/5

Which number is in the interval 30–39?

A. 34

B. 29

C. 40

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 46 • BACKWARD 3 • Q4/5

True or false?

19 is in the interval 20–29.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

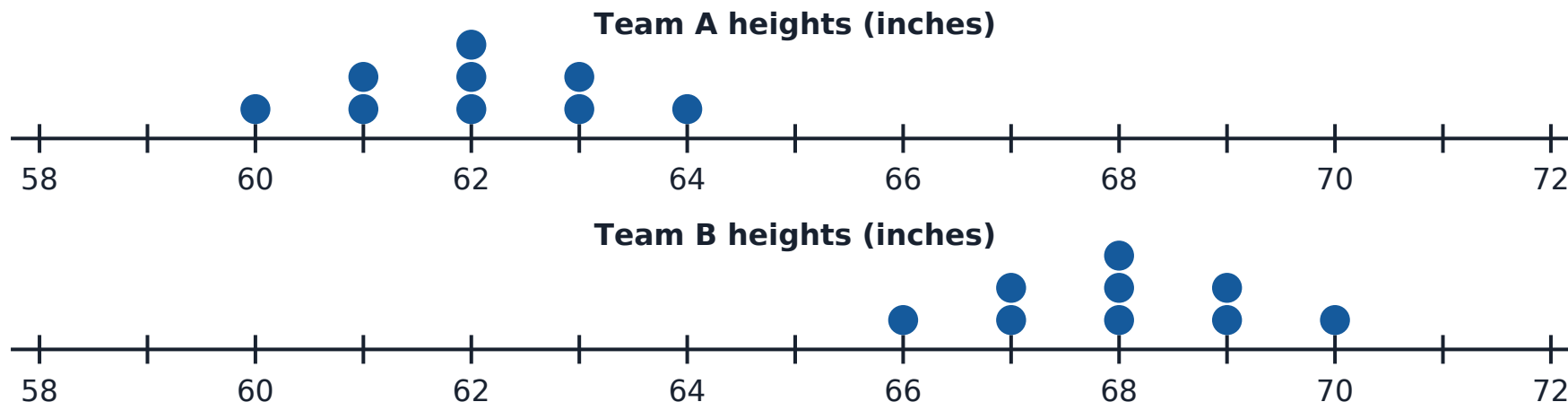
Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • 6.SP.B.4

FORWARD 1

Compare two dot plots • Grade 7 • Statistics & Probability • 7.SP.B.3

About how many inches greater is the center of Team B's heights?



Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 46 • FORWARD 1 • Q1/5

True or false?

The dot plots show that the two teams have about the same variability.



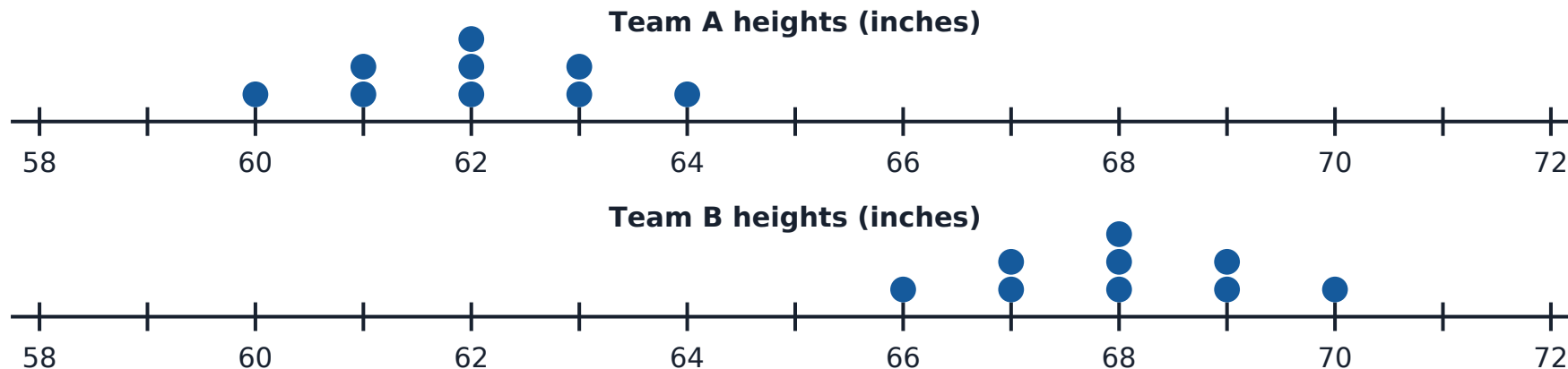
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 46 • FORWARD 1 • Q2/5

Which statement is true?



- A.** The distributions do not overlap. **B.** Team A is typically taller.
- C.** Team B has more variability. **D.** The medians are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 46 • FORWARD 1 • Q3/5

Team A's mean height is 62 in. Team B's mean height is 68 in. Each team's MAD is about 1 in.

The difference in means is about how many times the MAD?

Times: _____

True or false?

Two data sets with means of 50 and 56 and MADs of 3 have means that differ by 2 MADs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 6.SP.B.4

FORWARD 2

Construct a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

The table shows hours studied and test scores.

Which ordered pair would be plotted for Student C?

Student	A	B	C	D
Hours	1	3	4	6
Score	62	70	78	88

Ordered pair: _____

True or false?

To make a scatter plot of height and arm span, each person is shown as one point.

- A.** True
- B.** False

A scatter plot shows age on the x-axis and height on the y-axis.

A 10-year-old is 54 inches tall. Which point shows this person?

A. (10, 54)

B. (54, 10)

C. (10, 10)

D. (54, 54)

The table shows hours studied and test scores.

Which ordered pair would be plotted for Student D?

Student	A	B	C	D
Hours	1	3	4	6
Score	62	70	78	88

Ordered pair: _____

The ordered pairs (1, 62), (3, 70), (4, 78), and (6, 88) are plotted.

What association will the scatter plot show?

- A.** Positive
- B.** Negative
- C.** No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 46 • FORWARD 2 • Q5/5

Display data in box plots

MAIN	Grade 6 main question
BACKWARD 1	Ordering numbers
BACKWARD 2	Halfway between two numbers
BACKWARD 3	Reading a number line scale
FORWARD 1	Compare populations with box plots
FORWARD 2	Two-way tables

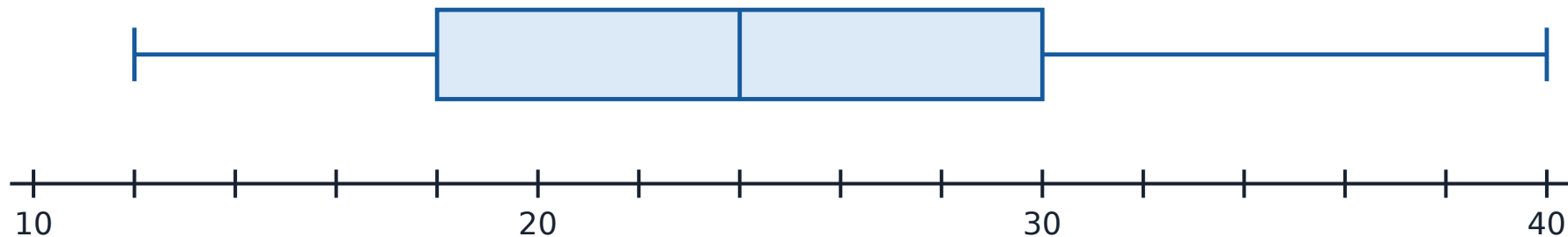
6.SP.B.4
2.NBT.A.4
2.MD.B.6
4.MD.A.2
7.SP.B.4
8.SP.A.4

SET 47 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

What is the median of the data shown in the box plot?

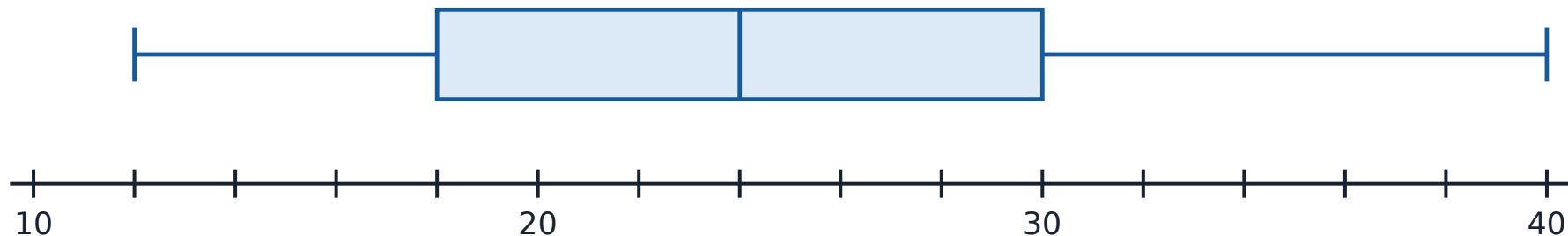


Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 47 • MAIN • Q1/5

What is the interquartile range of the data shown in the box plot?

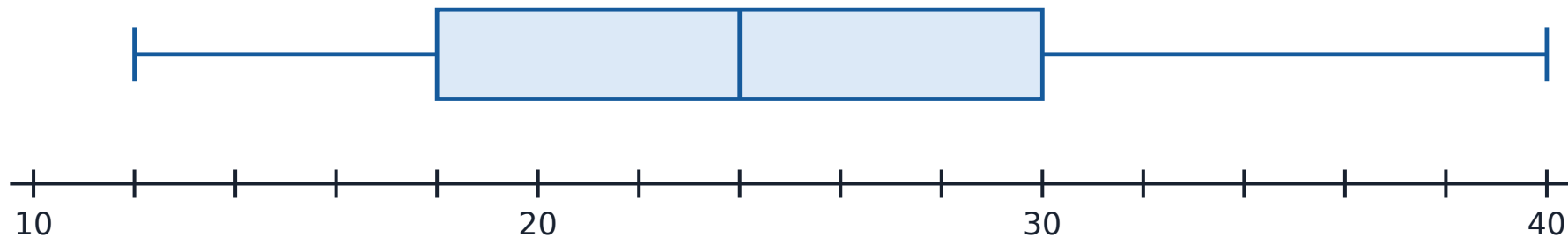


IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 47 • MAIN • Q2/5

What is the range of the data shown in the box plot?



A. 28

B. 12

C. 24

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 47 • MAIN • Q3/5

Data: 3, 5, 7, 8, 10, 12, 15

Find the five-number summary for a box plot.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

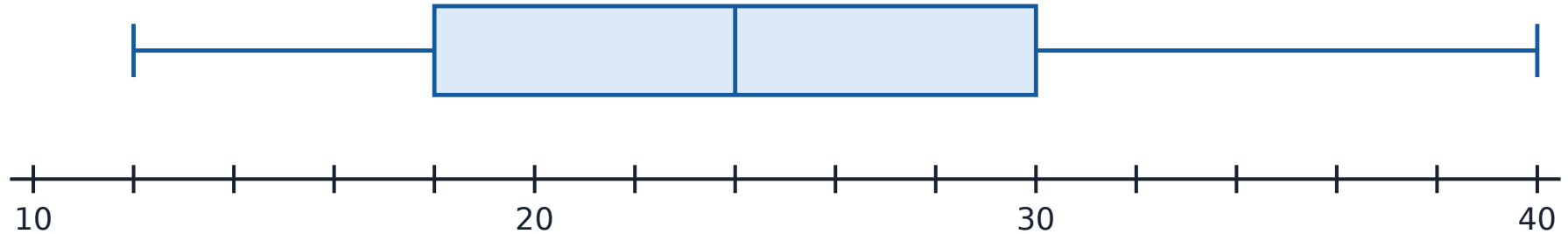
Maximum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 47 • MAIN • Q4/5

True or false?

In the box plot, the maximum value is 40.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 47 • MAIN • Q5/5

SET 47 • 6.SP.B.4

BACKWARD 1

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

15, 8, 23, 4, 16

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 47 • BACKWARD 1 • Q1/5

Which list is in order from least to greatest?

A. 7, 12, 21, 27

B. 12, 7, 21, 27

C. 7, 21, 12, 27

D. 27, 21, 12, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 47 • BACKWARD 1 • Q2/5

True or false?

These numbers are in order from least to greatest: 30, 33, 31, 35

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 47 • BACKWARD 1 • Q3/5

Order from least to greatest.

62, 26, 60, 20, 66

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 47 • BACKWARD 1 • Q4/5

Which number is least?

A. 108

B. 180

C. 118

D. 801

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 6.SP.B.4

BACKWARD 2

Halfway between two numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What number is halfway between 12 and 18?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 47 • BACKWARD 2 • Q1/5

What number is halfway between points A and B?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 47 • BACKWARD 2 • Q2/5

True or false?

The number halfway between 40 and 50 is 45.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 47 • BACKWARD 2 • Q3/5

What number is halfway between 6 and 10?

- A.** 8
- B.** 7
- C.** 16
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 47 • BACKWARD 2 • Q4/5

What number is halfway between 0 and 30?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 6.SP.B.4

BACKWARD 3

Reading a number line scale • Grade 4 • Measurement & Data • 4.MD.A.2

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 47 • BACKWARD 3 • Q1/5

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 47 • BACKWARD 3 • Q2/5

True or false?

Point C is at 12.



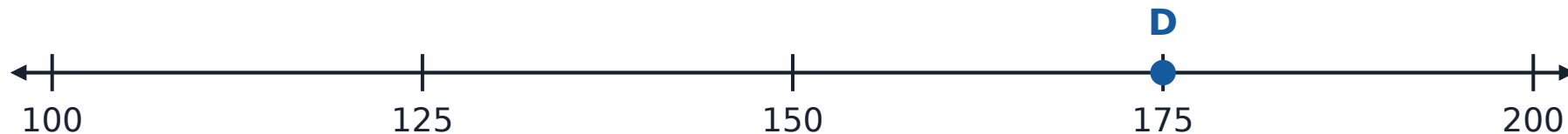
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 47 • BACKWARD 3 • Q3/5

What number is at point D?



A. 175

B. 170

C. 150

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 47 • BACKWARD 3 • Q4/5

What number is at point E?



E = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

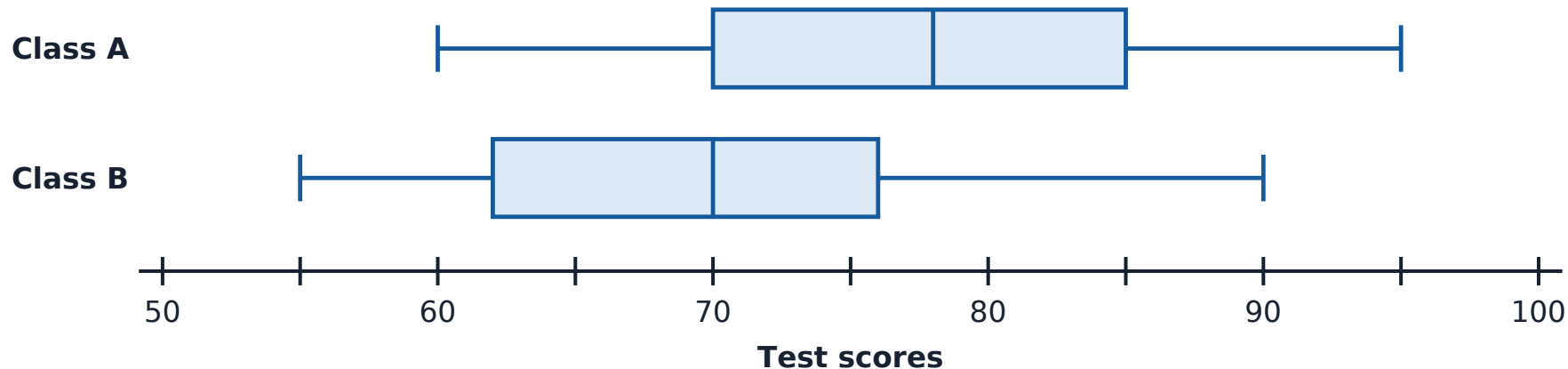
Grade 4 • Measurement & Data • 4.MD.A.2 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • 6.SP.B.4

FORWARD 1

Compare populations with box plots • Grade 7 • Statistics & Probability •
7.SP.B.4

Which class has the greater median score?



A. Class A

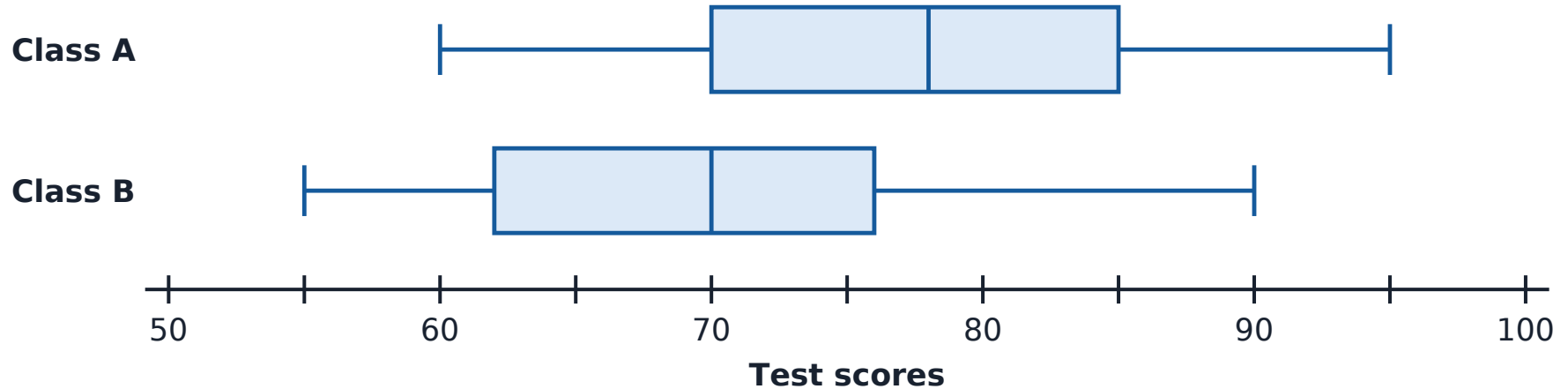
B. Class B

C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 47 • FORWARD 1 • Q1/5

By how many points do the medians differ?



A. 8

B. 5

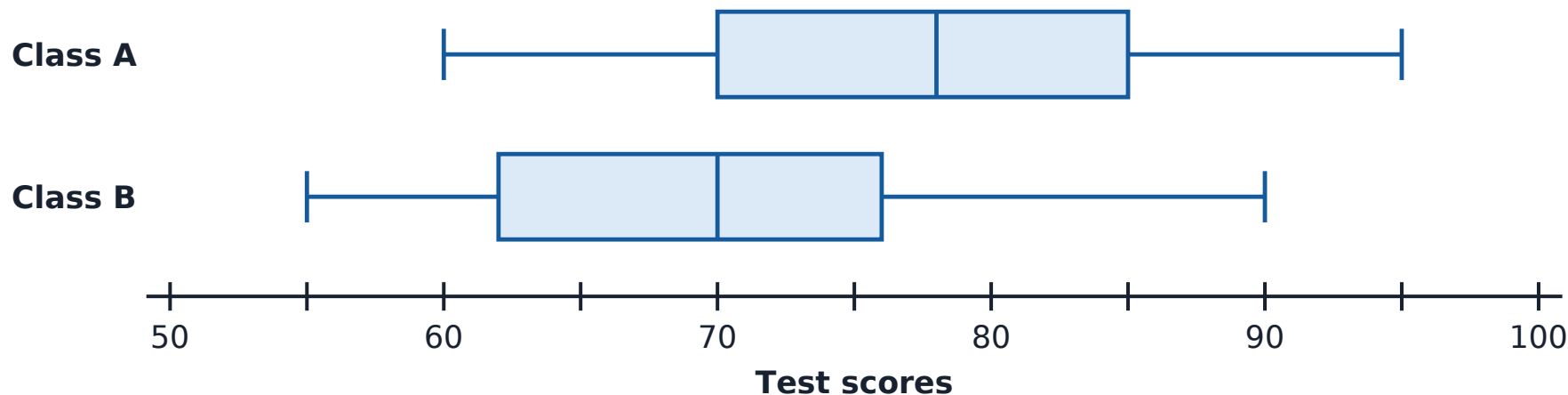
C. 15

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 47 • FORWARD 1 • Q2/5

What is the interquartile range of Class A's scores?



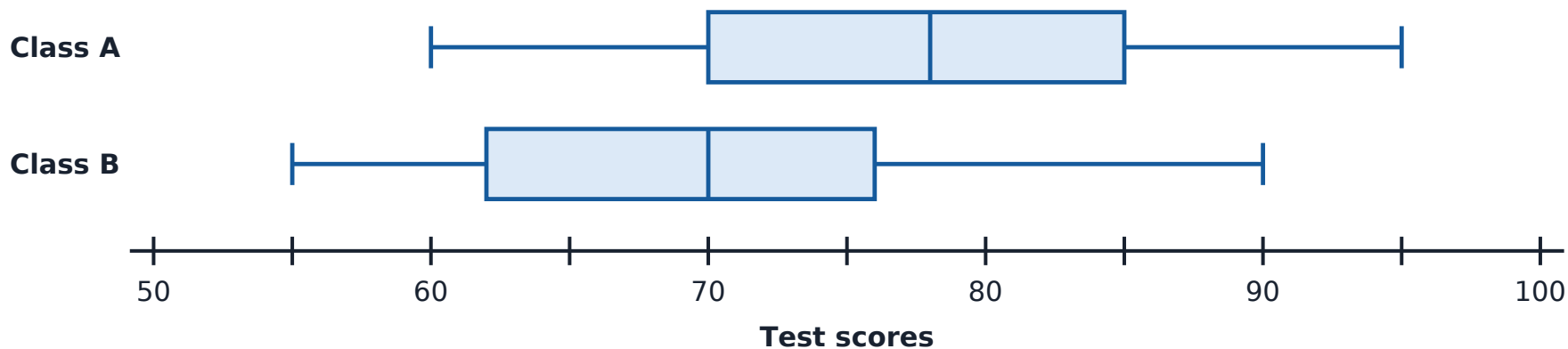
IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 47 • FORWARD 1 • Q3/5

True or false?

At least half of Class A scored 78 or higher.



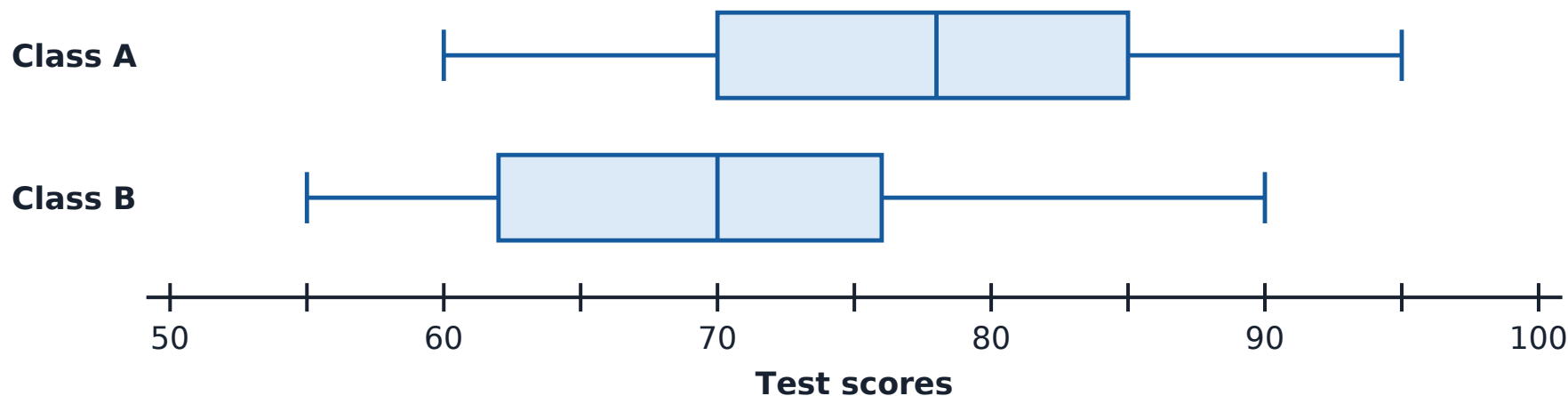
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 47 • FORWARD 1 • Q4/5

Which class had the highest single score?



Class: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 6.SP.B.4

FORWARD 2

Two-way tables • Grade 8 • Statistics & Probability • 8.SP.A.4

The table shows survey results.

How many 8th graders have a pet?

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 47 • FORWARD 2 • Q1/5

What fraction of the 7th graders have a pet?

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 47 • FORWARD 2 • Q2/5

What is the relative frequency of 8th graders who have no pet?

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{10}{22}$

D. $\frac{1}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 47 • FORWARD 2 • Q3/5

True or false?

60 students were surveyed in all.

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 47 • FORWARD 2 • Q4/5

What number belongs in place of the question mark?

	Walk	Bus	Total
Boys	12	?	30
Girls	15	10	25
Total	27	28	55

? = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 47 • FORWARD 2 • Q5/5

Report the number of observations

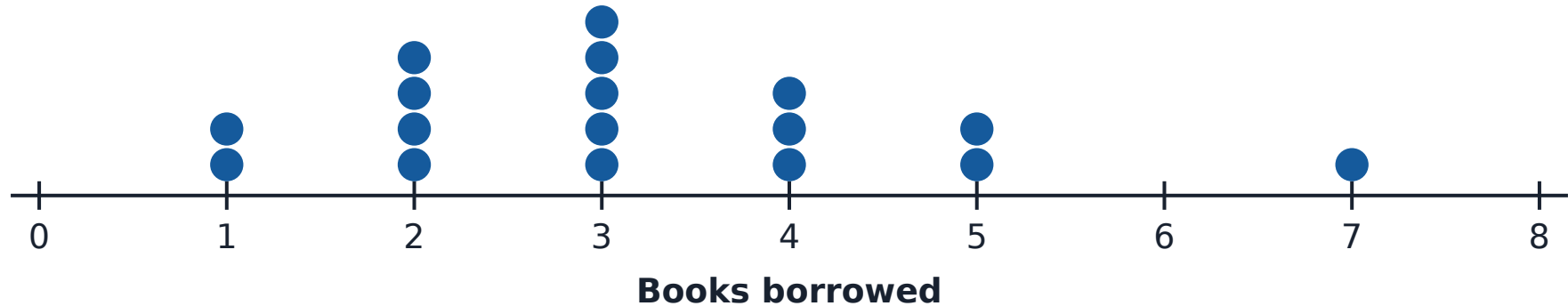
MAIN	Grade 6 main question	6.SP.B.5.a
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Totals from a bar graph	3.MD.B.3
BACKWARD 3	Adding several two-digit numbers	2.NBT.B.6
FORWARD 1	Use a random sample to make inferences	7.SP.A.2
FORWARD 2	Totals in two-way tables	8.SP.A.4

SET 48 • 6.SP.B.5.a

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.a

The dot plot shows how many books students borrowed.
How many observations are in the data set?

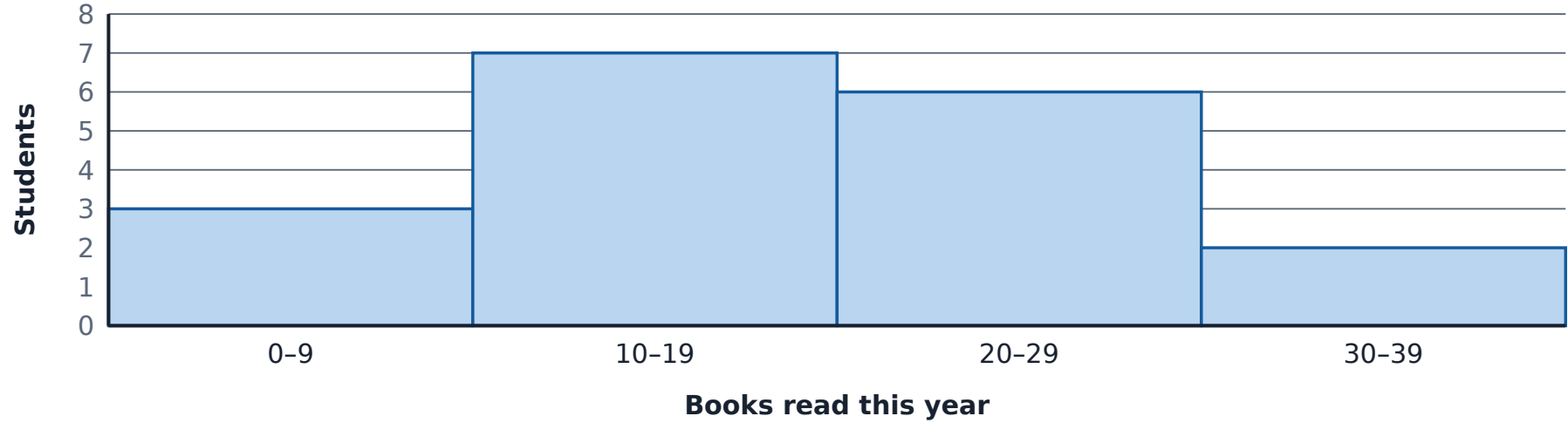


Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 48 • MAIN • Q1/5

How many students are represented in the histogram?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 48 • MAIN • Q2/5

The table shows how many siblings students have.
How many observations are in the data set?

Siblings	0	1	2	3
Students	4	9	6	2

Observations: _____

True or false?

A box plot shows how many observations are in a data set.

A. True

B. False

Data: 12, 15, 9, 15, 20, 11, 13, 18, 15, 10

How many observations are in the data set?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 48 • MAIN • Q5/5

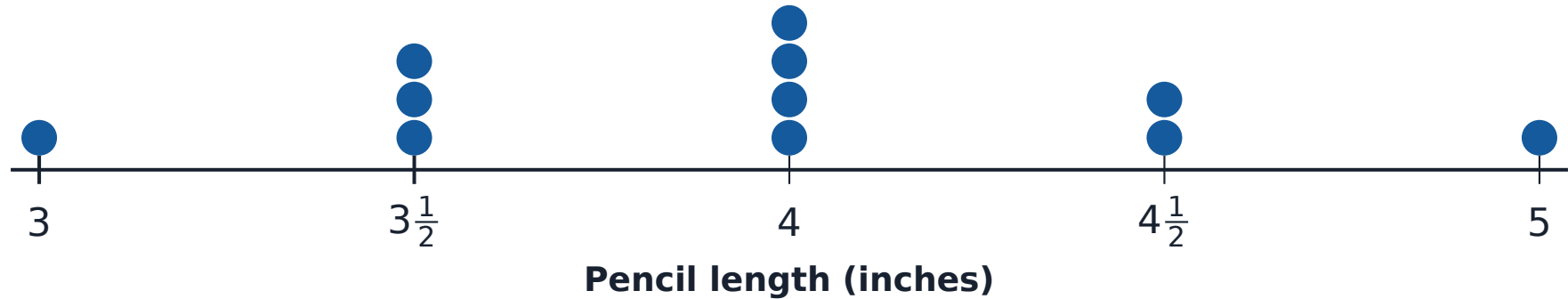
SET 48 • 6.SP.B.5.a

BACKWARD 1

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows pencil lengths.

How many pencils are 4 inches long?

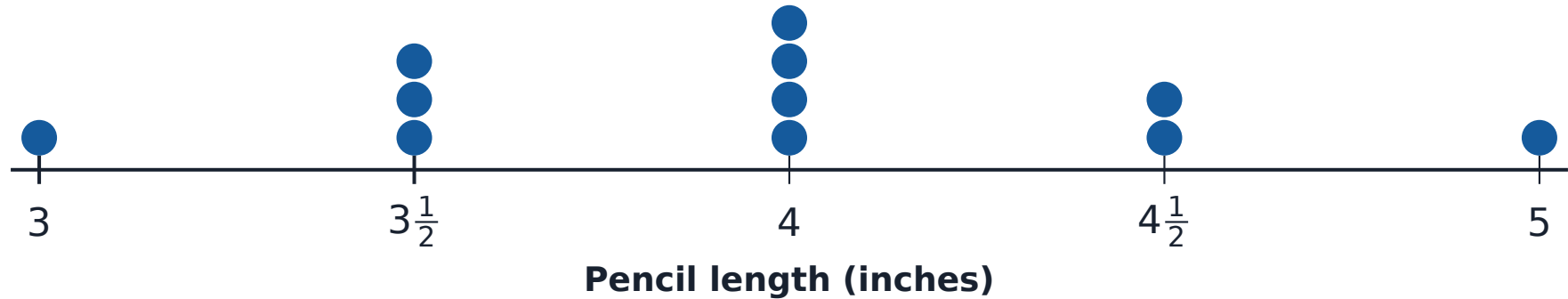


Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 48 • BACKWARD 1 • Q1/5

The line plot shows pencil lengths.
How many pencils were measured?



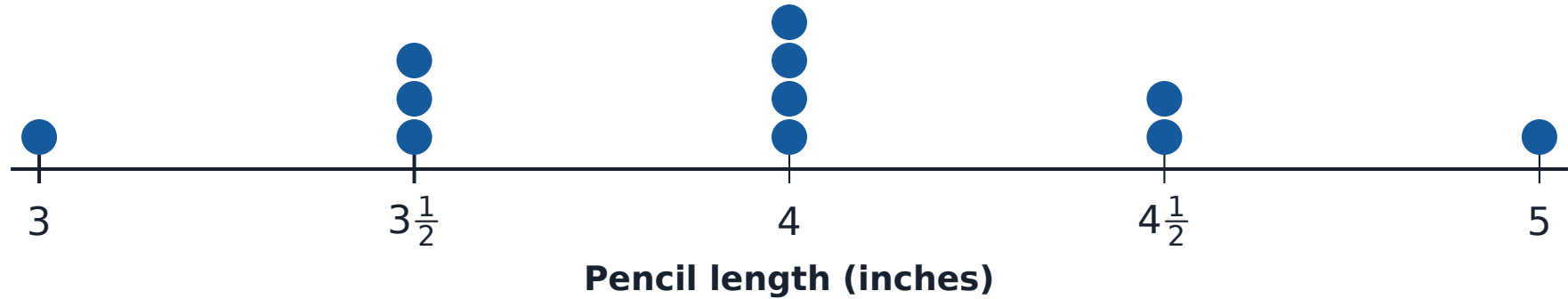
Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 48 • BACKWARD 1 • Q2/5

True or false?

Three pencils are $3\frac{1}{2}$ inches long.



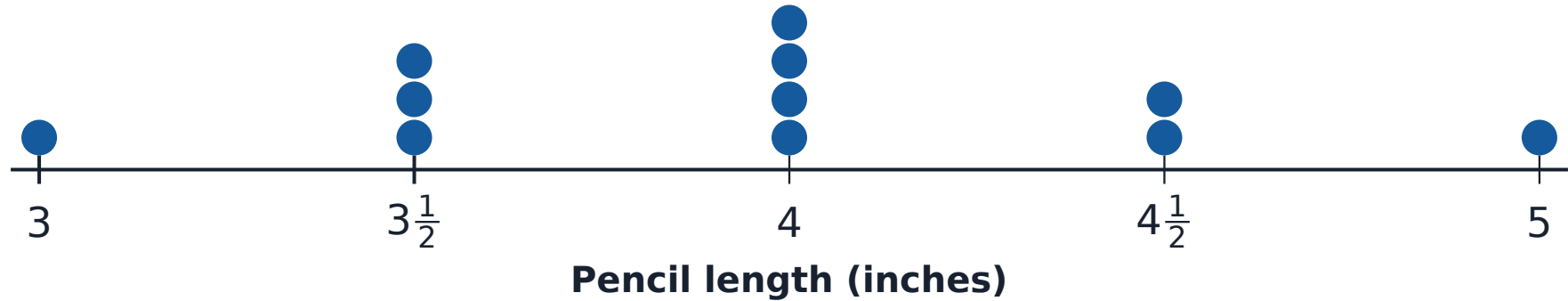
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 48 • BACKWARD 1 • Q3/5

How many pencils are longer than 4 inches?



A. 3

B. 4

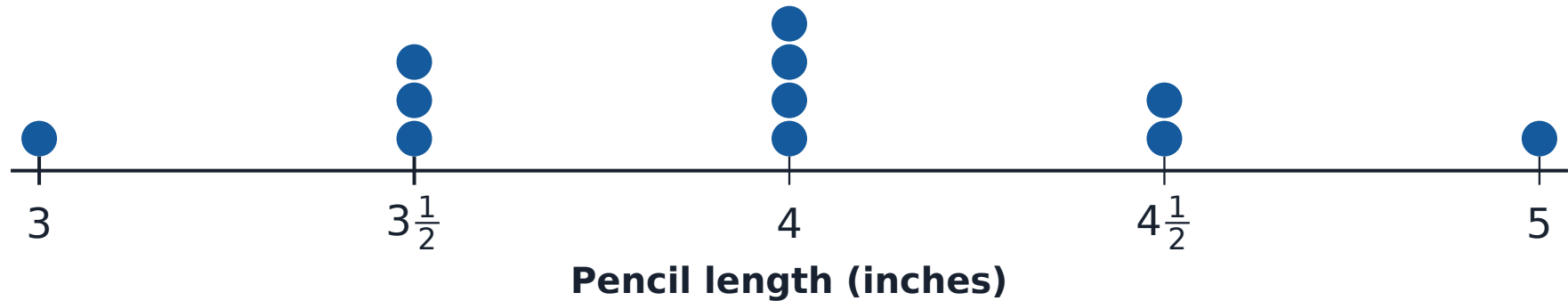
C. 2

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 48 • BACKWARD 1 • Q4/5

What is the difference between the longest and the shortest pencil?



Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

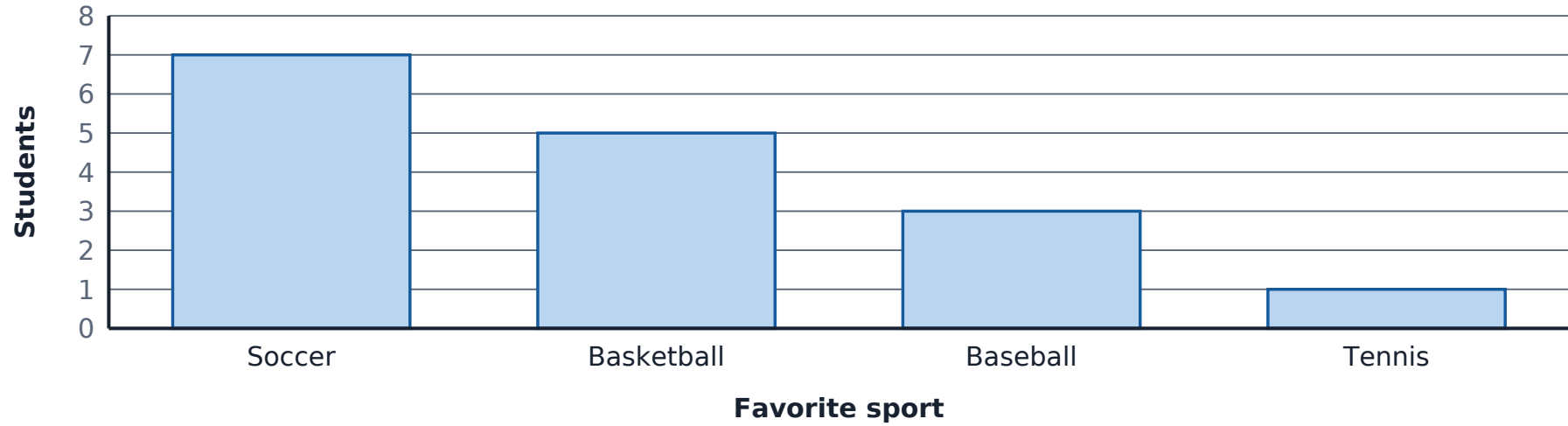
Grade 4 • Measurement & Data • 4.MD.B.4 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 6.SP.B.5.a

BACKWARD 2

Totals from a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many students answered the survey?

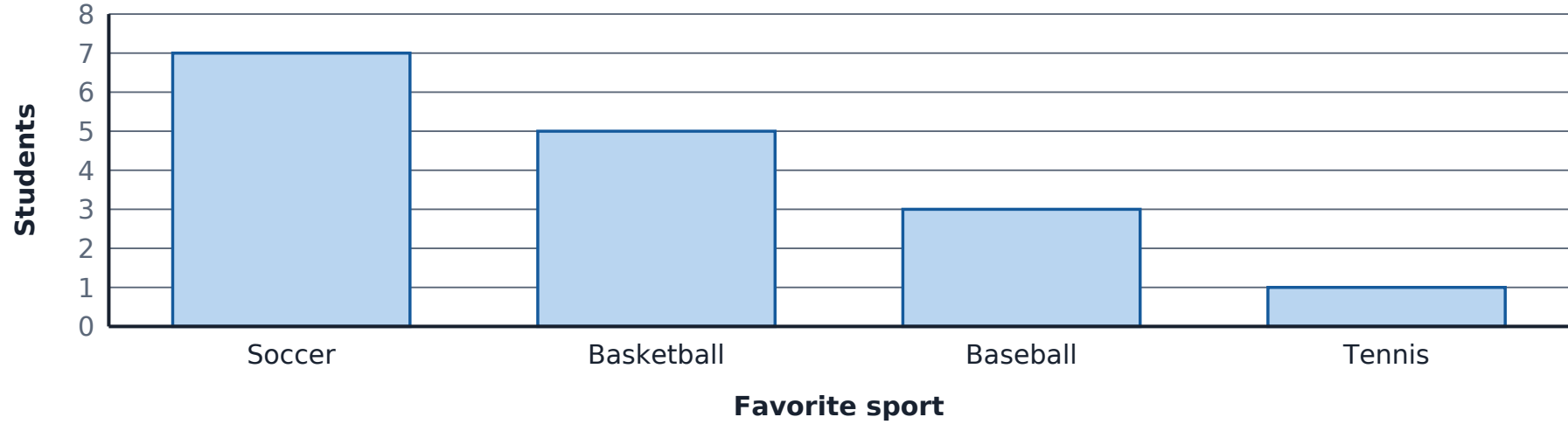


Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 48 • BACKWARD 2 • Q1/5

How many students chose soccer or tennis?



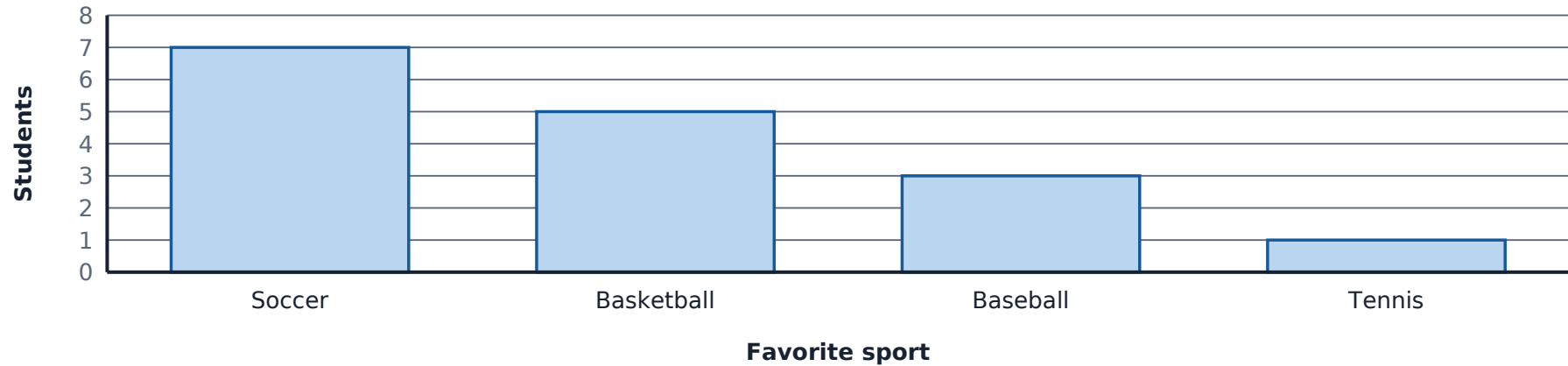
Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 48 • BACKWARD 2 • Q2/5

True or false?

12 students chose soccer or basketball.



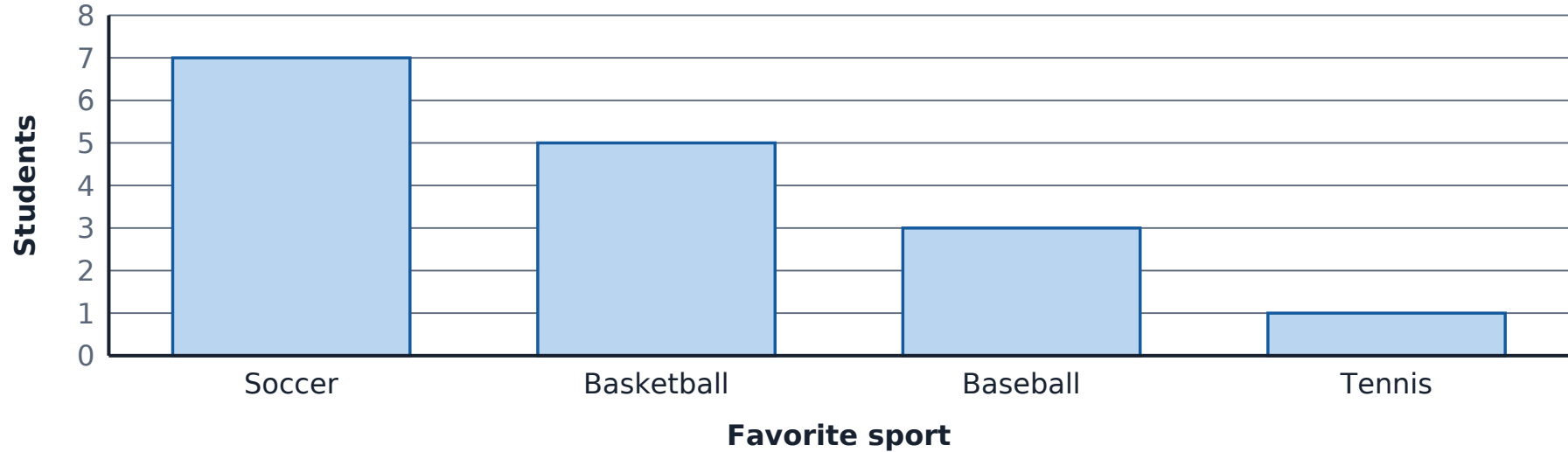
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 48 • BACKWARD 2 • Q3/5

How many students did NOT choose soccer?



A. 9

B. 7

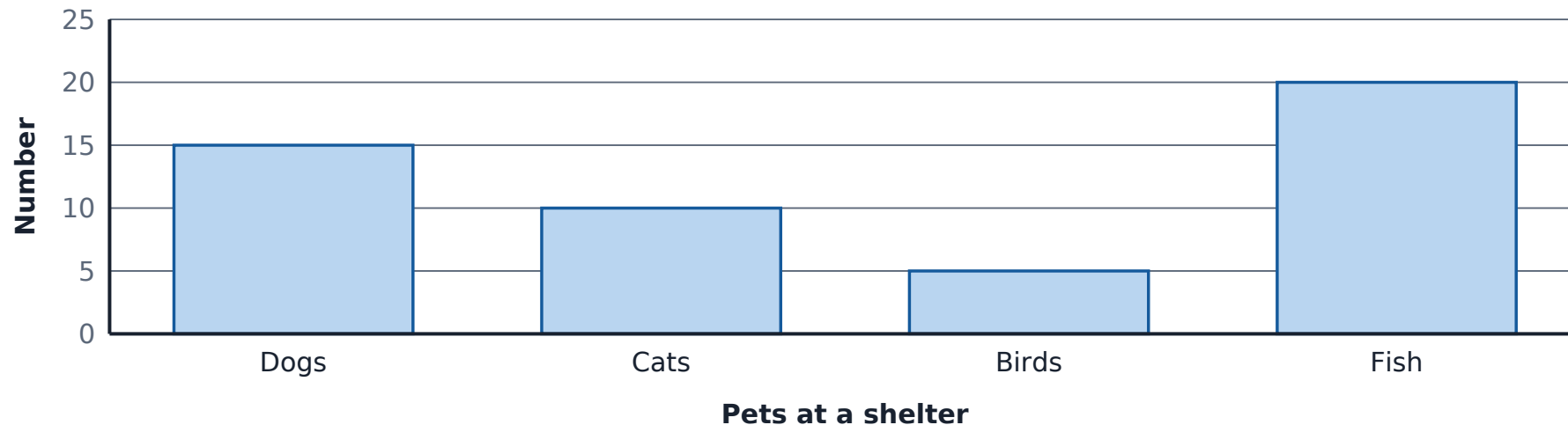
C. 16

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 48 • BACKWARD 2 • Q4/5

How many pets are at the shelter in all?



Pets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 6.SP.B.5.a

BACKWARD 3

Adding several two-digit numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6

Add.

$$14 + 23 + 31 + 12$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 48 • BACKWARD 3 • Q1/5

Add.

$$25 + 18 + 30$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 48 • BACKWARD 3 • Q2/5

True or false?

$$16 + 16 + 20 + 8 = 60$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 48 • BACKWARD 3 • Q3/5

$$41 + 27 + 13 + 9 = ?$$

- A.** 90
- B.** 80
- C.** 91
- D.** 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 48 • BACKWARD 3 • Q4/5

Add.

$$35 + 22 + 18 + 15$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • 6.SP.B.5.a

FORWARD 1

Use a random sample to make inferences • Grade 7 • Statistics & Probability •
7.SP.A.2

A researcher surveys 80 randomly chosen students out of 900.

What is the sample size?

A. 80

B. 900

C. 820

D. 980

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 48 • FORWARD 1 • Q1/5

True or false?

A larger random sample generally gives a better estimate about a population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 48 • FORWARD 1 • Q2/5

In a random sample of 25 students out of 500, 5 like jazz.

Estimate how many of the 500 students like jazz.

Estimate: _____

In a random sample of 50 voters, 30 support a new park. There are 2,000 voters.

About how many voters support the park?

A. 1,200

B. 600

C. 30

D. 1,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 48 • FORWARD 1 • Q4/5

Two random samples of 20 students each found 12 and 14 students who like pizza best.

What is the mean number from the two samples?

Mean: _____

SET 48 • 6.SP.B.5.a

FORWARD 2

Totals in two-way tables • Grade 8 • Statistics & Probability • 8.SP.A.4

How many students were surveyed in all?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 48 • FORWARD 2 • Q1/5

How many students play sports?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 48 • FORWARD 2 • Q2/5

True or false?

14 students play an instrument and play sports.

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 48 • FORWARD 2 • Q3/5

How many students play an instrument but do NOT play sports?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

A. 6

B. 14

C. 20

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 48 • FORWARD 2 • Q4/5

What number belongs in place of the question mark?

	Walk	Bus	Total
Boys	12	?	30
Girls	15	10	25
Total	27	28	55

? = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 48 • FORWARD 2 • Q5/5

Describe the attribute measured and its units

MAIN	Grade 6 main question	6.SP.B.5.b
BACKWARD 1	Choosing measurement units	4.MD.A.1
BACKWARD 2	Categories in a bar graph	3.MD.B.3
FORWARD 1	How data are collected: sampling methods	7.SP.A.1
FORWARD 2	Categorical data in two-way tables	8.SP.A.4

SET 49 • 6.SP.B.5.b

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.b

A data set lists the heights of 25 plants in centimeters.
What attribute is being measured, and in what unit?

Attribute: _____

Unit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 49 • MAIN • Q1/5

Which unit would be used to record the time students spend on homework?

- A.** Minutes
- B.** Inches
- C.** Pounds
- D.** Degrees

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

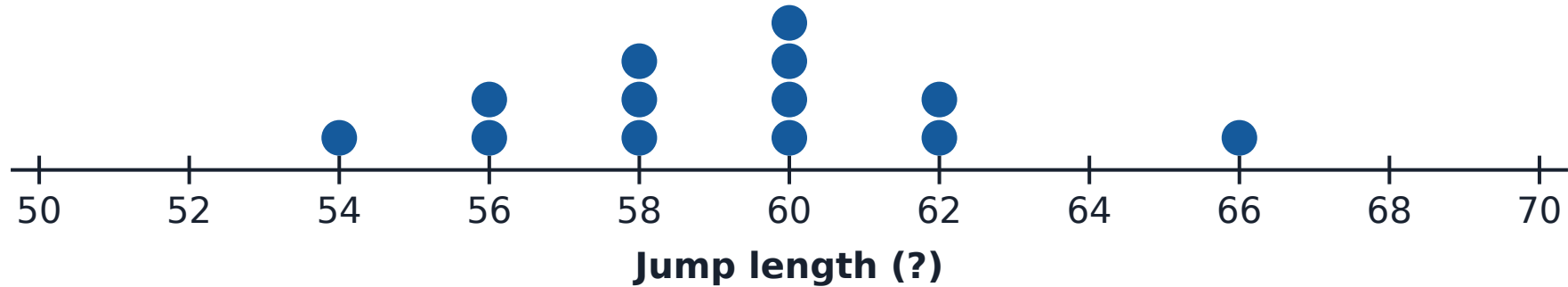
Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 49 • MAIN • Q2/5

True or false?

In a data set of "number of pets for each student," each observation is a count, not a length.

- A.** True
- B.** False

The dot plot shows how far students jumped.
Which unit is most likely?



- A.** Inches **B.** Miles **C.** Pounds **D.** Seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 49 • MAIN • Q4/5

A survey records how each student gets to school: bus, car, or walk.

Is this data numerical or categorical?

Answer: _____

SET 49 • 6.SP.B.5.b

BACKWARD 1

Choosing measurement units • Grade 4 • Measurement & Data • 4.MD.A.1

Which unit is best for measuring the length of a pencil?

- A.** Centimeters
- B.** Kilometers
- C.** Meters
- D.** Kilograms

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 49 • BACKWARD 1 • Q1/5

True or false?

A kilogram is heavier than a gram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 49 • BACKWARD 1 • Q2/5

Which unit measures liquid volume?

- A.** Liters
- B.** Grams
- C.** Meters
- D.** Seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 49 • BACKWARD 1 • Q3/5

Name one unit used to measure time.

Unit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 49 • BACKWARD 1 • Q4/5

True or false?

An inch is longer than a foot.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

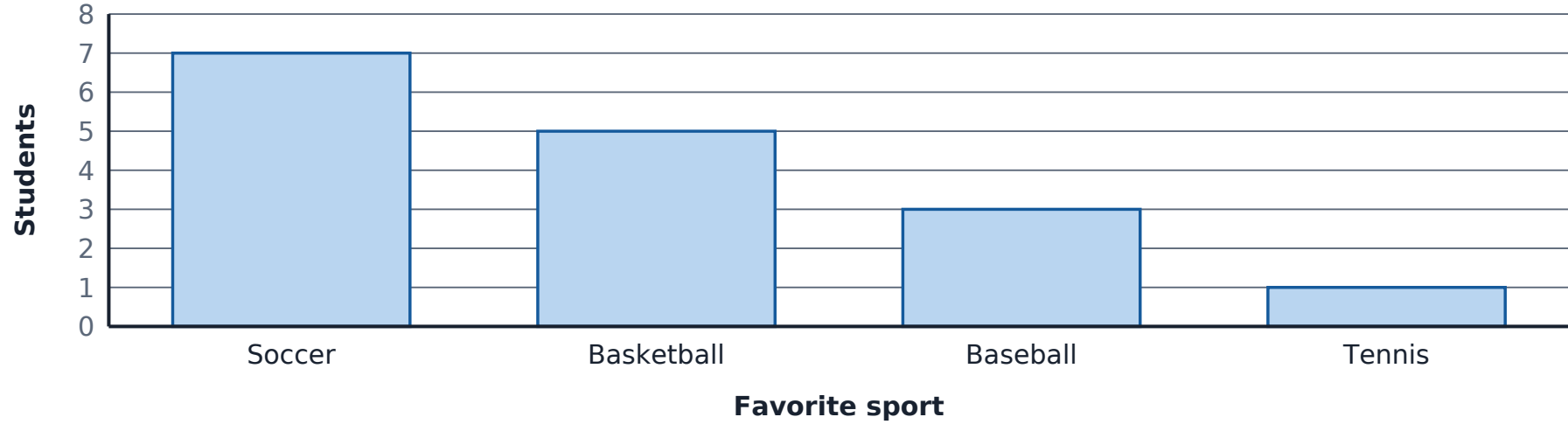
Grade 4 • Measurement & Data • 4.MD.A.1 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 6.SP.B.5.b

BACKWARD 2

Categories in a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many categories does the graph show?

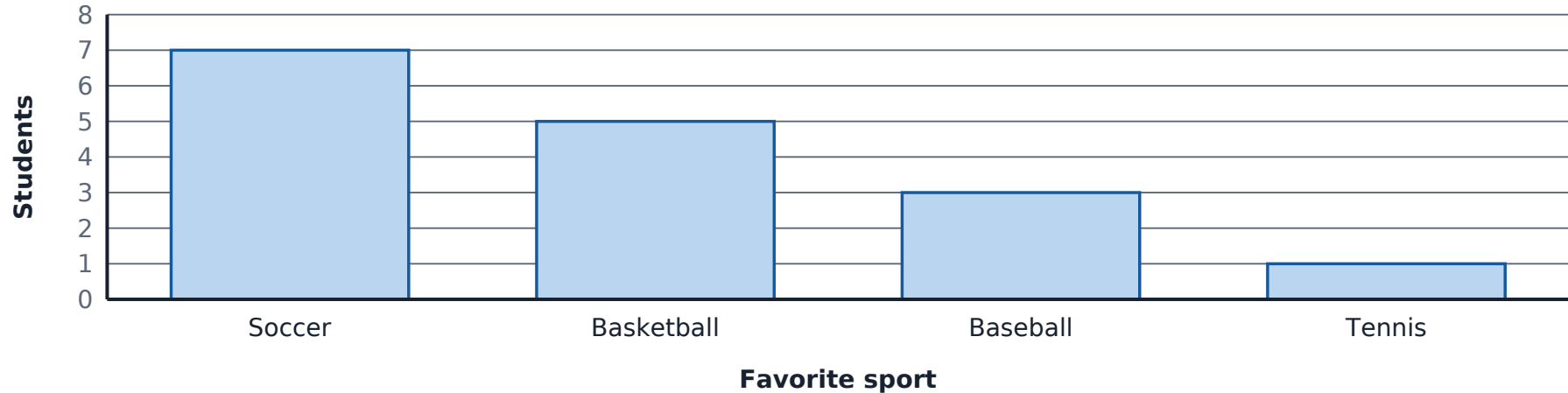


Categories: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 49 • BACKWARD 2 • Q1/5

What does the height of each bar show?



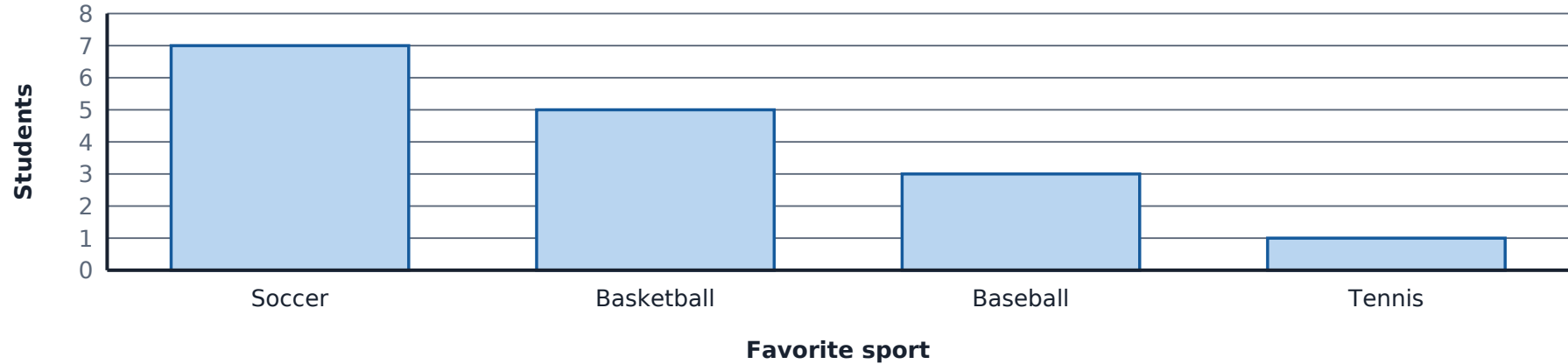
- A.** The number of students
- B.** The number of sports
- C.** The score of a game
- D.** The length of a game

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 49 • BACKWARD 2 • Q2/5

True or false?

"Tennis" is one of the categories in the graph.



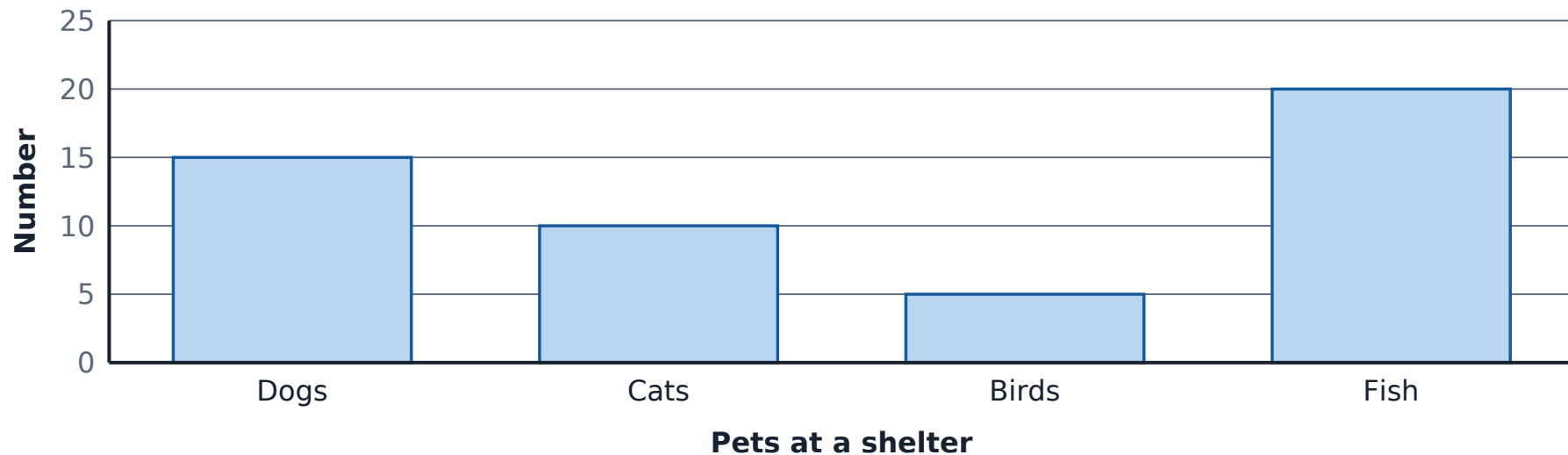
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 49 • BACKWARD 2 • Q3/5

Which is a category in the graph?

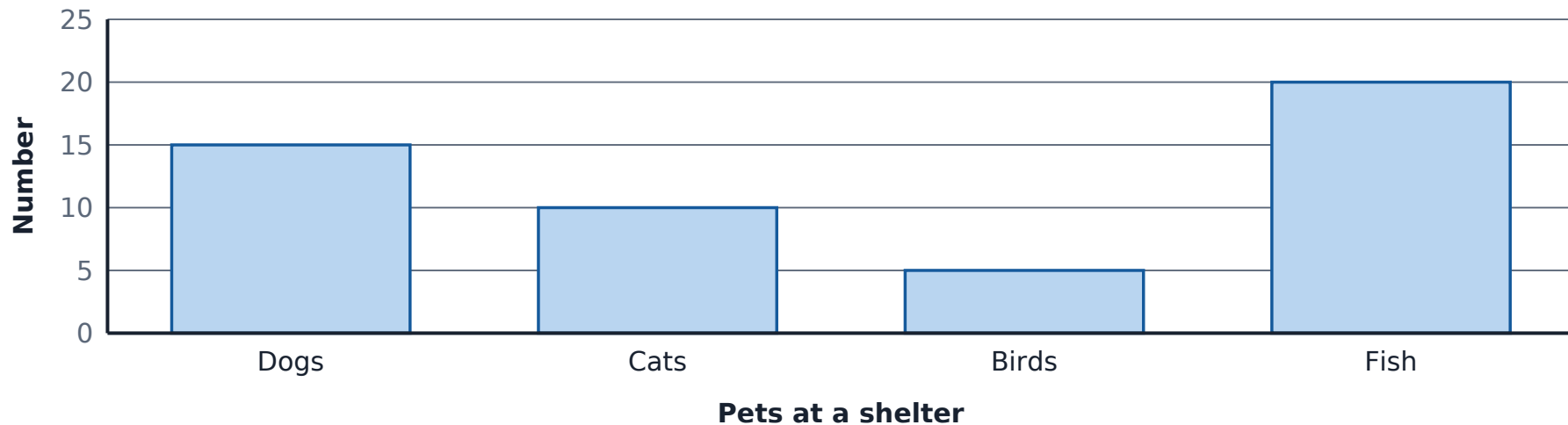


- A.** Birds **B.** Horses **C.** Soccer **D.** 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 49 • BACKWARD 2 • Q4/5

What is being counted in the graph?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 6.SP.B.5.b

FORWARD 1

How data are collected: sampling methods • Grade 7 • Statistics & Probability •
7.SP.A.1

Which method gives a random sample of the students in a school?

- A.** Draw names from a hat that holds every student's name
- B.** Ask your friends
- C.** Ask the students in the library
- D.** Ask the first 20 students to arrive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • FORWARD 1 • Q1/5

True or false?

Surveying people who walk out of a gym is a good way to learn how often all people in a town exercise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • FORWARD 1 • Q2/5

A town wants to know how residents feel about a new library.

Describe a way to choose a random sample.

Method: _____

Which sample is most likely to be biased?

- A.** Asking only people at a dog park whether they like dogs
- B.** Choosing 50 residents at random from a town list
- C.** Picking every 10th name from a school list
- D.** Drawing 30 names from a hat of all members

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • FORWARD 1 • Q4/5

True or false?

Every member of a population has an equal chance of being chosen in a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • FORWARD 1 • Q5/5

SET 49 • 6.SP.B.5.b

FORWARD 2

Categorical data in two-way tables • Grade 8 • Statistics & Probability •
8.SP.A.4

Which pair of variables would be shown in a two-way table?

- A.** Grade level and whether a student has a pet
- B.** Height and weight
- C.** Age and arm span
- D.** Hours slept and test score

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 2 • Q1/5

What fraction of students who play an instrument also play sports?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 2 • Q2/5

True or false?

The data in a two-way table are categorical.

- A. True
- B. False

What is the relative frequency of students who play sports?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

A. $\frac{3}{4}$

B. $\frac{1}{4}$

C. $\frac{1}{2}$

D. $\frac{3}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 2 • Q4/5

How many 7th graders do NOT have a pet?

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 2 • Q5/5

Find the mean and median of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Adding multi-digit numbers	4.NBT.B.4
BACKWARD 2	Dividing a total into equal groups	4.NBT.B.6
BACKWARD 3	Ordering numbers	2.NBT.A.4
BACKWARD 4	The number halfway between two numbers	5.NBT.B.7
FORWARD 1	Compare populations using means and medians	7.SP.B.4
FORWARD 2	Use a linear model to make predictions	8.SP.A.3

SET 50 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of the data.

4, 7, 9, 10, 15

Mean: _____

Find the median of the data.

12, 5, 9, 20, 7, 14

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 50 • MAIN • Q2/5

Five test scores are 82, 90, 75, 88, and 95.

What is the mean score?

Mean: _____

True or false?

The median of 3, 8, 8, 10, 21 is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 50 • MAIN • Q4/5

Heights (in inches): 58, 60, 61, 63, 63

What is the median height?

- A.** 61
- B.** 63
- C.** 60
- D.** 61.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 50 • MAIN • Q5/5

SET 50 • 6.SP.B.5.c

BACKWARD 1

Adding multi-digit numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Add.

$$82 + 90 + 75 + 88 + 95$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 50 • BACKWARD 1 • Q1/5

Add.

$$145 + 238 + 97$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 50 • BACKWARD 1 • Q2/5

True or false?

$$56 + 64 + 70 = 190$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 50 • BACKWARD 1 • Q3/5

$$125 + 175 + 200 = ?$$

- A.** 500
- B.** 400
- C.** 450
- D.** 550

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 50 • BACKWARD 1 • Q4/5

Add.

$$1,250 + 875$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 6.SP.B.5.c

BACKWARD 2

Dividing a total into equal groups • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Divide.

$$430 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 2 • Q1/5

Divide.

$$45 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 2 • Q2/5

True or false?

$$108 \div 4 = 27$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 2 • Q3/5

$$252 \div 6 = ?$$

A. 42

B. 41

C. 48

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 2 • Q4/5

Divide.

$$315 \div 7$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 6.SP.B.5.c

BACKWARD 3

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

12, 5, 9, 20, 7, 14

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 50 • BACKWARD 3 • Q1/5

Which list is in order from least to greatest?

A. 58, 60, 61, 63

B. 60, 58, 61, 63

C. 58, 61, 60, 63

D. 63, 61, 60, 58

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 50 • BACKWARD 3 • Q2/5

True or false?

These numbers are in order from least to greatest: 75, 82, 88, 90, 95

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 50 • BACKWARD 3 • Q3/5

Order from least to greatest.

47, 74, 44, 77

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 50 • BACKWARD 3 • Q4/5

Which number is in the middle when 3, 9, 6 are put in order?

- A.** 6
- B.** 3
- C.** 9
- D.** 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • 6.SP.B.5.c

BACKWARD 4

The number halfway between two numbers • Grade 5 • Number & Operations
in Base Ten • 5.NBT.B.7

Find the value.

$$(9 + 12) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 50 • BACKWARD 4 • Q1/5

Find the value.

$$(14 + 17) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 50 • BACKWARD 4 • Q2/5

True or false?

$$(6 + 9) \div 2 = 7.5$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 50 • BACKWARD 4 • Q3/5

$$(20 + 25) \div 2 = ?$$

- A.** 22.5
- B.** 22
- C.** 45
- D.** 23.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 50 • BACKWARD 4 • Q4/5

Find the value.

$$(3.5 + 4.5) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 50 • BACKWARD 4 • Q5/5

SET 50 • 6.SP.B.5.c

FORWARD 1

Compare populations using means and medians • Grade 7 • Statistics & Probability • 7.SP.B.4

A random sample of 6th graders has a mean height of 58 in. A random sample of 8th graders has a mean height of 64 in.

How much greater is the 8th graders' mean?

Difference: _____

Sample A has a median of 12 and Sample B has a median of 15. The samples have similar variability.

Which conclusion is best?

- A.** Values in population B tend to be greater.
- B.** Values in population A tend to be greater.
- C.** The populations are the same.
- D.** No conclusion is possible.

True or false?

To compare two populations, you can compare the means of random samples from each population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 50 • FORWARD 1 • Q3/5

Store A sample: daily sales mean \$520. Store B sample:
daily sales mean \$610.

Which store typically has greater daily sales?

Store: _____

Two random samples of plant heights have means of 24 cm and 30 cm. Each has a MAD of 3 cm.

How many MADs apart are the means?

A. 2

B. 6

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 50 • FORWARD 1 • Q5/5

SET 50 • 6.SP.B.5.c

FORWARD 2

Use a linear model to make predictions • Grade 8 • Statistics & Probability •
8.SP.A.3

The equation $y = 2.5x + 40$ models test score y after x hours of study.

Predict the score for 8 hours of study.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 2 • Q1/5

The equation $y = 2.5x + 40$ models test score y after x hours of study.

What does the slope 2.5 mean?

Meaning: _____

The model $y = 3x + 12$ gives a plant's height y (cm) after x weeks.

What does 12 represent?

- A.** The height at week 0
- B.** The growth per week
- C.** The number of weeks
- D.** The height after 12 weeks

True or false?

In the model $y = -0.5x + 30$, y decreases by 0.5 for each 1-unit increase in x .

A. True

B. False

The model $y = 15x + 100$ gives the cost y of x tickets.

Predict the cost of 20 tickets.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 2 • Q5/5

Find the interquartile range and mean absolute deviation

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	The number halfway between two numbers	5.NBT.B.7
BACKWARD 2	Sharing a total equally	5.MD.B.2
BACKWARD 3	Distance as absolute value	6.NS.C.7.c
FORWARD 1	Difference in centers as a multiple of variability	7.SP.B.3
FORWARD 2	Clusters and outliers in scatter plots	8.SP.A.1

SET 51 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the interquartile range of the data.

2, 4, 5, 7, 9, 11, 12, 15

IQR: _____

Find the mean absolute deviation of the data.

2, 4, 6, 8, 10

MAD: _____

What is the mean absolute deviation of the data?

3, 3, 5, 7, 7

A. 1.6

B. 2

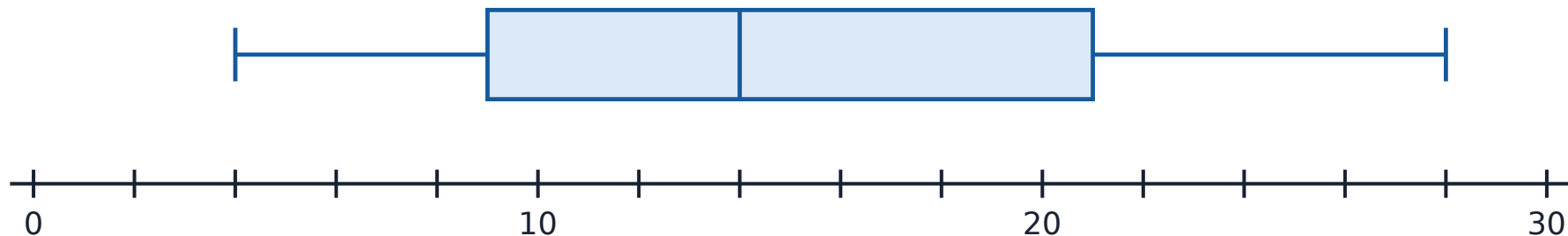
C. 4

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • MAIN • Q3/5

What is the interquartile range of the data shown in the box plot?



IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • MAIN • Q4/5

True or false?

A greater mean absolute deviation means the data values are more spread out from the mean.

A. True

B. False

SET 51 • 6.SP.B.5.c

BACKWARD 1

The number halfway between two numbers • Grade 5 • Number & Operations
in Base Ten • 5.NBT.B.7

Find the value.

$$(4 + 5) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 1 • Q1/5

Find the value.

$$(11 + 12) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 1 • Q2/5

True or false?

$$(7 + 10) \div 2 = 8.5$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 1 • Q3/5

$$(15 + 18) \div 2 = ?$$

A. 16.5

B. 16

C. 33

D. 17.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 1 • Q4/5

Find the value.

$$(2.5 + 3.5) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 6.SP.B.5.c

BACKWARD 2

Sharing a total equally • Grade 5 • Measurement & Data • 5.MD.B.2

Five jars hold 2, 4, 6, 8, and 10 cups of rice. The rice is shared equally among the jars.

How much is in each jar?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 51 • BACKWARD 2 • Q1/5

Four friends have 3, 3, 7, and 7 marbles. They share the marbles equally.

How many marbles does each friend get?

Marbles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 51 • BACKWARD 2 • Q2/5

True or false?

Bags with 1, 5, and 6 pounds of flour are shared equally. Each bag then has 4 pounds.

- A.** True
- B.** False

Six students have 2, 3, 3, 4, 5, and 7 pencils. They share equally.

How many pencils does each student get?

A. 4

B. 3

C. 24

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 51 • BACKWARD 2 • Q4/5

Three cups hold $\frac{1}{2}$, $\frac{1}{2}$, and 2 cups of water. The water is shared equally.

How much is in each cup?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • 6.SP.B.5.c

BACKWARD 3

Distance as absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|2 - 6|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 51 • BACKWARD 3 • Q1/5

How far is 9 from 6 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 51 • BACKWARD 3 • Q2/5

True or false?

$$|4 - 10| = 6$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 51 • BACKWARD 3 • Q3/5

What is the distance between 3 and 8 on a number line?

- A.** 5
- B.** 11
- C.** -5
- D.** 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 51 • BACKWARD 3 • Q4/5

Find the value.

$$|10 - 6|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • The Number System • 6.NS.C.7.c • Set 51 • BACKWARD 3 • Q5/5

SET 51 • 6.SP.B.5.c

FORWARD 1

Difference in centers as a multiple of variability • Grade 7 • Statistics & Probability • 7.SP.B.3

Class A's mean is 72 and Class B's mean is 80. Each class has a MAD of 4.

The difference in means is how many times the MAD?

Times: _____

Two data sets have means of 15 and 21. Each has a MAD of 2.

How many MADs apart are the means?

A. 3

B. 6

C. 2

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 51 • FORWARD 1 • Q2/5

True or false?

Two data sets have means that differ by 1 MAD.
The data sets probably overlap a lot.

- A. True
- B. False

Two data sets have medians of 40 and 55. Each has an IQR of 5.

The difference in medians is how many times the IQR?

Times: _____

Which pair of data sets has the LEAST overlap?

- A.** Means 10 and 30, each MAD 2
- B.** Means 10 and 12, each MAD 2
- C.** Means 10 and 14, each MAD 4
- D.** Means 10 and 11, each MAD 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 51 • FORWARD 1 • Q5/5

SET 51 • 6.SP.B.5.c

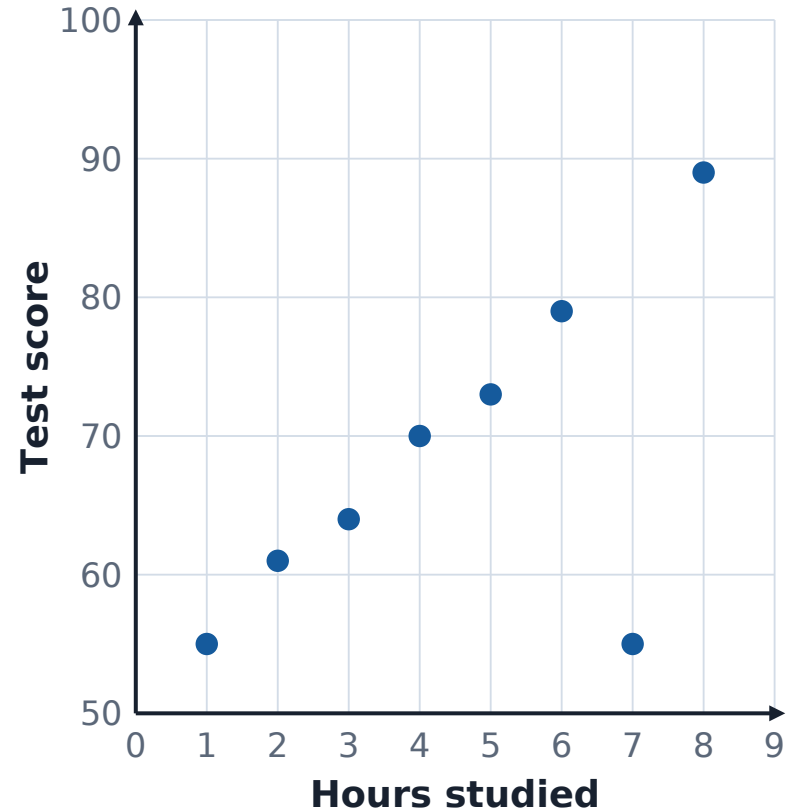
FORWARD 2

Clusters and outliers in scatter plots • Grade 8 • Statistics & Probability •
8.SP.A.1

Which point is an outlier?

Write its coordinates.

Outlier: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 2 • Q1/5

Which describes the scatter plot?

- A.** Two clusters
- B.** A strong positive association
- C.** No pattern
- D.** A single outlier



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

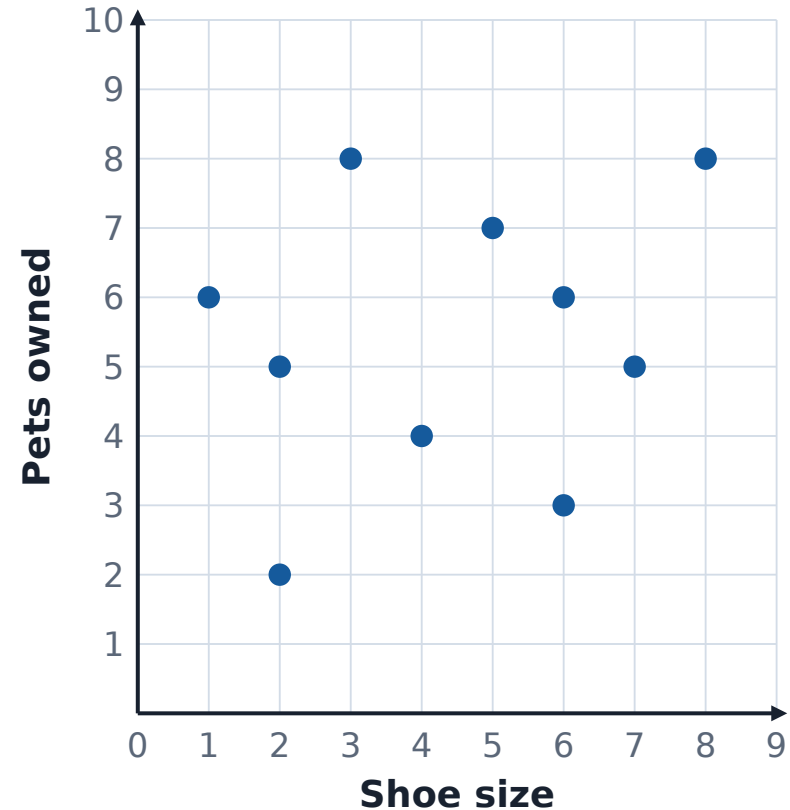
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 2 • Q2/5

True or false?

The scatter plot shows no association.

A. True

B. False

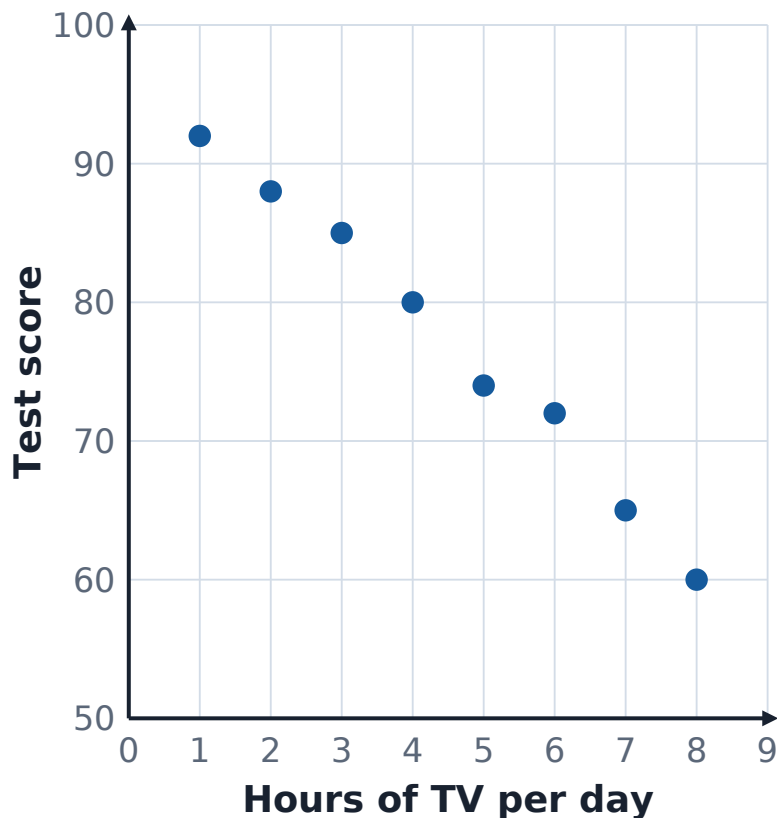


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 2 • Q3/5

Which describes the association in the scatter plot?

- A.** Negative linear
- B.** Positive linear
- C.** Nonlinear
- D.** No association



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 2 • Q4/5

True or false?

An outlier in a scatter plot is a point that is far from the overall pattern.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 2 • Q5/5

Choose measures of center and variability based on the shape of the data

MAIN	Grade 6 main question	6.SP.B.5.d
BACKWARD 1	Shape of a distribution	6.SP.A.2
BACKWARD 2	Computing the mean and median	6.SP.B.5.c
BACKWARD 3	Reading a dot plot	4.MD.B.4
FORWARD 1	Compare populations using appropriate measures	7.SP.B.4
FORWARD 2	Judge how well a line fits the data	8.SP.A.2

SET 52 • 6.SP.B.5.d

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.d

Data: 20, 22, 23, 25, 90

Which measure of center best describes a typical value?

A. Median

B. Mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 52 • MAIN • Q1/5

True or false?

For a symmetric distribution with no outliers, the mean and the median are about the same.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 52 • MAIN • Q2/5

A data set is skewed right and has an outlier.

Which pair of measures best describes its center and spread?

A. Median and IQR

B. Mean and MAD

C. Mean and range

D. Median and MAD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 52 • MAIN • Q3/5

Salaries: \$30,000; \$32,000; \$35,000; \$38,000;
\$250,000

Find the mean and the median. Which better describes
a typical salary?

Mean: _____

Median: _____

Better measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 52 • MAIN • Q4/5

Data: 2, 3, 3, 4, 40

Why is the median a better measure of center than the mean?

- A.** The outlier 40 pulls the mean up.
- B.** The median is always larger.
- C.** The mean ignores 40.
- D.** There is no median.

SET 52 • 6.SP.B.5.d

BACKWARD 1

Shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

Which word describes the shape of the data?

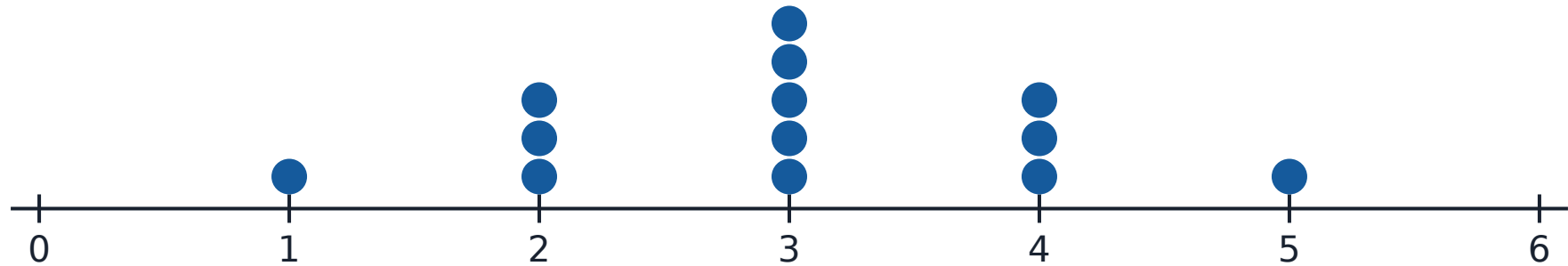


- A.** Skewed left
- B.** Skewed right
- C.** Symmetric
- D.** Uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 1 • Q1/5

Which word describes the shape of the data?



A. Symmetric

B. Skewed left

C. Skewed right

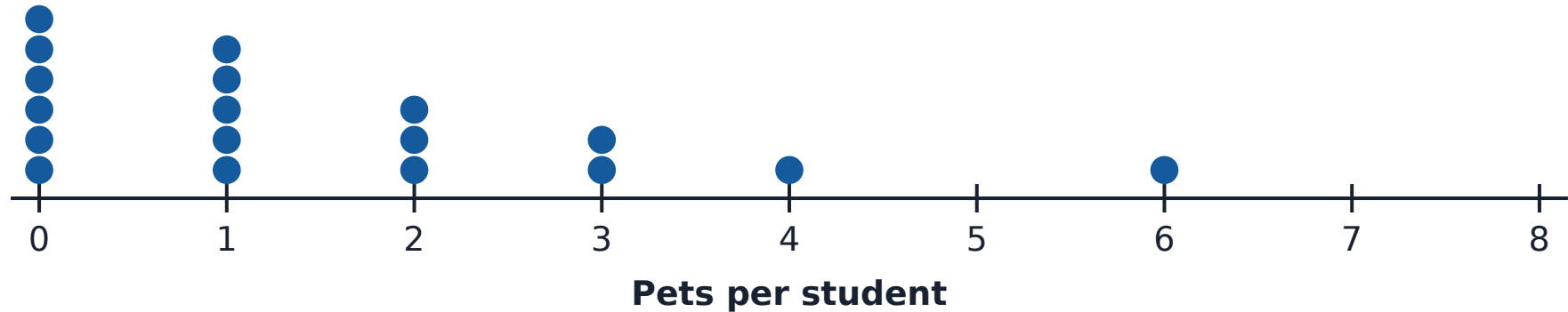
D. Uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 1 • Q2/5

True or false?

The data are skewed right.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 1 • Q3/5

True or false?

A data set with most values on the left and a long tail to the right is skewed right.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 1 • Q4/5

Which value in the data set is an outlier?

5, 6, 6, 7, 8, 30

A. 30

B. 5

C. 6

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 6.SP.B.5.d

BACKWARD 2

Computing the mean and median • Grade 6 • Statistics & Probability •
6.SP.B.5.c

Find the mean.

2, 3, 3, 4, 8

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 2 • Q1/5

Find the median.

2, 3, 3, 4, 8

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 2 • Q2/5

True or false?

The mean of 10, 20, 30 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 2 • Q3/5

What is the median of 1, 4, 6, 9?

- A.** 5
- B.** 6
- C.** 4
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 2 • Q4/5

Find the mean.

5, 5, 6, 8, 11

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 6.SP.B.5.d

BACKWARD 3

Reading a dot plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many students scored 9 on the quiz?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 52 • BACKWARD 3 • Q1/5

What is the lowest quiz score shown?



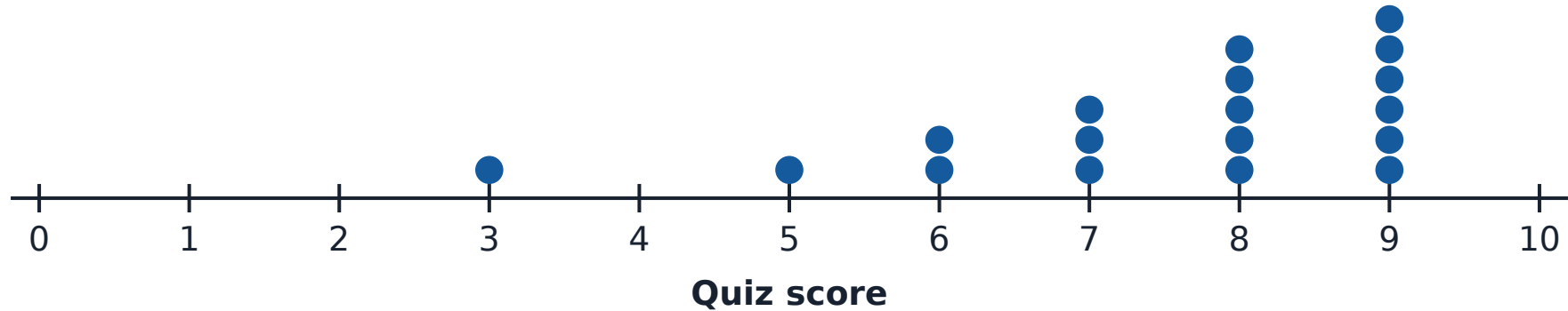
Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 52 • BACKWARD 3 • Q2/5

True or false?

Five students scored 8.



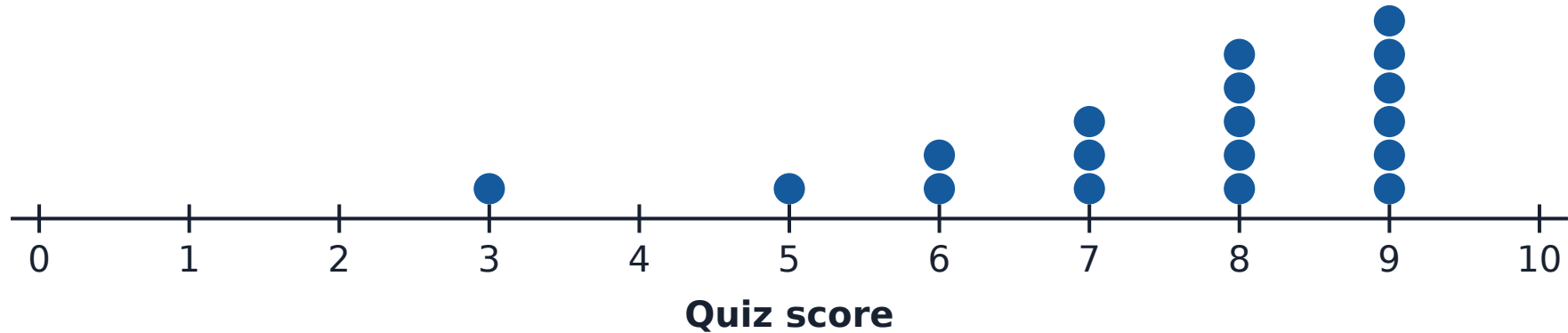
A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 52 • BACKWARD 3 • Q3/5

How many students took the quiz?



A. 18

B. 15

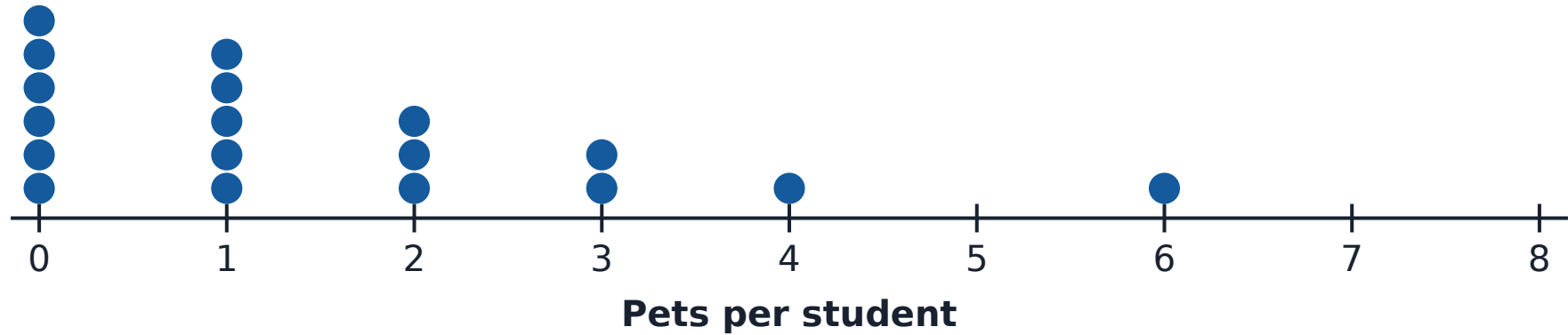
C. 9

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 52 • BACKWARD 3 • Q4/5

How many students had 0 pets?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • 6.SP.B.5.d

FORWARD 1

Compare populations using appropriate measures • Grade 7 • Statistics & Probability • 7.SP.B.4

Two random samples of house prices are both skewed right with outliers.

Which measures are best for comparing them?

- A.** Medians and IQRs
- B.** Means and MADs
- C.** Maximums
- D.** Minimums

True or false?

When two samples are symmetric with no outliers, comparing their means is appropriate.

- A. True
- B. False

Sample A: median 42, IQR 8. Sample B: median 50, IQR 8.

The difference in medians is how many times the IQR?

Times: _____

Class A test scores have mean 78 and MAD 5. Class B test scores have mean 78 and MAD 12.

Which statement is true?

- A.** Class B's scores vary more.
- B.** Class A's scores vary more.
- C.** Class B scored higher.
- D.** Class A scored higher.

Sample of commute times, Town X: median 25 min.
Town Y: median 18 min. Both have similar IQRs.
In which town are commutes typically longer?

Town: _____

SET 52 • 6.SP.B.5.d

FORWARD 2

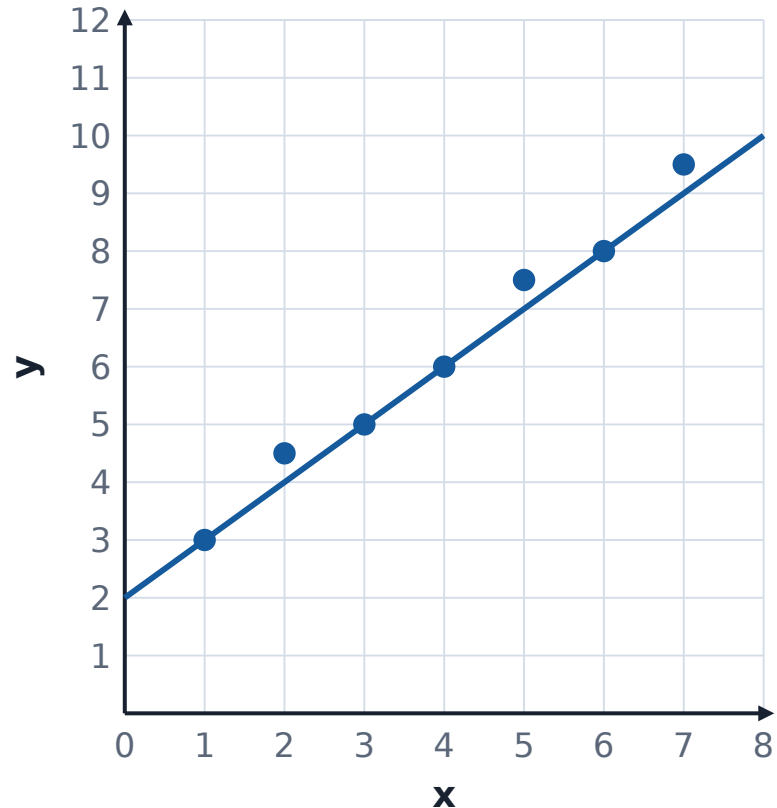
Judge how well a line fits the data • Grade 8 • Statistics & Probability • 8.SP.A.2

True or false?

The line is a good fit for the data.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

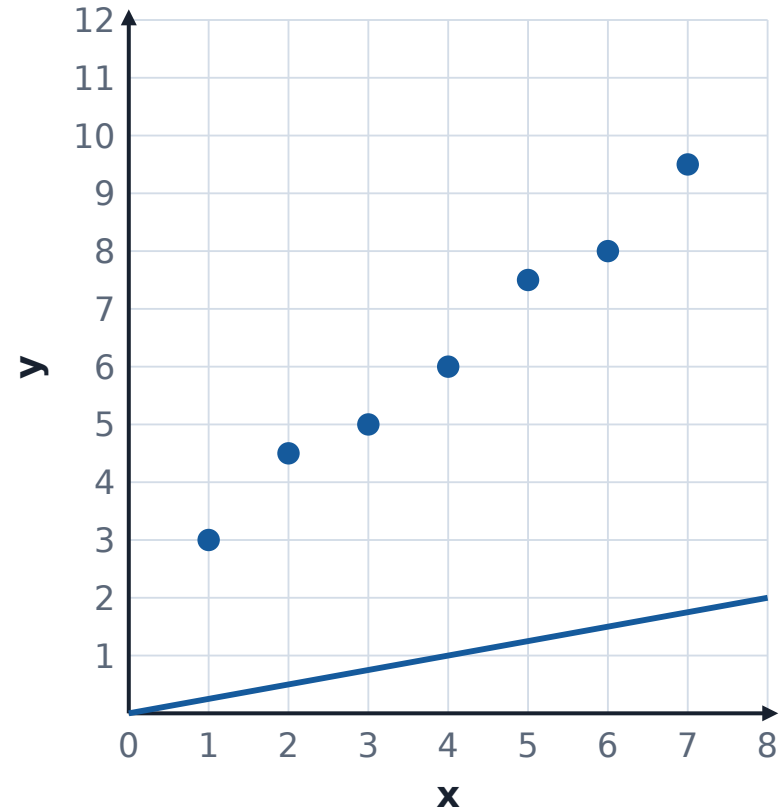
Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 52 • FORWARD 2 • Q1/5

True or false?

The line is a good fit for the data.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

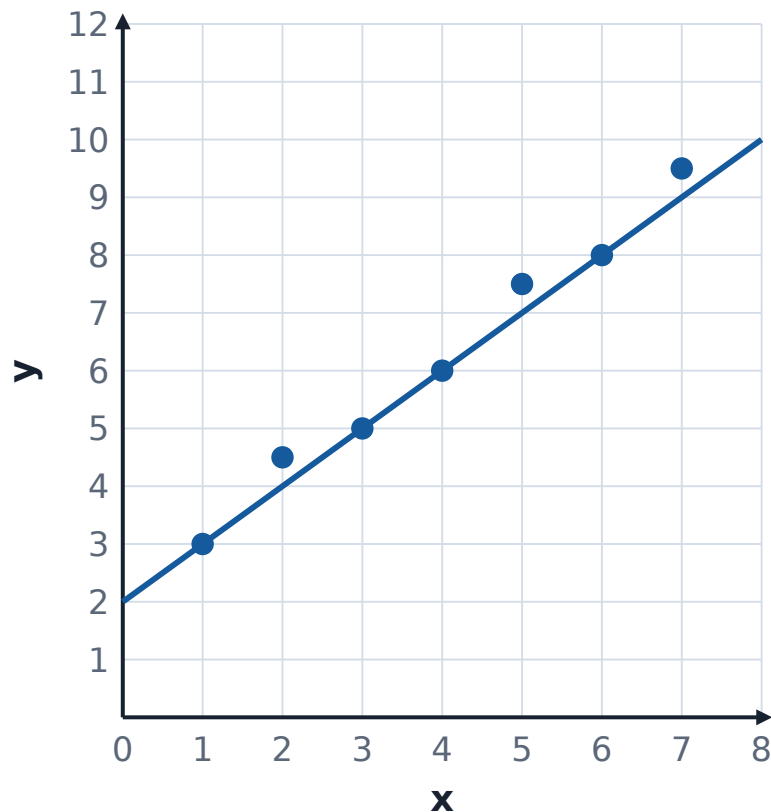
Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 52 • FORWARD 2 • Q2/5

Which is a sign of a good line of best fit?

- A.** Points are close to the line on both sides.
- B.** All points are above the line.
- C.** All points are below the line.
- D.** The line goes through only one point.

Use the line of best fit to
predict y when $x = 4$.

$y =$ _____

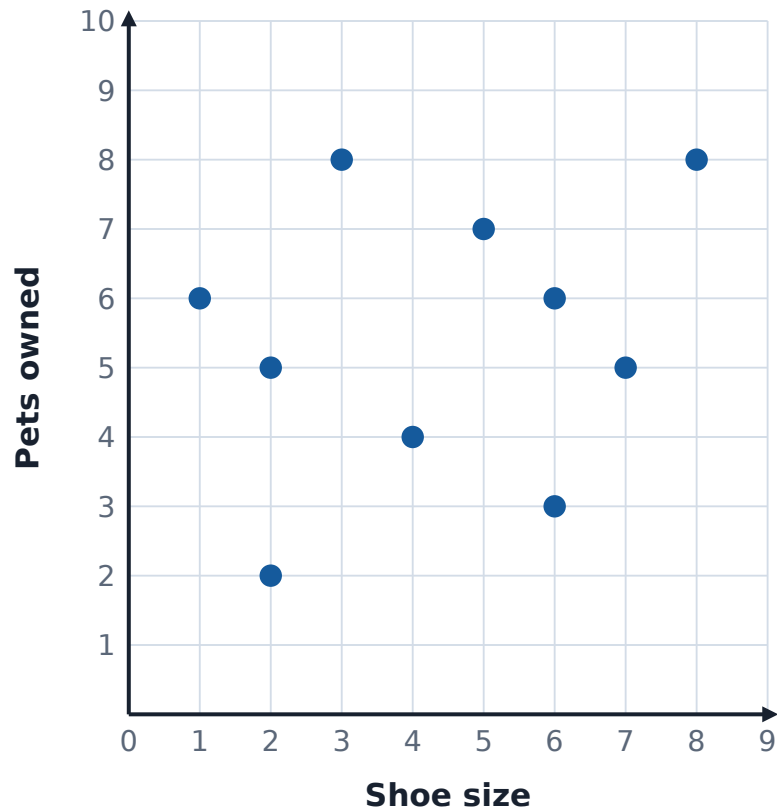


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 52 • FORWARD 2 • Q4/5

Why should a scatter plot with no association NOT be modeled with a straight line?

- A.** There is no linear trend to model.
- B.** Lines must pass through the origin.
- C.** There are too many points.
- D.** The axes are labeled.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 52 • FORWARD 2 • Q5/5